Johnson Controls Premier Rooftop Units 60 Ton to 80 Ton Technical Guide R-454B

Technical Guide 5466992-JTG-A-0324

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Contents

Product highlights and options

Efficiency

- R-454B refrigerant
- Optional integral energy recovery wheel (ERW)
- Optional variable speed drive compressors
- Multiple direct drive plenum (DDP) supply fans

Flexibility

- 208 VAC to 230 VAC, 460 VAC, or 575 VAC, 3 phase, 60 Hz
- Cooling only units with heating options including gas, electric, steam or hot water with modulation
- Exhaust or return fan options
- Various airflow path configurations for discharge and return and exhaust air
- Optional humidifier, sound attenuator, or air blender
- Optional low ambient temperature
- Variable frequency drive (VFD) options on all fans, see Figure 1

Figure 1: VFD displays

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Reliability and serviceability

- Multiple refrigeration circuits
- Refrigerant detection system
- Coil corrosion protection option
- Unit powered convenience outlet
- Optional door viewports
- Single point latching door option
- Internal air handler light option
- Replaceable core filter drier option
- Suction, liquid, and discharge line shutoff valve options
- Refrigeration pressure transducer options
- Start-up wizard

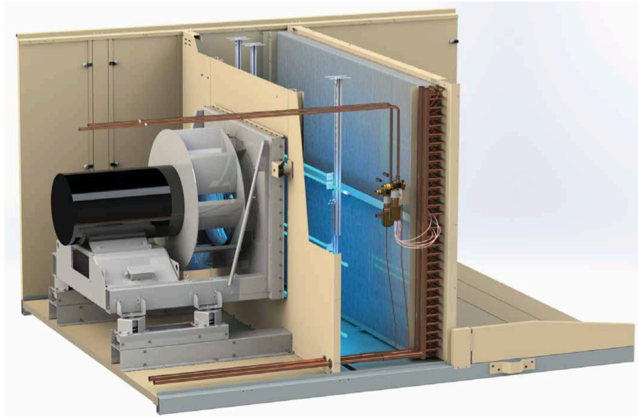
Indoor environmental quality

- Double wall construction with foam insulation
- Modulating hot gas reheat (HGRH) option
- Final filtration options, including high efficiency particulate air (HEPA) filters
- Ultraviolet (UV) lights option, see Figure 2
- Stainless steel drain pan
- Condensate overflow switch option
- Airflow measurement options for outside air, supply fan, and return fan

Controls

- 5.5 in., 5 row × 35 character (256 × 64 dot matrix) organic light-emitting diode (OLED) display with full numeric keypad and navigation buttons as standard
- 10 in., 1280 × 800 resolution, multi-touch capacitive, color touchscreen display available as a factory installed or field installed option
- Optional WiFi hotspot capability when it is not always possible to physically access the unit
- Smart Equipment technology enabling self-discovery on Verasys Building Automation Systems (BAS)
- The controller supports the BACnet® communication protocol and is designed and certified by BACnet Testing Laboratory (BTL) to meet the requirements of the advanced application control profile
- Twinning algorithms to allow multiple units to function as one on common supply and return duct shafts
- Variable air volume (VAV) and single zone VAV (SZVAV) control
- Building pressurization controls

Figure 2: UV lights



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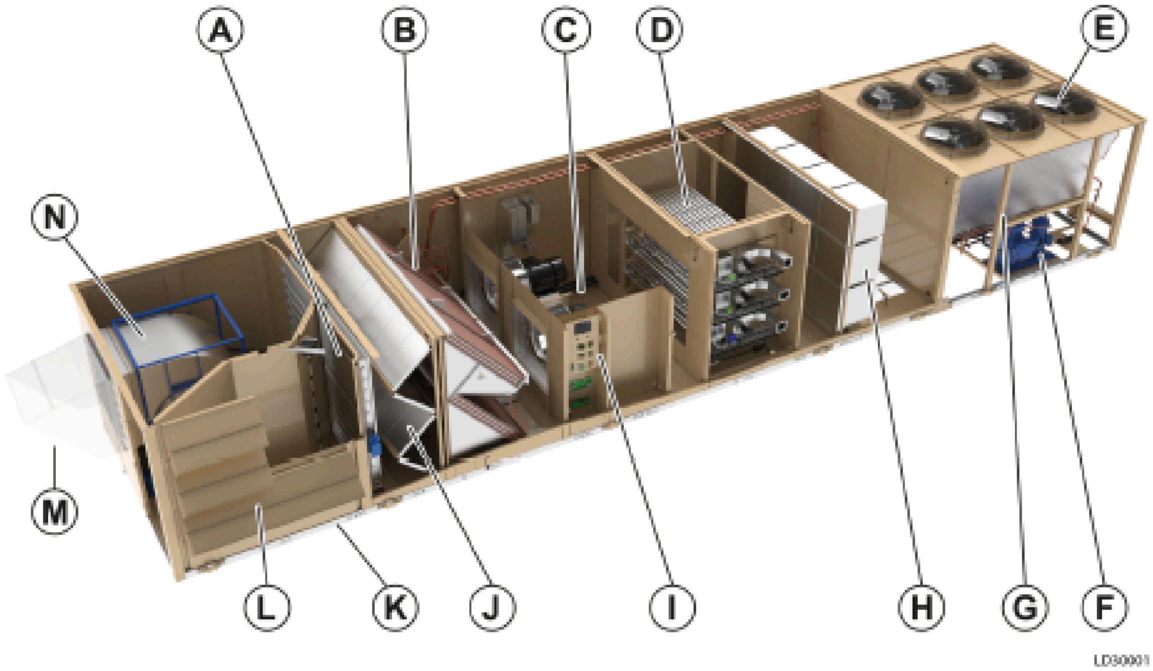
Patents

For access to Premier related patents, visit: https://jcipat.com.

Certifications

Component location

Figure 3: Packaged rooftop unit cabinet assembly



ⓘ **Note:** Some components listed are optional.

A	Economizer	H	Final filter
B	Evaporator coil	I	Unit controller and power
C	Direct drive plenum (DDP) supply fan	J	Filter section
D	Modulating or staged gas heat	K	Outside air (right side)
E	Condenser fans	L	Collapsible rain hoods
F	Scroll compressors	M	Exhaust air (front)
G	Condenser coil cleaning hatch - not shown	N	Exhaust and return fan

Features and benefits

General

The 60 ton to 80 ton packaged rooftop platform is designed with all the flexibility needed for today’s applications but with tomorrow's requirements in mind. Realizing that efficiency requirements are continuously pushing the envelope of technology, the Johnson Controls® Premier™ delivers today the energy efficiency levels exceeding those mandated by the U.S. Department of Energy for 2023. All cooling only and electric heat units have an integrated energy efficiency ratio (IEER) in excess of 13.2. All units with gas or hydronic heat have an IEER in excess of 13. For these particular rooftop units, when equipped with a variable speed drive compressor, they deliver Tier 2 efficiency levels of the Consortium of Energy Efficiency (CEE) for 2024.

Premier units are also designed for serviceability, with options such as a single handle latching mechanism for doors; a convenience outlet to power lights and tools; internal lights in the air handler section; viewports in doors of serviceable compartments to enable easier unit inspection; and extended grease lines to simplify fan bearing lubrication for belt driven fans. Standard direct drive supply fans, see item C in Figure 3, do not require lubricating fan bearings or changing belts. Numerous refrigeration options are also available, including replaceable core filter driers, liquid and suction isolation valves, as well as high and low pressure transducers in each circuit that enable easier sub-cooling and superheat measurements.

Besides options, there are also standard features to ensure straightforward and safe servicing of the units, including coil cleaning hatches in the condenser section to make cleaning condenser coils effortless, and door safety latches to keep doors safe when opened in a pressurized compartment. A discrete high and low voltage compartment minimizes the amount of safety equipment required when only the low voltage compartment is accessed.

Selection Navigator is an online program available to assist in providing unit selections, performance reports, unit drawings, lifting lug and connection drawings, BAS controls points list, controls sequence of operation, guide specification, and outputs to assist in design of the unit. Certain options may require the assistance of the local sales office.

In order to facilitate its application, Selection Navigator also provides building information modeling (BIM) files of the specific unit selected. This aids in modeling placement and integration of the rooftop unit into the overall building design.

This packaged rooftop product is a sophisticated, highly configurable rooftop unit. A controller is standard on each unit with specific sequences of operation correlated to the options selected. Selection Navigator provides the specific sequences of operation for the unit options selected, eliminating confusion of the unit's capability.

Although the many options make customizing unlikely, it is possible to have custom mechanical and sequences of operations designed into the unit from the factory for ultimate convenience and reliability.

1. Economizer

See item A in Figure 3. In order to deliver maximum efficiency, rooftop units need an efficient economizer. The rooftop unit offers various options for economizer control and fault detection, as well as damper leakage rates and cycle life to meet different regulatory requirements.

The amount of fresh air can also be optionally measured to ensure and record appropriate fresh air to the conditioned space. An air blender option is also available to deliver optimal comfort to the conditioned space.

2. Refrigeration System

See items B, E, and F in Figure 3. The refrigeration system is built to reliably deliver cooling through a variety of loads. Two independent refrigeration circuits ensure that in the unlikely event of a compressor failure, the second circuit can still deliver cooling to the space. An interlaced tube and fin evaporator provide maximum cooling performance even at part loads. The microchannel condenser reduces the risk of refrigerant leaks and minimizes the amount of refrigerant in the unit.

Standard fixed speed scroll compressors of different sizes deliver excellent part-load efficiency and control, which eliminates the need for inefficient hot gas bypass valves at low load conditions. Optional variable speed drive scroll compressor configurations are available for stable and efficient discharge air temperature (DAT) control regardless of the load.

For application flexibility, optional corrosion protection is available for the evaporator and condenser, as well as intrusion protection options for the condenser.

Realizing that the refrigeration system is the heart of the rooftop, when optionally equipped with transducers, the unit controller can display sub-cooling or superheat. Additionally, low ambient operation is an available option with a variable speed condenser fan.

Maximized serviceability is also available with options such as replaceable core filter driers as well as discharge, liquid, and suction line isolation valves. The standard unit is equipped with accessible sight glasses to verify proper refrigerant flow as well as conveniently located condenser coil cleaning hatches to facilitate maintenance.

3. Direct drive plenum (DDP) supply fan

See item C in Figure 3. Multiple DDP supply fans, see Figure 4, are used on 60 ton to 80 ton cabinets for an increased level of reliability and efficiency. Problems such as the interruption of conditioned air supply because of a broken belt, or the pollution of conditoned air with belt dust, are eliminated. The supply fan can be optionally equipped with an airflow measurement station to precisely measure the amount of air delivered to the conditioned space.

The speed of the supply fan is controlled by a variable frequency drive (VFD). VFD redundancy is optionally available to ensure uptime in the unlikely event of a VFD failure. For multiple supply fan units an optional VFD is provided with each motor.

4. Heating options

See item D in Figure 3. Gas heat options are available either in staged or modulating control. The flexibility of heater sizes meets the specific application heating needs. Electric heat is also available with size, staging and modulating options, see Figure 5. Either hot water or steam heat options are also available. All modulating heat options can be controlled precisely to temper the supply air, which is especially important when fresh air is needed in cold climates.

5. Double wall construction

The air handler section of the rooftop unit provides foam injected double wall construction for maximum unit rigidity and cleanability of the interior surfaces for long term indoor air quality (IAQ). This construction of the walls, roof, and floor provides an insulating value that minimizes unit sweating and contributes to the overall unit efficiency.

Figure 4: Dual DDP supply fan

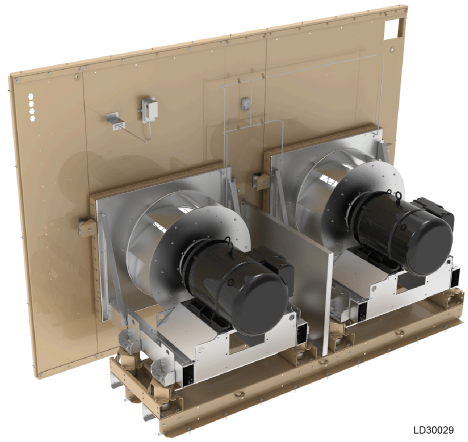
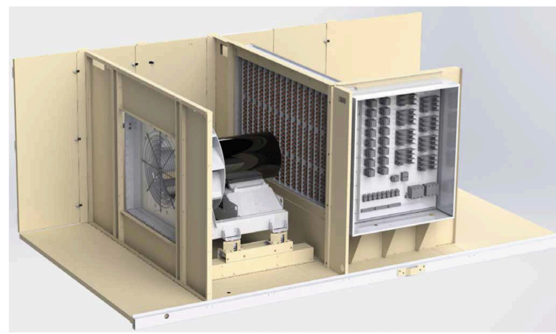


Figure 5: Electric heat



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6. Controls system

See item I in Figure 3. The rooftop units sophisticated options are intelligently controlled by a best-in-class controls platform built exclusively for this application. A 5.5 in., 5 row × 35 character (256 × 64 dot matrix) organic light-emitting diode (OLED) display, see Figure 6, with full numerical and optimized navigational keypad or a 10 in., 1280 × 800 touchscreen LED display, see Figure 7, each conveniently located in the low voltage compartment, act as the nerve center. The control system can be optionally augmented with a WiFi hotspot capability for local or line-of-sight smart device control. The touchscreen display shows a graphical representation of the Premier rooftop unit with real-time data, also enabling the ability to read documents and connect with multiple controllers.

Figure 6: OLED display

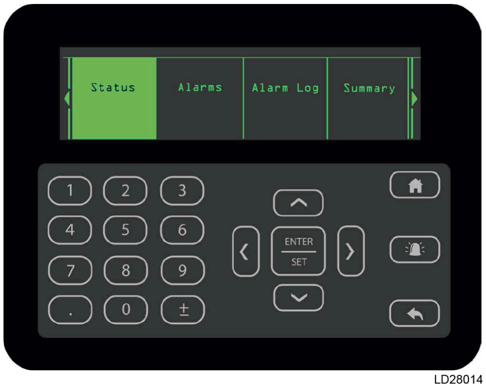
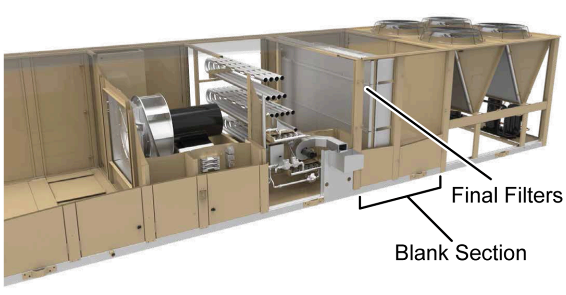


Figure 7: Touchscreen display

The control platform minimizes commissioning time when connected to the Verasys Building Automation Systems (BAS), with self-discovery of the rooftop unit and its points by the BAS.

The unit controller has sequences of operation for standalone applications. These sequences cover simple applications such as single zone variable air volume (SZVAV) control of the supply fan and compressors in accordance with ASHRAE 90.1-2019, as well as demand control ventilation (DCV) that ensures adequate fresh air to the building. Complex applications are also supported, like the twinning of multiple rooftops on a common supply and return duct shaft for the ultimate in redundancy for critical spaces, such as Medical Office Buildings (MOBs).

Figure 8: Final filter section



7. Blank section

See Figure 8. Blank sections can be provided for field installed accessories or to accommodate the installation of specific factory components including an air blender, final filters, humidifier, or sound attenuator.

The rooftop unit can be equipped with a final filter before the conditioned air is delivered from the rooftop into the supply duct. Various final filter options are available to meet critical needs including high efficiency particulate air (HEPA) filtration. Differential pressure measuring options can be supplied to meet the requirements for ASHRAE 170-2017.

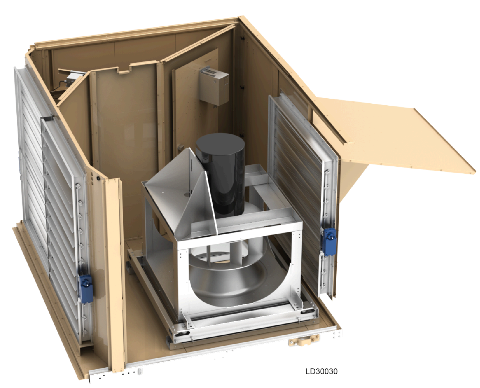
For cold northern climates where heat tends to dry out the supply air, a humidifier option with a stainless steel drain pan is also available.

In order to mitigate sound originating in the rooftop unit from reaching the conditioned space, optional sound attenuators in the rooftop unit can be factory-installed.

8. Exhaust and return fan

See item N in Figure 3. The rooftop units enable installation flexibility with the option of either exhaust or return fan to control the building static pressure. Return ductwork with shorter runs and lower static requirements usually only need an exhaust fan. A direct drive plenum return fan, see Figure 9, is available to overcome the static imposed by longer return ducts.

Figure 9: DDP return fan



Optional airflow measurement is available for units equipped with a return fan. This makes it possible to precisely understand and log the different airflows through the rooftop unit.

9. Hot gas reheat (HGRH)

Occupant comfort can be a challenge during shoulder months when low loads and high humidity occur. In many cases, the combined efforts of refrigeration system compressor multistep control, an interlaced evaporator coil, and the supply fan modulation of SZVAV control are sufficient. The rooftop unit provides an optional modulating hot gas reheat (HGRH) coil to further reduce the humidity level within the space.

10. Indoor air quality (IAQ)

To meet the critical needs of IAQ, the rooftop unit provides a stainless steel evaporator drain pan for longevity and to facilitate cleanability. The drain pan can be optionally equipped with an overflow switch to warn of improper drainage and minimize the potential for damage to the conditioned space. Ultraviolet (UV) light banks are also an available option that minimize mold and bacteria growth and assists in keeping the evaporator coil clean.

11. Energy recovery wheel (ERW)

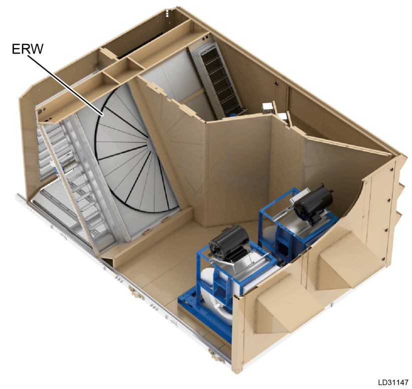
See Figure 10. The rooftop unit provides an option for an ERW as an integral part of the unit. This device uses exhaust air to condition the fresh air brought into the unit and the conditioned space, increasing the overall rooftop efficiency. As a factory-installed option, the ERW ensures reliability, minimizes field labor, and simplifies long-term maintenance of the device. On 60 ton to 80 ton units, an ERW is provided with dual exhaust fans.

12. Airflow flexibility

Whether a rooftop unit is mounted on a roof or on a grade, the orientation of airflow from the rooftop to the conditioned space and vice versa is important. Premier units are specifically designed to provide discharge airflow either at the bottom, top, left, or right.

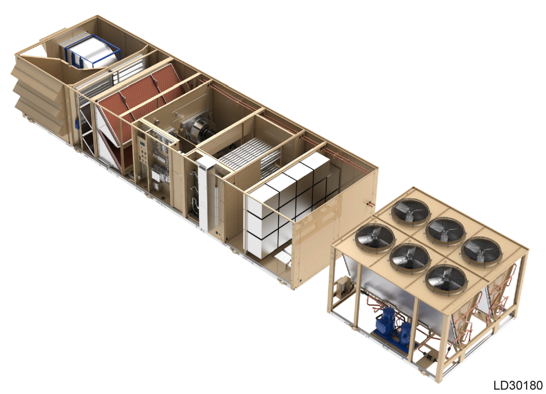
Similarly, the return airflow is designed for maximum flexibility vertically in either direction as well as horizontally in either direction. Depending on selected options, an end return is also possible. This airflow flexibility minimizes installation costs and maximizes possible locations for this flexible rooftop unit.

Figure 10: ERW 60 ton to 80 ton units



13. Split ship

Premier 60 ton to 80 ton units are provided with a split ship option, to meet specific applications where there is constraint for using a higher capacity crane due to unfavorable ground conditions or maneuvering space for the crane to make a safe lift. Split ship units are factory tested, pre-charged with refrigerant and delivered to the job site split in two segments, air handling and condenser section, for an easy lift and seamless integration between power and refrigeration connections. Units which are shipped split are intended to be reconnected into one on a common curb.



Nomenclature

Table 1: Example model number

GV	**F**	**A**	**C**	**2**	**B**	**5**	**G**	**A**	**1**	**A**	**6**	**0**	**D**	**D**	**2**	**0**	**2**	**B**	**F**	**2**	**A**	**2**	**G**	**2**	**C**	**0**	**E**	**0**	**D**	**M**	**A**	**A**	**3**	**1**	**0**	**0**	**1**	**A**	**0**	**0**	**0**	**L**	**E**	**0**	**0**	**0**	**1**
1,2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49

Note: The first row displays an example model number. The second row displays the corresponding digit.

Digit	Category	Option	Description
1,2	Product brand name	GV	Johnson Controls
3	Nominal capacity / Basic unit	F	60 ton
		H	70 ton
		J	80 ton
4	Efficiency level	A	Standard capacity, standard efficiency, with R-454B refrigerant
		B	Standard capacity, high efficiency, with R-454B refrigerant
		C	High capacity, standard efficiency, with R-454B refrigerant
		D	Standard capacity, standard efficiency, low sound, with R-454B refrigerant
		E	Standard capacity, high efficiency, low sound, with R-454B refrigerant
		F	High capacity, standard efficiency, low sound, with R-454B refrigerant
5	Heat source	A	Cooling only
		B	Staged gas aluminized burner
		C	Staged gas stainless steel burner
		G	Modulating gas stainless steel burner
		K	Steam coil
		L	Hot water coil
		M	Electric heat
6A	Electric heat capacity	0	None
		1	Low heat
		3	High heat
		4	Low heat with Silicon Controlled Rectifier (SCR)
		6	High heat with SCR
6B	Natural gas heat capacity	0	None
		2	500 MBH
		4	1000 MBH
		6	1500 MBH
6C	Hot water or steam heating coil option	0	None
		1	Low heat without valves
		2	Low heat with valves
		3	High heat without valves
		4	High heat with valves
7	Unit type	A	Single Zone VAV (SZVAV) no duct pressure transducer
		B	Variable Air Volume (VAV) duct pressure transducer

Digit	Category	Option	Description
8	Motor control options	1	Supply fan variable frequency drive (VFD)
		2	Supply fan VFD with line reactor
		3	Supply fan with dual VFD
		4	Supply fan with dual VFD with line reactor
		5	Supply fan VFD and return/exhaust fan VFD
			60 to 80 Ton with ERW—Supply fan VFD and dual exhaust fan option with individual VFD on each exhaust fan
		6	Supply fan VFDs with line reactor and return/exhaust fan VFD with line reactors
			60 to 80 ton with ERW—Supply fan VFDs with line reactor and dual exhaust fan option with individual VFD and line reactor on each exhaust fan
		7	Supply fan with dual VFD and redundant VFD for return fan / manual bypass for exhaust fan
			60 to 80 ton with ERW—Supply fan with dual VFD and dual exhaust fan option with individual VFD on each exhaust fan
		8	Supply fan with dual VFD with line reactors and redundant VFD for return fan / manual bypass for exhaust fan, and line reactors (for all fans)
			60 to 80 ton with ERW—Supply fan with dual VFD with line reactors and dual exhaust fan option with individual VFD and line reactor on each exhaust fan
9	Voltage	A	208-230 V 3Ph 60 Hz, single point terminal block
		B	208-230 V 3Ph 60 Hz, dual point terminal block
		C	208-230 V 3Ph 60 Hz, single point, non-fused disconnect
		D	208-230 V 3Ph 60 Hz, single point terminal block, 65KA short-circuit current rating (SCCR)
		E	208-230 V 3Ph 60 Hz, dual point terminal block, 65KA SCCR
		G	460 V 3Ph 60 Hz, single point terminal block
		H	460 V 3Ph 60 Hz, dual point terminal block
		J	460 V 3Ph 60 Hz, single point, non-fused disconnect
		K	460 V 3Ph 60 Hz, single point terminal block, 65KA SCCR
		L	460 V 3Ph 60 Hz, dual point terminal block, 65KA SCCR
		M	460 V 3Ph 60 Hz, single point fused disconnect, 65KA SCCR
		N	575 V 3Ph 60 Hz, single point terminal block
		P	575 V 3Ph 60 Hz, dual point terminal block
		Q	575 V 3Ph 60 Hz, single point non-fused disconnect
		R	575 V 3Ph 60 Hz, single point terminal block, 25KA SCCR
		S	575 V 3Ph 60 Hz, dual point terminal block, 25KA SCCR
		T	575 V 3Ph 60 Hz, single point fused disconnect, 25KA SCCR
10	Return configuration	A	Bottom return, right outside air (OA), side exhaust
		B	Bottom return, right OA, front exhaust
		C	Bottom return, left OA, side exhaust
		D	Bottom return, left OA, front exhaust
		E	Top return, right OA, side exhaust (no return fan available)
		F	Top return, right OA, front exhaust (no return fan available)
		G	Top return, left OA, side exhaust (no return fan available)
		H	Top return, left OA, front exhaust (no return fan available)
		J	Left return, right OA, front exhaust
		K	Right return, left OA, front exhaust
		L	Front return, left OA, right exhaust
		M	Front return, right OA, left exhaust
		N	Bottom return, no OA, no exhaust air (EA) (return fan or exhaust fan not available)
		P	Top return, no OA, no EA (return fan or exhaust fan not available)
		Q	Left return, no OA, no EA (return fan or exhaust fan not available)
		R	Right return, no OA, no EA (return fan or exhaust fan not available)
		S	Front return, no OA, no EA (return fan or exhaust fan not available)

Digit	Category	Option	Description
11	Discharge locations	1	Bottom discharge, from discharge plenum
		2	Bottom discharge, discharge through heat section
		3	Top discharge, from discharge plenum
		4	Right discharge, from discharge plenum
		5	Left discharge, from discharge plenum
		6	Left discharge, discharge through heat section
12	Supply configuration	A	None
		B	Small blank
		C	Large blank
		E	Large blank with final filter
		F	Small blank with humidifier and SST drain pan
		H	Large blank with humidifier and SST drain pan
		L	Small blank with sound attenuator
		M	Large blank with sound attenuator
		P	Small blank with final filter
		Q	Large blank with sound attenuator and final filter
		S	Large blank with sound attenuator and humidifier and SST drain pan
13	Final filter options	1	MERV 15 bag final filters with 2 in. MERV 8 filters
		2	MERV 14 rigid final filters with 2 in. MERV 8 filters
		3	MERV 17 HEPA final filters with 2 in. MERV 8 filters
		4	MERV 14/15 filter rack (no filters)
		5	High Efficiency Particulate Air (HEPA) filter rack (no filters)
		6	None
14	Final filter control options	0	None
		1	Combined pre and post filter transducer
		2	Separate pre and post filter transducer
		3	Combined pre and post filter transducer and combined magnehelic gauge
		4	Separate pre and post filter transducer and magnehelic gauge
		5	Combined pre and post filter magnehelic gauge
		6	Separate pre and post filter magnehelic gauge
		7	Combined pre and post filter transducer, separate pre and post filter magnehelic gauge
15	Supply fan	D	Dual direct drive plenum supply fans with 1 in. spring isolation
		E	Dual direct drive plenum supply fans with 2 in. spring isolation
		F	Dual direct drive plenum supply fans with 2 in. spring isolation and seismic restraint
16	Supply fan motor HP	A	5 HP x 2
		B	7.5 HP x 2
		C	10 HP x 2
		D	15 HP x 2
		E	20 HP x 2
		F	25 HP x 2
		G	30 HP x 2
		H	40 HP x 2
17	Supply fan motor type	2	ODP premium eff. 1800 RPM
		4	TEFC premium eff. 1800 RPM
18	Supply fan options	0	None
		1	Inlet guard
		2	Airflow measurement station
		3	Shaft grounding ring
		4	Inlet guard and airflow measurement station
		5	Inlet guard and shaft grounding
		6	Airflow measurement station and shaft grounding
		7	Shaft grounding, inlet guard and airflow measurement station

Digit	Category	Option	Description
19	Building pressure control	0	None
		1	Barometric damper
		2	Exhaust with VFD and backdraft damper
		3	Modulating damper (on/off exhaust fan only with no VFD)
		4	Modulating damper (return fan only with VFD)
20	Return/exhaust fan	A	None
		B	Exhaust fan with 1 in. spring isolation
		C	Exhaust fan with 2 in. spring isolation
		D	Exhaust fan with 2 in. spring isolation and seismic restraint
		H	DDP return fan with 1 in. spring isolation
		J	DDP return fan with 2 in. spring isolation
		K	DDP return fan with 2 in. spring isolation and seismic restraint
		L	Dual forward curved exhaust fan with 1 in. spring isolation
		M	Dual forward curved exhaust fan with 2 in. spring isolation
		N	Dual forward curved exhaust fan with 2 in. spring isolation and seismic restraint
21	Return/exhaust fan motor	A	None
		F	5 HP
		G	7.5 HP
		H	10 HP
		J	15 HP
		K	20 HP
		L	25 HP
		M	30 HP
		N	40 HP
		1	5 HP x 2
		2	7.5 HP x 2
		3	10 HP x 2
		4	15 HP x 2
		5	20 HP x 2
22	Return/exhaust fan motor type	0	None
		1	ODP premium eff. 1200 RPM
		2	ODP premiume eff. 1800 RPM
		3	TEFC premium eff. 1200 RPM
		4	TEFC premium eff. 1800 RPM
23	Return/exhaust fan options	A	None
		B	Shaft grounding ring
		C	Extended grease lines
		D	Extended grease lines and shaft grounding
		E	Belt guards
		F	Belt guards and shaft grounding
		G	Return fan airflow measurement station
		H	Return fan airflow measurement station and shaft grounding
		J	Extended grease lines and belt guards
		K	Extended grease lines and belt guards and shaft grounding
24	Return/exhaust fan drive	0	None
		1-9	RPM
		A-Z	Hz

Digit	Category	Option	Description
25	Evaporator options	G	Aluminum fin evaporator with stainless steel drain pan
		H	Aluminum fin evaporator with stainless steel drain pan with condensate overflow switch
		J	E-coat aluminum fin evaporator with stainless steel drain pan
		K	E-coat aluminum fin evaporator with stainless steel drain pan with condensate overflow switch
		L	Copper fin evaporator with stainless steel drain pan
		M	Copper fin evaporator with stainless steel drain pan with condensate overflow switch
26	Condenser Coil Options	2	With wire guards
		3	Full louvered panels
		4	Partial louvered panels, with wire guards
		6	E-coat condenser, with wire guards
		7	E-coat condenser, with full louvered panels
		8	E-coat condenser, partial louvered panels, with wire guards
27	Pre-evap filter options	A	Angled filter rack - no filters
		B	Angled filter rack - 2 in. throwaway filters
		C	Angled filter rack - 2 in. MERV 8 filters
		D	Rigid filter rack—no filters
		E	Rigid filter rack—MERV 15 bag filters with 2 in. MERV 8 pre filters
		F	Rigid filter rack—MERV 14 rigid filters with 2 in. MERV 8 pre filters
		G	Vertical filter rack—no filters
		H	Vertical filter rack—4 in. MERV 8 filters
28	Pre-evap filter control options	0	None
		1	Combined pre and post filter transducer
		2	Separate pre and post filter transducer
		3	Combined pre and post filter transducer and combined magnehelic gauge
		4	Separate pre and post filter transducer and magnehelic gauge
		5	Combined pre and post filter magnehelic gauge
		6	Separate pre and post filter magnehelic gauge
		7	Combined pre and post filter transducer, separate pre and post filter magnehelic gauge
29	Economizer options	A	None
		C	Dry bulb economizer, low leak dampers
		D	Single enthalpy economizer, low leak dampers
		E	Dual enthalpy economizer, low leak dampers
		F	Dry bulb economizer, low leak dampers with airflow measurement station
		G	Single enthalpy economizer, low leak dampers with airflow measurement station
		H	Dual enthalpy economizer, low leak dampers with airflow measurement station
		K	Dry bulb economizer, ultra low leak dampers
		L	Single enthalpy economizer, ultra low leak dampers
		S	Dual enthalpy economizer, ultra low leak dampers
		T	Dry bulb economizer, ultra low leak dampers with airflow measurement station
		U	Single enthalpy economizer, ultra low leak dampers with airflow measurement station
		V	Dual enthalpy economizer, ultra low leak dampers with airflow measurement station
30	Energy recovery options	0	None
		1	Low CFM ERW without VFD
		2	Low CFM ERW with VFD
		3	High CFM ERW without VFD
		4	High CFM ERW with VFD

Digit	Category	Option	Description
31	Refrigeration system piping options	A	None
		B	Suction and discharge valves
		C	Suction, discharge, and liquid valves
		D	Suction, discharge, and liquid valves with replaceable core filter driers
		E	Hot gas reheat (HGRH)
		F	Suction and discharge valves with standard filter driers with HGRH
		G	Suction, discharge, and liquid valves with HGRH
		H	Suction, discharge, and liquid valves with replaceable core filter driers with HGRH
		N	E-coat HGRH
		P	Suction and discharge valves with E-coat HGRH
		Q	Suction, discharge, and liquid valves with E-coat HGRH
		R	Suction, discharge, and liquid valves with replaceable core filter driers with E-coat HGRH
32	Lights/ detectors/convenience options	A	None
		B	Convenience outlet
		C	Convenience outlet and internal lights
		D	Supply smoke detector
		E	Return smoke detector
		F	Supply and return smoke detector
		G	Convenience outlet with supply smoke detector
		H	Convenience outlet with return smoke detector
		J	Convenience outlet with supply and return smoke detectors
		K	Convenience outlet and internal lights with supply smoke detector
		L	Convenience outlet and internal lights with return smoke detector
		M	Convenience outlet and internal lights with supply and return smoke detectors
33	Controls options	A	None
		B	Low ambient
		D	Subcool and superheat measurement
		E	Low ambient with subcool and superheat measurement
		F	With Verasys
		G	Low ambient with Verasys
		H	Subcool and superheat measurement with Verasys
		I	Low ambient with subcool and superheat measurement with Verasys
34	User interface options	A	BACnet MSTP, Modbus, N2
		B	BACnet IP
		G	BACnet MSTP, Modbus, N2 with mobile access portal (MAP)
		H	MAP with BACnet IP
		K	BACnet MSTP, Modbus, N2 with MAP and touch screen
		L	Touch screen with MAP with BACnet IP
		N	BACnet MSTP, Modbus, N2 with Customer Terminal Board (CTB)
		P	BACnet IP with CTB
		U	BACnet MSTP, Modbus, N2 with MAP and CTB
		V	MAP with BACnet IP and CTB
		X	BACnet MSTP, Modbus, N2 with Touch Screen and MAP and CTB
		Y	Touch Screen and MAP Gateway with BACnet IP with CTB
35	IAQ options	4	Refrigerant Detection System (RDS)
		5	UV lights and RDS
		6	CO2 sensors—demand controlled ventilation and RDS
		7	UV lights, CO2 sensors (demand controlled ventilation) and RDS

Digit	Category	Option	Description
36	Gas heat shipped loose kits	0	None
		1	Gas heat, side penetration
		2	Gas heat, bottom penetration
		3	Gas heat, high altitude kit natural gas (NG), side penetration
		4	Gas heat, high altitude kit NG, bottom penetration
		5	Gas heat, high altitude kit liquid propane (LP), side penetration
		6	Gas heat, high altitude kit LP, bottom penetration
		7	Gas heat, LP conversion kit, side penetration
		8	Gas heat, LP conversion kit, bottom penetration
37	Security options	0	None
		1	Supply and return opening burglar bars
		2	Supply opening burglar bars
		3	Return opening burglar bars
38	Door options	0	None
		1	Viewport
		2	Single handle with padlock
		3	Single handle with padlock and viewport
39	Cabinet shipping options	1	Single piece construction
		2	Two piece construction
40	Curb options	A	No roof curb
		B	Full perimeter roof curb
		C	Pedestal curb
41	Pre-evap options	0	None
		1	Blank pre-evap extension, no air blender
		3	Pre-evap extension with air blender
42	Shipped loose options	0	None
		1	Spare belts for return/exhaust
43	Construction standard	0	None
44	Supply fan VFD frequency	1	68 Hz
		2	70 Hz
		3	72 Hz
		A	66 Hz
		B	64 Hz
		C	62 Hz
		D	60 Hz
		E	58 Hz
		F	56 Hz
		G	54 Hz
		H	52 Hz
		J	50 Hz
		K	48 Hz
		L	46 Hz
		M	44 Hz
		N	42 Hz
		P	40 Hz
		Q	38 Hz
		R	36 Hz
		S	34 Hz
		T	32 Hz
		U	30 Hz
		V	28 Hz
		W	26 Hz

Digit	Category	Option	Description
45	Supply fan brake horsepower (each motor)	A	2 HP
		B	3 HP
		C	5 HP
		D	7.5 HP
		E	10 HP
		F	15 HP
		G	20 HP
		H	25 HP
		J	30 HP
		K	40 HP
46	Return fan brake horsepower	0	None
		C - L	Internal use only
47	Future 4	0	None
48	Testing and special requests	0	None
		T	Record test report
		M	Mechanical Special Request (SR)
		1	Mechanical SR and record test report
		S	Software SR
		3	Software SR and record test report
		B	Mechanical and software SR
		5	Mechanical and software SR and test report
		R	Rapid ship
		7	Rapid ship and test report
49	Generation/revision level	1	1st generation

Selection procedure examples

Given:	
Required cooling capacity	600,000 Btuh
Required sensible cooling	450,000 Btuh
Required heating	400,000 Btuh
Entering air on evaporator	80.0°F dry bulb (DB)/ 67.0°F wet bulb (WB)
Outside design temperature	95.0°F
Supply fan CFM	19,000 CFM
External static pressure (ESP)	2.00 iwg
Electrical supply voltage	460-3-60
Economizer required	
2 in. MERV 8 filters	
Variable air volume (VAV)	

Calculating cooling and heating capacity

1. Assume that the required cooling capacity and required sensible capacity include the space load requirements as well as the ventilation load requirements.
2. Determine the nominal unit size. Using Table 9 at design conditions of 80/67 and 19,000 CFM, we can determine capacity is 680,000 Btuh total and 471,000 Btuh sensible. Therefore, use 60 ton nominal unit size as a starting point.
3. Calculate the supply fan motor heat Btuh addition.
    a. See Select fan speed and horsepower requirements for supply fan to determine the horsepower (HP) of the supply fans. The example (using Table 72) is based on a 1397 RPM and 19.3 BHP (two fans at 15 nameplate motor HP each when we apply the UL60335 VFD limited direct drive fan requirements) fan requirement.
    b. Calculate sensible Btuh addition as a result of the supply fans HP:

       Supply fan sensible Btuh addition = 19.3 HP x 2,750 = 53,000 Btuh

       where 2,750 = constant for motor heat calculation
    c. Required capacity = Total load + supply fan motor heat:

       600 MBH + 53 MBH = 653 MBH (54.4 tons)

       Therefore our selection of 60 tons nominal from Table 9 is valid.
4. We can now use the capacity data from Table 9 to calculate the unit leaving air temperature.
    a. Unit sensible heat capacity corrected for supply air fan motor heat:

       471 MBH - 53 MBH = 418 MBH
    b. Supply air dry bulb temperature difference:

       $$\frac{Sensible\ \ Btu}{(1.085)\ (Supply\ \ CFM)} = \frac{418\ \ MBH}{(1.085)\ (19{,}000)} = 20.3°F$$

       Supply air dry bulb:

       80°F - 20.3°F = 59.7°F
    c. We can use the unit enthalpy delta to find the leaving wet bulb temperature:

       $$\frac{Total\ \ Btu}{(4.5)\ (Supply\ \ CFM)} = \frac{680\ \ MBH}{(4.5)\ (19{,}000)} = 8.0\ \ \text{BTU/lb}$$

   From a psychometric chart:
   - entering enthalpy of 80/67 is 31.5 Btu/lb
   - leaving air enthalpy = 31.5 Btu/lb- 8.0 Btu/lb = 23.5 Btu/lb

   From the psychometric chart, 23.5 Btu/lb enthalpy is a wet bulb of 55.5°F, so the unit LAT is 59.7°F DB and 55.5°F WB
5. For gas heating capacity, see Table 52:
    a. Trace down to the maximum output capacity (MBH) column.
    b. Find the output that exceeds the 400,000 Btuh requirement.

       The 500 MBH output meets this requirement.
    c. This option is available on the 60 ton unit.

       From the nomenclature, select digit 5, option B for staged gas, option C for staged gas with stainless steel burner, or option G for modulating gas with stainless steel burner.

       From digit 6B, select option 2 for 500 MBH. The resulting model number shows C2 for digits 5 and 6, assuming staged gas with stainless steel burner was selected.
6. For electric heating capacity, see Table 53.
    a. Trace down to the heat capacity (MBH) column.
    b. Find the output that exceeds the 400,000 Btuh requirement at the given 460-3-60 voltage.

       The 120 kW output exceeds this requirement.
    c. This option is available on the 60 ton unit.

From the nomenclature, select digit 5, option M for electric heat.

From digit 6A, select either option 1 with low staged electric heat or option 4 with low silicon controlled rectifier (SCR) electric heat. The resulting model number shows M4 for digits 5 and 6, assuming low SCR electric heat was selected.

Select fan speed and horsepower requirements for supply fan

1. See Table 72 for the 60 ton unit.
    a. Make any necessary additions to the static resistance for the ductwork. See Table 59 onwards for component static pressure drops (iwg):

Ductwork static resistance/ESP	2.00 iwg
+ return duct static pressure (assumed)	0.25 iwg
+ economizer static resistance addition	0.19 iwg
+ bottom supply air opening resistance addition	0.09 iwg
+ wet evaporator coil standard capacity static resistance addition	0.57 iwg
+ gas heat (500 MBH)	0.16 iwg
+ MERV 8 filters	0.13 iwg
Total Static Resistance (TSP)	3.38 iwg

    b. Enter Table 72 at 19,000 CFM and 3.38 iwg TSP:

    RPM = 1397

    BHP =19.3

2. From the nomenclature, select digit 15, option D to opt for dual direct drive plenum (DDP) supply fans with 1 in. spring isolation. Select digit 16, option E, for 20HP x 2 supply fan motor. Select digit 17, option 2 for ODP premium efficiency, 1,800 RPM.

Select fan speed and horsepower requirements of exhaust fan

1. See Table 76 for the 60 ton unit. In the following example, a unit is designed for exhaust air capacity of 12,000 CFM.
    a. Make any necessary additions to the static resistance. To find the exhaust air damper pressure drop, see Table 61:

Return duct static pressure (assumed)	0.60 iwg
+ exhaust air damper pressure drop	0.26 iwg
TSP	0.86 iwg

    b. Enter at 12,000 CFM and 0.86 iwg TSP:

    RPM = 290 (interpolated)

    BHP = 2.5 (interpolated)

2. From the nomenclature, select digit 19, option 2 for exhaust with variable frequency drive (VFD) and backdraft damper; digit 20, option B for the exhaust fan with 1 in. spring isolation; and digit 21, option F for 5 HP exhaust fan motor. The resulting model number shows 2BF for digits 19, 20, and 21.

Select energy recovery wheel (ERW), hot gas reheat (HGRH), sound attenuator, and humidifier

Refer to Selection Navigator or contact the local sales office for more information on available options and accessories.

Example model number

The following model number is an example based on the options selected in the previous sections.

Table 2: Example model number

GV	**F**	**1**	**C**	**2**	**B**	**5**	**G**	**A**	**1**	**A**	**6**	**0**	**D**	**E**	**2**	**0**	**2**	**B**	**F**	**2**	**A**	**2**	**G**	**2**	**C**	**0**	**E**	**0**	**D**	**M**	**A**	**A**	**3**	**1**	**0**	**0**	**1**	**A**	**0**	**0**	**0**	**L**	**E**	**0**	**0**	**0**	**2**
1,2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49

Note: The first row displays the example model number. The second row displays the corresponding digit.

Table 3: Model nomenclature description

Digit	Option	Description
3	F	Capacity 60 ton
4	1	Standard capacity, standard efficiency
5	C	Staged gas stainless steel
6	2	500 MBH
7	B	Variable air volume
8	5	Supply fan with VFD and return and exhaust fan with VFD
9	G	460 V 3Ph 60 Hz, single point terminal block
15	D	Dual direct drive plenum supply fans with 1 in. spring isolation
16	E	Supply fan motors, 20 HP x 2
17	2	Supply fan motor type, ODP premium efficiency 1,800 RPM
19	2	Exhaust fan with VFD and backdraft damper
20	B	Forward curved exhaust fan with 1 in. spring isolation
21	F	Exhaust fan motor, 5 HP
22	2	Exhaust fan motor type, ODP premium efficiency 1,800 RPM
27	C	Pre-evaporator filter, angled filter rack, 2 in. MERV 8 filters
29	E	Dual enthalpy economizer, low leak dampers
31	D	Suction, discharge, and liquid line valves with replaceable core filter driers
32	M	Convenience outlet and internal lights with supply and return smoke detectors
35	3	Ultraviolet (UV) lights and Carbon Dioxide (CO_2) sensors
49	2	Second generation

Physical data

Table 4: Physical data

Unit size	60 ton	70 ton	80 ton
Compressor data - standard capacity, standard efficiency			
Quantity/size (nominal HP)	1/12, 1/12, 1/15, 1/15	1/10, 1/15, 1/15, 1/20	1/15, 1/15, 1/15, 1/20
Type	Scroll	Scroll	Scroll
Capacity steps	8	9	7
Number of circuits	2	2	2
Compressor data - high capacity, standard efficiency			
Quantity/size (nominal HP)	1/13, 1/13, 1/15, 1/15	1/15, 1/15, 1/15, 1/20	1/15, 1/15, 1/20, 1/20
Type	Scroll	Scroll	Scroll
Capacity steps	8	7	8
Number of circuits	2	2	2
Compressor data - standard capacity, high efficiency			
Quantity/size (nominal HP)	1/10, 1/15, 1/15, 1/17.5	1/13, 1/15, 1/15, 1/27	1/15, 1/15, 1/15, 1/27
Type	Scroll	Scroll	Scroll
Capacity steps	15–100%	15–100%	15–100%
Number of circuits	2	2	2
Supply fan			
Quantity of fans/motors	2/2	2/2	2/2
Type	Direct-driven plenum		
Size (in.)	27	27	27
Motor size range each fan (min to max HP)	5-40	5-40	5-40
Airflow range (min to max CFM)	12,000-24,000	14,000-32,000	14,000-32,000
Total static pressure range (min to max iwg)	1-8.3	1-8.3	1-8.3
Return fan			
Quantity of fans/motors	1/1	1/1	1/1
Type	Direct-driven plenum		
Size (in.)	33	33	33
Motor size range (min to max HP)	7.5-40	7.5-40	7.5-40
Airflow range (min to max CFM)	12,000-24,000	14,000-32,000	14,000-32,000
Total static pressure range (min to max iwg)	0.5–3.0	0.5–3.0	0.5–3.0
Exhaust fan			
Quantity of fans/motors	1/1	1/1	1/1
Type	Belt-driven forward curved		
Size (in.)	30-30	30-30	30-30
Motor size range (min to max HP)	5-25	5-40	5-40
Airflow range (min to max CFM)	12,000-24,000	14,000-32,000	14,000-32,000
Total static pressure range (min to max iwg)	0.5–3.0	0.5–3.0	0.5–3.0
ERW dual exhaust fan			
Quantity of fans/motors	2/2	2/2	2/2
Type	Belt-driven forward curved		
Size (in.)	20-18	22-20	22-20
Motor size range each fan (min to max HP)	5-15	5-20	5-20
Airflow range (min to max CFM)	6,000-24,000	7,000-28,000	7,000-32,000
Total static pressure range (min to max iwg)	0.5-3.0	0.5-3.0	0.5-3.0
Condenser fan			
Quantity	4	6	
Type	Propeller		
Size (diameter, in.)	30		

Table 4: Physical data

Unit size	60 ton	70 ton	80 ton
Power (HP)	2		
Evaporator coil - standard capacity, standard efficiency			
Size (sq. ft.)	49.5	66	66
Number of rows/fins per in.	5/17	5/17	6/17
Size (tube diameter, in.)	3/8		
Evaporator coil - high capacity, standard efficiency			
Size (sq. ft.)	49.5	66	66
Number of rows/fins per in.	6/17	6/17	6/17
Size (tube diameter, in.)	3/8		
Evaporator coil - standard capacity, high efficiency			
Size (sq. ft.)	49.5	66	66
Number of rows/fins per in.	6/17	6/17	6/17
Size (tube diameter, in.)	3/8		
Hot gas reheat (HGRH) coil			
Coil type	Microchannel		
Control type	Modulating		
Size (area in sq. ft./thickness, in.)	28/1	33.4/1	33.4/1
Number of rows/fins per in.	1/23		
Minimum outside air (OA) temperature for mechanical cooling	45.0°F		
Low ambient option minimum OA temperature for mechanical cooling	-10.0°F		
Condenser coil - standard capacity, standard efficiency			
Size (sq. ft.)	153	191	191
Number of rows/fins per in.	1/23		
Size (thickness, in.)	1	1	1
Type	Microchannel		
Condenser coil - high capacity, standard efficiency			
Size (sq. ft.)	153	191	191
Number of rows/fins per in.	1/23		
Size (thickness, in.)	1		
Type	Microchannel		
Condenser coil - standard capacity, high efficiency			
Size (sq. ft.)	153	191	191
Number of rows/fins per in.	1/23		
Size (thickness, in.)	1		
Type	Microchannel		
ERW - high CFM			
Cassette dimensions (L x W x H, in.)	78 x 7.07 x 78	85 x 7.07 x 85	85 x 7.07 x 85
Wheel segments	8	8	8
Motor (V, ph, Hz)	208-230/3/60, 460/3/60, 575/3/60		
Horsepower (HP)	1/3	1/3	1/3
Filter type	2 in. MERV 8 pleated		
RA filters - size (number)	18 x 25 (3) and 20 x 25 (3)	20 x 24 (6) and 20 x 10 (2)	20 x 24 (6) and 20 x 10 (2)
OA filters - size (number)	18 x 25 (3) and 20 x 25 (3)	20 x 24 (6) and 20 x 10 (2)	20 x 24 (6) and 20 x 10 (2)
ERW - low CFM			
Cassette dimensions (L x W x H, in.)	68 x 6.07 x 68	72 x 7.07 x 72	72 x 7.07 x 72
Wheel segments	8	8	8
Motor (V/ph/Hz)	208-230/3/60, 460/3/60, 575/3/60		
Horsepower (HP)	1/4	1/4	1/4
Filter type	2 in. MERV 8 pleated		
RA filters - size (number)	16 x 24 (6)	18 x 24 (6)	18 x 24 (6)
OA filters - size (number)	16 x 24 (6)	18 x 24 (6)	18 x 24 (6)

Table 4: Physical data

Unit size		60 ton	70 ton	80 ton
Electric heat (208 V/230 V)				
Size range - low/high (kW)		90/120		
Heating steps - low/high		6/8		
SCR (modulating heat)		Available		
Electric heat (460 V/575 V)				
Size range - low/high (kW)		120/200		
Heating steps - low/high		6/10		
SCR (modulating heat)		Available		
Gas furnace				
Staged furnace sizes (input/output/stages)		500/405/2		
		1,000/810/4		
		1,500/1,215/6		
Airflow range (min to max CFM)		12,000-24,000	14,000-32,000	14,000-32,000
Gas heat steady state efficiency (SSE)		81%		
Modulating furnace sizes (input/output/turndown)		500/405/10:1		
		1,000/810/20:1		
		1,500/1,215/30:1		
Airflow range (min to max CFM)		12,000-24,000	14,000-32,000	14,000-32,000
Hot water coil				
Coil tube diameter (in.)		1/2		
Material		Copper tube/aluminum fin		
Fins per in. - low/high		14/10		
Size (H x L, in.)/rows low		52.5 x 84/1		
Size (H x L, in.)/rows high		52.5 x 84/2		
Steam coil				
Coil tube diameter (in.)		1		
Material		Copper tube/aluminum fin		
Fins per in. - low/high		8/14	8/14	10/14
Size (H x L, in.)/rows low		48 x 84/1		
Size (H x L, in.)/rows high		48 x 84/1		
Draw-through - 2 in. throwaway angled filters				
Quantity		28		
Size (L x W, in.)		16 x 24		
Total filter face area (sq. ft.)		70.83		
Draw-through - 2 in. MERV 8 angled filters				
Quantity		28		
Size (L x W, in.)		16 x 24		
Total filter face area (sq. ft.)		70.83		
Draw-through - MERV 15 bag filters with 2 in. MERV 8 pre-filters				
Pre-filters	Quantity	9/6		
	Size (L x W, in.)	24 x 24/20 x 24		
	Total filter face area (sq. ft.)	53.61		
Bag filters	Quantity	9/6		
	Size (L x W, in.)	24 x 24/20 x 24		
	Total filter face area (sq. ft.)	53.02		
Draw-through - MERV 14 rigid filters with 2 in. MERV 8 pre-filters				
Pre-filters	Quantity	9/6		
	Size (L x W, in.)	24 x 24/20 x 24		
	Total filter face area (sq. ft.)	53.61		

Table 4: Physical data

Unit size		60 ton	70 ton	80 ton
Rigid filters	Quantity	9/6		
	Size (L x W, in.)	24 x 24/20 x 24		
	Total filter face area (sq. ft.)	53.02		
Draw-through - vertical 4 in. MERV 8 filters				
Quantity		9/6		
Size (L x W, in.)		24 x 24/20 x 24		
Total filter face area (sq. ft.)		53.02		
Final filters - MERV 15 bag filters with 2 in. pre-filters				
Pre-filters	Quantity	6/8		
	Size (L x W, in.)	24 x 24/20 x 24		
	Total filter face area (sq. ft.)	48.47		
Bag filters	Quantity	6/4/4		
	Size (L x W, in.)	24 x 24/24 x 20/20 x 24		
	Total filter face area (sq. ft.)	47.93		
Final filters - MERV 14 rigid filters with 2 in. pre-filters				
Pre-filters	Quantity	6/8		
	Size (L x W, in.)	24 x 24/20 x 24		
	Total filter face area (sq. ft.)	48.47		
Rigid filters	Quantity	9/6		
	Size (L x W, in.)	24 x 24/20 x 24		
	Total filter face area (sq. ft.)	53.02		
Final filters - MERV 17 HEPA® filters with 2 in. pre-filters				
Pre-filters	Quantity	6/8		
	Size (L x W, in.)	24 x 24/20 x 24		
	Total filter face area (sq. ft.)	48.47		
HEPA® filters	Quantity	8/8/2		
	Size* (L x W, in.)	23-3/8 x 23-3/8/24 x 12/12 x 12		
	Total filter face area (sq. ft.)	48.36		

Note: HEPA filter sizes shown are actual, all other filters are nominal.

Capacity performance

Energy efficiency ratio and integrated energy efficiency ratio (EER/IEER) ratings

Table 5: EER/IEER ratings

Capacity	Efficiency	Heat source	EER	IEER
60 ton	Standard capacity/standard efficiency	Cooling only/electric heat	10.5	15.4
		Gas/steam/hot water	10.3	15.3
	Standard capacity/high efficiency	Cooling only/electric heat	10.8	16.6
		Gas/steam/hot water	10.6	16.5
	High capacity/standard efficiency	Cooling only/electric heat	10.3	15.1
		Gas/steam/hot water	10.1	15.0

Altitude and temperature correction factors

Table 6: Altitude and temperature correction factors

Air temp °F	Altitude, ft										
	0*	**1,000**	**2,000**	**3,000**	**4,000**	**5,000**	**6,000**	**7,000**	**8,000**	**9,000**	**10,000**
40	1.060	1.022	0.986	0.95	0.916	0.822	0.849	0.818	0.788	0.758	0.729
50	1.039	1.002	0.966	0.931	0.898	0.864	0.832	0.802	0.772	0.743	0.715
60	1.019	0.982	0.948	0.913	0.880	0.848	0.816	0.787	0.757	0.729	0.701
70	1.000	0.964	0.930	0.896	0.864	0.832	0.801	0.772	0.743	0.715	0.688
80	0.982	0.947	0.913	0.88	0.848	0.817	0.787	0.758	0.73	0.702	0.676
90	0.964	0.929	0.897	0.864	0.833	0.802	0.772	0.744	0.716	0.689	0.663
100	0.946	0.912	0.880	0.848	0.817	0.787	0.758	0.730	0.703	0.676	0.651

Note:

* Correction factors at sea level to calculate for actual temperature conditions.

The information below must be used to assist in application of product when applied at altitudes equal to or above 1000 ft above sea level.

The following examples assist in determining the airflow performance of Premier units at specific altitudes.

Example 1: What are the corrected cubic ft per minute (CFM), static pressure, and brake horse power (BHP) at an elevation of 5,000 ft if the airflow performance data is 19,000 CFM, 1.4 in. of water gauge (iwg), and 15.0 BHP?

Solution: At an elevation of 5,000 ft, the supply fan still delivers 19,000 CFM if the revolutions per minute (RPM) is unchanged. However, the altitude correction must be used to determine the static pressure and BHP. Since no temperature data is given, we assume an air temperature of 70.0°F. Table 6 shows the correction factor to be 0.832.

Corrected static pressure = 1.4 x 0.832 = 1.16 iwg

Corrected BHP = 15.0 x 0.832 = 12.48

Example 2: A system, located at 5,000 ft of elevation is to deliver 19,000 CFM at a static pressure of 1.4 iwg. Use the unit blower tables to select the RPM, blower speed, and the BHP requirement.

Solution: As in the example above, no temperature information is given, so 70.0°F is assumed.

The 1.4 iwg static pressure given is at an elevation of 5,000 ft. The first step is to convert this static pressure to equivalent sea level conditions.

Sea level static pressure = 1.4 iwg / 0.832 = 1.68 iwg

Enter Table 72 at 19,000 CFM and static pressure of 1.68 iwg. The RPM listed is the same RPM needed at 5,000 ft.

Using interpolation, the corresponding BHP listed in the table is 10.8. This value must be corrected for elevation.

BHP at 5,000 ft = 10.8 x 0.832 = 8.98

Example 3: Plot fan performance using Table 72.

Plot the fan performance at cooling sea level (0 ft) elevation. Design conditions are a 60 ton unit producing 20,000 CFM at 2.0 external static pressure (ESP) with additional static losses for a wet evaporator coil, bottom return air, bottom supply air, and angled filter rack with 2 in. MERV 8 filters. Refer to Table 59 onwards for component static pressure drops (iwg)

Wet evaporator coil standard capacity additional static loss = 0.62 iwg

Bottom return air additional static loss = 0.10 iwg

Bottom supply air additional static loss = 0.11 iwg

Air filter additional static loss = 0.13 iwg

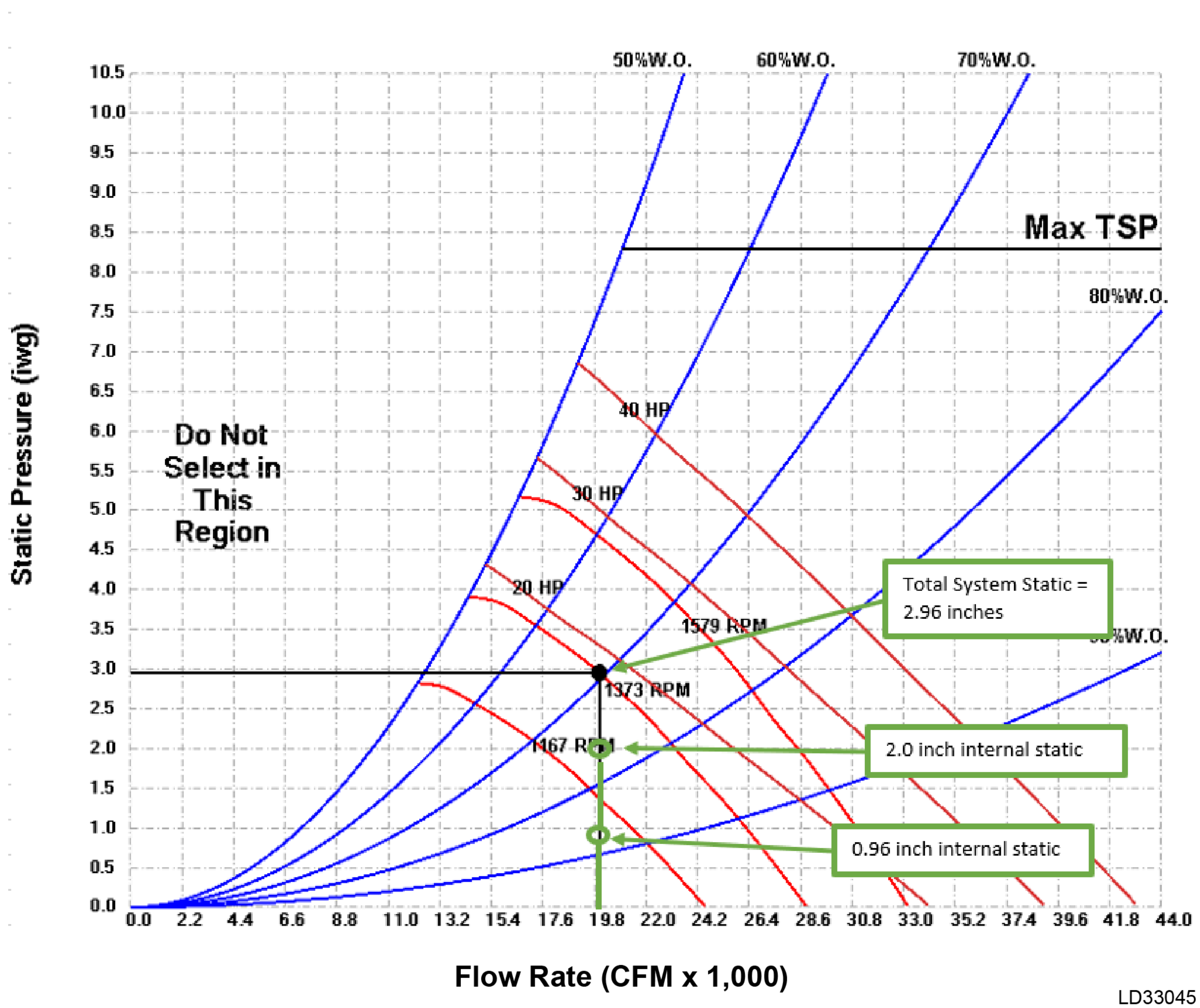
Add the ESP and all additional losses (internal static loss):

ESP loss = 2.0 iwg

Total internal static loss = 0.96 iwg

Total system static = 2.96 iwg

Figure 11: 60 ton unit fan performance



Cooling performance data

Cooling performance data 60 ton — standard capacity, standard efficiency

Table 7: 60 ton standard efficiency — 75°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	813	449	45	813	383	45	813	317	45	813	251	45	—	—	—	—	—	—	—	—	—
	72	745	526	45	745	460	45	745	394	45	745	328	45	745	262	45	—	—	—	—	—	—
	67	681	602	44	681	536	44	681	469	44	681	403	44	681	337	44	681	271	44	—	—	—
	62	642	637	44	623	610	44	623	543	44	623	477	44	623	410	44	623	344	44	623	278	44
	57	642	641	44	613	610	44	585	581	44	—	—	—	—	—	—	—	—	—	—	—	—
15,000	77	860	508	45	860	426	45	860	344	45	860	263	45	—	—	—	—	—	—	—	—	—
	72	790	603	45	790	521	45	790	439	45	790	357	45	790	275	45	—	—	—	—	—	—
	67	724	697	45	724	614	45	724	532	45	724	449	45	724	367	45	724	286	45	—	—	—
	62	708	702	45	675	668	44	663	623	44	663	540	44	663	458	44	663	376	44	663	294	44
	57	708	706	45	675	672	44	643	639	44	611	606	44	607	547	44	607	464	44	607	382	44
19,000	77	901	585	46	901	482	46	901	379	46	901	277	46	—	—	—	—	—	—	—	—	—
	72	829	702	45	829	599	45	829	496	45	829	393	45	829	290	45	—	—	—	—	—	—
	67	778	766	45	762	715	45	762	611	45	762	508	45	762	404	45	762	302	45	—	—	—
	62	778	771	45	740	732	45	703	694	44	699	621	44	699	517	44	699	414	44	699	311	44
	57	778	776	45	740	737	45	703	698	44	667	661	44	640	628	44	640	525	44	640	421	44
21,000	77	916	623	46	916	509	46	916	396	46	916	283	46	—	—	—	—	—	—	—	—	—
	72	844	752	45	844	637	45	844	524	45	844	410	45	844	297	45	—	—	—	—	—	—
	67	806	794	45	776	764	45	776	650	45	776	536	45	776	422	45	776	309	45	—	—	—
	62	806	799	45	767	758	45	728	719	45	712	660	45	712	546	45	712	432	45	712	318	45
	57	806	804	45	767	763	45	728	723	45	690	684	44	653	646	44	652	554	44	652	439	44
24,000	77	933	680	46	933	550	46	933	421	46	933	293	46	—	—	—	—	—	—	—	—	—
	72	861	825	45	861	695	45	861	565	45	861	436	45	861	307	45	—	—	—	—	—	—
	67	843	831	45	801	788	45	793	708	45	793	578	45	793	449	45	793	319	45	—	—	—
	62	843	836	45	801	793	45	760	750	45	728	719	45	728	589	45	728	459	45	728	329	45
	57	843	841	45	801	797	45	760	755	45	720	713	45	680	673	44	667	597	44	667	467	44

Table 8: 60 ton standard efficiency — 85°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	784	439	51	784	373	51	784	307	51	784	242	51	—	—	—	—	—	—	—	—	—
	72	718	516	50	718	450	50	718	384	50	718	318	50	718	252	50	—	—	—	—	—	—
	67	657	592	50	657	525	50	657	459	50	657	393	50	657	327	50	657	261	50	—	—	—
	62	625	620	50	601	599	49	601	533	49	601	466	49	601	400	49	601	334	49	601	268	49
	57	625	623	50	596	593	49	568	564	49	549	538	49	549	471	49	549	405	49	549	339	49
15,000	77	827	498	51	827	416	51	827	334	51	827	253	51	—	—	—	—	—	—	—	—	—
	72	760	592	51	760	510	51	760	428	51	760	346	51	760	265	51	—	—	—	—	—	—
	67	697	686	50	697	603	50	697	521	50	697	439	50	697	357	50	697	275	50	—	—	—
	62	688	682	50	655	649	50	638	612	50	638	529	50	638	447	50	638	365	50	638	283	50
	57	688	686	50	655	652	50	624	620	49	593	587	49	584	535	49	584	453	49	584	371	49
19,000	77	865	574	51	865	472	51	865	369	51	865	266	51	—	—	—	—	—	—	—	—	—
	72	796	692	51	796	588	51	796	485	51	796	382	51	796	279	51	—	—	—	—	—	—
	67	753	742	50	732	704	50	732	600	50	732	496	50	732	393	50	732	290	50	—	—	—
	62	753	747	50	717	709	50	681	672	50	671	609	50	671	506	50	671	402	50	671	299	50
	57	753	752	50	717	714	50	681	676	50	646	640	50	615	616	49	615	513	49	615	409	49
21,000	77	878	612	51	878	499	51	878	386	51	878	273	51	—	—	—	—	—	—	—	—	—
	72	809	741	51	809	627	51	809	513	51	809	399	51	809	286	51	—	—	—	—	—	—
	67	780	769	51	744	753	50	744	639	50	744	525	50	744	411	50	744	297	50	—	—	—
	62	780	774	51	742	734	50	704	695	50	683	649	50	683	534	50	683	420	50	683	307	50
	57	780	778	51	742	738	50	704	699	50	667	661	50	631	624	50	626	542	50	626	428	50
24,000	77	894	669	52	894	540	52	894	411	52	894	282	52	—	—	—	—	—	—	—	—	—
	72	825	814	51	825	684	51	825	554	51	825	425	51	825	296	51	—	—	—	—	—	—
	67	815	803	51	774	761	51	760	697	50	760	567	50	760	437	50	760	308	50	—	—	—
	62	815	808	51	774	766	51	734	725	50	698	707	50	698	577	50	698	447	50	698	318	50
	57	815	813	51	774	771	51	734	729	50	695	689	50	657	650	50	640	585	50	640	455	50

Table 9: 60 ton standard efficiency — 95°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	753	428	57	753	363	57	753	297	57	753	231	57	—	—	—	—	—	—	—	—	—
	72	690	506	56	690	439	56	690	373	56	690	307	56	690	242	56	—	—	—	—	—	—
12,000	67	632	581	55	632	515	55	632	448	55	632	382	55	632	316	55	632	250	55	—	—	—
	62	606	601	55	578	572	55	578	522	55	578	455	55	578	389	55	578	323	55	578	257	55
	57	606	604	55	578	575	55	551	547	54	528	527	54	528	460	54	528	394	54	528	328	54
	77	792	487	57	792	405	57	792	324	57	792	242	57	—	—	—	—	—	—	—	—	—
	72	728	582	56	728	499	56	728	417	56	728	335	56	728	254	56	—	—	—	—	—	—
15,000	67	668	675	56	668	592	56	668	510	56	668	427	56	668	345	56	668	264	56	—	—	—
	62	666	660	56	634	628	55	612	600	55	612	518	55	612	435	55	612	353	55	612	271	55
	57	666	664	56	634	631	55	603	599	55	573	568	55	561	524	55	561	442	55	561	359	55
	77	826	564	57	826	461	57	826	358	57	826	256	57	—	—	—	—	—	—	—	—	—
	72	761	680	57	761	577	57	761	474	57	761	371	57	761	268	57	—	—	—	—	—	—
19,000	67	728	717	56	700	692	56	700	588	56	700	485	56	700	382	56	700	279	56	—	—	—
	62	728	722	56	692	685	56	657	649	56	642	598	55	642	494	55	642	391	55	642	287	55
	57	728	726	56	692	689	56	657	653	56	623	617	55	589	583	55	589	501	55	589	397	55
	77	838	602	57	838	488	57	838	375	57	838	262	57	—	—	—	—	—	—	—	—	—
	72	773	729	57	773	615	57	773	501	57	773	388	57	773	275	57	—	—	—	—	—	—
21,000	67	753	742	57	716	704	56	711	627	56	711	513	56	711	399	56	711	286	56	—	—	—
	62	753	747	57	716	708	56	679	671	56	653	637	56	653	522	56	653	408	56	653	295	56
	57	753	751	57	716	713	56	679	675	56	643	637	55	608	601	55	598	529	55	598	415	55
	77	852	658	58	852	529	58	852	400	58	852	271	58	—	—	—	—	—	—	—	—	—
	72	787	803	57	787	673	57	787	543	57	787	414	57	787	285	57	—	—	—	—	—	—
24,000	67	786	774	57	746	734	56	725	685	56	725	555	56	725	425	56	725	296	56	—	—	—
	62	786	779	57	746	738	56	707	698	56	669	659	56	667	565	56	667	435	56	667	305	56
	57	786	784	57	746	743	56	707	703	56	669	663	56	632	625	55	611	572	55	611	442	55

Table 10: 60 ton standard efficiency — 105°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	720	418	63	720	352	63	720	286	63	720	221	63	—	—	—	—	—	—	—	—	—
	72	660	495	62	660	428	62	660	362	62	660	296	62	660	231	62	—	—	—	—	—	—
12,000	67	605	570	61	605	503	61	605	437	61	605	371	61	605	305	61	605	239	61	—	—	—
	62	586	581	61	559	553	61	553	510	60	553	444	60	553	378	60	553	311	60	553	245	60
	57	586	584	61	559	556	61	532	528	60	506	501	60	505	449	60	505	382	60	505	316	60
	77	755	476	63	755	394	63	755	313	63	755	232	63	—	—	—	—	—	—	—	—	—
	72	695	570	62	695	488	62	695	406	62	695	324	62	695	243	62	—	—	—	—	—	—
15,000	67	642	633	62	638	580	62	638	498	62	638	416	62	638	334	62	638	252	62	—	—	—
	62	642	637	62	612	605	61	584	588	61	584	506	61	584	424	61	584	341	61	584	259	61
	57	642	641	62	612	609	61	582	578	61	552	547	60	535	512	60	535	429	60	535	347	60
	77	786	553	64	786	450	64	786	347	64	786	245	64	—	—	—	—	—	—	—	—	—
	72	725	669	63	725	565	63	725	462	63	725	359	63	725	257	63	—	—	—	—	—	—
19,000	67	701	690	62	667	680	62	667	576	62	667	473	62	667	370	62	667	267	62	—	—	—
	62	701	695	62	666	659	62	632	624	62	612	585	61	612	482	61	612	378	61	612	275	61
	57	701	699	62	666	663	62	632	628	62	599	594	61	566	560	61	561	488	60	561	385	60
	77	797	590	64	797	477	64	797	364	64	797	251	64	—	—	—	—	—	—	—	—	—
	72	735	718	63	735	604	63	735	490	63	735	376	63	735	263	63	—	—	—	—	—	—
21,000	67	724	714	63	688	676	62	677	615	62	677	501	62	677	387	62	677	274	62	—	—	—
	62	724	718	63	688	681	62	652	644	62	621	624	61	621	510	61	621	396	61	621	282	61
	57	724	723	63	688	685	62	652	648	62	618	612	61	584	577	61	569	517	61	569	403	61
	77	810	647	64	810	518	64	810	389	64	810	260	64	—	—	—	—	—	—	—	—	—
	72	755	738	63	748	661	63	748	531	63	748	402	63	748	273	63	—	—	—	—	—	—
24,000	67	755	744	63	716	704	63	689	673	62	689	543	62	689	413	62	689	284	62	—	—	—
	62	755	748	63	716	709	63	679	670	62	642	633	62	633	552	62	633	422	62	633	293	62
	57	755	753	63	716	713	63	679	674	62	642	636	62	606	599	61	581	559	61	581	429	61

Table 11: 60 ton standard efficiency — 115°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	685	407	69	685	341	69	685	275	69	685	210	69	—	—	—	—	—	—	—	—	—
	72	628	483	68	628	417	68	628	351	68	628	285	68	628	219	68	—	—	—	—	—	—
	67	576	558	67	576	492	67	576	425	67	576	359	67	576	293	67	576	227	67	—	—	—
	62	564	559	67	538	532	66	526	498	66	526	432	66	526	366	66	526	299	66	526	233	66
	57	564	563	67	538	535	66	512	508	66	486	482	65	481	437	65	481	370	65	481	304	65
15,000	77	717	465	70	717	383	70	717	302	70	717	220	70	—	—	—	—	—	—	—	—	—
	72	659	558	69	659	476	69	659	394	69	659	312	69	659	231	69	—	—	—	—	—	—
	67	617	608	68	605	568	68	605	486	68	605	404	68	605	322	68	605	240	68	—	—	—
	62	617	612	68	587	581	67	558	551	67	555	493	67	555	411	67	555	329	67	555	247	67
	57	617	616	68	587	585	67	558	554	67	529	525	66	508	499	66	508	417	66	508	335	66
19,000	77	744	541	70	744	438	70	744	336	70	744	233	70	—	—	—	—	—	—	—	—	—
	72	686	657	69	686	553	69	686	450	69	686	347	69	686	245	69	—	—	—	—	—	—
	67	672	662	69	638	627	68	631	564	68	631	460	68	631	357	68	631	254	68	—	—	—
	62	672	666	69	638	631	68	605	598	68	579	572	67	579	469	67	579	365	67	579	262	67
	57	672	670	69	638	635	68	605	601	68	573	568	67	542	536	67	531	475	66	531	372	66
21,000	77	753	579	70	753	466	70	753	352	70	753	239	70	—	—	—	—	—	—	—	—	—
	72	695	705	69	695	591	69	695	478	69	695	364	69	695	251	69	—	—	—	—	—	—
	67	693	683	69	658	647	69	640	602	68	640	488	68	640	375	68	640	261	68	—	—	—
	62	693	688	69	658	651	69	624	616	68	590	582	67	588	497	67	588	383	67	588	269	67
	57	693	692	69	658	655	69	624	620	68	590	585	67	558	551	67	539	503	66	539	389	66
24,000	77	765	636	71	765	506	71	765	378	71	765	249	71	—	—	—	—	—	—	—	—	—
	72	721	706	70	707	649	70	707	519	70	707	390	70	707	261	70	—	—	—	—	—	—
	67	721	711	70	684	673	69	652	660	69	652	530	69	652	401	69	652	271	69	—	—	—
	62	721	715	70	684	677	69	648	640	69	613	604	68	599	539	68	599	409	68	599	280	68
	57	721	720	70	684	681	69	648	644	69	613	607	68	578	572	67	549	546	67	549	416	67

Notes:

Rated performance is at sea level. Cooling capacities are gross cooling capacities.

CFM = airflow, EWB = entering wet bulb air, EDB = entering dry bulb air, TMBH = total cooling capacity (MBH)

SMBH = sensible cooling capacity (MBH), kW = total input

Cooling performance data 60 ton - standard capacity, high efficiency

Table 12: 60 ton high efficiency - 75°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	800	444	43	800	378	43	800	312	43	800	247	43	—	—	—	—	—	—	—	—	—
	72	736	523	43	736	457	43	736	390	43	736	325	43	736	259	43	—	—	—	—	—	—
	67	676	600	43	676	533	43	676	467	43	676	401	43	676	335	43	676	269	43	—	—	—
	62	642	637	42	620	608	42	620	542	42	620	475	42	620	409	42	620	343	42	620	277	42
	57	642	641	42	615	612	42	587	583	42	569	548	42	569	482	42	569	415	42	569	349	42
15,000	77	847	504	44	847	422	44	847	340	44	847	259	44	—	—	—	—	—	—	—	—	—
	72	781	600	43	781	518	43	781	436	43	781	354	43	781	272	43	—	—	—	—	—	—
	67	719	695	43	719	612	43	719	530	43	719	447	43	719	365	43	719	283	43	—	—	—
	62	708	702	43	677	669	43	661	622	43	661	539	43	661	457	43	661	375	43	661	293	43
	57	708	706	43	677	673	43	645	641	43	615	609	42	607	547	42	607	464	42	607	382	42
19,000	77	891	582	44	891	479	44	891	376	44	891	274	44	—	—	—	—	—	—	—	—	—
	72	823	700	43	823	597	43	823	494	43	823	391	43	823	288	43	—	—	—	—	—	—
	67	779	767	43	759	714	43	759	610	43	759	506	43	759	403	43	759	300	43	—	—	—
	62	779	772	43	742	734	43	707	698	43	698	620	43	698	517	43	698	413	43	698	310	43
	57	779	777	43	742	739	43	707	702	43	672	666	43	641	629	43	641	525	43	641	421	43
21,000	77	907	620	44	907	507	44	907	394	44	907	281	44	—	—	—	—	—	—	—	—	—
	72	839	750	44	839	636	44	839	522	44	839	409	44	839	295	44	—	—	—	—	—	—
	67	809	797	43	773	764	43	773	649	43	773	535	43	773	421	43	773	308	43	—	—	—
	62	809	802	43	770	762	43	732	723	43	711	660	43	711	546	43	711	432	43	711	318	43
	57	809	806	43	770	766	43	732	727	43	695	689	43	659	652	43	653	554	43	653	440	43
24,000	77	928	678	44	928	549	44	928	420	44	928	291	44	—	—	—	—	—	—	—	—	—
	72	859	824	44	859	694	44	859	565	44	859	435	44	859	306	44	—	—	—	—	—	—
	67	848	836	44	807	793	43	792	708	43	792	578	43	792	448	43	792	319	43	—	—	—
	62	848	841	44	807	798	43	766	756	43	729	720	43	729	589	43	729	459	43	729	330	43
	57	848	846	44	807	803	43	766	761	43	726	720	43	688	680	43	669	598	43	669	468	43

Table 13: 60 ton high efficiency - 85°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	777	436	49	777	370	49	777	305	49	777	239	49	—	—	—	—	—	—	—	—	—
	72	714	515	48	714	448	48	714	382	48	714	316	48	714	251	48	—	—	—	—	—	—
	67	656	591	48	656	525	48	656	458	48	656	392	48	656	326	48	656	260	48	—	—	—
	62	628	623	47	601	599	47	601	533	47	601	466	47	601	400	47	601	334	47	601	268	47
	57	628	626	47	601	598	47	574	570	47	552	539	47	552	472	47	552	406	47	552	340	47
15,000	77	821	496	49	821	414	49	821	332	49	821	251	49	—	—	—	—	—	—	—	—	—
	72	757	591	48	757	509	48	757	427	48	757	345	48	757	264	48	—	—	—	—	—	—
	67	696	685	48	696	603	48	696	521	48	696	438	48	696	356	48	696	274	48	—	—	—
	62	691	686	48	660	653	48	639	612	48	639	530	48	639	447	48	639	365	48	639	283	48
	57	691	690	48	660	657	48	629	625	48	599	594	47	587	537	47	587	454	47	587	372	47
19,000	77	861	573	49	861	471	49	861	368	49	861	265	49	—	—	—	—	—	—	—	—	—
	72	796	691	49	796	588	49	796	485	49	796	382	49	796	279	49	—	—	—	—	—	—
	67	759	748	48	733	704	48	733	600	48	733	497	48	733	394	48	733	291	48	—	—	—
	62	759	753	48	723	715	48	688	679	48	674	610	48	674	507	48	674	404	48	674	300	48
	57	759	757	48	723	720	48	688	683	48	653	648	48	620	613	47	619	515	47	619	411	47
21,000	77	876	612	49	876	498	49	876	385	49	876	272	49	—	—	—	—	—	—	—	—	—
	72	810	741	49	810	627	49	810	513	49	810	399	49	810	286	49	—	—	—	—	—	—
	67	787	776	49	749	737	48	747	640	48	747	526	48	747	412	48	747	298	48	—	—	—
	62	787	781	49	749	741	48	712	703	48	687	650	48	687	536	48	687	422	48	687	308	48
	57	787	786	49	749	746	48	712	707	48	676	670	48	640	633	48	630	544	47	630	430	47
24,000	77	895	669	49	895	540	49	895	411	49	895	282	49	—	—	—	—	—	—	—	—	—
	72	828	815	49	828	685	49	828	555	49	828	426	49	828	297	49	—	—	—	—	—	—
	67	826	813	49	784	771	49	764	698	48	764	568	48	764	439	48	764	309	48	—	—	—
	62	826	819	49	784	776	49	745	735	48	706	695	48	703	579	48	703	449	48	703	319	48
	57	826	823	49	784	781	49	745	739	48	706	699	48	668	660	48	645	587	48	645	457	48

Table 14: 60 ton high efficiency - 95°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	751	428	54	751	362	54	751	296	54	751	231	54	—	—	—	—	—	—	—	—	—
	72	691	506	54	691	440	54	691	373	54	691	308	54	691	242	54	—	—	—	—	—	—
	67	634	582	53	634	515	53	634	449	53	634	383	53	634	317	53	634	251	53	—	—	—
	62	612	607	53	586	579	53	581	523	53	581	457	53	581	390	53	581	324	53	581	258	53
	57	612	611	53	586	583	53	559	555	52	533	528	52	533	462	52	533	396	52	533	330	52
15,000	77	792	487	55	792	405	55	792	324	55	792	242	55	—	—	—	—	—	—	—	—	—
	72	730	582	54	730	500	54	730	418	54	730	336	54	730	254	54	—	—	—	—	—	—
	67	673	663	53	671	593	53	671	511	53	671	429	53	671	347	53	671	265	53	—	—	—
	62	673	668	53	643	636	53	616	602	53	616	520	53	616	437	53	616	355	53	616	273	53
	57	673	672	53	643	640	53	612	608	53	583	577	53	566	526	52	566	444	52	566	362	52
19,000	77	829	564	55	829	462	55	829	359	55	829	256	55	—	—	—	—	—	—	—	—	—
	72	766	682	54	766	578	54	766	475	54	766	372	54	766	269	54	—	—	—	—	—	—
	67	738	727	54	705	694	54	705	590	54	705	487	54	705	384	54	705	281	54	—	—	—
	62	738	732	54	703	695	54	668	659	53	648	600	53	648	496	53	648	393	53	648	290	53
	57	738	736	54	703	699	54	668	663	53	634	628	53	601	594	53	595	504	53	595	400	53
21,000	77	843	603	55	843	489	55	843	376	55	843	263	55	—	—	—	—	—	—	—	—	—
	72	779	731	54	779	617	54	779	503	54	779	390	54	779	276	54	—	—	—	—	—	—
	67	765	753	54	727	715	54	718	629	54	718	515	54	718	401	54	718	288	54	—	—	—
	62	765	758	54	727	720	54	691	682	54	660	639	53	660	525	53	660	411	53	660	297	53
	57	765	763	54	727	724	54	691	686	54	655	649	53	620	613	53	606	533	53	606	419	53
24,000	77	859	660	55	859	531	55	859	402	55	859	273	55	—	—	—	—	—	—	—	—	—
	72	800	783	55	795	675	55	795	545	55	795	416	55	795	287	55	—	—	—	—	—	—
	67	800	789	55	760	747	54	733	688	54	733	558	54	733	428	54	733	299	54	—	—	—
	62	800	794	55	760	752	54	721	712	54	683	673	53	674	568	53	674	438	53	674	308	53
	57	800	798	55	760	757	54	721	716	54	683	677	53	646	639	53	619	576	53	619	446	53

Table 15: 60 ton high efficiency - 105°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	724	419	60	724	353	60	724	288	60	724	222	60	—	—	—	—	—	—	—	—	—
	72	665	496	59	665	430	59	665	364	59	665	298	59	665	232	59	—	—	—	—	—	—
	67	610	572	59	610	506	59	610	439	59	610	373	59	610	307	59	610	241	59	—	—	—
	62	596	591	59	569	563	58	559	513	58	559	446	58	559	380	58	559	314	58	559	248	58
	57	596	594	59	569	566	58	543	539	58	517	512	58	512	452	58	512	386	58	512	320	58
15,000	77	761	478	61	761	396	61	761	315	61	761	233	61	—	—	—	—	—	—	—	—	—
	72	701	572	60	701	490	60	701	408	60	701	326	60	701	245	60	—	—	—	—	—	—
	67	654	644	59	645	583	59	645	501	59	645	418	59	645	336	59	645	255	59	—	—	—
	62	654	648	59	623	617	59	593	586	59	592	509	59	592	427	59	592	344	59	592	263	59
	57	654	652	59	623	620	59	593	589	59	564	559	58	543	515	58	543	433	58	543	351	58
19,000	77	795	555	61	795	452	61	795	350	61	795	247	61	—	—	—	—	—	—	—	—	—
	72	734	672	60	734	568	60	734	465	60	734	362	60	734	259	60	—	—	—	—	—	—
	67	715	704	60	680	668	60	676	580	60	676	476	60	676	373	60	676	270	60	—	—	—
	62	715	709	60	680	673	60	646	638	59	621	589	59	621	485	59	621	382	59	621	279	59
	57	715	713	60	680	677	60	646	641	59	613	607	59	580	574	58	570	492	58	570	389	58
21,000	77	807	593	61	807	480	61	807	367	61	807	254	61	—	—	—	—	—	—	—	—	—
	72	746	721	60	746	607	60	746	493	60	746	380	60	746	266	60	—	—	—	—	—	—
	67	740	729	60	703	691	60	687	619	60	687	505	60	687	391	60	687	277	60	—	—	—
	62	740	734	60	703	696	60	667	659	59	632	623	59	631	514	59	631	400	59	631	286	59
	57	740	738	60	703	700	60	667	663	59	632	627	59	598	592	59	579	521	58	579	407	58
24,000	77	822	650	61	822	521	61	822	392	61	822	263	61	—	—	—	—	—	—	—	—	—
	72	774	757	61	761	665	61	761	535	61	761	406	61	761	277	61	—	—	—	—	—	—
	67	774	762	61	734	722	60	701	677	60	701	547	60	701	417	60	701	288	60	—	—	—
	62	774	767	61	734	726	60	696	687	60	659	649	59	645	557	59	645	427	59	645	297	59
	57	774	772	61	734	731	60	696	691	60	659	653	59	622	615	59	591	564	59	591	434	59

Table 16: 60 ton high efficiency - 115°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	694	410	67	694	344	67	694	278	67	694	213	67	—	—	—	—	—	—	—	—	—
	72	637	486	66	637	420	66	637	354	66	637	288	66	637	222	66	—	—	—	—	—	—
	67	584	562	65	584	495	65	584	429	65	584	363	65	584	297	65	584	231	65	—	—	—
	62	576	571	65	550	544	64	535	502	64	535	436	64	535	369	64	535	303	64	535	237	64
	57	576	575	65	550	547	64	524	520	64	499	494	64	490	441	63	490	375	63	490	308	63
15,000	77	728	468	67	728	386	67	728	305	67	728	223	67	—	—	—	—	—	—	—	—	—
	72	670	562	66	670	480	66	670	398	66	670	316	66	670	234	66	—	—	—	—	—	—
	67	631	622	66	616	572	65	616	490	65	616	407	65	616	325	65	616	244	65	—	—	—
	62	631	626	66	602	595	65	572	565	65	565	497	65	565	415	65	565	333	65	565	251	65
	57	631	630	66	602	599	65	572	568	65	544	539	64	518	504	64	518	421	64	518	339	64
19,000	77	758	545	68	758	442	68	758	339	68	758	237	68	—	—	—	—	—	—	—	—	—
	72	700	661	67	700	558	67	700	454	67	700	351	67	700	249	67	—	—	—	—	—	—
	67	689	679	66	655	644	66	644	568	66	644	465	66	644	362	66	644	259	66	—	—	—
	62	689	683	66	655	648	66	622	614	65	591	577	65	591	473	65	591	370	65	591	267	65
	57	689	687	66	655	652	66	622	618	65	589	584	65	558	552	64	542	480	64	542	377	64
21,000	77	769	583	68	769	470	68	769	356	68	769	243	68	—	—	—	—	—	—	—	—	—
	72	713	697	67	710	596	67	710	482	67	710	369	67	710	255	67	—	—	—	—	—	—
	67	713	702	67	677	665	66	654	607	66	654	493	66	654	379	66	654	266	66	—	—	—
	62	713	707	67	677	670	66	642	634	66	608	599	65	601	502	65	601	388	65	601	274	65
	57	713	711	67	677	674	66	642	638	66	608	602	65	575	568	65	551	509	64	551	394	64
24,000	77	782	640	68	782	511	68	782	382	68	782	253	68	—	—	—	—	—	—	—	—	—
	72	744	728	67	723	654	67	723	524	67	723	395	67	723	266	67	—	—	—	—	—	—
	67	744	733	67	706	694	67	668	656	66	666	535	66	666	405	66	666	276	66	—	—	—
	62	744	738	67	706	698	67	668	660	66	632	623	66	612	544	65	612	414	65	612	285	65
	57	744	742	67	706	703	67	668	664	66	632	627	66	597	590	65	563	555	65	561	421	65

Notes:

Rated performance is at sea level. Cooling capacities are gross cooling capacities.

CFM = airflow, EWB = entering wet bulb air, EDB = entering dry bulb air, TMBH = total cooling capacity (MBH)

SMBH = sensible cooling capacity (MBH), kW = total input

Cooling performance data 60 ton - high capacity, standard efficiency

Table 17: 60 ton high capacity - 75°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	854	463	49	854	397	49	854	331	49	854	265	49	—	—	—	—	—	—	—	—	—
	72	782	541	48	782	475	48	782	408	48	782	343	48	782	277	48	—	—	—	—	—	—
	67	715	617	47	715	551	47	715	484	47	715	418	47	715	352	47	715	286	47	—	—	—
	62	670	664	47	654	625	47	654	558	47	654	492	47	654	425	47	654	359	47	654	293	47
	57	670	668	47	640	637	47	611	607	46	—	—	—	—	—	—	—	—	—	—	—	—
15,000	77	905	522	49	905	440	49	905	359	49	905	277	49	—	—	—	—	—	—	—	—	—
	72	831	618	48	831	536	48	831	454	48	831	372	48	831	290	48	—	—	—	—	—	—
	67	762	712	48	762	630	48	762	547	48	762	465	48	762	383	48	762	301	48	—	—	—
	62	741	735	48	707	699	47	698	639	47	698	556	47	698	474	47	698	392	47	698	310	47
	57	741	739	48	707	704	47	674	669	47	641	635	47	639	563	47	639	480	47	639	398	47
19,000	77	950	599	50	950	496	50	950	393	50	950	291	50	—	—	—	—	—	—	—	—	—
	72	875	718	49	875	614	49	875	511	49	875	408	49	875	305	49	—	—	—	—	—	—
	67	815	803	48	804	731	48	804	627	48	804	523	48	804	420	48	804	317	48	—	—	—
	62	815	809	48	777	768	48	738	729	48	738	637	48	738	534	48	738	430	48	738	327	48
	57	815	813	48	777	773	48	738	733	48	701	695	47	676	645	47	676	541	47	676	438	47
21,000	77	966	637	50	966	523	50	966	410	50	966	297	50	—	—	—	—	—	—	—	—	—
	72	890	767	49	890	652	49	890	539	49	890	425	49	890	312	49	—	—	—	—	—	—
	67	846	833	49	819	780	48	819	666	48	819	552	48	819	438	48	819	324	48	—	—	—
	62	846	839	49	805	796	48	765	755	48	752	677	48	752	562	48	752	448	48	752	335	48
	57	846	844	49	805	801	48	765	760	48	726	719	47	689	685	47	689	570	47	689	456	47
24,000	77	984	694	50	984	564	50	984	435	50	984	306	50	—	—	—	—	—	—	—	—	—
	72	909	840	49	909	710	49	909	580	49	909	451	49	909	322	49	—	—	—	—	—	—
	67	885	872	49	842	828	49	837	724	49	837	594	49	837	464	49	837	335	49	—	—	—
	62	885	878	49	842	833	49	799	789	48	770	736	48	770	605	48	770	476	48	770	346	48
	57	885	883	49	842	838	49	799	794	48	757	751	48	717	709	47	706	614	47	706	484	47

Table 18: 60 ton high capacity - 85°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	823	452	55	823	386	55	823	320	55	823	255	55	—	—	—	—	—	—	—	—	—
	72	754	530	54	754	464	54	754	397	54	754	332	54	754	266	54	—	—	—	—	—	—
	67	690	606	53	690	539	53	690	473	53	690	407	53	690	341	53	690	275	53	—	—	—
	62	651	646	52	631	614	52	631	547	52	631	480	52	631	414	52	631	348	52	631	282	52
	57	651	650	52	622	619	52	593	589	52	577	552	51	577	486	51	577	419	51	577	353	51
15,000	77	870	511	55	870	429	55	870	347	55	870	266	55	—	—	—	—	—	—	—	—	—
	72	799	606	54	799	524	54	799	442	54	799	360	54	799	279	54	—	—	—	—	—	—
	67	733	700	53	733	618	53	733	535	53	733	453	53	733	371	53	733	289	53	—	—	—
	62	719	713	53	686	679	53	672	627	53	672	544	53	672	462	53	672	380	53	672	298	53
	57	719	718	53	686	683	53	653	649	52	621	615	52	615	551	52	615	468	52	615	386	52
19,000	77	911	588	56	911	485	56	911	382	56	911	280	56	—	—	—	—	—	—	—	—	—
	72	839	706	55	839	602	55	839	499	55	839	396	55	839	293	55	—	—	—	—	—	—
	67	790	778	54	772	719	54	772	615	54	772	511	54	772	408	54	772	305	54	—	—	—
	62	790	783	54	752	744	54	715	706	53	708	625	53	708	521	53	708	418	53	708	315	53
	57	790	788	54	752	748	54	715	710	53	678	672	53	649	633	52	649	529	52	649	425	52
21,000	77	926	626	56	926	512	56	926	399	56	926	286	56	—	—	—	—	—	—	—	—	—
	72	854	755	55	854	641	55	854	527	55	854	413	55	854	300	55	—	—	—	—	—	—
	67	819	806	54	786	768	54	786	654	54	786	540	54	786	426	54	786	312	54	—	—	—
	62	819	812	54	779	771	54	740	730	53	721	664	53	721	550	53	721	436	53	721	322	53
	57	819	817	54	779	775	54	740	735	53	701	695	53	664	656	52	661	558	52	661	444	52
24,000	77	943	682	56	943	553	56	943	424	56	943	295	56	—	—	—	—	—	—	—	—	—
	72	871	828	55	871	698	55	871	568	55	871	439	55	871	310	55	—	—	—	—	—	—
	67	856	843	55	814	800	54	802	712	54	802	582	54	802	452	54	802	323	54	—	—	—
	62	856	849	55	814	805	54	772	762	54	738	723	53	738	593	53	738	463	53	738	333	53
	57	856	854	55	814	810	54	772	767	54	732	725	53	692	684	53	677	601	53	677	471	53

Table 19: 60 ton high capacity - 95°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	791	441	61	791	375	61	791	309	61	791	244	61	—	—	—	—	—	—	—	—	—
	72	725	518	60	725	452	60	725	386	60	725	320	60	725	254	60	—	—	—	—	—	—
	67	663	594	59	663	528	59	663	462	59	663	395	59	663	329	59	663	263	59	—	—	—
	62	632	626	58	606	602	58	606	535	58	606	469	58	606	402	58	606	336	58	606	270	58
	57	632	630	58	603	600	58	575	571	57	555	540	57	555	474	57	555	408	57	555	341	57
15,000	77	834	500	61	834	418	61	834	336	61	834	255	61	—	—	—	—	—	—	—	—	—
	72	766	595	60	766	513	60	766	431	60	766	349	60	766	267	60	—	—	—	—	—	—
	67	703	688	59	703	606	59	703	523	59	703	441	59	703	359	59	703	277	59	—	—	—
	62	696	690	59	664	657	59	644	615	58	644	532	58	644	450	58	644	368	58	644	286	58
	57	696	694	59	664	661	59	632	628	58	601	595	58	590	538	57	590	456	57	590	374	57
19,000	77	871	576	62	871	473	62	871	371	62	871	268	62	—	—	—	—	—	—	—	—	—
	72	803	694	61	803	590	61	803	487	61	803	384	61	803	281	61	—	—	—	—	—	—
	67	763	752	60	738	706	60	738	602	60	738	499	60	738	396	60	738	293	60	—	—	—
	62	763	756	60	726	718	60	690	681	59	678	612	59	678	508	59	678	405	59	678	302	59
	57	763	761	60	726	723	60	690	685	59	655	649	58	621	619	58	621	516	58	621	412	58
21,000	77	884	614	62	884	501	62	884	387	62	884	274	62	—	—	—	—	—	—	—	—	—
	72	816	743	61	816	629	61	816	515	61	816	401	61	816	288	61	—	—	—	—	—	—
	67	790	778	61	751	738	60	751	641	60	751	527	60	751	413	60	751	300	60	—	—	—
	62	790	783	61	751	743	60	713	704	59	689	651	59	689	537	59	689	423	59	689	309	59
	57	790	788	61	751	748	60	713	709	59	676	670	59	640	633	58	632	544	58	632	430	58
24,000	77	899	671	63	899	541	63	899	412	63	899	284	63	—	—	—	—	—	—	—	—	—
	72	831	816	61	831	686	61	831	556	61	831	427	61	831	298	61	—	—	—	—	—	—
	67	825	812	61	784	771	61	766	699	60	766	569	60	766	439	60	766	310	60	—	—	—
	62	825	818	61	784	776	61	744	734	60	705	694	59	704	580	59	704	450	59	704	320	59
	57	825	823	61	784	780	61	744	739	60	705	698	59	666	659	59	646	587	58	646	457	58

Table 20: 60 ton high capacity - 105°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
12,000	77	756	429	67	756	364	67	756	298	67	756	232	67	—	—	—	—	—	—	—	—	—
	72	693	507	66	693	440	66	693	374	66	693	308	66	693	243	66	—	—	—	—	—	—
	67	635	582	65	635	516	65	635	449	65	635	383	65	635	317	65	635	251	65	—	—	—
	62	611	606	65	583	577	64	580	523	64	580	456	64	580	390	64	580	324	64	580	258	64
	57	611	610	65	583	580	64	556	552	63	531	528	63	531	462	63	531	395	63	531	329	63
15,000	77	795	488	68	795	406	68	795	325	68	795	243	68	—	—	—	—	—	—	—	—	—
	72	731	583	67	731	500	67	731	418	67	731	336	67	731	255	67	—	—	—	—	—	—
	67	672	662	66	671	593	66	671	511	66	671	429	66	671	347	66	671	265	66	—	—	—
	62	672	666	66	640	633	65	615	602	65	615	519	65	615	437	65	615	355	65	615	273	65
	57	672	670	66	640	637	65	609	605	64	578	573	64	563	525	64	563	443	64	563	361	64
19,000	77	829	564	69	829	461	69	829	359	69	829	256	69	—	—	—	—	—	—	—	—	—
	72	764	681	68	764	578	68	764	475	68	764	372	68	764	269	68	—	—	—	—	—	—
	67	734	723	67	703	693	66	703	589	66	703	486	66	703	383	66	703	280	66	—	—	—
	62	734	728	67	699	691	66	664	655	66	645	599	65	645	495	65	645	392	65	645	289	65
	57	734	732	67	699	695	66	664	659	66	629	624	65	595	589	64	592	502	64	592	399	64
21,000	77	840	602	69	840	489	69	840	375	69	840	263	69	—	—	—	—	—	—	—	—	—
	72	776	730	68	776	616	68	776	502	68	776	389	68	776	276	68	—	—	—	—	—	—
	67	759	748	68	722	710	67	714	628	67	714	514	67	714	400	67	714	287	67	—	—	—
	62	759	753	68	722	715	67	685	677	66	656	638	65	656	524	65	656	410	65	656	296	65
	57	759	758	68	722	719	67	685	681	66	649	644	65	614	607	65	601	531	64	601	417	64
24,000	77	854	659	69	854	529	69	854	401	69	854	272	69	—	—	—	—	—	—	—	—	—
	72	792	774	68	790	673	68	790	544	68	790	414	68	790	285	68	—	—	—	—	—	—
	67	792	780	68	752	740	67	728	686	67	728	556	67	728	426	67	728	297	67	—	—	—
	62	792	785	68	752	744	67	714	705	67	676	666	66	669	566	66	669	436	66	669	307	66
	57	792	790	68	752	749	67	714	709	67	676	670	66	639	631	65	614	574	65	614	444	65

Table 21: 60 ton high capacity - 115°F ambient air temperature

CFM	EWB	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
	(°F)	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	720	418	75	720	352	75	720	286	75	720	221	75	—	—	—	—	—	—	—	—	—
	72	660	494	73	660	428	73	660	362	73	660	296	73	660	231	73	—	—	—	—	—	—
12,000	67	605	570	72	605	503	72	605	437	72	605	371	72	605	305	72	605	239	72	—	—	—
	62	588	583	72	561	556	71	553	510	71	553	444	71	553	378	71	553	312	71	553	246	71
	57	588	587	72	561	559	71	535	531	70	509	504	70	506	449	70	506	383	70	506	316	70
	77	754	476	75	754	394	75	754	313	75	754	231	75	—	—	—	—	—	—	—	—	—
	72	694	570	74	694	488	74	694	406	74	694	324	74	694	242	74	—	—	—	—	—	—
15,000	67	645	636	73	638	580	73	638	498	73	638	416	73	638	334	73	638	252	73	—	—	—
	62	645	640	73	615	608	72	585	577	71	585	506	71	585	424	71	585	342	71	585	260	71
	57	645	644	73	615	612	72	585	581	71	555	550	71	536	512	70	536	430	70	536	347	70
	77	784	552	76	784	449	76	784	347	76	784	244	76	—	—	—	—	—	—	—	—	—
	72	724	668	75	724	565	75	724	462	75	724	359	75	724	256	75	—	—	—	—	—	—
19,000	67	704	694	74	669	658	74	666	576	73	666	473	73	666	370	73	666	267	73	—	—	—
	62	704	698	74	669	662	74	635	627	73	612	585	72	612	482	72	612	378	72	612	275	72
	57	704	702	74	669	666	74	635	631	73	602	597	72	570	563	71	561	488	71	561	385	71
	77	794	590	76	794	476	76	794	363	76	794	250	76	—	—	—	—	—	—	—	—	—
	72	734	717	75	734	603	75	734	489	75	734	376	75	734	263	75	—	—	—	—	—	—
21,000	67	728	717	75	691	679	74	676	615	74	676	501	74	676	387	74	676	273	74	—	—	—
	62	728	722	75	691	684	74	655	647	73	621	624	72	621	510	72	621	396	72	621	282	72
	57	728	726	75	691	688	74	655	651	73	621	615	72	587	580	72	570	517	71	570	403	71
	77	807	646	77	807	517	77	807	388	77	807	260	77	—	—	—	—	—	—	—	—	—
	72	758	742	76	746	660	75	746	531	75	746	401	75	746	272	75	—	—	—	—	—	—
24,000	67	758	747	76	719	707	75	688	672	74	688	542	74	688	413	74	688	283	74	—	—	—
	62	758	752	76	719	712	75	682	673	74	645	635	73	633	552	73	633	422	73	633	293	73
	57	758	756	76	719	716	75	682	677	74	645	639	73	609	602	72	580	559	71	580	429	71

Notes:

Rated performance is at sea level. Cooling capacities are gross cooling capacities.

CFM = airflow, EWB = entering wet bulb air, EDB = entering dry bulb air, TMBH = total cooling capacity (MBH)

SMBH = sensible cooling capacity (MBH), kW = total input

Cooling performance data 70 ton - standard capacity, standard efficiency

Table 22: 70 ton standard efficiency - 75°F ambient air temperature

CFM	EWB	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
	(°F)	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	964	530	51	964	452	51	964	375	51	964	299	51	—	—	—	—	—	—	—	—	—
	72	882	620	51	882	543	51	882	465	51	882	388	51	882	311	51	—	—	—	—	—	—
14,000	67	806	709	51	806	631	51	806	553	51	806	476	51	806	399	51	806	321	51	—	—	—
	62	759	752	51	736	717	51	736	639	51	736	561	51	736	484	51	736	406	51	736	329	51
	57	759	757	51	724	721	51	690	685	51	672	645	51	672	567	51	672	489	51	672	412	51
	77	1,035	619	51	1035	518	51	1035	417	51	1,035	317	51	—	—	—	—	—	—	—	—	—
	72	950	736	51	950	635	51	950	533	51	950	432	51	950	331	51	—	—	—	—	—	—
18,500	67	870	851	51	870	749	51	870	647	51	870	546	51	870	444	51	870	343	51	—	—	—
	62	859	852	51	818	810	51	796	760	51	796	658	51	796	556	51	796	454	51	796	353	51
	57	859	857	51	818	814	51	778	773	51	739	732	51	728	665	51	728	563	51	728	461	51
	77	1,080	706	51	1,080	581	51	1,080	457	51	1,080	332	51	—	—	—	—	—	—	—	—	—
	72	993	848	51	993	723	51	993	597	51	993	472	51	993	348	51	—	—	—	—	—	—
23,000	67	936	922	51	911	863	51	911	737	51	911	611	51	911	486	51	911	361	51	—	—	—
	62	936	928	51	890	881	51	845	834	51	834	748	51	834	622	51	834	497	51	834	372	51
	57	936	934	51	890	886	51	845	839	51	801	794	51	763	756	51	763	630	51	763	505	51
	77	1,110	791	51	1,110	643	51	1,110	494	51	1,110	346	51	—	—	—	—	—	—	—	—	—
	72	1,022	958	51	1,022	809	51	1,022	660	51	1,022	511	51	1,022	363	51	—	—	—	—	—	—
27,500	67	997	982	51	947	930	51	939	824	51	939	674	51	939	525	51	939	377	51	—	—	—
	62	997	988	51	947	937	51	897	886	51	860	836	51	860	686	51	860	537	51	860	388	51
	57	997	994	51	947	942	51	897	891	51	849	841	51	802	793	51	786	695	51	786	545	51
	77	1,133	877	51	1,133	704	51	1,133	532	51	1,133	361	51	—	—	—	—	—	—	—	—	—
	72	1,047	1,024	51	1,046	896	51	1,046	723	51	1,046	550	51	1,046	378	51	—	—	—	—	—	—
32,000	67	1,047	1,032	51	994	977	51	963	912	51	963	739	51	963	566	51	963	393	51	—	—	—
	62	1,047	1,038	51	994	983	51	941	929	51	890	877	50	884	751	51	884	578	51	884	405	51
	57	1,047	1,045	51	994	989	51	941	935	51	890	882	50	840	831	51	809	761	51	809	587	51

Table 23: 70 ton standard efficiency - 85°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	930	518	57	930	441	57	930	364	57	930	287	57	—	—	—	—	—	—	—	—	—
	72	851	608	57	851	531	57	851	454	57	851	376	57	851	299	57	—	—	—	—	—	—
14,000	67	778	697	57	778	619	57	778	541	57	778	464	57	778	387	57	778	309	57	—	—	—
	62	738	732	57	711	705	57	711	627	57	711	549	57	711	472	57	711	394	57	711	317	57
	57	738	736	57	704	701	57	671	666	56	649	633	56	649	555	56	649	477	56	649	400	56
	77	996	607	57	996	506	57	996	405	57	996	305	57	—	—	—	—	—	—	—	—	—
	72	914	724	57	914	622	57	914	521	57	914	420	57	914	319	57	—	—	—	—	—	—
18,500	67	838	839	57	838	737	57	838	635	57	838	533	57	838	432	57	838	331	57	—	—	—
	62	834	827	57	794	786	57	767	746	57	767	644	57	767	543	57	767	441	57	767	340	57
	57	834	832	57	794	791	57	755	750	57	716	710	57	700	651	56	700	550	56	700	448	56
	77	1,038	694	57	1,038	569	57	1,038	445	57	1,038	320	57	—	—	—	—	—	—	—	—	—
	72	955	836	57	955	710	57	955	585	57	955	460	57	955	335	57	—	—	—	—	—	—
23,000	67	908	894	57	876	849	57	876	724	57	876	598	57	876	473	57	876	348	57	—	—	—
	62	908	900	57	863	854	57	819	808	57	802	734	57	802	609	57	802	483	57	802	358	57
	57	908	906	57	863	859	57	819	813	57	775	768	57	733	725	57	733	616	57	733	491	57
	77	1,065	779	57	1,065	630	57	1,065	482	57	1,065	334	57	—	—	—	—	—	—	—	—	—
	72	981	946	57	981	796	57	981	647	57	981	498	57	981	350	57	—	—	—	—	—	—
27,500	67	965	951	57	916	901	57	901	811	57	901	661	57	901	512	57	901	364	57	—	—	—
	62	965	957	57	916	907	57	868	857	57	826	822	57	826	673	57	826	523	57	826	374	57
	57	965	963	57	916	912	57	868	862	57	821	813	57	775	766	57	754	681	57	754	531	57
	77	1,086	865	57	1,086	692	57	1,086	520	57	1,086	349	57	—	—	—	—	—	—	—	—	—
	72	1,013	991	57	1,003	883	57	1,003	710	57	1,003	538	57	1,003	365	57	—	—	—	—	—	—
32,000	67	1,013	998	57	961	944	57	923	898	57	923	725	57	923	552	57	923	380	57	—	—	—
	62	1,013	1,004	57	961	950	57	910	898	57	860	847	57	848	737	57	848	564	57	848	391	57
	57	1,013	1,010	57	961	956	57	910	903	57	860	852	57	811	802	57	775	746	57	775	573	57

Table 24: 70 ton standard efficiency - 95°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	895	506	64	895	429	64	895	352	64	895	275	64	—	—	—	—	—	—	—	—	—
	72	819	596	63	819	519	63	819	441	63	819	364	63	819	287	63	—	—	—	—	—	—
14,000	67	748	684	63	748	606	63	748	529	63	748	451	63	748	374	63	748	297	63	—	—	—
	62	716	710	63	683	676	63	683	614	63	683	536	63	683	458	63	683	381	63	683	304	63
	57	716	714	63	683	680	63	650	646	63	623	619	63	623	541	63	623	464	63	623	386	63
	77	955	595	64	955	494	64	955	393	64	955	293	64	—	—	—	—	—	—	—	—	—
	72	877	711	64	877	609	64	877	508	64	877	407	64	877	306	64	—	—	—	—	—	—
18,500	67	808	796	63	803	723	63	803	621	63	803	520	63	803	418	63	803	317	63	—	—	—
	62	808	801	63	769	761	63	735	732	63	735	631	63	735	529	63	735	427	63	735	326	63
	57	808	806	63	769	765	63	730	725	63	692	686	63	671	637	63	671	535	63	671	434	63
	77	993	681	64	993	557	64	993	432	64	993	308	64	—	—	—	—	—	—	—	—	—
	72	913	822	64	913	697	64	913	572	64	913	447	64	913	322	64	—	—	—	—	—	—
23,000	67	878	864	64	838	836	64	838	710	64	838	584	64	838	459	64	838	334	64	—	—	—
	62	878	870	64	833	825	63	790	780	63	767	720	63	767	594	63	767	469	63	767	344	63
	57	878	875	64	833	830	63	790	785	63	748	741	63	706	698	63	701	602	63	701	476	63
	77	1,017	766	64	1,017	618	64	1,017	470	64	1,017	322	64	—	—	—	—	—	—	—	—	—
	72	937	932	64	937	783	64	937	634	64	937	485	64	937	337	64	—	—	—	—	—	—
27,500	67	931	918	64	883	868	64	861	797	64	861	647	64	861	498	64	861	349	64	—	—	—
	62	931	924	64	883	874	64	837	826	63	791	779	63	789	658	63	789	509	63	789	360	63
	57	931	929	64	883	879	64	837	831	63	791	783	63	746	737	63	721	666	63	721	516	63
	77	1,036	852	65	1,036	680	65	1,036	508	65	1,036	336	65	—	—	—	—	—	—	—	—	—
	72	976	955	64	957	869	64	957	696	64	957	524	64	957	352	64	—	—	—	—	—	—
32,000	67	976	962	64	925	910	64	881	884	64	881	711	64	881	538	64	881	366	64	—	—	—
	62	976	968	64	925	916	64	876	865	64	827	815	63	809	723	63	809	550	63	809	377	63
	57	976	974	64	925	921	64	876	870	64	827	820	63	780	771	63	740	731	63	740	558	63

Table 25: 70 ton standard efficiency - 105°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	857	494	71	857	417	71	857	340	71	857	263	71	—	—	—	—	—	—	—	—	—
	72	784	583	70	784	506	70	784	428	70	784	351	70	784	274	70	—	—	—	—	—	—
14,000	67	716	671	70	716	593	70	716	515	70	716	438	70	716	360	70	716	283	70	—	—	—
	62	693	687	70	660	653	70	653	600	70	653	522	70	653	445	70	653	367	70	653	290	70
	57	693	691	70	660	657	70	628	624	69	596	591	69	596	527	69	596	450	69	596	372	69
	77	912	582	71	912	481	71	912	381	71	912	280	71	—	—	—	—	—	—	—	—	—
	72	837	698	71	837	596	71	837	495	71	837	394	71	837	293	71	—	—	—	—	—	—
18,500	67	780	768	70	767	709	70	767	607	70	767	506	70	767	404	70	767	303	70	—	—	—
	62	780	773	70	741	733	70	704	695	70	701	616	70	701	514	70	701	413	70	701	312	70
	57	780	778	70	741	738	70	704	699	70	667	661	70	640	623	69	640	521	69	640	419	69
	77	946	669	72	946	544	72	946	419	72	946	295	72	—	—	—	—	—	—	—	—	—
	72	870	809	71	870	683	71	870	558	71	870	433	71	870	309	71	—	—	—	—	—	—
23,000	67	845	833	71	802	789	71	798	696	71	798	570	71	798	445	71	798	320	71	—	—	—
	62	845	838	71	802	794	71	760	750	70	730	705	70	730	580	70	730	454	70	730	329	70
	57	845	843	71	802	799	71	760	755	70	719	712	70	678	671	70	667	587	70	667	461	70
	77	968	753	72	968	605	72	968	457	72	968	309	72	—	—	—	—	—	—	—	—	—
	72	896	876	71	891	769	71	891	620	71	891	471	71	891	323	71	—	—	—	—	—	—
27,500	67	896	883	71	849	835	71	819	782	71	819	633	71	819	484	71	819	335	71	—	—	—
	62	896	889	71	849	840	71	804	793	71	759	748	70	750	643	70	750	494	70	750	345	70
	57	896	894	71	849	845	71	804	798	71	759	752	70	715	707	70	685	650	70	685	501	70
	77	985	839	72	985	667	72	985	495	72	985	323	72	—	—	—	—	—	—	—	—	—
	72	938	917	72	910	856	71	910	683	71	910	510	71	910	338	71	—	—	—	—	—	—
32,000	67	938	924	72	888	873	71	840	824	71	837	696	71	837	523	71	837	351	71	—	—	—
	62	938	930	72	888	879	71	840	829	71	793	781	70	768	707	70	768	534	70	768	361	70
	57	938	936	72	888	884	71	840	834	71	793	786	70	747	739	70	702	715	70	702	542	70

Table 26: 70 ton standard efficiency - 115°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	817	481	79	817	404	79	817	327	79	817	250	79	—	—	—	—	—	—	—	—	—
	72	747	570	78	747	492	78	747	415	78	747	338	78	747	261	78	—	—	—	—	—	—
14,000	67	682	657	77	682	579	77	682	501	77	682	424	77	682	347	77	682	269	77	—	—	—
	62	668	662	77	636	629	77	622	586	77	622	508	77	622	431	77	622	353	77	622	276	77
	57	668	666	77	636	633	77	604	600	77	573	568	76	567	513	76	567	435	76	567	358	76
	77	866	569	79	866	468	79	866	367	79	866	267	79	—	—	—	—	—	—	—	—	—
	72	795	684	79	795	582	79	795	481	79	795	380	79	795	279	79	—	—	—	—	—	—
18,500	67	749	738	78	728	695	78	728	593	78	728	491	78	728	390	78	728	289	78	—	—	—
	62	749	743	78	712	704	78	675	666	77	665	601	77	665	500	77	665	398	77	665	297	77
	57	749	747	78	712	708	78	675	670	77	639	633	77	607	607	77	607	506	77	607	404	77
	77	897	655	80	897	530	80	897	406	80	897	282	80	—	—	—	—	—	—	—	—	—
	72	824	795	79	824	669	79	824	544	79	824	419	79	824	294	79	—	—	—	—	—	—
23,000	67	811	799	79	769	756	78	756	681	78	756	555	78	756	430	78	756	305	78	—	—	—
	62	811	804	79	769	761	78	728	719	78	692	690	78	692	564	78	692	439	78	692	314	78
	57	811	809	79	769	765	78	728	723	78	688	681	77	648	641	77	631	571	77	631	445	77
	77	916	740	80	916	591	80	916	443	80	916	295	80	—	—	—	—	—	—	—	—	—
	72	858	839	79	843	755	79	843	606	79	843	457	79	843	309	79	—	—	—	—	—	—
27,500	67	858	846	79	813	799	79	774	767	78	774	618	78	774	469	78	774	320	78	—	—	—
	62	858	851	79	813	804	79	768	759	78	725	714	78	709	627	78	709	478	78	709	329	78
	57	858	856	79	813	809	79	768	763	78	725	718	78	682	675	77	647	634	77	647	485	77
	77	932	826	80	932	653	80	932	481	80	932	310	80	—	—	—	—	—	—	—	—	—
	72	897	878	80	860	841	79	860	668	79	860	496	79	860	324	79	—	—	—	—	—	—
32,000	67	897	884	80	849	835	79	802	787	79	791	681	78	791	508	78	791	336	78	—	—	—
	62	897	890	80	849	840	79	802	792	79	757	745	78	725	692	78	725	518	78	725	346	78
	57	897	895	80	849	845	79	802	797	79	757	750	78	712	704	78	669	660	77	663	526	77

Notes:

Rated performance is at sea level. Cooling capacities are gross cooling capacities.

CFM = airflow, EWB = entering wet bulb air, EDB = entering dry bulb air, TMBH = total cooling capacity (MBH)

SMBH = sensible cooling capacity (MBH), kW = total input

Cooling performance data 70 ton - standard capacity, high efficiency

Table 27: 70 ton high efficiency - 75°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	959	528	48	959	451	48	959	374	48	959	297	48	—	—	—	—	—	—	—	—	—
	72	877	618	48	877	541	48	877	464	48	877	386	48	877	309	48	—	—	—	—	—	—
	67	802	707	48	802	629	48	802	552	48	802	474	48	802	397	48	802	319	48	—	—	—
	62	759	753	48	732	715	48	732	637	48	732	560	48	732	482	48	732	404	48	732	327	48
	57	759	757	48	724	721	48	690	685	48	668	643	48	668	565	48	668	487	48	668	410	48
18,500	77	1,028	617	48	1,028	516	48	1,028	415	48	1,028	315	48	—	—	—	—	—	—	—	—	—
	72	944	734	48	944	633	48	944	531	48	944	430	48	944	329	48	—	—	—	—	—	—
	67	865	849	48	865	747	48	865	645	48	865	544	48	865	442	48	865	341	48	—	—	—
	62	860	852	48	818	810	48	791	757	48	791	655	48	791	553	48	791	452	48	791	351	48
	57	860	857	48	818	815	48	778	773	48	738	732	48	723	662	48	723	560	48	723	459	48
23,000	77	1,071	703	48	1,071	579	48	1,071	454	48	1,071	330	48	—	—	—	—	—	—	—	—	—
	72	985	846	48	985	720	48	985	595	48	985	470	48	985	345	48	—	—	—	—	—	—
	67	937	923	48	904	860	48	904	734	48	904	609	48	904	483	48	904	358	48	—	—	—
	62	937	929	48	890	881	48	844	834	48	828	745	48	828	619	48	828	494	48	828	369	48
	57	937	934	48	890	886	48	844	839	48	800	793	48	756	753	48	756	627	48	756	502	48
27,500	77	1,099	788	48	1,099	640	48	1,099	491	48	1,099	343	48	—	—	—	—	—	—	—	—	—
	72	1,012	955	48	1,012	806	48	1,012	657	48	1,012	508	48	1,012	360	48	—	—	—	—	—	—
	67	997	982	48	946	930	48	930	821	48	930	671	48	930	522	48	930	374	48	—	—	—
	62	997	988	48	946	936	48	896	885	48	852	833	48	852	683	48	852	534	48	852	385	48
	57	997	994	48	946	942	48	896	890	48	848	840	48	800	791	48	779	691	48	779	542	48
32,000	77	1,121	874	48	1,121	701	48	1,121	529	48	1,121	358	48	—	—	—	—	—	—	—	—	—
	72	1,047	1,024	48	1,035	893	48	1,035	720	48	1,035	547	48	1,035	375	48	—	—	—	—	—	—
	67	1,047	1,031	48	993	976	48	953	909	48	953	735	48	953	562	48	953	390	48	—	—	—
	62	1,047	1,038	48	993	982	48	940	928	48	889	876	48	875	748	48	875	575	48	875	402	48
	57	1,047	1,044	48	993	988	48	940	934	48	889	881	48	839	829	48	801	757	48	801	584	48

Table 28: 70 ton high efficiency - 85°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	926	516	54	926	439	54	926	362	54	926	286	54	—	—	—	—	—	—	—	—	—
	72	847	607	54	847	529	54	847	452	54	847	375	54	847	298	54	—	—	—	—	—	—
	67	773	695	54	773	617	54	773	539	54	773	462	54	773	384	54	773	307	54	—	—	—
	62	738	732	54	706	703	54	706	625	54	706	547	54	706	469	54	706	392	54	706	315	54
	57	738	736	54	705	701	54	671	667	54	644	630	54	644	552	54	644	475	54	644	397	54
18,500	77	990	605	54	990	504	54	990	404	54	990	303	54	—	—	—	—	—	—	—	—	—
	72	908	722	54	908	620	54	908	519	54	908	418	54	908	317	54	—	—	—	—	—	—
	67	835	822	54	832	734	54	832	632	54	832	531	54	832	429	54	832	328	54	—	—	—
	62	835	828	54	795	786	54	761	744	54	761	642	54	761	540	54	761	439	54	761	337	54
	57	835	832	54	795	791	54	756	750	54	717	710	54	695	649	54	695	547	54	695	445	54
23,000	77	1,029	692	54	1,029	567	54	1,029	442	54	1,029	318	54	—	—	—	—	—	—	—	—	—
	72	947	833	54	947	708	54	947	582	54	947	457	54	947	333	54	—	—	—	—	—	—
	67	908	894	54	868	847	54	868	721	54	868	595	54	868	470	54	868	345	54	—	—	—
	62	908	900	54	863	854	54	819	808	54	795	731	54	795	606	54	795	480	54	795	355	54
	57	908	905	54	863	859	54	819	813	54	775	768	54	733	725	54	726	613	54	726	488	54
27,500	77	1,054	776	54	1,054	628	54	1,054	479	54	1,054	332	54	—	—	—	—	—	—	—	—	—
	72	971	943	54	971	793	54	971	644	54	971	496	54	971	347	54	—	—	—	—	—	—
	67	965	950	54	916	900	54	892	808	54	892	658	54	892	509	54	892	360	54	—	—	—
	62	965	957	54	916	906	54	868	857	54	821	808	54	817	669	54	817	520	54	817	371	54
	57	965	962	54	916	912	54	868	862	54	821	813	54	774	766	54	747	677	54	747	528	54
32,000	77	1,075	862	54	1,075	690	54	1,075	518	54	1,075	346	54	—	—	—	—	—	—	—	—	—
	72	1,012	990	54	993	880	54	993	707	54	993	535	54	993	362	54	—	—	—	—	—	—
	67	1,012	997	54	960	944	54	914	895	54	914	722	54	914	549	54	914	377	54	—	—	—
	62	1,012	1,004	54	960	950	54	909	898	54	859	847	54	839	734	54	839	561	54	839	388	54
	57	1,012	1,010	54	960	956	54	909	903	54	859	852	54	811	801	54	768	743	54	768	570	54

Table 29: 70 ton high efficiency - 95°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	890	505	61	890	427	61	890	351	61	890	274	61	—	—	—	—	—	—	—	—	—
	72	814	594	60	814	517	60	814	439	60	814	362	60	814	285	60	—	—	—	—	—	—
	67	743	682	60	743	604	60	743	527	60	743	449	60	743	372	60	743	295	60	—	—	—
	62	716	710	60	683	676	60	678	612	60	678	534	60	678	456	60	678	379	60	678	302	60
	57	716	714	60	683	680	60	650	646	60	619	617	59	619	539	59	619	462	59	619	384	59
18,500	77	949	593	61	949	492	61	949	391	61	949	291	61	—	—	—	—	—	—	—	—	—
	72	871	709	60	871	607	60	871	506	60	871	405	60	871	304	60	—	—	—	—	—	—
	67	808	796	60	798	721	60	798	619	60	798	518	60	798	416	60	798	315	60	—	—	—
	62	808	801	60	769	761	60	730	721	60	729	628	60	729	527	60	729	425	60	729	324	60
	57	808	806	60	769	765	60	730	725	60	692	686	60	666	635	60	666	533	60	666	432	60
23,000	77	985	679	61	985	554	61	985	430	61	985	305	61	—	—	—	—	—	—	—	—	—
	72	906	820	61	906	694	61	906	569	61	906	444	61	906	320	61	—	—	—	—	—	—
	67	878	865	60	833	819	60	831	707	60	831	582	60	831	457	60	831	332	60	—	—	—
	62	878	870	60	833	825	60	790	780	60	760	717	60	760	592	60	760	466	60	760	341	60
	57	878	875	60	833	830	60	790	785	60	748	741	60	706	698	60	695	599	60	695	473	60
27,500	77	1,008	764	61	1,008	615	61	1,008	467	61	1,008	319	61	—	—	—	—	—	—	—	—	—
	72	932	911	61	928	780	61	928	631	61	928	482	61	928	334	61	—	—	—	—	—	—
	67	932	918	61	883	868	60	853	794	60	853	644	60	853	495	60	853	347	60	—	—	—
	62	932	924	61	883	874	60	836	826	60	790	779	60	781	655	60	781	506	60	781	357	60
	57	932	929	61	883	879	60	836	831	60	790	783	60	745	737	60	714	663	60	714	513	60
32,000	77	1,027	849	61	1,027	677	61	1,027	505	61	1,027	334	61	—	—	—	—	—	—	—	—	—
	72	976	955	61	948	867	61	948	694	61	948	521	61	948	349	61	—	—	—	—	—	—
	67	976	962	61	925	909	61	875	859	60	873	708	60	873	535	60	873	363	60	—	—	—
	62	976	968	61	925	915	61	875	864	60	827	814	60	801	720	60	801	547	60	801	374	60
	57	976	974	61	925	921	61	875	869	60	827	819	60	779	770	60	733	723	60	732	555	60

Table 30: 70 ton high efficiency - 105°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	852	492	67	852	415	67	852	338	67	852	261	67	—	—	—	—	—	—	—	—	—
	72	780	582	67	780	504	67	780	427	67	780	350	67	780	273	67	—	—	—	—	—	—
	67	712	669	66	712	591	66	712	514	66	712	436	66	712	359	66	712	282	66	—	—	—
	62	694	689	66	661	654	66	649	599	66	649	521	66	649	443	66	649	366	66	649	288	66
	57	694	693	66	661	658	66	628	624	66	596	591	66	592	526	66	592	448	66	592	371	66
18,500	77	906	581	68	906	479	68	906	379	68	906	278	68	—	—	—	—	—	—	—	—	—
	72	832	696	67	832	594	67	832	493	67	832	392	67	832	291	67	—	—	—	—	—	—
	67	781	769	67	762	707	67	762	605	67	762	504	67	762	403	67	762	301	67	—	—	—
	62	781	775	67	742	734	67	704	695	66	696	614	66	696	513	66	696	411	66	696	310	66
	57	781	779	67	742	739	67	704	699	66	667	661	66	636	621	66	636	519	66	636	417	66
23,000	77	938	666	68	938	542	68	938	417	68	938	293	68	—	—	—	—	—	—	—	—	—
	72	864	807	68	864	681	68	864	556	68	864	431	68	864	306	68	—	—	—	—	—	—
	67	846	834	67	803	789	67	792	694	67	792	568	67	792	443	67	792	318	67	—	—	—
	62	846	839	67	803	794	67	760	751	67	725	703	67	725	578	67	725	452	67	725	327	67
	57	846	844	67	803	799	67	760	755	67	719	712	67	678	671	66	662	584	66	662	459	66
27,500	77	959	751	68	959	603	68	959	454	68	959	307	68	—	—	—	—	—	—	—	—	—
	72	896	876	68	884	767	68	884	618	68	884	469	68	884	321	68	—	—	—	—	—	—
	67	896	883	68	849	835	67	813	780	67	813	631	67	813	482	67	813	333	67	—	—	—
	62	896	889	68	849	840	67	803	793	67	759	747	67	744	641	67	744	492	67	744	343	67
	57	896	894	68	849	845	67	803	798	67	759	752	67	715	707	66	679	648	66	679	498	66
32,000	77	976	837	68	976	664	68	976	492	68	976	321	68	—	—	—	—	—	—	—	—	—
	72	938	917	68	902	853	68	902	680	68	902	508	68	902	336	68	—	—	—	—	—	—
	67	938	924	68	889	873	68	840	824	67	831	694	67	831	521	67	831	349	67	—	—	—
	62	938	930	68	889	879	68	840	830	67	793	781	67	762	705	67	762	532	67	762	359	67
	57	938	936	68	889	884	68	840	834	67	793	786	67	747	739	67	702	693	66	695	539	66

Table 31: 70 ton high efficiency - 115°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	813	480	75	813	403	75	813	326	75	813	249	75	—	—	—	—	—	—	—	—	—
	72	743	568	74	743	491	74	743	414	74	743	337	74	743	260	74	—	—	—	—	—	—
14,000	67	679	656	74	679	578	74	679	500	74	679	423	74	679	345	74	679	268	74	—	—	—
	62	668	662	74	636	630	73	619	585	73	619	507	73	619	429	73	619	352	73	619	275	73
	57	668	666	74	636	633	73	605	601	73	574	569	73	564	512	73	564	434	73	564	357	73
	77	861	568	75	861	467	75	861	366	75	861	265	75	—	—	—	—	—	—	—	—	—
	72	790	682	75	790	581	75	790	479	75	790	378	75	790	277	75	—	—	—	—	—	—
18,500	67	750	739	74	724	693	74	724	591	74	724	490	74	724	388	74	724	287	74	—	—	—
	62	750	744	74	713	705	74	676	667	74	661	600	73	661	498	73	661	396	73	661	295	73
	57	750	748	74	713	709	74	676	671	74	640	634	73	604	597	73	604	504	73	604	402	73
	77	890	653	76	890	529	76	890	404	76	890	280	76	—	—	—	—	—	—	—	—	—
	72	819	793	75	819	667	75	819	542	75	819	417	75	819	293	75	—	—	—	—	—	—
23,000	67	812	800	75	770	757	74	751	679	74	751	554	74	751	428	74	751	303	74	—	—	—
	62	812	805	75	770	762	74	728	719	74	688	678	74	687	562	74	687	437	74	687	312	74
	57	812	810	75	770	766	74	728	724	74	688	682	74	649	641	73	627	569	73	627	443	73
	77	908	738	76	908	590	76	908	441	76	908	293	76	—	—	—	—	—	—	—	—	—
	72	859	840	75	837	753	75	837	604	75	837	455	75	837	307	75	—	—	—	—	—	—
27,500	67	859	846	75	813	800	75	769	765	74	769	616	74	769	467	74	769	318	74	—	—	—
	62	859	852	75	813	805	75	769	759	74	725	714	74	704	625	74	704	476	74	704	327	74
	57	859	857	75	813	810	75	769	764	74	725	719	74	683	675	74	642	632	73	642	483	73
	77	923	824	76	923	651	76	923	479	76	923	308	76	—	—	—	—	—	—	—	—	—
	72	898	878	76	853	839	75	853	666	75	853	494	75	853	322	75	—	—	—	—	—	—
32,000	67	898	884	76	849	835	75	802	787	75	785	679	75	785	506	75	785	334	75	—	—	—
	62	898	890	76	849	841	75	802	792	75	757	745	74	719	689	74	719	516	74	719	344	74
	57	898	896	76	849	845	75	802	797	75	757	750	74	712	704	74	669	660	74	656	523	73

Notes:

Rated performance is at sea level. Cooling capacities are gross cooling capacities.

CFM = airflow, EWB = entering wet bulb air, EDB = entering dry bulb air, TMBH = total cooling capacity (MBH)

SMBH = sensible cooling capacity (MBH), kW = total input

Cooling performance data 70 ton - high capacity, standard efficiency

Table 32: 70 ton high capacity - 75°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	1,018	548	55	1,018	471	55	1,018	394	55	1,018	317	55	—	—	—	—	—	—	—	—	—
	72	931	640	55	931	562	55	931	485	55	931	408	55	931	331	55	—	—	—	—	—	—
14,000	67	851	729	55	851	651	55	851	574	55	851	496	55	851	419	55	851	341	55	—	—	—
	62	792	786	54	778	738	55	778	660	55	778	582	55	778	504	55	778	427	55	778	350	55
	57	792	790	54	757	754	54	722	718	54	711	666	54	711	588	54	711	510	54	711	433	54
	77	1,098	639	56	1,098	538	56	1,098	437	56	1,098	336	56	—	—	—	—	—	—	—	—	—
	72	1,008	757	55	1,008	655	55	1,008	554	55	1,008	453	55	1,008	352	55	—	—	—	—	—	—
18,500	67	924	873	55	924	771	55	924	669	55	924	567	55	924	466	55	924	365	55	—	—	—
	62	901	894	55	859	850	55	846	782	55	846	680	55	846	578	55	846	477	55	846	375	55
	57	901	899	55	859	855	55	818	813	55	778	771	54	775	688	54	775	586	54	775	484	54
	77	1,150	726	56	1,150	601	56	1,150	477	56	1,150	352	56	—	—	—	—	—	—	—	—	—
	72	1,058	870	55	1,058	744	55	1,058	619	55	1,058	494	55	1,058	369	55	—	—	—	—	—	—
23,000	67	986	972	55	972	885	55	972	759	55	972	634	55	972	509	55	972	384	55	—	—	—
	62	986	978	55	939	929	55	893	881	55	891	772	55	891	646	55	891	520	55	891	395	55
	57	986	984	55	939	935	55	893	887	55	847	840	55	815	780	55	815	655	55	815	529	55
	77	1,183	811	56	1,183	663	56	1,183	514	56	1,183	366	56	—	—	—	—	—	—	—	—	—
	72	1,090	980	56	1,090	830	56	1,090	681	56	1,090	533	56	1,090	384	56	—	—	—	—	—	—
27,500	67	1,053	1,038	55	1,002	996	55	1,002	847	55	1,002	697	55	1,002	548	55	1,002	399	55	—	—	—
	62	1,053	1,045	55	1,001	991	55	950	938	55	919	859	55	919	710	55	919	561	55	919	412	55
	57	1,053	1,051	55	1,001	997	55	950	944	55	900	892	55	852	842	55	841	719	55	841	570	55
	77	1,206	896	56	1,206	724	56	1,206	552	56	1,206	380	56	—	—	—	—	—	—	—	—	—
	72	1,113	1,089	56	1,113	916	56	1,113	743	56	1,113	570	56	1,113	398	56	—	—	—	—	—	—
32,000	67	1,107	1,091	56	1,052	1,034	55	1,024	933	55	1,024	760	55	1,024	587	55	1,024	414	55	—	—	—
	62	1,107	1,098	56	1,052	1,040	55	997	984	55	943	929	55	940	773	55	940	600	55	940	427	55
	57	1,107	1,105	56	1,052	1,047	55	997	990	55	943	935	55	891	881	55	860	783	55	860	610	55

Table 33: 70 ton high capacity - 85°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	982	536	62	982	459	62	982	382	62	982	305	62	—	—	—	—	—	—	—	—	—
	72	899	627	61	899	549	61	899	472	61	899	395	61	899	318	61	—	—	—	—	—	—
	67	821	716	61	821	638	61	821	560	61	821	483	61	821	405	61	821	328	61	—	—	—
	62	772	766	61	750	724	61	750	646	61	750	568	61	750	491	61	750	413	61	750	336	61
	57	772	770	61	738	734	61	704	699	61	685	652	61	685	574	61	685	497	61	685	419	61
18,500	77	1,056	626	62	1,056	525	62	1,056	424	62	1,056	323	62	—	—	—	—	—	—	—	—	—
	72	970	743	62	970	642	62	970	540	62	970	439	62	970	338	62	—	—	—	—	—	—
	67	889	859	61	889	757	61	889	655	61	889	553	61	889	452	61	889	351	61	—	—	—
	62	876	869	61	835	827	61	814	768	61	814	666	61	814	564	61	814	462	61	814	361	61
	57	876	874	61	835	832	61	795	790	61	756	749	61	745	673	61	745	571	61	745	470	61
23,000	77	1,104	713	63	1,104	588	63	1,104	463	63	1,104	339	63	—	—	—	—	—	—	—	—	—
	72	1,016	856	62	1,016	730	62	1,016	605	62	1,016	480	62	1,016	355	62	—	—	—	—	—	—
	67	957	943	62	933	871	62	933	745	62	933	619	62	933	494	62	933	369	62	—	—	—
	62	957	949	62	911	901	61	866	855	61	855	757	61	855	631	61	855	505	61	855	380	61
	57	957	955	62	911	907	61	866	860	61	821	814	61	783	765	61	783	639	61	783	514	61
27,500	77	1,135	798	63	1,135	649	63	1,135	501	63	1,135	353	63	—	—	—	—	—	—	—	—	—
	72	1,046	966	62	1,046	816	62	1,046	667	62	1,046	518	62	1,046	370	62	—	—	—	—	—	—
	67	1,020	1,005	62	969	953	62	961	832	62	961	682	62	961	533	62	961	385	62	—	—	—
	62	1,020	1,012	62	969	959	62	920	908	62	881	844	61	881	695	61	881	545	61	881	396	61
	57	1,020	1,018	62	969	965	62	920	913	62	871	863	61	823	814	61	807	704	61	807	554	61
32,000	77	1,155	883	63	1,155	710	63	1,155	538	63	1,155	367	63	—	—	—	—	—	—	—	—	—
	72	1,071	1,047	62	1,066	902	62	1,066	729	62	1,066	556	62	1,066	384	62	—	—	—	—	—	—
	67	1,071	1,055	62	1,016	999	62	981	918	62	981	745	62	981	572	62	981	399	62	—	—	—
	62	1,071	1,062	62	1,016	1,005	62	963	951	62	911	897	61	901	758	61	901	585	61	901	412	61
	57	1,071	1,068	62	1,016	1,011	62	963	956	62	911	903	61	860	850	61	824	767	61	824	594	61

Table 34: 70 ton high capacity - 95°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	945	523	69	945	446	69	945	369	69	945	292	69	—	—	—	—	—	—	—	—	—
	72	864	613	69	864	536	69	864	458	69	864	381	69	864	304	69	—	—	—	—	—	—
	67	790	702	68	790	624	68	790	546	68	790	469	68	790	392	68	790	314	68	—	—	—
	62	749	743	68	721	710	68	721	632	68	721	554	68	721	477	68	721	399	68	721	322	68
	57	749	747	68	715	712	68	682	677	67	659	638	67	659	560	67	659	482	67	659	405	67
18,500	77	1,012	612	70	1,012	511	70	1,012	410	70	1,012	310	70	—	—	—	—	—	—	—	—	—
	72	930	729	69	930	628	69	930	526	69	930	425	69	930	324	69	—	—	—	—	—	—
	67	852	844	69	852	742	69	852	640	69	852	539	69	852	437	69	852	336	69	—	—	—
	62	848	841	68	808	800	68	781	753	68	781	651	68	781	549	68	781	447	68	781	346	68
	57	848	846	68	808	804	68	769	763	68	730	723	68	714	658	68	714	556	68	714	455	68
23,000	77	1,056	699	70	1,056	574	70	1,056	450	70	1,056	325	70	—	—	—	—	—	—	—	—	—
	72	972	841	69	972	716	69	972	591	69	972	466	69	972	341	69	—	—	—	—	—	—
	67	925	911	69	893	856	69	893	730	69	893	604	69	893	479	69	893	354	69	—	—	—
	62	925	917	69	880	870	69	835	825	68	818	741	68	818	615	68	818	490	68	818	365	68
	57	925	923	69	880	875	69	835	829	68	792	785	68	749	741	68	749	624	68	749	498	68
27,500	77	1,084	784	70	1,084	636	70	1,084	487	70	1,084	339	70	—	—	—	—	—	—	—	—	—
	72	999	951	70	999	802	70	999	653	70	999	504	70	999	356	70	—	—	—	—	—	—
	67	985	970	69	935	919	69	918	817	69	918	667	69	918	518	69	918	369	69	—	—	—
	62	985	976	69	935	925	69	886	875	69	842	829	68	842	679	68	842	530	68	842	381	68
	57	985	982	69	935	931	69	886	880	69	839	831	68	792	783	68	770	688	68	770	538	68
32,000	77	1,102	869	71	1,102	696	71	1,102	524	71	1,102	353	71	—	—	—	—	—	—	—	—	—
	72	1,032	1,010	70	1,017	887	70	1,017	714	70	1,017	542	70	1,017	370	70	—	—	—	—	—	—
	67	1,032	1,017	70	979	963	69	936	903	69	936	730	69	936	557	69	936	384	69	—	—	—
	62	1,032	1,024	70	979	969	69	927	915	69	876	863	69	859	742	69	859	569	69	859	396	69
	57	1,032	1,030	70	979	975	69	927	921	69	876	868	69	827	817	68	786	751	68	786	578	68

Table 35: 70 ton high capacity - 105°F ambient air temperature

CFM	EWB	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
	(°F)	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	905	509	77	905	432	77	905	355	77	905	279	77	—	—	—	—	—	—	—	—	—
	72	828	599	76	828	522	76	828	445	76	828	367	76	828	290	76	—	—	—	—	—	—
14,000	67	756	688	75	756	610	75	756	532	75	756	454	75	756	377	75	756	300	75	—	—	—
	62	725	719	75	691	684	75	690	617	75	690	540	75	690	462	75	690	385	75	690	307	75
	57	725	723	75	691	688	75	658	654	75	630	623	74	630	545	74	630	468	74	630	390	74
	77	966	598	78	966	497	78	966	397	78	966	296	78	—	—	—	—	—	—	—	—	—
	72	888	715	77	888	613	77	888	512	77	888	411	77	888	310	77	—	—	—	—	—	—
18,500	67	819	806	76	813	727	76	813	625	76	813	524	76	813	422	76	813	321	76	—	—	—
	62	819	812	76	779	771	76	745	737	75	745	635	75	745	533	75	745	432	75	745	330	75
	57	819	817	76	779	776	76	741	736	75	703	697	75	681	642	75	681	540	75	681	439	75
	77	1,006	685	78	1,006	560	78	1,006	436	78	1,006	311	78	—	—	—	—	—	—	—	—	—
	72	926	826	77	926	701	77	926	576	77	926	451	77	926	326	77	—	—	—	—	—	—
23,000	67	891	877	77	850	840	76	850	714	76	850	589	76	850	464	76	850	339	76	—	—	—
	62	891	883	77	847	838	76	803	793	76	779	725	76	779	599	76	779	474	76	779	349	76
	57	891	888	77	847	843	76	803	798	76	761	754	75	720	712	75	713	607	75	713	482	75
	77	1,031	770	78	1,031	621	78	1,031	473	78	1,031	325	78	—	—	—	—	—	—	—	—	—
	72	950	936	77	950	787	77	950	638	77	950	489	77	950	341	77	—	—	—	—	—	—
27,500	67	946	932	77	898	883	77	873	801	77	873	652	77	873	502	77	873	354	77	—	—	—
	62	946	938	77	898	889	77	851	840	76	805	793	76	800	663	76	800	513	76	800	364	76
	57	946	944	77	898	894	77	851	845	76	805	798	76	760	751	75	732	671	75	732	522	75
	77	1,047	855	78	1,047	682	78	1,047	510	78	1,047	339	78	—	—	—	—	—	—	—	—	—
	72	991	969	78	967	872	78	967	699	78	967	527	78	967	355	78	—	—	—	—	—	—
32,000	67	991	976	78	939	923	77	890	887	77	890	714	77	890	541	77	890	368	77	—	—	—
	62	991	983	78	939	930	77	889	878	77	840	827	76	816	726	76	816	552	76	816	380	76
	57	991	988	78	939	935	77	889	883	77	840	832	76	792	783	76	747	734	75	747	561	75

Table 36: 70 ton high capacity - 115°F ambient air temperature

CFM	EWB	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
	(°F)	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	862	495	85	862	418	85	862	341	85	862	265	85	—	—	—	—	—	—	—	—	—
	72	789	585	84	789	507	84	789	430	84	789	353	84	789	276	84	—	—	—	—	—	—
14,000	67	721	673	83	721	595	83	721	517	83	721	440	83	721	362	83	721	285	83	—	—	—
	62	698	693	83	665	658	83	658	602	83	658	525	83	658	447	83	658	370	83	658	292	83
	57	698	697	83	665	662	83	633	629	82	602	596	82	601	530	82	601	452	82	601	375	82
	77	918	584	86	918	483	86	918	382	86	918	282	86	—	—	—	—	—	—	—	—	—
	72	843	700	85	843	598	85	843	497	85	843	396	85	843	295	85	—	—	—	—	—	—
18,500	67	787	775	84	773	711	84	773	610	84	773	508	84	773	407	84	773	306	84	—	—	—
	62	787	780	84	748	741	84	711	702	83	707	619	83	707	517	83	707	416	83	707	314	83
	57	787	785	84	748	745	84	711	706	83	674	668	83	647	626	82	647	524	82	647	422	82
	77	953	670	86	953	546	86	953	421	86	953	297	86	—	—	—	—	—	—	—	—	—
	72	877	811	85	877	686	85	877	560	85	877	435	85	877	311	85	—	—	—	—	—	—
23,000	67	855	842	85	812	798	85	806	698	85	806	573	85	806	448	85	806	323	85	—	—	—
	62	855	848	85	812	803	85	769	760	84	738	708	84	738	583	84	738	457	84	738	332	84
	57	855	853	85	812	808	85	769	764	84	728	722	83	688	680	83	675	590	83	675	465	83
	77	975	755	87	975	607	87	975	459	87	975	311	87	—	—	—	—	—	—	—	—	—
	72	907	887	86	899	771	86	899	622	86	899	474	86	899	325	86	—	—	—	—	—	—
27,500	67	907	893	86	860	845	85	826	785	85	826	635	85	826	486	85	826	338	85	—	—	—
	62	907	899	86	860	851	85	814	804	85	769	758	84	757	646	84	757	497	84	757	348	84
	57	907	904	86	860	856	85	814	808	85	769	762	84	725	717	83	692	654	83	692	504	83
	77	989	840	87	989	668	87	989	496	87	989	324	87	—	—	—	—	—	—	—	—	—
	72	948	927	86	914	857	86	914	684	86	914	511	86	914	339	86	—	—	—	—	—	—
32,000	67	948	934	86	898	883	86	849	833	85	841	697	85	841	525	85	841	352	85	—	—	—
	62	948	940	86	898	888	86	849	838	85	801	790	84	771	708	84	771	535	84	771	362	84
	57	948	945	86	898	894	86	849	843	85	801	794	84	755	746	84	709	700	83	704	543	83

Notes:

Rated performance is at sea level. Cooling capacities are gross cooling capacities.

CFM = airflow, EWB = entering wet bulb air, EDB = entering dry bulb air, TMBH = total cooling capacity (MBH)

SMBH = sensible cooling capacity (MBH), kW = total input

Cooling performance data 80 ton - standard capacity, standard efficiency

Table 37: 80 ton standard efficiency - 75°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	1,018	548	54	1,018	471	54	1,018	394	54	1,018	317	54	—	—	—	—	—	—	—	—	—
	72	931	640	54	931	562	54	931	485	54	931	408	54	931	331	54	—	—	—	—	—	—
14,000	67	851	729	54	851	651	54	851	574	54	851	496	54	851	419	54	851	341	54	—	—	—
	62	792	786	53	778	738	53	778	660	53	778	582	53	778	504	53	778	427	53	778	350	53
	57	792	790	53	757	754	53	722	718	53	711	666	53	711	588	53	711	510	53	711	433	53
	77	1,098	639	54	1,098	538	54	1,098	437	54	1,098	336	54	—	—	—	—	—	—	—	—	—
	72	1,008	757	54	1,008	655	54	1,008	554	54	1,008	453	54	1,008	352	54	—	—	—	—	—	—
18,500	67	924	873	54	924	771	54	924	669	54	924	567	54	924	466	54	924	365	54	—	—	—
	62	901	894	54	859	850	54	846	782	54	846	680	54	846	578	54	846	477	54	846	375	54
	57	901	899	54	859	855	54	818	813	53	778	771	53	775	688	53	775	586	53	775	484	53
	77	1,150	726	55	1,150	601	55	1,150	477	55	1,150	352	55	—	—	—	—	—	—	—	—	—
	72	1,058	870	54	1,058	744	54	1,058	619	54	1,058	494	54	1,058	369	54	—	—	—	—	—	—
23,000	67	986	972	54	972	885	54	972	759	54	972	634	54	972	509	54	972	384	54	—	—	—
	62	986	978	54	939	929	54	893	881	54	891	772	54	891	646	54	891	520	54	891	395	54
	57	986	984	54	939	935	54	893	887	54	847	840	54	815	780	53	815	655	53	815	529	53
	77	1,183	811	55	1,183	663	55	1,183	514	55	1,183	366	55	—	—	—	—	—	—	—	—	—
	72	1,090	980	54	1,090	830	54	1,090	681	54	1,090	533	54	1,090	384	54	—	—	—	—	—	—
27,500	67	1,053	1,038	54	1,002	996	54	1,002	847	54	1,002	697	54	1,002	548	54	1,002	399	54	—	—	—
	62	1,053	1,045	54	1,001	991	54	950	938	54	919	859	54	919	710	54	919	561	54	919	412	54
	57	1,053	1,051	54	1,001	997	54	950	944	54	900	892	54	852	842	54	841	719	54	841	570	54
	77	1,206	896	55	1,206	724	55	1,206	552	55	1,206	380	55	—	—	—	—	—	—	—	—	—
	72	1,113	1,089	55	1,113	916	55	1,113	743	55	1,113	570	55	1,113	398	55	—	—	—	—	—	—
32,000	67	1,107	1,091	54	1,052	1,034	54	1,024	933	54	1,024	760	54	1,024	587	54	1,024	414	54	—	—	—
	62	1,107	1,098	54	1,052	1,040	54	997	984	54	943	929	54	940	773	54	940	600	54	940	427	54
	57	1,107	1,105	54	1,052	1,047	54	997	990	54	943	935	54	891	881	54	860	783	54	860	610	54

Table 38: 80 ton standard efficiency - 85°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	982	536	61	982	459	61	982	382	61	982	305	61	—	—	—	—	—	—	—	—	—
	72	899	627	60	899	549	60	899	472	60	899	395	60	899	318	60	—	—	—	—	—	—
14,000	67	821	716	60	821	638	60	821	560	60	821	483	60	821	405	60	821	328	60	—	—	—
	62	772	766	60	750	724	60	750	646	60	750	568	60	750	491	60	750	413	60	750	336	60
	57	772	770	60	738	734	60	704	699	59	685	652	59	685	574	59	685	497	59	685	419	59
	77	1,056	626	61	1,056	525	61	1,056	424	61	1,056	323	61	—	—	—	—	—	—	—	—	—
	72	970	743	61	970	642	61	970	540	61	970	439	61	970	338	61	—	—	—	—	—	—
18,500	67	889	859	60	889	757	60	889	655	60	889	553	60	889	452	60	889	351	60	—	—	—
	62	876	869	60	835	827	60	814	768	60	814	666	60	814	564	60	814	462	60	814	361	60
	57	876	874	60	835	832	60	795	790	60	756	749	60	745	673	60	745	571	60	745	470	60
	77	1,104	713	61	1,104	588	61	1,104	463	61	1,104	339	61	—	—	—	—	—	—	—	—	—
	72	1,016	856	61	1,016	730	61	1,016	605	61	1,016	480	61	1,016	355	61	—	—	—	—	—	—
23,000	67	957	943	60	933	871	60	933	745	60	933	619	60	933	494	60	933	369	60	—	—	—
	62	957	949	60	911	901	60	866	855	60	855	757	60	855	631	60	855	505	60	855	380	60
	57	957	955	60	911	907	60	866	860	60	821	814	60	783	765	60	783	639	60	783	514	60
	77	1,135	798	62	1,135	649	62	1,135	501	62	1,135	353	62	—	—	—	—	—	—	—	—	—
	72	1,046	966	61	1,046	816	61	1,046	667	61	1,046	518	61	1,046	370	61	—	—	—	—	—	—
27,500	67	1,020	1,005	61	969	953	61	961	832	61	961	682	61	961	533	61	961	385	61	—	—	—
	62	1,020	1,012	61	969	959	61	920	908	60	881	844	60	881	695	60	881	545	60	881	396	60
	57	1,020	1,018	61	969	965	61	920	913	60	871	863	60	823	814	60	807	704	60	807	554	60
	77	1,155	883	62	1,155	710	62	1,155	538	62	1,155	367	62	—	—	—	—	—	—	—	—	—
	72	1,071	1,047	61	1,066	902	61	1,066	729	61	1,066	556	61	1,066	384	61	—	—	—	—	—	—
32,000	67	1,071	1,055	61	1,016	999	61	981	918	61	981	745	61	981	572	61	981	399	61	—	—	—
	62	1,071	1,062	61	1,016	1,005	61	963	951	61	911	897	60	901	758	60	901	585	60	901	412	60
	57	1,071	1,068	61	1,016	1,011	61	963	956	61	911	903	60	860	850	60	824	767	60	824	594	60

Table 39: 80 ton standard efficiency - 95°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	945	523	68	945	446	68	945	369	68	945	292	68	—	—	—	—	—	—	—	—	—
	72	864	613	67	864	536	67	864	458	67	864	381	67	864	304	67	—	—	—	—	—	—
14,000	67	790	702	67	790	624	67	790	546	67	790	469	67	790	392	67	790	314	67	—	—	—
	62	749	743	66	721	710	66	721	632	66	721	554	66	721	477	66	721	399	66	721	322	66
	57	749	747	66	715	712	66	682	677	66	659	638	66	659	560	66	659	482	66	659	405	66
	77	1,012	612	68	1,012	511	68	1,012	410	68	1,012	310	68	—	—	—	—	—	—	—	—	—
	72	930	729	68	930	628	68	930	526	68	930	425	68	930	324	68	—	—	—	—	—	—
18,500	67	852	844	67	852	742	67	852	640	67	852	539	67	852	437	67	852	336	67	—	—	—
	62	848	841	67	808	800	67	781	753	67	781	651	67	781	549	67	781	447	67	781	346	67
	57	848	846	67	808	804	67	769	763	67	730	723	66	714	658	66	714	556	66	714	455	66
	77	1,056	699	69	1,056	574	69	1,056	450	69	1,056	325	69	—	—	—	—	—	—	—	—	—
	72	972	841	68	972	716	68	972	591	68	972	466	68	972	341	68	—	—	—	—	—	—
23,000	67	925	911	68	893	856	67	893	730	67	893	604	67	893	479	67	893	354	67	—	—	—
	62	925	917	68	880	870	67	835	825	67	818	741	67	818	615	67	818	490	67	818	365	67
	57	925	923	68	880	875	67	835	829	67	792	785	67	749	741	66	749	624	66	749	498	66
	77	1,084	784	69	1,084	636	69	1,084	487	69	1,084	339	69	—	—	—	—	—	—	—	—	—
	72	999	951	68	999	802	68	999	653	68	999	504	68	999	356	68	—	—	—	—	—	—
27,500	67	985	970	68	935	919	68	918	817	68	918	667	68	918	518	68	918	369	68	—	—	—
	62	985	976	68	935	925	68	886	875	67	842	829	67	842	679	67	842	530	67	842	381	67
	57	985	982	68	935	931	68	886	880	67	839	831	67	792	783	67	770	688	67	770	538	67
	77	1,102	869	69	1,102	696	69	1,102	524	69	1,102	353	69	—	—	—	—	—	—	—	—	—
	72	1,032	1,010	69	1,017	887	68	1,017	714	68	1,017	542	68	1,017	370	68	—	—	—	—	—	—
32,000	67	1,032	1,017	69	979	963	68	936	903	68	936	730	68	936	557	68	936	384	68	—	—	—
	62	1,032	1,024	69	979	969	68	927	915	68	876	863	67	859	742	67	859	569	67	859	396	67
	57	1,032	1,030	69	979	975	68	927	921	68	876	868	67	827	817	67	786	751	67	786	578	67

Table 40: 80 ton standard efficiency - 105°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	905	509	75	905	432	75	905	355	75	905	279	75	—	—	—	—	—	—	—	—	—
	72	828	599	75	828	522	75	828	445	75	828	367	75	828	290	75	—	—	—	—	—	—
14,000	67	756	688	74	756	610	74	756	532	74	756	454	74	756	377	74	756	300	74	—	—	—
	62	725	719	74	691	684	73	690	617	73	690	540	73	690	462	73	690	385	73	690	307	73
	57	725	723	74	691	688	73	658	654	73	630	623	73	630	545	73	630	468	73	630	390	73
	77	966	598	76	966	497	76	966	397	76	966	296	76	—	—	—	—	—	—	—	—	—
	72	888	715	75	888	613	75	888	512	75	888	411	75	888	310	75	—	—	—	—	—	—
18,500	67	819	806	75	813	727	75	813	625	75	813	524	75	813	422	75	813	321	75	—	—	—
	62	819	812	75	779	771	74	745	737	74	745	635	74	745	533	74	745	432	74	745	330	74
	57	819	817	75	779	776	74	741	736	74	703	697	73	681	642	73	681	540	73	681	439	73
	77	1,006	685	77	1,006	560	77	1,006	436	77	1,006	311	77	—	—	—	—	—	—	—	—	—
	72	926	826	76	926	701	76	926	576	76	926	451	76	926	326	76	—	—	—	—	—	—
23,000	67	891	877	75	850	840	75	850	714	75	850	589	75	850	464	75	850	339	75	—	—	—
	62	891	883	75	847	838	75	803	793	74	779	725	74	779	599	74	779	474	74	779	349	74
	57	891	888	75	847	843	75	803	798	74	761	754	74	720	712	74	713	607	74	713	482	74
	77	1,031	770	77	1,031	621	77	1,031	473	77	1,031	325	77	—	—	—	—	—	—	—	—	—
	72	950	936	76	950	787	76	950	638	76	950	489	76	950	341	76	—	—	—	—	—	—
27,500	67	946	932	76	898	883	75	873	801	75	873	652	75	873	502	75	873	354	75	—	—	—
	62	946	938	76	898	889	75	851	840	75	805	793	74	800	663	74	800	513	74	800	364	74
	57	946	944	76	898	894	75	851	845	75	805	798	74	760	751	74	732	671	74	732	522	74
	77	1,047	855	77	1,047	682	77	1,047	510	77	1,047	339	77	—	—	—	—	—	—	—	—	—
	72	991	969	76	967	872	76	967	699	76	967	527	76	967	355	76	—	—	—	—	—	—
32,000	67	991	976	76	939	923	76	890	887	75	890	714	75	890	541	75	890	368	75	—	—	—
	62	991	983	76	939	930	76	889	878	75	840	827	75	816	726	75	816	552	75	816	380	75
	57	991	988	76	939	935	76	889	883	75	840	832	75	792	783	74	747	734	74	747	561	74

Table 41: 80 ton standard efficiency - 115°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	862	495	84	862	418	84	862	341	84	862	265	84	—	—	—	—	—	—	—	—	—
	72	789	585	83	789	507	83	789	430	83	789	353	83	789	276	83	—	—	—	—	—	—
14,000	67	721	673	82	721	595	82	721	517	82	721	440	82	721	362	82	721	285	82	—	—	—
	62	698	693	81	665	658	81	658	602	81	658	525	81	658	447	81	658	370	81	658	292	81
	57	698	697	81	665	662	81	633	629	81	602	596	80	601	530	80	601	452	80	601	375	80
	77	918	584	84	918	483	84	918	382	84	918	282	84	—	—	—	—	—	—	—	—	—
	72	843	700	83	843	598	83	843	497	83	843	396	83	843	295	83	—	—	—	—	—	—
18,500	67	787	775	83	773	711	82	773	610	82	773	508	82	773	407	82	773	306	82	—	—	—
	62	787	780	83	748	741	82	711	702	82	707	619	82	707	517	82	707	416	82	707	314	82
	57	787	785	83	748	745	82	711	706	82	674	668	81	647	626	81	647	524	81	647	422	81
	77	953	670	85	953	546	85	953	421	85	953	297	85	—	—	—	—	—	—	—	—	—
	72	877	811	84	877	686	84	877	560	84	877	435	84	877	311	84	—	—	—	—	—	—
23,000	67	855	842	83	812	798	83	806	698	83	806	573	83	806	448	83	806	323	83	—	—	—
	62	855	848	83	812	803	83	769	760	82	738	708	82	738	583	82	738	457	82	738	332	82
	57	855	853	83	812	808	83	769	764	82	728	722	82	688	680	81	675	590	81	675	465	81
	77	975	755	85	975	607	85	975	459	85	975	311	85	—	—	—	—	—	—	—	—	—
	72	907	887	84	899	771	84	899	622	84	899	474	84	899	325	84	—	—	—	—	—	—
27,500	67	907	893	84	860	845	84	826	785	83	826	635	83	826	486	83	826	338	83	—	—	—
	62	907	899	84	860	851	84	814	804	83	769	758	82	757	646	82	757	497	82	757	348	82
	57	907	904	84	860	856	84	814	808	83	769	762	82	725	717	82	692	654	81	692	504	81
	77	989	840	85	989	668	85	989	496	85	989	324	85	—	—	—	—	—	—	—	—	—
	72	948	927	85	914	857	84	914	684	84	914	511	84	914	339	84	—	—	—	—	—	—
32,000	67	948	934	85	898	883	84	849	833	83	841	697	83	841	525	83	841	352	83	—	—	—
	62	948	940	85	898	888	84	849	838	83	801	790	83	771	708	82	771	535	82	771	362	82
	57	948	945	85	898	894	84	849	843	83	801	794	83	755	746	82	709	700	82	704	543	82

Notes:

Rated performance is at sea level. Cooling capacities are gross cooling capacities.

CFM = airflow, EWB = entering wet bulb air, EDB = entering dry bulb air, TMBH = total cooling capacity (MBH)

SMBH = sensible cooling capacity (MBH), kW = total input

Cooling performance data 80 ton standard capacity, high efficiency

Table 42: 80 ton high efficiency - 75°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	1,032	553	55	1,032	476	55	1,032	399	55	1,032	322	55	—	—	—	—	—	—	—	—	—
	72	944	645	54	944	567	54	944	490	54	944	413	54	944	336	54	—	—	—	—	—	—
14,000	67	862	734	54	862	656	54	862	578	54	862	501	54	862	423	54	862	346	54	—	—	—
	62	802	796	54	787	743	54	787	665	54	787	587	54	787	509	54	787	432	54	787	354	54
	57	802	800	54	767	763	54	732	727	54	—	—	—	—	—	—	—	—	—	—	—	—
	77	1,112	643	55	1,112	542	55	1,112	441	55	1,112	340	55	—	—	—	—	—	—	—	—	—
	72	1,020	761	55	1,020	660	55	1,020	558	55	1,020	457	55	1,020	356	55	—	—	—	—	—	—
18,500	67	935	878	54	935	776	54	935	674	54	935	572	54	935	471	54	935	369	54	—	—	—
	62	912	904	54	870	861	54	856	787	54	856	685	54	856	583	54	856	481	54	856	380	54
	57	912	909	54	870	866	54	829	823	54	788	781	54	784	693	54	784	591	54	784	489	54
	77	1,165	731	55	1,165	606	55	1,165	481	55	1,165	357	55	—	—	—	—	—	—	—	—	—
	72	1,072	874	55	1,072	749	55	1,072	623	55	1,072	498	55	1,072	374	55	—	—	—	—	—	—
23,000	67	998	983	54	984	890	54	984	764	54	984	639	54	984	513	54	984	388	54	—	—	—
	62	998	989	54	950	940	54	903	892	54	902	776	54	902	651	54	902	525	54	902	400	54
	57	998	995	54	950	946	54	903	897	54	858	850	54	826	785	54	826	660	54	826	534	54
	77	1,200	816	55	1,200	667	55	1,200	519	55	1,200	371	55	—	—	—	—	—	—	—	—	—
	72	1,105	985	55	1,105	835	55	1,105	686	55	1,105	537	55	1,105	389	55	—	—	—	—	—	—
27,500	67	1,066	1,050	55	1,016	1,001	55	1,016	852	55	1,016	702	55	1,016	553	55	1,016	404	55	—	—	—
	62	1,066	1,057	55	1,013	1,003	54	962	950	54	932	865	54	932	715	54	932	566	54	932	417	54
	57	1,066	1,063	55	1,013	1,008	54	962	955	54	911	903	54	862	853	54	853	725	54	853	575	54
	77	1,224	901	56	1,224	728	56	1,224	556	56	1,224	385	56	—	—	—	—	—	—	—	—	—
	72	1,129	1,094	55	1,129	921	55	1,129	748	55	1,129	575	55	1,129	403	55	—	—	—	—	—	—
32,000	67	1,121	1,104	55	1,064	1,046	55	1,039	938	55	1,039	765	55	1,039	592	55	1,039	419	55	—	—	—
	62	1,121	1,111	55	1,064	1,053	55	1,009	996	54	955	941	54	954	779	54	954	605	54	954	432	54
	57	1,121	1,118	55	1,064	1,059	55	1,009	1,002	54	955	946	54	902	892	54	873	788	54	873	615	54

Table 43: 80 ton high efficiency - 85°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	997	541	61	997	464	61	997	387	61	997	310	61	—	—	—	—	—	—	—	—	—
	72	912	632	61	912	555	61	912	477	61	912	400	61	912	323	61	—	—	—	—	—	—
14,000	67	833	721	60	833	643	60	833	565	60	833	488	60	833	411	60	833	333	60	—	—	—
	62	781	775	60	761	729	60	761	651	60	761	574	60	761	496	60	761	419	60	761	341	60
	57	781	779	60	747	743	60	712	707	60	695	657	60	695	579	60	695	502	60	695	424	60
	77	1,071	631	62	1,071	529	62	1,071	429	62	1,071	328	62	—	—	—	—	—	—	—	—	—
	72	984	748	61	984	647	61	984	545	61	984	444	61	984	343	61	—	—	—	—	—	—
18,500	67	902	864	61	902	762	61	902	660	61	902	558	61	902	457	61	902	356	61	—	—	—
	62	886	878	60	845	836	60	825	773	60	825	671	60	825	569	60	825	467	60	825	366	60
	57	886	884	60	845	841	60	804	799	60	765	758	60	755	678	60	755	576	60	755	475	60
	77	1,121	718	62	1,121	593	62	1,121	468	62	1,121	344	62	—	—	—	—	—	—	—	—	—
	72	1,031	861	61	1,031	735	61	1,031	610	61	1,031	485	61	1,031	360	61	—	—	—	—	—	—
23,000	67	968	953	61	947	876	61	947	750	61	947	624	61	947	499	61	947	374	61	—	—	—
	62	968	960	61	921	912	61	876	865	60	868	762	60	868	636	60	868	511	60	868	385	60
	57	968	965	61	921	917	61	876	870	60	831	823	60	794	771	60	794	645	60	794	519	60
	77	1,152	803	62	1,152	654	62	1,152	506	62	1,152	358	62	—	—	—	—	—	—	—	—	—
	72	1,062	971	62	1,062	821	62	1,062	672	62	1,062	524	62	1,062	375	62	—	—	—	—	—	—
27,500	67	1,032	1,017	61	981	965	61	976	837	61	976	688	61	976	539	61	976	390	61	—	—	—
	62	1,032	1,024	61	981	971	61	931	919	61	895	850	61	895	700	61	895	551	61	895	402	61
	57	1,032	1,030	61	981	977	61	931	925	61	882	874	60	834	825	60	819	709	60	819	560	60
	77	1,174	888	62	1,174	715	62	1,174	543	62	1,174	371	62	—	—	—	—	—	—	—	—	—
	72	1,084	1,061	62	1,084	907	62	1,084	734	62	1,084	562	62	1,084	389	62	—	—	—	—	—	—
32,000	67	1,084	1,068	62	1,030	1,012	61	997	924	61	997	750	61	997	577	61	997	405	61	—	—	—
	62	1,084	1,075	62	1,030	1,019	61	976	963	61	923	910	61	915	764	61	915	590	61	915	418	61
	57	1,084	1,082	62	1,030	1,025	61	976	969	61	923	915	61	872	862	60	838	773	60	838	600	60

Table 44: 80 ton high efficiency - 95°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	961	528	68	961	451	68	961	374	68	961	297	68	—	—	—	—	—	—	—	—	—
	72	878	619	67	878	541	67	878	464	67	878	387	67	878	310	67	—	—	—	—	—	—
14,000	67	802	707	67	802	630	67	802	552	67	802	474	67	802	397	67	802	320	67	—	—	—
	62	761	754	67	732	716	66	732	638	66	732	560	66	732	482	66	732	405	66	732	327	66
	57	761	759	67	726	723	66	692	687	66	669	643	66	669	565	66	669	488	66	669	410	66
	77	1,029	617	69	1,029	516	69	1,029	415	69	1,029	315	69	—	—	—	—	—	—	—	—	—
	72	944	734	68	944	633	68	944	531	68	944	430	68	944	329	68	—	—	—	—	—	—
18,500	67	866	850	67	866	747	67	866	646	67	866	544	67	866	443	67	866	341	67	—	—	—
	62	860	853	67	820	811	67	792	758	67	792	656	67	792	554	67	792	453	67	792	351	67
	57	860	858	67	820	816	67	780	775	67	741	734	66	725	663	66	725	562	66	725	460	66
	77	1,073	704	69	1,073	579	69	1,073	455	69	1,073	330	69	—	—	—	—	—	—	—	—	—
	72	988	846	68	988	721	68	988	596	68	988	471	68	988	346	68	—	—	—	—	—	—
23,000	67	938	924	68	907	861	68	907	735	68	907	610	68	907	484	68	907	359	68	—	—	—
	62	938	930	68	892	883	68	847	836	67	832	747	67	832	621	67	832	496	67	832	370	67
	57	938	935	68	892	888	68	847	841	67	803	796	67	761	755	67	761	629	67	761	504	67
	77	1,102	789	69	1,102	641	69	1,102	492	69	1,102	344	69	—	—	—	—	—	—	—	—	—
	72	1,016	956	69	1,016	807	69	1,016	658	69	1,016	509	69	1,016	361	69	—	—	—	—	—	—
27,500	67	998	983	68	948	932	68	934	822	68	934	673	68	934	524	68	934	375	68	—	—	—
	62	998	990	68	948	938	68	899	888	68	857	834	67	857	685	67	857	536	67	857	387	67
	57	998	996	68	948	944	68	899	893	68	851	843	67	804	795	67	784	694	67	784	544	67
	77	1,122	874	70	1,122	702	70	1,122	530	70	1,122	358	70	—	—	—	—	—	—	—	—	—
	72	1,047	1,024	69	1,036	893	69	1,036	720	69	1,036	547	69	1,036	375	69	—	—	—	—	—	—
32,000	67	1,047	1,031	69	993	976	68	953	909	68	953	735	68	953	562	68	953	390	68	—	—	—
	62	1,047	1,038	69	993	983	68	941	929	68	889	876	68	875	748	67	875	575	67	875	402	67
	57	1,047	1,044	69	993	989	68	941	934	68	889	881	68	839	830	67	800	757	67	800	584	67

Table 45: 80 ton high efficiency - 105°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	922	515	76	922	438	76	922	361	76	922	284	76	—	—	—	—	—	—	—	—	—
	72	843	605	75	843	528	75	843	450	75	843	373	75	843	296	75	—	—	—	—	—	—
	67	770	693	74	770	615	74	770	538	74	770	460	74	770	383	74	770	306	74	—	—	—
	62	736	730	74	702	701	73	702	623	73	702	545	73	702	468	73	702	390	73	702	313	73
	57	736	735	74	702	699	73	668	664	73	641	629	73	641	551	73	641	473	73	641	396	73
18,500	77	984	604	76	984	503	76	984	402	76	984	301	76	—	—	—	—	—	—	—	—	—
	72	903	720	75	903	618	75	903	517	75	903	416	75	903	315	75	—	—	—	—	—	—
	67	831	819	75	828	733	75	828	631	75	828	529	75	828	428	75	828	327	75	—	—	—
	62	831	824	75	791	783	74	757	742	74	757	640	74	757	539	74	757	437	74	757	336	74
	57	831	829	75	791	787	74	752	747	74	714	707	73	693	648	73	693	546	73	693	444	73
23,000	77	1,024	690	77	1,024	565	77	1,024	441	77	1,024	316	77	—	—	—	—	—	—	—	—	—
	72	943	832	76	943	706	76	943	581	76	943	456	76	943	331	76	—	—	—	—	—	—
	67	904	891	75	866	846	75	866	720	75	866	594	75	866	469	75	866	344	75	—	—	—
	62	904	897	75	859	850	75	816	805	74	793	731	74	793	605	74	793	480	74	793	355	74
	57	904	902	75	859	855	75	816	810	74	773	766	74	731	723	74	726	613	74	726	488	74
27,500	77	1,050	775	77	1,050	627	77	1,050	478	77	1,050	330	77	—	—	—	—	—	—	—	—	—
	72	968	942	76	968	792	76	968	643	76	968	495	76	968	346	76	—	—	—	—	—	—
	67	961	947	76	912	897	76	890	807	75	890	657	75	890	508	75	890	360	75	—	—	—
	62	961	953	76	912	903	76	865	854	75	818	806	75	816	669	75	816	519	75	816	371	75
	57	961	959	76	912	908	76	865	859	75	818	811	75	772	764	74	747	677	74	747	528	74
32,000	77	1,068	860	77	1,068	688	77	1,068	516	77	1,068	344	77	—	—	—	—	—	—	—	—	—
	72	1,007	985	77	986	878	76	986	705	76	986	532	76	986	360	76	—	—	—	—	—	—
	67	1,007	992	77	955	938	76	907	893	76	907	720	76	907	547	76	907	374	76	—	—	—
	62	1,007	998	77	955	945	76	904	892	75	854	841	75	832	732	75	832	559	75	832	386	75
	57	1,007	1,004	77	955	950	76	904	898	75	854	846	75	806	797	74	762	741	74	762	567	74

Table 46: 80 ton high efficiency - 115°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	880	501	84	880	424	84	880	347	84	880	270	84	—	—	—	—	—	—	—	—	—
	72	805	591	83	805	513	83	805	436	83	805	359	83	805	282	83	—	—	—	—	—	—
	67	735	679	82	735	601	82	735	523	82	735	446	82	735	368	82	735	291	82	—	—	—
	62	710	704	81	677	670	81	671	608	81	671	531	81	671	453	81	671	375	81	671	298	81
	57	710	708	81	677	674	81	645	640	80	613	608	80	612	536	80	612	458	80	612	381	80
18,500	77	936	589	84	936	488	84	936	387	84	936	287	84	—	—	—	—	—	—	—	—	—
	72	860	705	83	860	604	83	860	502	83	860	401	83	860	300	83	—	—	—	—	—	—
	67	799	788	83	788	717	82	788	615	82	788	514	82	788	412	82	788	311	82	—	—	—
	62	799	793	83	761	753	82	723	714	81	721	625	81	721	523	81	721	421	81	721	320	81
	57	799	797	83	761	757	82	723	718	81	686	680	81	660	632	81	660	530	81	660	428	81
23,000	77	973	676	85	973	551	85	973	426	85	973	302	85	—	—	—	—	—	—	—	—	—
	72	895	817	84	895	691	84	895	566	84	895	441	84	895	316	84	—	—	—	—	—	—
	67	869	856	83	826	812	83	822	704	83	822	579	83	822	454	83	822	329	83	—	—	—
	62	869	862	83	826	817	83	783	773	82	753	715	82	753	589	82	753	464	82	753	338	82
	57	869	867	83	826	822	83	783	778	82	741	735	82	701	693	81	689	597	81	689	471	81
27,500	77	996	761	85	996	612	85	996	464	85	996	316	85	—	—	—	—	—	—	—	—	—
	72	923	903	84	918	777	84	918	628	84	918	479	84	918	331	84	—	—	—	—	—	—
	67	923	910	84	876	861	84	844	791	83	844	642	83	844	493	83	844	344	83	—	—	—
	62	923	916	84	876	866	84	829	819	83	784	772	82	774	652	82	774	503	82	774	354	82
	57	923	921	84	876	872	84	829	823	83	784	777	82	739	731	82	708	660	81	708	511	81
32,000	77	1,012	846	85	1,012	673	85	1,012	501	85	1,012	330	85	—	—	—	—	—	—	—	—	—
	72	966	945	85	934	863	84	934	690	84	934	517	84	934	345	84	—	—	—	—	—	—
	67	966	952	85	915	900	84	865	849	83	860	704	83	860	531	83	860	359	83	—	—	—
	62	966	958	85	915	906	84	865	854	83	817	805	83	789	715	82	789	542	82	789	369	82
	57	966	964	85	915	911	84	865	859	83	817	810	83	770	761	82	724	714	81	721	550	81

Notes:

Rated performance is at sea level. Cooling capacities are gross cooling capacities.

CFM = airflow, EWB = entering wet bulb air, EDB = entering dry bulb air, TMBH = total cooling capacity (MBH)

SMBH = sensible cooling capacity (MBH), kW = total input

Cooling performance data 80 ton - high capacity, standard efficiency

Table 47: 80 ton high capacity - 75°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	1,068	566	59	1,068	489	59	1,068	412	59	1,068	335	59	—	—	—	—	—	—	—	—	—
	72	976	658	59	976	581	59	976	503	59	976	426	59	976	349	59	—	—	—	—	—	—
	67	892	748	59	892	670	59	892	592	59	892	515	59	892	437	59	892	360	59	—	—	—
	62	822	816	58	814	756	58	814	678	58	814	601	58	814	523	58	814	445	58	814	368	58
	57	822	820	58	786	782	58	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
18,500	77	1,154	657	60	1,154	556	60	1,154	455	60	1,154	354	60	—	—	—	—	—	—	—	—	—
	72	1,059	776	59	1,059	674	59	1,059	573	59	1,059	471	59	1,059	370	59	—	—	—	—	—	—
	67	971	892	59	971	790	59	971	688	59	971	587	59	971	485	59	971	384	59	—	—	—
	62	937	929	59	893	884	59	889	802	59	889	700	59	889	598	59	889	496	59	889	395	59
	57	937	934	59	893	889	59	851	845	58	813	810	58	813	708	58	813	606	58	813	504	58
23,000	77	1,212	744	60	1,212	620	60	1,212	495	60	1,212	370	60	—	—	—	—	—	—	—	—	—
	72	1,114	889	60	1,114	763	60	1,114	638	60	1,114	513	60	1,114	388	60	—	—	—	—	—	—
	67	1,027	1,012	59	1,023	905	59	1,023	779	59	1,023	654	59	1,023	528	59	1,023	403	59	—	—	—
	62	1,027	1,019	59	978	968	59	938	918	59	938	792	59	938	666	59	938	541	59	938	415	59
	57	1,027	1,025	59	978	973	59	930	924	59	883	875	58	859	801	58	859	675	58	859	550	58
27,500	77	1,249	830	60	1,249	681	60	1,249	533	60	1,249	385	60	—	—	—	—	—	—	—	—	—
	72	1,151	999	60	1,151	850	60	1,151	701	60	1,151	552	60	1,151	403	60	—	—	—	—	—	—
	67	1,099	1,082	60	1,058	1,017	59	1,058	867	59	1,058	717	59	1,058	568	59	1,058	419	59	—	—	—
	62	1,099	1,090	60	1,045	1,034	59	992	979	59	970	880	59	970	731	59	970	581	59	970	432	59
	57	1,099	1,096	60	1,045	1,040	59	992	985	59	940	932	59	890	880	59	888	741	59	888	591	59
32,000	77	1,275	915	60	1,275	742	60	1,275	570	60	1,275	398	60	—	—	—	—	—	—	—	—	—
	72	1,176	1,109	60	1,176	936	60	1,176	763	60	1,176	590	60	1,176	418	60	—	—	—	—	—	—
	67	1,157	1,140	60	1,098	1,080	60	1,082	953	59	1,082	780	59	1,082	607	59	1,082	435	59	—	—	—
	62	1,157	1,147	60	1,098	1,087	60	1,042	1,028	59	993	968	59	993	794	59	993	621	59	993	448	59
	57	1,157	1,154	60	1,098	1,093	60	1,042	1,035	59	986	977	59	932	922	59	910	805	59	910	631	59

Table 48: 80 ton high capacity - 85°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	1,031	553	67	1,031	476	67	1,031	399	67	1,031	322	67	—	—	—	—	—	—	—	—	—
	72	943	644	66	943	567	66	943	490	66	943	412	66	943	335	66	—	—	—	—	—	—
	67	861	734	65	861	656	65	861	578	65	861	500	65	861	423	65	861	346	65	—	—	—
	62	801	794	65	786	742	65	786	664	65	786	586	65	786	509	65	786	431	65	786	354	65
	57	801	799	65	765	762	64	731	726	64	—	—	—	—	—	—	—	—	—	—	—	—
18,500	77	1,111	643	67	1,111	542	67	1,111	441	67	1,111	340	67	—	—	—	—	—	—	—	—	—
	72	1,019	761	66	1,019	660	66	1,019	558	66	1,019	457	66	1,019	356	66	—	—	—	—	—	—
	67	934	877	66	934	775	66	934	673	66	934	572	66	934	470	66	934	369	66	—	—	—
	62	910	902	66	868	859	65	855	786	65	855	684	65	855	582	65	855	481	65	855	379	65
	57	910	908	66	868	864	65	827	821	65	787	780	65	783	692	65	783	590	65	783	489	65
23,000	77	1,164	730	68	1,164	605	68	1,164	481	68	1,164	356	68	—	—	—	—	—	—	—	—	—
	72	1,070	874	67	1,070	748	67	1,070	623	67	1,070	498	67	1,070	373	67	—	—	—	—	—	—
	67	996	981	66	983	889	66	983	764	66	983	638	66	983	513	66	983	388	66	—	—	—
	62	996	987	66	948	938	66	902	890	66	901	776	66	901	650	66	901	525	66	901	400	66
	57	996	993	66	948	944	66	902	896	66	856	848	65	825	785	65	825	659	65	825	534	65
27,500	77	1,198	815	68	1,198	667	68	1,198	518	68	1,198	370	68	—	—	—	—	—	—	—	—	—
	72	1,103	984	67	1,103	835	67	1,103	685	67	1,103	537	67	1,103	388	67	—	—	—	—	—	—
	67	1,063	1,048	67	1,014	1,001	66	1,014	851	66	1,014	701	66	1,014	552	66	1,014	404	66	—	—	—
	62	1,063	1,054	67	1,011	1,000	66	960	948	66	930	864	66	930	714	66	930	565	66	930	416	66
	57	1,063	1,061	67	1,011	1,006	66	960	953	66	910	901	66	860	851	65	851	724	65	851	574	65
32,000	77	1,221	900	68	1,221	728	68	1,221	556	68	1,221	384	68	—	—	—	—	—	—	—	—	—
	72	1,127	1,094	67	1,127	920	67	1,127	747	67	1,127	575	67	1,127	403	67	—	—	—	—	—	—
	67	1,118	1,102	67	1,062	1,044	67	1,037	937	67	1,037	764	67	1,037	591	67	1,037	419	67	—	—	—
	62	1,118	1,109	67	1,062	1,051	67	1,007	994	66	953	939	66	952	778	66	952	605	66	952	432	66
	57	1,118	1,115	67	1,062	1,057	67	1,007	1,000	66	953	944	66	900	890	66	871	788	65	871	615	65

Table 49: 80 ton high capacity - 95°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	992	539	74	992	462	74	992	385	74	992	308	74	—	—	—	—	—	—	—	—	—
	72	907	630	73	907	552	73	907	475	73	907	398	73	907	321	73	—	—	—	—	—	—
14,000	67	828	719	72	828	641	72	828	563	72	828	486	72	828	408	72	828	331	72	—	—	—
	62	778	772	72	756	727	71	756	649	71	756	571	71	756	494	71	756	416	71	756	339	71
	57	778	776	72	743	740	71	709	704	71	690	655	71	690	577	71	690	499	71	690	422	71
	77	1,065	628	75	1,065	527	75	1,065	426	75	1,065	326	75	—	—	—	—	—	—	—	—	—
	72	978	746	74	978	644	74	978	543	74	978	442	74	978	341	74	—	—	—	—	—	—
18,500	67	896	862	73	896	760	73	896	658	73	896	556	73	896	455	73	896	353	73	—	—	—
	62	882	874	73	841	832	72	820	770	72	820	668	72	820	566	72	820	465	72	820	364	72
	57	882	879	73	841	837	72	800	795	72	761	754	72	750	676	71	750	574	71	750	472	71
	77	1,113	715	76	1,113	591	76	1,113	466	76	1,113	342	76	—	—	—	—	—	—	—	—	—
	72	1,024	858	75	1,024	733	75	1,024	608	75	1,024	483	75	1,024	358	75	—	—	—	—	—	—
23,000	67	962	948	74	940	873	74	940	748	74	940	622	74	940	497	74	940	372	74	—	—	—
	62	962	954	74	916	906	73	870	859	73	862	759	73	862	634	73	862	508	73	862	383	73
	57	962	960	74	916	911	73	870	864	73	826	818	72	789	768	72	789	642	72	789	517	72
	77	1,144	800	76	1,144	652	76	1,144	504	76	1,144	356	76	—	—	—	—	—	—	—	—	—
	72	1,054	968	75	1,054	819	75	1,054	670	75	1,054	521	75	1,054	373	75	—	—	—	—	—	—
27,500	67	1,025	1,010	75	974	958	74	969	835	74	969	685	74	969	536	74	969	387	74	—	—	—
	62	1,025	1,016	75	974	964	74	924	913	73	889	847	73	889	698	73	889	548	73	889	399	73
	57	1,025	1,022	75	974	970	74	924	918	73	876	868	73	828	819	72	814	707	72	814	557	72
	77	1,165	885	76	1,165	713	76	1,165	541	76	1,165	369	76	—	—	—	—	—	—	—	—	—
	72	1,076	1,052	75	1,075	905	75	1,075	732	75	1,075	559	75	1,075	387	75	—	—	—	—	—	—
32,000	67	1,076	1,060	75	1,022	1,004	75	990	921	74	990	748	74	990	575	74	990	402	74	—	—	—
	62	1,076	1,067	75	1,022	1,011	75	968	956	74	916	903	73	908	761	73	908	588	73	908	415	73
	57	1,076	1,073	75	1,022	1,017	75	968	962	74	916	908	73	865	855	73	832	771	72	832	597	72

Table 50: 80 ton high capacity - 105°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
	77	950	524	82	950	447	82	950	370	82	950	294	82	—	—	—	—	—	—	—	—	—
	72	868	615	81	868	537	81	868	460	81	868	383	81	868	306	81	—	—	—	—	—	—
14,000	67	793	703	80	793	625	80	793	548	80	793	470	80	793	393	80	793	316	80	—	—	—
	62	752	746	79	723	711	79	723	633	79	723	556	79	723	478	79	723	400	79	723	323	79
	57	752	750	79	718	714	79	684	679	78	661	639	78	661	561	78	661	483	78	661	406	78
	77	1,016	613	83	1,016	512	83	1,016	412	83	1,016	311	83	—	—	—	—	—	—	—	—	—
	72	933	730	82	933	629	82	933	527	82	933	426	82	933	325	82	—	—	—	—	—	—
18,500	67	855	845	81	855	743	81	855	641	81	855	540	81	855	438	81	855	337	81	—	—	—
	62	850	843	81	810	802	80	782	753	80	782	651	80	782	550	80	782	448	80	782	347	80
	57	850	848	81	810	806	80	771	765	80	732	725	79	716	659	79	716	557	79	716	455	79
	77	1,060	700	84	1,060	575	84	1,060	451	84	1,060	326	84	—	—	—	—	—	—	—	—	—
	72	975	842	83	975	717	83	975	592	83	975	467	83	975	342	83	—	—	—	—	—	—
23,000	67	927	913	82	896	857	81	896	731	81	896	605	81	896	480	81	896	355	81	—	—	—
	62	927	919	82	882	872	81	837	827	81	821	742	80	821	616	80	821	491	80	821	366	80
	57	927	925	82	882	878	81	837	832	81	794	787	80	751	743	79	751	625	79	751	499	79
	77	1,088	785	84	1,088	637	84	1,088	488	84	1,088	341	84	—	—	—	—	—	—	—	—	—
	72	1,003	952	83	1,003	803	83	1,003	654	83	1,003	505	83	1,003	357	83	—	—	—	—	—	—
27,500	67	987	972	83	937	921	82	922	818	82	922	669	82	922	519	82	922	371	82	—	—	—
	62	987	979	83	937	927	82	889	877	81	845	830	81	845	680	81	845	531	81	845	382	81
	57	987	984	83	937	933	82	889	883	81	841	834	81	795	786	80	774	689	80	774	540	80
	77	1,107	870	85	1,107	698	85	1,107	526	85	1,107	354	85	—	—	—	—	—	—	—	—	—
	72	1,035	1,012	84	1,022	889	83	1,022	716	83	1,022	543	83	1,022	371	83	—	—	—	—	—	—
32,000	67	1,035	1,020	84	982	965	83	941	904	82	941	731	82	941	558	82	941	386	82	—	—	—
	62	1,035	1,027	84	982	972	83	930	918	82	879	866	81	863	743	81	863	570	81	863	397	81
	57	1,035	1,033	84	982	977	83	930	924	82	879	871	81	829	820	80	790	752	80	790	579	80

Table 51: 80 ton high capacity - 115°F ambient air temperature

CFM	EWB (°F)	95°F EDB			90°F EDB			85°F EDB			80°F EDB			75°F EDB			70°F EDB			65°F EDB		
		TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW	TMBH	SMBH	kW
14,000	77	905	510	91	905	433	91	905	356	91	905	279	91	—	—	—	—	—	—	—	—	—
	72	828	600	89	828	522	89	828	445	89	828	368	89	828	291	89	—	—	—	—	—	—
	67	756	688	88	756	610	88	756	532	88	756	455	88	756	377	88	756	300	88	—	—	—
	62	725	719	88	692	684	87	690	617	87	690	540	87	690	462	87	690	385	87	690	307	87
	57	725	723	88	692	689	87	659	655	86	630	623	86	630	545	86	630	467	86	630	390	86
18,500	77	965	598	92	965	497	92	965	396	92	965	296	92	—	—	—	—	—	—	—	—	—
	72	887	714	90	887	613	90	887	511	90	887	410	90	887	309	90	—	—	—	—	—	—
	67	817	805	89	812	727	89	812	625	89	812	523	89	812	422	89	812	321	89	—	—	—
	62	817	810	89	778	770	89	744	736	88	744	634	88	744	533	88	744	431	88	744	330	88
	57	817	815	89	778	775	89	740	735	88	703	696	87	680	642	87	680	540	87	680	438	87
23,000	77	1,004	685	93	1,004	560	93	1,004	435	93	1,004	311	93	—	—	—	—	—	—	—	—	—
	72	925	826	91	925	700	91	925	575	91	925	450	91	925	326	91	—	—	—	—	—	—
	67	889	876	91	849	840	90	849	714	90	849	588	90	849	463	90	849	338	90	—	—	—
	62	889	882	91	846	837	90	802	792	89	778	725	89	778	599	89	778	473	89	778	348	89
	57	889	887	91	846	842	90	802	797	89	760	753	88	719	711	87	712	607	87	712	481	87
27,500	77	1,029	770	93	1,029	621	93	1,029	473	93	1,029	325	93	—	—	—	—	—	—	—	—	—
	72	949	936	92	949	786	92	949	637	92	949	489	92	949	340	92	—	—	—	—	—	—
	67	946	932	92	898	882	91	872	801	90	872	651	90	872	502	90	872	353	90	—	—	—
	62	946	938	92	898	888	91	850	840	90	804	792	89	800	662	89	800	513	89	800	364	89
	57	946	943	92	898	893	91	850	845	90	804	797	89	759	750	88	731	671	88	731	521	88
32,000	77	1,046	854	93	1,046	682	93	1,046	510	93	1,046	338	93	—	—	—	—	—	—	—	—	—
	72	991	969	92	966	872	92	966	699	92	966	527	92	966	354	92	—	—	—	—	—	—
	67	991	976	92	939	923	92	889	887	91	889	714	91	889	541	91	889	368	91	—	—	—
	62	991	983	92	939	929	92	888	877	91	839	827	90	815	725	89	815	552	89	815	379	89
	57	991	989	92	939	935	92	888	882	91	839	832	90	791	782	89	746	734	88	746	560	88

Notes:

Rated performance is at sea level. Cooling capacities are gross cooling capacities.

CFM = airflow, EWB = entering wet bulb air, EDB = entering dry bulb air, TMBH = total cooling capacity (MBH)

SMBH = sensible cooling capacity (MBH), kW = total input

Staged and modulating gas heat data

Table 52: Staged and modulating gas heat

Staged and modulating gas heat					Staged gas heat	Modulating gas heat
Model	Gas input capacity (MBH)	Maximum output capacity (MBH)	Minimum airflow (CFM)	Gas connection size (in.)	Steps	Turndown
60 ton	500	405	12,000	1.00	2	10:1
	1000	810	12,000	1.25	4	20:1
	1,500	1,215	12,000	1.50	6	30:1
70 ton to 80 ton	500	405	14,000	1.00	2	10:1
	1,000	810	14,000	1.25	4	20:1
	1,500	1,215	14,000	1.50	6	30:1

Note:

- Minimum airflow (CFM) is based on maximum output capacity (MBH).
- For proper operation, the building supplied natural gas pressure must be 7 iwg to 14 iwg and liquid propane pressure must be 12 iwg to 14 iwg.
- Temperature rise can be calculated where temperature rise = (gas input capacity (Btu) x 0.81) / (CFM x 1.085).
- Maximum leaving air temperature cannot exceed 120.0°F

Electric heat data

Table 53: Electric heat performance data

Voltage	Size (kW)	Heat capacity (MBH)	Minimum airflow (CFM)	Maximum temp. rise (°F)*
Low heat				
208/230	90	307	10,000	28
460	120	410	12,000	31
575	120	410	12,000	31
High heat				
208/230	120	410	10,000	38
460	200	683	12,000	52
575	200	683	12,000	52

Note:

* Maximum temperature rise is based on unit operation at the minimum airflow (CFM).

Hot water coil data

Table 54: Hot water coil performance data - 60 ton to 80 ton (1 row, low heat option)

GPM	Water pressure drop	FPS	CFM	Capacity (MBH) at entering water			
				140.0°F	160.0°F	180.0°F	200.0°F
25	2.4	2.1	12,000	304.7	389.1	474.0	559.3
			19,000	354.2	452.7	551.6	651.0
			26,000	386.5	494.2	602.4	711.0
			32,000	406.9	520.5	634.5	749.0
50	5.2	4.1	12,000	359.3	457.7	556.6	656.0
			19,000	432.1	550.7	669.8	789.6
			26,000	482.8	615.5	748.8	882.7
			32,000	516.2	658.2	800.9	944.2
80	9.8	6.6	12,000	384.6	489.3	594.7	700.4
			19,000	469.9	598.0	726.8	856.1
			26,000	530.9	675.8	821.4	967.7
			32,000	572.0	728.3	885.2	1042.8
90	12.3	7.4	12,000	389.6	495.6	602.3	709.3
			19,000	477.5	607.6	738.3	869.6
			26,000	540.8	688.3	836.3	985.1
			32,000	583.6	742.8	902.6	1063.1

Table 55: Hot water coil performance data - 60 ton to 80 ton (2 rows, high heat option)

GPM	Water pressure drop	FPS	CFM	Capacity (MBH) at entering water			
				140.0°F	160.0°F	180.0°F	200.0°F
50	3	2.1	12,000	457.3	583.1	709.5	836.3
			19,000	568.9	726.2	884.3	1043.2
			26,000	645.5	824.6	1004.4	1185.2
			32,000	694.9	888	1082	1276.9
90	5.4	3.7	12,000	502.2	639.1	776.3	914
			19,000	644.7	821	998.1	1175.8
			26,000	748.6	953.8	1159.9	1367
			32,000	818.5	1043.2	1268.8	1495.5
130	8	5.3	12,000	521.3	662.9	804.7	946.9
			19,000	678.3	862.9	1048.4	1234.1
			26,000	795.8	1012.8	1230.7	1449.4
			32,000	876.4	1115.5	1355.7	1596.8

Table 55: Hot water coil performance data - 60 ton to 80 ton (2 rows, high heat option)

GPM	Water pressure drop	FPS	CFM	Capacity (MBH) at entering water			
				140.0°F	160.0°F	180.0°F	200.0°F
160	12	6.6	12,000	529.7	673.3	817.2	961.4
			19,000	693.4	881.7	1070.8	1260.1
			26,000	817.3	1039.5	1262.7	1486.6
			32,000	903	1148.7	1395.4	1643.1

Note:

For Glycol solution performance access Selection Navigator.

Steam coil data

Table 56: Steam coil performance data - 60 ton to 70 ton (low heat option)

CFM	Capacity (MBH) at steam pressure (PSI)				
	2	4	6	10	15
12,000	538.1	558.1	576.7	609.6	645.0
17,000	607.9	631.5	652.2	689.3	729.4
22,000	657.1	680.7	704.1	744.0	787.1
26,000	685.9	710.2	734.2	775.4	820.3
28,000	697.1	721.9	746.2	787.9	833.5

Table 57: Steam coil performance data - 80 ton (low heat option)

CFM	Capacity (MBH) at steam pressure (PSI)				
	2	4	6	10	15
16,000	727.2	751.7	775.9	820.2	869.8
22,000	818.5	829.5	854.0	901.4	954.0
26,000	854.4	865.0	889.0	937.4	991.7
32,000	890.0	900.4	923.3	972.1	1028.1

Table 58: Steam coil performance data - 60 ton to 80 ton (high heat option)

CFM	Capacity (MBH) at steam pressure (PSI)				
	2	4	6	10	15
12,000	867.5	920.9	940.2	987.8	1044.6
17,000	1013.7	1317.4	1110.2	1131.2	1188.8
22,000	1182.0	1330.15	1478.3	1232.9	1282.2
32,000	—			1343.8	1365.4

Airflow performance

Component static pressure drops (iwg)

Table 59: Component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	Evaporator coil—standard capacity, standard efficiency	0.29	0.37	0.45	0.53	0.61	0.81	—	—	—	—
	Evaporator coil—high capacity, standard capacity, high efficiency	0.39	0.49	0.60	0.71	0.82	1.06	—	—	—	—
70 ton	Evaporator coil—standard capacity, standard efficiency	—	0.31	0.43	0.56	0.70	0.98	1.13	1.29	1.45	1.62
	Evaporator coil—high capacity, standard capacity, high efficiency	—	0.34	0.47	0.61	0.76	1.07	1.23	1.40	1.58	1.77
80 ton	Evaporator coil—standard capacity, standard efficiency	—	0.34	0.47	0.61	0.76	1.07	1.23	1.40	1.58	1.77
	Evaporator coil—high capacity, standard capacity, high efficiency	—	0.34	0.47	0.61	0.76	1.07	1.23	1.40	1.58	1.77
60 ton hot gas reheat (HGRH) coil		0.08	0.10	0.13	0.15	0.17	0.22	—	—	—	—
70 ton to 80 ton HGRH coil		—	0.07	0.09	0.11	0.13	0.18	0.20	0.22	0.25	0.27

Table 60: Return air opening - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	Bottom return basic econ and barometric econ	0.07	0.07	0.08	0.09	0.10	0.11	—	—	—	—
	Bottom return exhaust fan	0.07	0.07	0.07	0.08	0.09	0.10	—	—	—	—
	Bottom return return fan	0.11	0.13	0.15	0.18	0.21	0.28	—	—	—	—
	Damper	0.01	0.01	0.01	0.02	0.02	0.03	—	—	—	—
	Left and right return—basic econ and barometric econ	0.07	0.08	0.08	0.09	0.10	0.12	—	—	—	—
	Front return—basic econ—basic econ and barometric econ	0.07	0.07	0.07	0.08	0.08	0.10	—	—	—	—
	Top return—basic econ—basic econ and barometric econ	0.07	0.07	0.08	0.09	0.10	0.12	—	—	—	—
	Left and right return—exhaust fan	0.06	0.07	0.07	0.07	0.08	0.08	—	—	—	—
	Front return—basic econ—exhaust fan	0.07	0.07	0.07	0.08	0.08	0.10	—	—	—	—
	Top return—basic econ—exhaust fan	0.07	0.07	0.07	0.08	0.09	0.10	—	—	—	—
	Left, right, front return—return fan	0.11	0.13	0.15	0.18	0.21	0.28	—	—	—	—
	Bottom return—ERW	0.08	0.08	0.09	0.10	0.11	0.14	—	—	—	—

Table 60: Return air opening - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
70 ton to 80 ton	Bottom return basic econ and barometric econ	—	0.07	0.08	0.09	0.10	0.11	0.13	0.14	0.15	0.17
	Bottom return exhaust fan	—	0.07	0.07	0.08	0.09	0.10	0.11	0.12	0.13	0.14
	Bottom return return fan	—	0.13	0.15	0.18	0.21	0.28	0.32	0.37	0.42	0.47
	Damper	—	0.01	0.01	0.01	0.02	0.02	0.03	0.03	0.04	0.04
	Left and right return—basic econ and barometric econ	—	0.08	0.08	0.09	0.10	0.12	0.13	0.15	0.16	0.18
	Front return—basic econ—basic econ and barometric econ	—	0.07	0.07	0.08	0.08	0.10	0.11	0.11	0.12	0.13
	Top return—basic econ—basic econ and barometric econ	—	0.07	0.08	0.09	0.10	0.12	0.13	0.14	0.15	0.17
	Left and right return—exhaust fan	—	0.07	0.07	0.07	0.08	0.08	0.09	0.10	0.10	0.11
	Front return—basic econ—exhaust fan	—	0.07	0.07	0.08	0.08	0.10	0.11	0.11	0.12	0.13
	Top return—basic econ—exhaust fan	—	0.07	0.07	0.08	0.09	0.10	0.11	0.12	0.13	0.14
	Left, right, front return—return fan	—	0.11	0.12	0.14	0.17	0.22	0.25	0.28	0.32	0.36
	Bottom return—ERW	—	0.07	0.08	0.08	0.09	0.10	0.11	0.12	0.13	0.14

Table 61: Exhaust air opening - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	Barometric Econ Hood and Damper	0.17	0.22	0.29	0.37	0.47	0.69	—	—	—	—
	Econ with Exhaust Fan - Hood and Backdraft Damper	0.26	0.30	0.35	0.40	0.47	0.62	—	—	—	—
	Econ with Exhaust Fan - Hood and Modulating Damper	0.06	0.08	0.10	0.13	0.16	0.23	—	—	—	—
	Econ with Return Fan - Hood and Modulating Damper	0.04	0.06	0.08	0.10	0.12	0.17	—	—	—	—
	Barometric Econ Damper	0.14	0.18	0.23	0.29	0.37	0.55	—	—	—	—
	Econ with Exhaust Fan Backdraft Damper	0.02	0.03	0.04	0.05	0.07	0.09	—	—	—	—
	Econ with Exhaust Fan Modulating Damper	0.22	0.25	0.29	0.33	0.37	0.48	—	—	—	—
	Econ with Return Fan Modulating Damper	0.01	0.01	0.01	0.02	0.02	0.03	—	—	—	—
	Econ with Fan - Hood	0.04	0.05	0.06	0.08	0.10	0.14	—	—	—	—
	Econ with ERW Fan - Hood	0.07	0.10	0.12	0.16	0.20	0.28	—	—	—	—
	Econ With ERW - Hood and Backdraft Damper	0.32	0.39	0.47	0.56	0.66	0.89	1.03	—	—	—

Table 61: Exhaust air opening - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
70 ton to 80 ton	Barometric Econ Hood and Damper	—	0.04	0.05	0.06	0.08	0.11	0.13	0.15	0.17	0.19
	Econ with Exhaust Fan - Hood and Backdraft Damper	—	0.28	0.33	0.38	0.43	0.56	0.64	0.72	0.81	0.91
	Econ with Exhaust Fan - Hood and Modulating Damper	—	0.06	0.08	0.10	0.13	0.18	0.22	0.25	0.29	0.33
	Econ with Return Fan - Hood and Modulating Damper	—	0.04	0.05	0.06	0.08	0.11	0.13	0.15	0.2	0.19
	Barometric Econ Damper	—	0.01	0.01	0.01	0.01	0.02	0.02	0.03	0.03	0.03
	Econ with Exhaust Fan Backdraft Damper	—	0.03	0.04	0.05	0.07	0.09	0.11	0.13	0.15	0.17
	Econ with Exhaust Fan Modulating Damper	—	0.25	0.29	0.33	0.37	0.48	0.54	0.60	0.67	0.75
	Econ with Return Fan Modulating Damper	—	0.01	0.01	0.01	0.01	0.02	0.02	0.03	0.03	0.03
	Econ with Fan - Hood	—	0.03	0.04	0.05	0.06	0.09	0.11	0.12	0.14	0.16
	Econ with ERW Fan - Hood	—	0.06	0.08	0.10	0.12	0.18	0.21	0.24	0.28	0.32
	Econ With ERW - Hood and Backdraft Damper	—	0.31	0.37	0.43	0.50	0.65	0.75	0.85	0.95	1.07

Table 62: Supply air opening - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	Discharge from blank section (left and right)	0.05	0.07	0.09	0.11	0.13	0.19	—	—	—	—
	Discharge from blank section (top and bottom)	0.04	0.06	0.07	0.09	0.11	0.16	—	—	—	—
	Discharge from heating section (bottom and left)	0.04	0.05	0.06	0.08	0.10	0.14	—	—	—	—
70 ton to 80 ton	Discharge from blank section (left and right)	—	0.04	0.05	0.06	0.07	0.11	0.12	0.14	0.16	0.18
	Discharge from blank section (top and bottom)	—	0.03	0.04	0.05	0.07	0.09	0.11	0.12	0.14	0.16
	Discharge from heating section (bottom and left)	—	0.05	0.06	0.08	0.10	0.14	0.16	0.19	0.21	0.24

Table 63: Outside air - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	Barometric Econ - Hood and Dampers - Low Leak	0.14	0.18	0.22	0.27	0.32	0.44	—	—	—	—
	Barometric Econ - Hood and Dampers - Ultra Low Leak	0.14	0.18	0.22	0.26	0.32	0.43	—	—	—	—
	Econ with Fan- Hood and Dampers - Low Leak	0.10	0.12	0.15	0.17	0.20	0.27	—	—	—	—
	Econ with Fan - Hood and Dampers - Ultra Low Leak	0.10	0.12	0.14	0.17	0.20	0.27	—	—	—	—
	Barometric Econ - Mist and Damper - Low Leak	0.09	0.10	0.12	0.14	0.17	0.22	—	—	—	—
	Barometric Econ - Mist and Damper - Ultra Low Leak	0.09	0.10	0.12	0.14	0.16	0.22	—	—	—	—
	Econ with Fan- Mist and Dampers - Low Leak	0.07	0.08	0.09	0.11	0.12	0.16	—	—	—	—
	Econ with Fan - Mist and Dampers - Ultra Low Leak	0.07	0.08	0.09	0.10	0.12	0.15	—	—	—	—
	Barometric Econ - Hood	0.05	0.07	0.10	0.12	0.15	0.22	—	—	—	—
	Econ with Fan - Hood	0.03	0.04	0.05	0.07	0.08	0.12	—	—	—	—
70 ton to 80 ton	Barometric Econ - Hood and Dampers - Low Leak	—	0.17	0.21	0.26	0.31	0.43	0.49	0.57	0.64	0.72
	Barometric Econ - Hood and Dampers - Ultra Low Leak	—	0.17	0.21	0.26	0.31	0.43	0.49	0.56	0.64	0.72
	Econ with Fan- Hood and Dampers - Low Leak	—	0.12	0.14	0.17	0.20	0.26	0.30	0.34	0.39	0.43
	Econ with Fan - Hood and Dampers - Ultra Low Leak	—	0.12	0.14	0.17	0.19	0.26	0.30	0.34	0.38	0.43
	Barometric Econ - Mist and Damper - Low Leak	—	0.10	0.12	0.14	0.16	0.21	0.24	0.27	0.30	0.33
	Barometric Econ - Mist and Damper - Ultra Low Leak	—	0.10	0.12	0.14	0.16	0.21	0.23	0.26	0.30	0.33
	Econ with Fan- Mist and Dampers - Low Leak	—	0.08	0.09	0.10	0.11	0.15	0.16	0.18	0.20	0.22
	Econ with Fan - Mist and Dampers - Ultra Low Leak	—	0.08	0.09	0.10	0.11	0.14	0.16	0.18	0.20	0.22
	Barometric Econ - Hood	—	0.07	0.10	0.12	0.15	0.22	0.26	0.30	0.34	0.39
	Econ with Fan - Hood	—	0.04	0.05	0.07	0.08	0.12	0.14	0.16	0.18	0.21

Table 64: ERW - low CFM - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	ERW and filter OA	1.40	1.66	1.94	2.22	2.51	3.11	—	—	—	—
	ERW and filter RA	1.35	1.62	1.89	2.17	2.45	3.05	—	—	—	—
	Bypass damper (OA or RA)	0.17	0.19	0.21	0.23	0.25	0.29	—	—	—	—
70 ton to 80 ton	ERW and filter OA	1.22	1.46	1.69	1.94	2.19	2.71	2.98	3.25	3.53	3.82
	ERW and filter RA	1.19	1.41	1.65	1.89	2.13	2.64	2.90	3.17	3.44	3.72
	Bypass damper (OA or RA)	0.12	0.14	0.17	0.20	0.23	0.30	0.33	0.37	0.42	0.46

Table 65: ERW - high CFM - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	ERW and filter OA	1.07	1.27	1.48	1.70	1.92	2.39	—	—	—	—
	ERW and filter RA	1.03	1.23	1.44	1.66	1.88	2.35	—	—	—	—
	Bypass damper (OA or RA)	0.21	0.24	0.28	0.31	0.35	0.44	—	—	—	—
70 ton to 80 ton	ERW and filter OA	1.05	1.24	1.44	1.64	1.84	2.27	2.49	2.72	2.95	3.19
	ERW and filter RA	1.01	1.20	1.40	1.59	1.80	2.22	2.44	2.67	2.90	3.14
	Bypass damper (OA or RA)	0.13	0.16	0.19	0.22	0.26	0.35	0.39	0.44	0.50	0.55

Table 66: Draw-through filters - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	2 in. throwaway angled filters	0.06	0.08	0.09	0.10	0.11	0.14	—	—	—	—
	2 in. MERV 8 angled filters	0.06	0.08	0.09	0.11	0.13	0.17	—	—	—	—
	MERV 15 bag filters with 2 in. MERV 8 pre—filters	0.52	0.60	0.67	0.75	0.83	0.98	—	—	—	—
	MERV 14 rigid filters with 2 in. MERV 8 pre—filters	0.26	0.32	0.37	0.43	0.49	0.61	—	—	—	—
	Vertical 4 in. MERV 8 filters	0.04	0.05	0.07	0.08	0.10	0.14	—	—	—	—
70 ton to 80 ton	2 in. throwaway angled filters	—	0.08	0.09	0.10	0.11	0.14	0.15	0.17	0.18	0.20
	2 in. MERV 8 angled filters	—	0.08	0.09	0.11	0.13	0.17	0.19	0.21	0.23	0.25
	MERV 15 bag filters with 2 in. MERV 8 pre—filters	—	0.60	0.67	0.75	0.83	0.98	1.06	1.13	1.20	1.28
	MERV 14 rigid filters with 2 in. MERV 8 pre—filters	—	0.32	0.37	0.43	0.49	0.61	0.67	0.74	0.80	0.86
	Vertical 4 in. MERV 8 filters	—	0.05	0.07	0.08	0.10	0.14	0.16	0.18	0.20	0.23

Table 67: Final filters - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	MERV 15 bag filters with 2 in. pre—filters	0.52	0.60	0.67	0.75	0.83	0.98	—	—	—	—
	MERV 14 rigid filters with 2 in. pre—filters	0.26	0.32	0.37	0.43	0.49	0.61	—	—	—	—
	MERV 17 HEPA filters with 2 in. pre—filters	0.34	0.36	0.39	0.42	0.45	0.51	—	—	—	—
70 ton to 80 ton	MERV 15 bag filters with 2 in. pre—filters	—	0.60	0.67	0.75	0.83	0.98	1.06	1.13	1.20	1.28
	MERV 14 rigid filters with 2 in. pre—filters	—	0.32	0.37	0.43	0.49	0.61	0.67	0.74	0.80	0.86
	MERV 17 HEPA filters with 2 in. pre—filters	—	0.36	0.39	0.42	0.45	0.51	0.55	0.58	0.62	0.66

Table 68: Gas heat - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton to 80 ton	500 MBH	—	0.07	0.10	0.14	0.18	0.27	0.33	0.38	0.45	0.52
	1000 MBH	0.08	0.14	0.19	0.25	0.30	0.38	0.42	0.46	0.50	0.53
	1500 MBH	0.17	0.21	0.26	0.32	0.39	0.59	0.71	0.84	0.98	1.14

Table 69: Electric heat - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton to 80 ton	Low and high heat	0.03	0.03	0.03	0.03	0.03	0.04	0.04	0.04	0.04	0.04

Table 70: Hot water and steam coil heat - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	1 row hot water, low heat	0.05	0.07	0.08	0.1	0.12	0.16	—	—	—	—
	2 rows hot water, high heat	0.08	0.11	0.13	0.16	0.19	0.26	—	—	—	—
	Steam, low heat	0.04	0.05	0.07	0.08	0.1	0.14	—	—	—	—
	Steam, high heat	0.08	0.1	0.12	0.14	0.17	0.23	—	—	—	—
70 ton	1 row hot water, low heat	—	0.07	0.08	0.1	0.12	0.16	0.18	0.2	—	—
	2 rows hot water, high heat	—	0.11	0.13	0.16	0.19	0.26	0.3	0.34	—	—
	Steam, low heat	—	0.05	0.07	0.08	0.1	0.14	0.16	0.18	—	—
	Steam, high heat	—	0.1	0.12	0.14	0.17	0.23	0.26	0.29	—	—
80 ton	1 row hot water, low heat	—	—	0.08	0.1	0.12	0.16	0.18	0.2	0.23	0.25
	2 rows hot water, high heat	—	—	0.13	0.16	0.19	0.26	0.3	0.34	0.38	0.43
	Steam, low heat	—	—	0.09	0.1	0.13	0.17	0.2	0.23	0.26	0.29
	Steam, high heat	—	—	0.12	0.14	0.17	0.23	0.26	0.29	0.32	0.36

Table 71: Options - component static pressure drops (iwg)

Component		SCFM									
		12,000	14,000	16,000	18,000	20,000	24,000	26,000	28,000	30,000	32,000
60 ton	Humidifier	0.00	0.00	0.01	0.01	—	—	—	—	—	—
	Sound attenuator	0.01	0.02	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
	Inlet guard—supply fan	0.25	0.34	0.43	0.53	0.65	0.91	—	—	—	—
	Air blender	0.02	0.05	0.08	0.1	0.13	0.18	—	—	—	—
70 ton to 80 ton	Humidifier	—	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.02
	Sound attenuator	0.01	0.02	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
	Inlet guard—supply fan	—	0.34	0.43	0.53	0.65	0.91	1.05	1.21	1.38	1.56
	Air blender	—	0.03	0.05	0.07	0.09	0.14	0.16	0.19	0.21	0.23

Supply fan data

Table 72: 60 ton total static pressure

CFM	1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP		3.5 TSP		4.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
12,000	805	3.72	905	5.40	996	7.18	1087	—	—	—	—	—	—	—
13,000	839	4.15	933	5.89	1021	7.79	1103	9.74	1190	11.80	—	—	—	—
14,000	873	4.61	963	6.42	1049	8.41	1128	10.47	1203	12.59	1285	14.84	1369	17.25
15,000	910	5.13	996	7.02	1077	9.06	1155	11.24	1227	13.47	1297	15.75	1373	18.16
16,000	950	5.71	1030	7.68	1107	9.77	1182	12.03	1253	14.38	1321	16.76	1386	19.20
17,000	989	6.33	1064	8.37	1139	10.55	1211	12.86	1280	15.31	1346	17.81	1409	20.35
18,000	1028	7.00	1100	9.13	1173	11.40	1241	13.75	1308	16.27	1373	18.90	1434	21.55
19,000	1067	7.73	1139	9.96	1207	12.29	1273	14.74	1337	17.29	1400	20.00	1461	22.79
20,000	1107	8.51	1179	10.85	1242	13.24	1307	15.79	1368	18.40	1429	21.16	1488	24.04
21,000	1147	9.35	1219	11.80	1278	14.28	1341	16.90	1401	19.60	1459	22.40	1517	25.34
22,000	1188	10.26	1257	12.81	1318	15.40	1376	18.06	1435	20.88	1491	23.73	1546	26.71
23,000	1230	11.23	1296	13.89	1358	16.59	1411	19.33	1469	22.21	1524	25.16	1577	28.19
24,000	1272	12.28	1336	15.03	1398	17.85	1450	20.68	1504	23.60	1558	26.68	1610	29.77

Table 73: 60 ton total static pressure

CFM	4.5 TSP		5.0 TSP		5.5 TSP		6.0 TSP		6.5 TSP		7.0 TSP		7.5 TSP		8.0 TSP		8.3 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
12,000	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
13,000	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
14,000	1451	19.81	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
15,000	1452	20.73	1529	23.44	—	—	—	—	—	—	—	—	—	—	—	—	—	—
16,000	1457	21.76	1531	24.49	1604	27.34	1675	30.31	—	—	—	—	—	—	—	—	—	—
17,000	1470	22.94	1537	25.66	1607	28.52	1676	31.51	1744	34.62	—	—	—	—	—	—	—	—
18,000	1494	24.25	1552	26.99	1614	29.83	1680	32.83	1745	35.95	1810	39.19	1873	42.52	—	—	—	—
19,000	1519	25.59	1576	28.43	1630	31.32	1688	34.30	1750	37.42	1812	40.67	1874	44.03	1934	47.49	1970	49.61
20,000	1546	26.98	1601	29.92	1654	32.92	1706	35.96	1760	39.06	1818	42.30	1877	45.67	1936	49.14	1971	51.28
21,000	1573	28.38	1627	31.47	1679	34.56	1731	37.71	1780	40.89	1830	44.12	1884	47.47	1941	50.95	1975	53.09
22,000	1601	29.83	1654	33.03	1706	36.26	1756	39.50	1805	42.79	1852	46.12	1899	49.48	1949	52.94	—	—
23,000	1630	31.35	1682	34.62	1733	37.98	1782	41.36	1830	44.75	1877	48.17	1923	51.64	—	—	—	—
24,000	1660	32.96	1711	36.29	1761	39.72	1809	43.23	1857	46.76	1903	50.29	1948	53.86	—	—	—	—

Note:

The CFM, SP, and BHP values are based on two fans for the DDP supply fans (GEN II 270—9—120)

Table 74: 70 ton to 80 ton total static pressure

CFM	1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP		3.5 TSP		4.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
14,000	873	4.61	963	6.42	1049	8.41	1128	10.47	1203	12.59	1285	14.84	1369	17.25
15,000	910	5.13	996	7.02	1077	9.06	1155	11.24	1227	13.47	1297	15.75	1373	18.16
16,000	950	5.71	1030	7.68	1107	9.77	1182	12.03	1253	14.38	1321	16.76	1386	19.20
17,000	989	6.33	1064	8.37	1139	10.55	1211	12.86	1280	15.31	1346	17.81	1409	20.35
18,000	1028	7.00	1100	9.13	1173	11.40	1241	13.75	1308	16.27	1373	18.90	1434	21.55
19,000	1067	7.73	1139	9.96	1207	12.29	1273	14.74	1337	17.29	1400	20.00	1461	22.79
20,000	1107	8.51	1179	10.85	1242	13.24	1307	15.79	1368	18.40	1429	21.16	1488	24.04
21,000	1147	9.35	1219	11.80	1278	14.28	1341	16.90	1401	19.60	1459	22.40	1517	25.34
22,000	1188	10.26	1257	12.81	1318	15.40	1376	18.06	1435	20.88	1491	23.73	1546	26.71
23,000	1230	11.23	1296	13.89	1358	16.59	1411	19.33	1469	22.21	1524	25.16	1577	28.19
24,000	1272	12.28	1336	15.03	1398	17.85	1450	20.68	1504	23.60	1558	26.68	1610	29.77
25,000	1315	13.40	1376	16.25	1436	19.18	1490	22.11	1539	25.10	1592	28.24	1643	31.45

Table 74: 70 ton to 80 ton total static pressure

CFM	1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP		3.5 TSP		4.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
26,000	1358	14.60	1417	17.55	1475	20.57	1530	23.63	1577	26.70	1627	29.87	1677	33.20
27,000	1401	15.89	1458	18.92	1514	22.05	1569	25.22	1616	28.39	1662	31.62	1712	35.00
28,000	1445	17.25	1499	20.38	1554	23.60	1608	26.88	1657	30.17	1700	33.48	1746	36.90
29,000	1489	18.70	1542	21.93	1594	25.24	1646	28.63	1696	32.04	1739	35.45	1782	38.92
30,000	1533	20.25	1584	23.56	1635	26.97	1685	30.46	1735	33.98	1780	37.51	1820	41.06
31,000	1577	21.88	1627	25.29	1676	28.80	1725	32.38	1774	36.01	1819	39.66	1859	43.30
32,000	1622	23.61	1670	27.12	1718	30.72	1765	34.40	1812	38.13	1858	41.89	1900	45.65

Table 75: 70 ton to 80 ton total static pressure

CFM	4.5 TSP		5.0 TSP		5.5 TSP		6.0 TSP		6.5 TSP		7.0 TSP		7.5 TSP		8.0 TSP		8.3 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
14,000	1451	19.81	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
15,000	1452	20.73	1529	23.44	—	—	—	—	—	—	—	—	—	—	—	—	—	—
16,000	1457	21.76	1531	24.49	1604	27.34	1675	30.31	—	—	—	—	—	—	—	—	—	—
17,000	1470	22.94	1537	25.66	1607	28.52	1676	31.51	1744	34.62	—	—	—	—	—	—	—	—
18,000	1494	24.25	1552	26.99	1614	29.83	1680	32.83	1745	35.95	1810	39.19	1873	42.52	—	—	—	—
19,000	1519	25.59	1576	28.43	1630	31.32	1688	34.30	1750	37.42	1812	40.67	1874	44.03	1934	47.49	1970	49.61
20,000	1546	26.98	1601	29.92	1654	32.92	1706	35.96	1760	39.06	1818	42.30	1877	45.67	1936	49.14	1971	51.28
21,000	1573	28.38	1627	31.47	1679	34.56	1731	37.71	1780	40.89	1830	44.12	1884	47.47	1941	50.95	1975	53.09
22,000	1601	29.83	1654	33.03	1706	36.26	1756	39.50	1805	42.79	1852	46.12	1899	49.48	1949	52.94	—	—
23,000	1630	31.35	1682	34.62	1733	37.98	1782	41.36	1830	44.75	1877	48.17	1923	51.64	—	—	—	—
24,000	1660	32.96	1711	36.29	1761	39.72	1809	43.23	1857	46.76	1903	50.29	1948	53.86	—	—	—	—
25,000	1692	34.69	1741	38.06	1789	41.55	1837	45.13	1884	48.78	1929	52.46	—	—	—	—	—	—
26,000	1725	36.53	1772	39.94	1819	43.47	1866	47.11	1911	50.85	1956	54.64	—	—	—	—	—	—
27,000	1759	38.47	1805	41.94	1850	45.51	1895	49.19	1940	52.99	—	—	—	—	—	—	—	—
28,000	1794	40.46	1839	44.05	1883	47.68	1926	51.39	—	—	—	—	—	—	—	—	—	—
29,000	1828	42.52	1873	46.24	1916	49.96	1958	53.73	—	—	—	—	—	—	—	—	—	—
30,000	1863	44.70	1907	48.48	1950	52.34	—	—	—	—	—	—	—	—	—	—	—	—
31,000	1899	47.00	1942	50.82	1984	54.78	—	—	—	—	—	—	—	—	—	—	—	—
32,000	1937	49.44	1977	53.30	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Note:

The CFM, SP, and BHP values are based on two fans for the DDP supply fans (GEN II 270—9—120)

Exhaust fan data

Table 76: 60 ton total static pressure

CFM	0.5 TSP		1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
12,000	257	1.97	335	3.21	—	—	—	—	—	—	—	—
13,000	264	2.27	340	3.65	—	—	—	—	—	—	—	—
14,000	271	2.60	345	4.09	407	5.54	—	—	—	—	—	—
15,000	280	2.98	350	4.55	411	6.09	—	—	—	—	—	—
16,000	289	3.41	357	5.05	416	6.76	470	8.42	—	—	—	—
17,000	297	3.88	364	5.60	421	7.42	473	9.12	—	—	—	—
18,000	306	4.38	371	6.19	427	8.10	478	9.97	525	11.85	—	—
19,000	315	4.93	378	6.82	433	8.82	483	10.87	529	12.73	—	—
20,000	323	5.53	385	7.50	440	9.60	489	11.76	534	13.80	576	15.86
21,000	332	6.19	394	8.26	447	10.43	494	12.68	539	14.92	580	16.96
22,000	342	6.92	402	9.09	454	11.33	501	13.64	544	16.03	585	18.26

Table 76: 60 ton total static pressure

CFM	0.5 TSP		1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
23,000	351	7.71	411	9.98	461	12.26	507	14.68	550	17.16	591	19.61
24,000	361	8.57	420	10.93	468	13.26	514	15.78	556	18.34	596	20.94

Table 77: 70 ton to 80 ton total static pressure

CFM	0.5 TSP		1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
14,000	271	2.60	345	4.09	407	5.54	—	—	—	—	—	—
15,000	280	2.98	350	4.55	411	6.09	—	—	—	—	—	—
16,000	289	3.41	357	5.05	416	6.76	470	8.42	—	—	—	—
17,000	297	3.88	364	5.60	421	7.42	473	9.12	—	—	—	—
18,000	306	4.38	371	6.19	427	8.10	478	9.97	525	11.85	—	—
19,000	315	4.93	378	6.82	433	8.82	483	10.87	529	12.73	—	—
20,000	323	5.53	385	7.50	440	9.60	489	11.76	534	13.80	576	15.86
21,000	332	6.19	394	8.26	447	10.43	494	12.68	539	14.92	580	16.96
22,000	342	6.92	402	9.09	454	11.33	501	13.64	544	16.03	585	18.26
23,000	351	7.71	411	9.98	461	12.26	507	14.68	550	17.16	591	19.61
24,000	361	8.57	420	10.93	468	13.26	514	15.78	556	18.34	596	20.94
25,000	371	9.50	429	11.92	476	14.34	521	16.95	562	19.58	601	22.29
26,000	381	10.50	437	12.98	484	15.52	528	18.17	569	20.90	607	23.69
27,000	390	11.58	446	14.10	493	16.78	535	19.45	576	22.29	614	25.15
28,000	400	12.74	455	15.29	502	18.11	542	20.81	583	23.76	620	26.69
29,000	410	13.98	464	16.57	511	19.51	551	22.29	590	25.31	627	28.32
30,000	420	15.30	473	17.94	520	20.97	559	23.86	597	26.87	634	30.04
31,000	430	16.71	482	19.40	528	22.49	568	25.53	604	28.55	641	31.83
32,000	440	18.20	492	20.95	537	23.73	577	27.28	612	30.35	648	33.68

Table 78: 60 ton total static pressure table for dual exhaust fans

CFM	0.5 TSP		1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
12,000	371	2.09	465	3.01	569	4.23	669	5.63	—	—	—	—
13,000	389	2.52	471	3.42	569	4.69	662	6.09	754	7.68	838	9.3
14,000	407	3.01	481	3.91	568	5.15	657	6.6	742	8.17	829	9.96
15,000	432	3.64	493	4.49	571	5.7	657	7.21	738	8.81	816	10.51
16,000	461	4.42	509	5.16	578	6.34	656	7.83	734	9.46	811	11.25
17,000	490	5.31	525	5.92	587	7.07	658	8.53	735	10.26	805	12
18,000	518	6.3	543	6.77	599	7.91	664	9.35	734	11.03	804	12.88
19,000	547	7.41	561	7.7	614	8.87	672	10.25	736	11.91	805	13.84
20,000	576	8.64	579	8.72	629	9.93	682	11.29	742	12.94	804	14.77
21,000	605	10	625	10.55	646	11.1	695	12.45	749	14.04	807	15.87
22,000	634	11.5	634	11.5	664	12.37	709	13.73	758	15.29	813	17.12
23,000	662	13.14	672	13.45	682	13.76	725	15.15	770	16.68	820	18.45
24,000	691	14.93	696	15.09	700	15.24	742	16.68	784	18.21	830	19.94

Table 79: 70 ton to 80 ton total static pressure table for dual exhaust fans

CFM	0.5 TSP		1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
14,000	335	2.26	443	3.62	539	5.19	629	6.97	—	—	—	—
15,000	346	2.61	444	3.96	538	5.58	629	7.5	703	9.32	—	—
16,000	358	3.02	450	4.39	541	6.08	624	7.92	704	9.98	—	—
17,000	371	3.48	457	4.89	543	6.58	620	8.39	703	10.62	770	12.69
18,000	384	3.98	463	5.39	543	7.08	623	9.03	697	11.12	771	13.5

Table 79: 70 ton to 80 ton total static pressure table for dual exhaust fans

CFM	0.5 TSP		1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
19,000	398	4.55	468	5.91	547	7.66	625	9.71	694	11.73	768	14.19
20,000	412	5.17	476	6.52	554	8.37	627	10.38	696	12.52	762	14.8
21,000	426	5.86	487	7.22	561	9.13	628	11.05	699	13.36	760	15.58
22,000	443	6.65	498	8.01	567	9.87	632	11.85	700	14.21	763	16.55
23000	463	7.6	511	8.87	572	10.62	639	12.79	701	15.04	765	17.56
24,000	483	8.64	524	9.8	579	11.5	646	13.8	703	15.93	767	18.57
25,000	503	9.76	537	10.79	588	12.48	653	14.8	706	17.01	768	19.57
26,000	524	10.98	550	11.86	599	13.58	657	15.78	716	18.23	769	20.61
27,000	544	12.3	564	13.02	611	14.78	663	16.85	724	19.49	774	21.84
28,000	564	13.71	578	14.26	624	16.06	671	18.06	730	20.74	781	23.25
29,000	584	15.24	593	15.6	637	17.43	681	19.41	734	21.96	789	24.74
30,000	604	16.87	607	17.01	649	18.88	692	20.87	740	23.28	796	26.27
31,000	624	18.61	642	19.52	663	20.42	704	22.46	747	24.76	801	27.74
32,000	644	20.47	656	20.98	676	22.06	716	24.15	757	26.39	805	29.22

Return fan data

Table 80: 60 ton total static pressure

CFM	0.5 TSP		1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
12,000	667	1.96	745	3.17	812	4.59	876	6.03	936	7.51	998	9.09
13,000	711	2.26	786	3.58	847	5.06	908	6.55	966	8.16	1,022	9.77
14,000	755	2.61	825	4.02	884	5.48	942	7.16	998	8.80	1,051	10.53
15,000	799	2.99	863	4.48	925	6.01	978	7.79	1,031	9.48	1,083	11.31
16,000	844	3.43	903	4.99	966	6.63	1,015	8.34	1,065	10.29	1,115	12.09
17,000	889	3.92	945	5.55	1,006	7.30	1,055	9.00	1,101	11.04	1,149	13.01
18,000	934	4.45	988	6.15	1,043	7.96	1,096	9.80	1,139	11.73	1,184	13.98
19,000	979	5.03	1,032	6.81	1,082	8.69	1,137	10.68	1,179	12.57	1,220	14.82
20,000	1,025	5.65	1,076	7.52	1,123	9.49	1,175	11.55	1,220	13.69	1,259	15.68
21,000	1,071	6.31	1,121	8.29	1,165	10.35	1,212	12.46	1,261	14.65	1,299	16.74
22,000	1,116	7.01	1,165	9.13	1,208	11.26	1,252	13.45	1,300	15.76	1,340	17.95
23,000	1,162	7.75	1,210	10.05	1,252	12.23	1,293	14.51	1,337	16.85	1,381	19.25
24,000	1,208	8.54	1,255	11.04	1,296	13.27	1,335	15.65	1,376	18.04	1,420	20.58

Table 81: 70 ton to 80 ton total static pressure

CFM	0.5 TSP		1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
14,000	755	2.61	825	4.02	884	5.48	942	7.16	998	8.80	1,051	10.53
15,000	799	2.99	863	4.48	925	6.01	978	7.79	1,031	9.48	1,083	11.31
16,000	844	3.43	903	4.99	966	6.63	1,015	8.34	1,065	10.29	1,115	12.09
17,000	889	3.92	945	5.55	1,006	7.30	1,055	9.00	1,101	11.04	1,149	13.01
18,000	934	4.45	988	6.15	1,043	7.96	1,096	9.80	1,139	11.73	1,184	13.98
19,000	979	5.03	1,032	6.81	1,082	8.69	1,137	10.68	1,179	12.57	1,220	14.82
20,000	1,025	5.65	1,076	7.52	1,123	9.49	1,175	11.55	1,220	13.69	1,259	15.68
21,000	1,071	6.31	1,121	8.29	1,165	10.35	1,212	12.46	1,261	14.65	1,299	16.74
22,000	1,116	7.01	1,165	9.13	1,208	11.26	1,252	13.45	1,300	15.76	1,340	17.95
23,000	1,162	7.75	1,210	10.05	1,252	12.23	1,293	14.51	1,337	16.85	1,381	19.25
24,000	1,208	8.54	1,255	11.04	1,296	13.27	1,335	15.65	1,376	18.04	1,420	20.58
25,000	1,254	9.38	1,300	12.09	1,340	14.38	1,378	16.84	1,416	19.32	1,458	21.89
26,000	1,300	10.26	1,346	13.20	1,385	15.57	1,421	18.10	1,457	20.68	1,496	23.29
27,000	1,347	11.20	1,391	14.38	1,430	16.85	1,465	19.43	1,500	22.11	1,535	24.79

Table 81: 70 ton to 80 ton total static pressure

CFM	0.5 TSP		1.0 TSP		1.5 TSP		2.0 TSP		2.5 TSP		3.0 TSP	
	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP	RPM	BHP
28,000	1,393	12.19	1,436	15.62	1,475	18.23	1,509	20.84	1,543	23.61	1,576	26.39
29,000	1,440	13.25	1,482	16.91	1,520	19.68	1,554	22.34	1,586	25.18	1,618	28.07
30,000	1,486	14.37	1,528	18.26	1,565	21.22	1,598	23.94	1,630	26.84	1,661	29.82
31,000	1,533	15.55	1,574	19.67	1,610	22.83	1,643	25.64	1,674	28.58	1,704	31.64
32,000	1,580	16.80	1,619	21.14	1,655	24.53	1,688	27.44	1718	30.42	1,747	33.55

Electrical data

Electrical service sizing

In order to determine the electrical service required for the Premier rooftop unit, use the appropriate calculations listed below from U.L. 60335-2-40. Based on the configuration of the unit, the calculations yield different minimum circuit ampacity (MCA) and maximum overcurrent protection (MOP).

Using the following load definitions and calculations, determine the correct electrical sizing for the unit. All concurrent load conditions must be considered in the calculations, and the highest value for any combination of loads must be used.

Load definitions:

- **LOAD1** is the current of the largest motor – compressor or fan motor.
- **LOAD2** is the sum of the remaining motor currents that may run concurrently with LOAD1.
- **LOAD3** is the current of the electric heaters – zero for cooling only units.
- **LOAD4** is the sum of any remaining currents greater than or equal to 1.0 amp.

Use the following calculations to determine MCA and MOP for units supplied with a single-point power connection:

Note: For electric heat applications, the nameplate MCA/MOP value is the larger of the cooling mode or the heating mode value calculated.

MCA = (1.25 x Load1) + (1.25 x Load3) + Load2 + Load4

Exception: Load3 may be multiplied by 1 if the electric heater is greater than 50 kW

MOP = (2.25 x Load1) + Load2 + Load3 + Load4

Use the following calculations to determine MCA and MOP for units supplied with a dual-point power connection:

Note: For electric heat applications, the nameplate MCA 1/MOP 1 value is the larger of the cooling mode or the heating mode value calculated.

Electrical circuit 1: compressors and crankcase heaters, condenser fan motors, electric heat (MCA 1/MOP 1).

Electrical circuit 2: supply fan, return and exhaust fan, ERW, 120 V transformer for lights, UV lights, convenience outlet, and 24 VAC control transformers (MCA 2/MOP 2).

Use the formulas above to calculate MCA and MOP for electrical circuits 1 and 2.

For single or dual point power connection, if the MOP does not equal a standard current rating of an overcurrent protective device, then the marked maximum rating is to be the next lower standard rating. However, if the device selected for MOP is less than the MCA, then select the lowest standard maximum fuse size greater than or equal to the MCA.

Power supply voltage limits

Table 82: Power supply voltage limits

Power supply	Minimum voltage	Maximum voltage
208-230V/3Ph/60Hz	187	253
460V/3Ph/60Hz	414	506
575V/3Ph/60Hz	518	632

Fixed speed compressor data

Table 83: Standard efficiency and standard capacity compressor electrical data

Model	Compressor	Nominal voltage					
		208-230/3/60*		460/3/60		575/3/60	
		RLA	LRA	RLA	LRA	RLA	LRA
60 ton	1A	41	304	19.2	147	16.7	122
	1B	48.1	351	24.7	197	22.4	135
	2A	41	304	19.2	147	16.7	122
	2B	48.1	351	24.7	197	22.4	135
70 ton	1A	39.1	267	18.6	142	15.4	103
	1B	67.3	485	32.7	215	26.3	175
	2A	48.1	351	24.7	197	22.4	135
	2B	48.1	351	24.7	197	22.4	135
80 ton	1A	48.1	351	24.7	197	22.4	135
	1B	48.1	351	24.7	197	22.4	135
	2A	48.1	351	24.7	197	22.4	135
	2B	67.3	485	32.7	215	26.3	175

Note:

* 208 V to 230 V amps are based on 230 V operation.

Table 84: Standard efficiency and high capacity compressor electrical data

Model	Compressor	Nominal voltage					
		208-230/3/60*		460/3/60		575/3/60	
		RLA	LRA	RLA	LRA	RLA	LRA
60 ton	1A	44.2	315	22.4	158	18.5	136
	1B	48.1	351	24.7	197	22.4	135
	2A	44.2	315	22.4	158	18.5	136
	2B	48.1	351	24.7	197	22.4	135
70 ton	1A	48.1	351	24.7	197	22.4	135
	1B	48.1	351	24.7	197	22.4	135
	2A	48.1	351	24.7	197	22.4	135
	2B	67.3	485	32.7	215	26.3	175
80 ton	1A	48.1	351	24.7	197	22.4	135
	1B	67.3	485	32.7	215	26.3	175
	2A	48.1	351	24.7	197	22.4	135
	2B	67.3	485	32.7	215	26.3	175

Note:

* 208 V to 230 V amps are based on 230 V operation.

Variable speed drive compressor data

Table 85: High efficiency and standard capacity compressor electrical data

Model	Compressor	Nominal voltage								
		208—230/3/60*			460/3/60			575/3/60		
		RLA	Input A	LRA	RLA	Input A	LRA	RLA	Input A	LRA
60 ton	VSD[2,3]	—	80	—	—	40	—	—	31	—
	1B	48.1	—	351	24.7	—	197	22.4	—	135
	2A	39.1	—	267	18.6	—	142	15.4	—	103
	2B	48.1	—	351	24.7	—	197	22.4	—	135
70 ton	VSD[2,3]	—	104	—	—	47	—	—	37	
	1B	48.1	—	351	24.7	—	197	22.4	—	135
	2A	44.2	—	315	22.4	—	158	18.6	—	136
	2B	48.1	—	351	24.7	—	197	22.4	—	135
80 ton	VSD[2,3]	—	104	—	—	47	—	—	37	—
	1B	48.1	—	351	24.7	—	197	22.4	—	135
	2A	48.1	—	351	24.7	—	197	22.4	—	135
	2B	48.1	—	351	24.7	—	197	22.4	—	135

Note:

1. 208 V to 230 V amps are based on 230 V operation.
2. Variable speed drive compressor on circuit 1A.
3. Input A supplied for maximum input current to compressor VFD.

Condenser fan motor

Table 86: Condenser fan motor 60 ton to 80 ton

Unit configurations					Each motor data	208-230V/3Ph/60Hz *		460V/3Ph/60Hz		575V/3Ph/60Hz	
						SS FLA	VFD IN A	SS FLA	VFD IN A	SS FLA	VFD IN A
						7.2	9.5	3.4	4.3	1.7	3.9
Condenser fan option	Unit size	Comp config	Total quantity of fans	Quantity SS fans	Quantity VFD fans	Total fan amps					
Standard ambient	60 ton	All configurations	4	4	0	28.8		13.6		6.8	
	70 ton to 80 ton	All configurations	6	6	0	43.2		20.4		10.2	
Low ambient	60 ton	All configurations	4	2	2	33.4		15.4		11.2	
	70 ton to 80 ton	All configurations	6	4	2	47.8		22.2		14.6	

Note:

* 208 V to 230 V amps are based on 230 V operation.

Electric heat

Table 87: Electric heat amp draw

Capacity	Type of heat	Heater nominal kW	Each stage kW	Steps	208-230V/3Ph/60Hz*	460V/3Ph/60Hz	575V/3Ph/60Hz
					AMPS	AMPS	AMPS
60 ton to 80 ton	Low heat	90	15	6	207.49	—	—
	Low heat with SCR**			6***			
	Low heat	120	20	6	—	137.63	110.55
	Low heat with SCR**			6***			
	High heat	120	15	8	276.65	—	—
	High heat with SCR**			8***			
	High heat	200	20	10	—	229.39	184.43
	High heat with SCR**			10***			

Note:

* 208 V to 230 V amps are based on 230 V operation.

** Silicon Controlled Rectifier (SCR) is a modulating electric heat option.

*** SCR electric heat is installed on the first stage of the unit and provides modulation from 0–100%

Supply fan and return and exhaust fan motor data

Table 88: Premium efficiency - ODP

Motor HP		208-230/3/60* FLA	460/3/60 FLA	575/3/60 FLA
Supply fan motor (each) **	5	13.2	6.6	5.3
	7.5	19.4	9.7	8
	10	25	12.5	10
	15	36	18	14.2
	20	48	24	19.1
	25	60	30	24.5
	30	72	36	29
	40	98	49	40
Return fan motor**	7.5	19.7	9.9	7.88
	10	25.2	12.6	10.1
	15	36.9	18.5	14.8
	20	48.5	24.3	19.4
	25	57.5	28.7	23
	30	68.6	34.3	27.4
	40	92.6	46.3	37
Exhaust fan motor	5	13.2	6.6	5.3
	7.5	19.4	9.7	8.01
	10	25	12.5	10
	15	36	18	14.2
	20	48	24	19.1
	25	60	30	24.5
	30	72	36	29
	40	98	49	40

Table 88: Premium efficiency - ODP

Motor HP		Nominal voltage		
		208-230/3/60*	460/3/60	575/3/60
		FLA	FLA	FLA
Exhaust fan motor with VFD	5	14.1	7.5	6.2
	7.5	21	10.3	9.2
	10	31.1	13.9	12.5
	15	45.1	20.2	17.3
	20	58	27.2	23.6
	25	71.5	32.3	28.3
	30	83.7	38.1	34.6
	40	114.7	54.1	47.2

Table 89: Premium efficiency - TEFC

Motor HP		Nominal voltage		
		208-230/3/60*	460/3/60	575/3/60
		FLA	FLA	FLA
Supply fan motor (each) **	5	13.4	6.7	5.3
	7.5	19	9.5	7.6
	10	24	12	9.6
	15	36.2	18.1	14.6
	20	48	24	19.2
	25	62	31	24
	30	76	38	29
	40	96	48	39
Return fan motor**	7.5	19.1	9.59	7.67
	10	25.6	12.8	10.2
	15	36.4	18.2	14.6
	20	48.6	24.3	19.5
	25	58.2	29.1	23.3
	30	68.6	34.3	27.4
	40	92.6	46.3	37.0
Exhaust fan motor	5	13.4	6.7	5.3
	7.5	19	9.5	7.6
	10	24	12	9.6
	15	36.2	18.1	14.6
	20	48	24	19.2
	25	62	31	24
	30	76	36	29
	40	96	48	39
Exhaust fan motor with VFD	5	14.1	7.5	6.2
	7.5	21	10.3	9.2
	10	31.1	13.9	12.5
	15	45.1	20.2	17.3
	20	58	27.2	23.6
	25	71.5	32.3	28.3
	30	83.7	38.1	34.6
	40	114.7	54.1	47.2

Note:

* 208 V to 230 V amps are based on 230 V operation.

** Actual unit nameplate amps are based on operating conditions and are calculated based on a supply fan amp value that may be less than the maximum amp values shown in the table. See the selection program output for supply fan amps. See the selection program output for return fan amps.

ERW motor data

Table 90: ERW motor

Model	Type	Nominal voltage		
		208-230/3/60*	460/3/60	575/3/60
		FLA	FLA	FLA
60 ton	Low CFM ERW	2.6	1.3	1.2
	High CFM ERW	2.6	1.3	1.2
70 ton	Low CFM ERW	2.6	1.3	1.2
	High CFM ERW	2.6	1.3	1.2
80 ton	Low CFM ERW	2.6	1.3	1.2
	High CFM ERW	2.6	1.3	1.2

Note:

* 208 V to 230 V amps are based on 230 V operation.

Miscellaneous data

Table 91: Miscellaneous data

Description	Nominal voltage		
	208-230/3/60*	460/3/60	575/3/60
	AMPS	AMPS	AMPS
Control transformer 0.2 kVA	1.0	0.5	0.5
Convenience outlet	9.0	4.5	3.5
Gas heat (per module)	0.92	0.46	0.37
500 MBH	0.92	0.46	0.37
1000 MBH	1.84	0.92	0.74
1500 MBH	2.76	1.38	1.10
Crankcase heater 90W	0.3	0.2	0.1
Ultraviolet (UV) lights 60 ton to 80 ton	9.7	4.4	3.6
Internal lights (per light)	0.05	0.03	0.02

Note:

* 208 V to 230 V amps are based on 230 V operation.

Controls

General

The control system for the Premier unit is fully self-contained. To aid in unit setup, maintenance, and operation, the unit controller is equipped with a standard user interface (UI) that has a 5 row × 35 character OLED display. The OLED display presents plain language text in a menu-driven format to facilitate use.

The unit controller can be connected to and operated by a building automation system (BAS). In addition, hard wired control options to the unit controller are provided using a customer terminal board.

The unit controller uses the latest technology and provides complete control for the unit along with standard BACnet® MS/TP and BACnet IP, Modbus™, and N2 communications. The unit controller also has a USB flash drive that can be used to capture historic data on unit operation and to update firmware.

Unit mode

Unit type

The controller supports two rooftop types: multi zone variable air volume (MZVAV) and single zone VAV (SZVAV). Both types can be configured to operate in several different modes of occupancy, temperature control, warm-up/cool down, and automated demand response.

Occupancy modes

Depending on the application, the controller can be indexed between occupied, unoccupied, optimal start, or coast modes of operation. These occupancy modes are initiated automatically via an internal schedule or externally from a BAS command, hard wired input, or tenant override via zone sensor.

Sensors

Temperature control sensor

The controller compares the discharge air temperature (DAT) against a setpoint to determine the heating or cooling demand. On multi zone VAV units, the DAT setpoint can be reset by different methods: fixed, outside air temperature (OAT), supply fan speed, external control, zone, or return air temperature (RAT). See also DAT setpoint reset.

Changeover sensor

The controller compares this temperature against a setpoint to determine whether to place the unit in heating, cooling, or idle mode. Zone temperature or RAT—or a BAS input—can be used as the changeover sensor.

Mandatory sensors

The following sensors are required to make the unit operate and are included as standard on every rooftop.

- Discharge air temperature
- Return air temperature
- Outside air temperature
- Mixed air temperature

Optional sensors

Depending on the application, additional sensors may be required.

- Zone air temperature
- Zone humidity
- Enthalpy sensor (for use with economizer)
- Dual enthalpy sensors (measures enthalpy of outdoor air and return air; used with economizer)
- Supply duct pressure (VAV units)
- Building pressure (required for exhaust/return fan)
- Carbon dioxide (CO_2) (required for demand ventilation)
- Smoke detector

Warm-up and cool down methods

The warm-up and cool down feature supports three methods of initiating the warm-up and cool down mode: occupied command, schedule, optimal start.

Occupied command

When the occupancy status for a unit switches to occupied mode, the supply fan turns ON for a 5-minute stabilization period, after which the RAT sensor compares to the warm-up setpoint. If the value is equal to or below the warm-up setpoint, heating is controlled to the warm-up DAT setpoint. If the value is above the warm-up setpoint, the control goes into occupied mode. Warm-up stays active until the RAT sensor is above the warm-up setpoint for 5 minutes or the early start period expires. The control goes into occupied mode thereafter. Cool down operates opposite of warm-up, using cooling instead of heating stages.

Schedule

At the early start period, prior to the scheduled occupancy, the supply fan turns ON for a 5-minute stabilization period and then the RAT sensor is compared to the warm-up set point. If the value is equal to or below the warm-up setpoint, heating is controlled to the warm-up DAT setpoint. If the value is above the warm-up setpoint, the control goes into occupied mode. Cool down operates opposite of warm-up, using cooling instead of heating stages.

Optimal start

The optimal start control algorithm is a function of the difference between zone temperature and occupied temperature setpoint, the OAT, and the amount of time prior to scheduled occupancy.

When unoccupied, the controller calculates an early start time based on control temperature deviation from occupied temperature setpoint. This ensures the space is at conditioned levels when the occupied period starts.

Historical data determines when optimal start begins, but the early start time can be user limited. The optimal start time is always within the same calendar day. The warm-up or cool down DAT setpoints function until the demand is satisfied. The control uses the heating or cooling setpoint to determine when the demand is satisfied. The control goes into occupied mode thereafter.

Coast

When enabled, 1 hour (adjustable) prior to the transition from occupied to unoccupied, the operational cooling and heating setpoints begin to increase or decrease linearly at 1.0°F per hour (adjustable) towards unoccupied cooling and heating setpoints. The operational cooling and heating DAT setpoints also begin to increase or decrease linearly at 1.0°F per hour (adjustable). During this time, the fans continue to run and the ventilation sequence remains active.

Multiple unit staggered start

When enabled, the start-up delay utilizes the minimum OFF time plus a delay time. This is used to prevent all units on a site from restarting simultaneously after a power failure.

The delay time can be manually set or randomized between 0 and 60 seconds. If staggered, start override time is set to 0 seconds.

Additional functionality includes applying a start-up delay when switching into the occupied mode. This only happens if the supply fan command is currently OFF when the occupancy mode switches into occupied. This staggers start-up for multiple rooftop units if all are commanded from the same occupied command. After the timer has expires, the rooftop unit can start.

Automated demand response methods

The controller is capable of three automated demand response methods: demand shed, load shed, and capacity limit.

Demand shed

This strategy operates by offsetting the operating cooling setpoint higher and offsetting the operating heating setpoint lower when the unit controller receives a demand shed signal from a binary hard wired input or BAS command. The offset is adjustable.

Load shed

This function shuts down all cooling and electric heat for an adjustable time period in response to a BAS command.

Capacity limit

This strategy operates by limiting total allowed mechanical cooling or electric heat capacity in proportion to the value of an analog BAS cooling/heating capacity limit signal.

Supply air system

The supply fan is used to circulate air to condition the temperature of the space and to provide ventilation to the space. Supply fan starts and operating hours are measured under all operating conditions. Two user-selectable supply air modes are available: MZVAV and SZVAV.

Multizone variable air volume (MZVAV)

Duct pressure control

When the fan energizes, the output from the controller maintains the supply duct pressure to the duct pressure setpoint. If the duct pressure is greater than the duct pressure setpoint, the supply fan output decreases.

If the duct pressure is below the duct pressure setpoint, the supply fan output increases. If the duct pressure reaches the duct pressure shutdown setpoint, the fan and all other outputs of the unit de-energize.

Note: If the unit is in a heating mode, the controller continues to vary supply fan output to control duct pressure to the duct static pressure setpoint. Therefore, in any VAV heating mode, all VAV boxes must be commanded open far enough to get adequate airflow to support the heating function and to prevent the heat section high temperature limit switches from opening. A VAV box binary output on the customer terminal board energizes any time a MZVAV unit is in any heating mode.

DAT setpoint reset

The DAT setpoint reset feature provides five different methods of varying the DAT setpoint:

- fixed
- OAT
- zone and return air temperature (RAT)
- supply fan speed
- external control

Fixed

- The DAT setpoint is a fixed value and does not change in response to another signal. It can be user adjusted from the default value over a BAS or local UI.

OAT

- **Cooling -**The DAT setpoint is reset from cooling DAT upper setpoint to cooling DAT lower setpoint as the operating OAT varies from OAT lower setpoint to OAT upper setpoint.
- **Heating -** The DAT setpoint is reset from heating DAT lower setpoint to heating DAT upper setpoint as the operating OAT varies from OAT upper setpoint to OAT lower setpoint.

Zone and return air temperature

- **Cooling -** The DAT setpoint resets from cooling DAT upper setpoint to the cooling DAT lower setpoint as the zone and return air temperature varies from the zone and return air temperature lower setpoint to the upper zone and return air temperature setpoint.
- **Heating -** The DAT setpoint resets from heating DAT lower setpoint to the heating DAT upper setpoint as the zone and return air temperature varies from the zone and return air temperature upper setpoint to the lower zone and return air temperature setpoint.

  Note: If zone temperature sensor becomes unreliable or is not installed, the operating zone temperature uses the RAT instead.

Supply fan speed

- **Cooling –** The cooling DAT setpoint is reset from DAT cooling upper setpoint to lower setpoint as the supply fan output varies from supply fan DAT lower threshold setpoint to supply fan DAT upper threshold setpoint.
- **Heating -** The heating DAT setpoint is reset from DAT heating lower setpoint to upper setpoint as the supply fan output varies from supply fan DAT lower threshold setpoint to supply fan DAT upper threshold setpoint.

External control

- The DAT setpoint is controlled directly by a third party - BACnet communication or voltage signal. For cooling mode, the hard-wired DAT setpoint varies from DAT upper setpoint to DAT lower setpoint as the input varies from low value to high value. For heating mode, the hard-wired DAT setpoint varies from DAT lower setpoint to DAT upper setpoint as the input varies from low value to high value.

Single zone variable air volume (SZVAV)

Mechanical cooling

Zone temperature control - The DAT setpoint resets between high and low cooling setpoints based on the zone cooling demand. Staged cooling command is based on the time and temperature differential that operating zone temperature has been operating above cooling setpoint. The unit controls the DAT by staging or modulating the compressors. The supply fan runs at the minimum cooling speed setpoint.

As the zone cooling demand increases and the DAT controls to the low cooling setpoint, the supply fan output resets between minimum cooling speed and 100% to maintain the zone temperature at cooling zone temperature setpoint.

Staged temperature control - The unit can be controlled by a standard two-stage cooling/heating thermostat. Thermostat connections are hard wired through the customer terminal board: G, Y1, Y2, W1, W2. Thermostat commands can also be communicated by the BAS. The fan operates at minimum or maximum speed in occupied/unoccupied mode based on the thermostat input.

Economizer

When free cooling is available, the outside air (OA) damper modulates to maintain the mixed air temperature (MAT) at the cooling DAT setpoint. During the free cooling mode, if the OA damper modulates to 100% open and cannot maintain the cooling DAT setpoint, the compressors are staged or modulated to maintain the cooling DAT setpoint. The supply fan runs at the minimum cooling speed setpoint.

If the zone temperature exceeds the cooling zone temperature setpoint, the OA damper is 100% open, and all cooling stages are staged ON (or locked out due to compressor lockout), then the supply fan resets between minimum cooling speed and 100% to maintain zone temperature at the cooling zone temperature setpoint.

Satisfied

If the occupancy mode is occupied without a heating or cooling demand, the supply fan runs at the fan only minimum fan speed.

Modulated heating

The DAT setpoint resets between low and high heating setpoints based on the zone heating demand. The heating command is based on the time and temperature differential that the operating zone temperature has been operating below operating heating setpoint. The unit controls the DAT by modulating the heat (electric, gas, or hydronic). The supply fan runs at the minimum speed setpoint.

As the zone heating demand increases and the DAT controls to the high heating setpoint, the supply fan resets between heating minimum speed and 100% to maintain the zone temperature at heating zone temperature setpoint.

Staged heating

The DAT setpoint resets between low and high heating setpoints based on the zone heating demand. Staged heating command is based on the time and temperature differential that the operating zone temperature has been operating below operating heating setpoint. The unit controls the DAT by staging the heat (electric or gas). The supply fan runs at maximum capacity.

Unoccupied mode

If the occupancy mode is unoccupied, the unit does not operate in SZVAV mode but instead operates in the following manner:

A. Unoccupied cooling

- The unit controls to the unoccupied zone cooling setpoint.
- The cooling DAT setpoint is set to the cooling setpoint. The unit controls the DAT to the operating cooling DAT setpoint by staging or modulating the compressors. The supply fan runs at maximum capacity.

B. Unoccupied heating

- The unit controls the zone temperature to the unoccupied zone heating setpoint.
- The heating DAT setpoint is set to the high heating DAT setpoint. The unit controls the DAT to the operating heating DAT setpoint by staging or modulating the heat (electric, gas or hydronic). The supply fan runs at maximum capacity.

Exhaust air (EA) system

To control building pressure, three power exhaust control options are available: none, barometric damper, modulating exhaust fan, modulating EA damper with fan, and external control.

None

The unit does not have an exhaust option.

Barometric damper

A mechanical barometric damper relief damper is installed, however this damper is not controlled by the units controller.

Modulating exhaust fan VFD

If the building pressure is greater than the building pressure setpoint, the exhaust fan analog output increases. If the building pressure is less than the building pressure setpoint, the exhaust fan analog output decreases.

Variable speed exhaust fan command is energized any time the exhaust fan speed output is greater than exhaust fan speed on (adjustable) and the fan command is de-energized any time the exhaust fan speed output is less than exhaust fan speed off (adjustable). A minimum deadband between the on and off setpoints is enforced.

Dual powered exhaust with VFD

Dual DWDI forward-curved centrifugal exhaust fans are provided to exhaust building return air to relieve building static pressure on 60 ton to 80 ton units with an ERW. If the building pressure is greater than the building pressure setpoint, the exhaust fan analog output increases. If the building pressure is less than the building pressure setpoint, the exhaust fan analog output decreases. If ERW command is off, variable speed exhaust fans 1 and 2 commands are energized any time the exhaust fan speed output is greater than exhaust fan speed on (adjustable) and the fan commands are de-energized any time the exhaust fan speed output is less than exhaust fan speed off (adjustable). If ERW command is on, variable speed exhaust fan 1 command is energized any time the exhaust fan speed output is greater than exhaust fan speed on (adjustable) and the fan command is de-energized any time the exhaust fan speed output is less than exhaust fan speed off (adjustable). Exhaust fan 2 command is off during this time.

Modulating EA damper with fan (on/off)

The controller modulates the opening of the EA damper to maintain the building pressure setpoint. If the building pressure is greater than the building pressure setpoint, the EA damper output increases to open the EA damper. If the building pressure is less than the building pressure setpoint, the EA damper output decreases to close the EA damper.

Exhaust fan is energized when the exhaust damper is greater than exhaust damper position for exhaust fan to turn on (adjustable), the controller energizes the exhaust fan. If the exhaust damper is less than exhaust damper position for exhaust fan to turn off (adjustable), the controller de-energizes the exhaust fan. To reduce fan cycling, a minimum deadband is enforced between the fan start and fan stop setpoints.

External control

Exhaust fan speed or EA damper position is controlled directly by a third party (BACnet communication or voltage signal).

Return air (RA) system

Two return fan control options are available: return fan discharge pressure control and supply fan airflow tracking.

In either return fan mode, the controller modulates the opening of the EA damper to maintain the building pressure at setpoint. If the building pressure is above the building pressure setpoint, the EA damper output increases. If the building pressure is below the building pressure setpoint, the EA damper output decreases. In this mode, the EA damper position can alternatively be controlled directly by a third party (BAS communication or voltage signal).

Return fan discharge pressure

The controller modulates the speed of the return fan to maintain the return fan discharge pressure to a setpoint. As the supply fan speed increases or the RA or EA dampers open, the return fan discharge pressure decreases, and the return fan speed increases to maintain the pressure setpoint. As the supply fan speed decreases or the RA or EA dampers close, the return fan discharge pressure increases and the fan speed decreases to maintain the operating pressure setpoint. The operating pressure setpoint is a calculated value based on OA damper position and supply fan speed, fan pressure high and low setpoint.

Supply fan airflow tracking

When equipped with optional supply and return fan airflow measuring stations, the controller monitors the supply fan airflow and modulates the speed of the return fan to a return airflow setpoint. The return airflow setpoint is a calculated value based on the supply fan airflow, minus a return airflow differential (adjustable). If the return fan airflow decreases below the operating flow setpoint, the return fan speed increases. If the return fan airflow increases above the operating flow setpoint, the return fan speed decreases.

Cooling system

The controller supports both staged and variable speed direct expansion (DX) cooling.

DX cooling

Capacity control

Constant speed compressors stage up or down with the staged cooling command parameter to control the DAT to operating cooling setpoint. During dehumidification control, the compressors switch to evaporator temperature control.

When equipped with a variable speed drive compressor, the capacity modulates via vernier control. The first stage on refrigeration circuit one is variable speed, while all other stages are constant speed. Start count equalization does not apply to the variable speed compressor. The variable speed compressor is always the first compressor ON and last compressor OFF, except in dehumidification mode.

Compressor staging algorithm

The controller employs patented algorithms to sequence the compressor stages as a method of quickly meeting cooling demand while reducing overshoot as DAT reaches operating cooling setpoint.

The controller excludes staging combinations that result in one refrigeration circuit having two or more compressors energized than the other refrigeration circuit.

Rapid start

To shorten the time it takes to get compressors running, the optimum stage up algorithm is used. It estimates the number of compressor steps required to meet the initial cooling demand and quickly starts the appropriate number of compressors. When the unit first starts up or after an unoccupied to occupied transition, if there are no compressors running and the economizer is not suitable, rapid start is active for the first 5 minutes (adjustable) of cooling operation. During rapid start, the inter-stage delay time between compressors reduces to 15 seconds.

Variable speed drive compressors

Envelope control keeps the variable speed drive compressor within the allowable range of saturated suction and saturated discharge temperature for the compressor. Oil management algorithms are designed to keep the oil level above the minimum on variable speed drive compressors. Overall cooling demand, oil management, saturated DAT, and saturated suction temperature all influence the allowable operational speed of the variable speed drive compressor. The controller monitors each of those requirements to provide the appropriate minimum and maximum speeds to the compressor.

Condensate overflow alarm

When the condensate overflow switch alarm is ON, the unit disables mechanical cooling. Supply fan operation remains enabled during this period. The alarm must be manually reset at the controller.

Condenser system

The rooftop unit condenser fan control is designed to control staged and modulating condenser fans based on the saturated liquid line temperature. Constant speed condenser fans are staged up or down based on the saturated liquid line temperature. Condenser fan start count equalization is used for fixed speed fans.

Saturated liquid line temperature control

After a stabilization period of 60 seconds during saturated liquid temperature control and after the first compressor is energized, condenser fans are staged ON or OFF to achieve stage up or down setpoint. If saturated liquid line temperature is above the stage up setpoint, an additional condenser fan is energized. If saturated liquid line temperature falls below the stage down setpoint, a condenser fan is de-energized.

The variable speed fan is the first ON and the last OFF when optionally equipped on the unit. The variable speed fan modulates condenser fan speed between 80.0°F and 110.0°F.

If saturated liquid line temperature becomes unreliable, the unit reverts to fail-safe mode.

Fail-safe mode

In fail-safe mode, compressor operation maybe limited below a user selected ambient temperature. Condenser fan operation is controlled by ambient temperature and compressor operation. If the saturated liquid line temperature is not reliable, one condenser fan is energized for every compressor that energizes, and the variable speed condenser fan runs at 100%.

When the saturated liquid line temperature becomes reliable, the unit reverts to saturated liquid line temperature control.

Low ambient mode

This functionality is only available during saturated liquid line temperature control. It is automatically enabled on a circuit with a modulating condenser fan.

When low ambient mode is NOT enabled on a circuit, the lower range of the OAT cooling enable setpoint is 45.0°F.

When low ambient mode IS enabled on a circuit, the lower range of the OAT cooling enable setpoint is -10.0°F.

Low ambient mode cannot be automatically enabled on a circuit unless there is a variable speed condenser fan and liquid line pressure sensor installed on that circuit.

Heating system

The controller supports multiple heating types including electric heat (staged, modulated), gas heat (staged, modulated), and a hydronic heating coil. The supported heating-related features include: warm-up, discharge air (DA) tempering, load shedding (electric heat), and runtime equalization. If the DAT is lower than the DAT setpoint, the heating demand increases and a heating stage is energized. If the DAT is greater than the DAT setpoint, the heating demand decreases and a heating stage is de-engergized.

Electric heat

The application supports up to ten stages of electric heat as well as modulated electric heat.

Staged electric heat

If the DAT is lower than the DAT setpoint, the heating demand increases and a heating stage is energized. If the DAT is greater than the DAT setpoint, the heating demand decreases and a heating stage is de-engergized.

Modulating electric heat

If the DAT is lower than the DAT heating setpoint, the heating demand increases. If the DAT is greater than the DAT heating setpoint, the heating demand decreases. The modulating heating stage is connected to a SCR controller. Additional stages are sequenced ON and OFF while the SCR stage automatically fills the gap between the step-controlled stages, providing full proportional control over the entire heater range. Both the SCR stage and the step-controlled stages are controlled by the heating demand. The modulating stage is the first ON and last OFF.

Runtime equalization

Runtime equalization only applies to equally sized, non-modulating heating stages. The modulating stage is first ON and last OFF. The controller records the number of starts and runtime statistics for each stage of electric heat.

At the initiation of each heating demand, the stage with the lowest total runtime energizes first. When runtime is equal, the staging is in numerical order. The stage with the next lowest total runtime energizes next and so on. At the termination of heating demand, the stages with the highest runtime stage OFF in reverse order.

Gas heating

The application supports up to six stages in three furnace modules. The concept behind the staged and modulating furnace options is to offer multiple furnace modules, with each module being its own self-contained, fully functioning furnace.

Staged gas heat

If the DAT is lower than the DAT heating setpoint, the staged heating demand increases. If the DAT is greater than the heating setpoint, the staged heating demand decreases. The gas heat modules are energized in response to the heating demand. Each gas heat module has a low fire and a high fire mode, which are individually energized in response to the heating demand. Low fire is always energized prior to high fire.

Modulating gas heat

To accomplish this, one furnace module is adapted for modulation. The selected furnace module is started and modulated, while the other ON/OFF furnace modules are staged, which allows a much broader modulation range. If the DAT is lower than the DAT heating setpoint, the heating demand increases. If the DAT is greater than the DAT heating setpoint, the heating demand decreases. Each gas heat module has a low fire and a high fire mode, which are individually energized in response to the heating demand. Low fire is always energized prior to high fire.

Modulating hydronic heating coil

If the DAT is lower than the DAT heating setpoint, the heating valve modulates open. If the DAT is greater than the DAT heating setpoint, the heating valve modulates closed. The application setup of either direct or reverse action allows support for either normally open or normally closed valves.

Freezestat alarm

The application supports a freezestat input that is normally closed. The input opens if:

- the OAT is greater than 40.0°F (adjustable), no action is taken and the unit operates normally.
- the OAT is 40.0°F or less (adjustable), the hot water valve opens 100%, the supply fan de-energizes, the economizer OA damper fully closes, and all other fans de-energize.

The control returns to normal if either

- Freezestat input closes
- the OAT rises above 40.0°F (adjustable).

Discharge air tempering

When the discharge air tempering is enabled, the unit must be in fan only or cooling mode when the OAT being brought in to ventilate the space causes the DAT to decrease below the DAT tempering setpoint. The air must be heated to prevent low temperature air from being delivered to the space. Discharge air tempering is available on both staged and modulated heating types. The compressors cannot start when discharge air tempering is active.

Economizer

When the economizer is installed, economizer suitability is true, and there is a call for cooling, then the controller modulates the OA damper between the minimum position and 100% to maintain the MAT to the operating cooling DAT setpoint. Economizer suitability is determined based on the changeover option selected and the available sensors. If the economizer is unable to meet cooling demand after the OA damper has been 100% open for a period of time, the unit controller allows mechanical cooling to operate. When economizer and mechanical cooling are both operating, the OA damper is 100% open.

Changeover method

The unit can be equipped with one of three types of optional economizers: dry bulb, single enthalpy, or dual enthalpy. When the economizer selection variable is set to auto (default), the unit controller selects which economizer method to use based on which temperature and humidity sensors are present and reliable (see Table 92).

Table 92: Free cooling sensor requirements

	OAT	OAH	RAT	RAH
Dry bulb	X			
Single enthalpy	X	X		
Dual enthalpy	X	X	X	X

The order of economizer selection is dual enthalpy (all sensors reliable), single enthalpy (only OA sensors reliable), and dry bulb (only OAT sensor). The user can also manually select the economizer type when all required sensors are available. If economizer cooling alone is insufficient for the cooling load, the controller stages up compressors, one at a time, to meet demand.

Dry bulb

With the dry bulb economizer, the controller monitors the OAT only and compares it to a reference temperature setting. Outside air is deemed suitable for economizing when the OAT is less than the reference temperature setting. This method of economizing is effective, but it is prone to some changeover inefficiencies since it is based on sensible temperatures only and does not take outside air moisture content into consideration.

Single enthalpy

With the optional single enthalpy economizer, the controller monitors the OA enthalpy in addition to the OAT and compares it to a reference enthalpy setting and a reference temperature setting. Outside air is deemed suitable for economizing when the OA enthalpy is determined to be less than the reference enthalpy setting and the OAT is less than the reference temperature setting. This method of economizing allows the reference temperature setting to be set higher than the dry bulb economizer and is consequently a more efficient rooftop economizer.

Dual enthalpy

With the optional dual enthalpy economizer, the controller monitors and compares the OA and RA enthalpies in addition to comparing the OAT to the reference temperature setting. Outside air is deemed suitable for economizing when the OA enthalpy is determined to be less than the RA enthalpy and the OAT is less than the reference temperature setting. This method of economizing is the most accurate and provides the highest degree of energy efficiency for a rooftop economizer.

Auto

Auto selects the changeover method based on which sensors are present and reliable. If the OAT, RAT, OA humidity, and RA humidity values are reliable, then the dual enthalpy changeover method is used. If either of the RAT or RA humidity sensors is unreliable, then the single enthalpy changeover method is used. If the OA humidity sensor is unreliable, then the dry bulb changeover method is used.

Table 93: Manufacturer fault detection and diagnostics (FDD)

Manufacturer fault	Description
Economizer - damper not modulating	Economizer damper actuator is inoperable or not responding to commands.
Economizer - not econ when available	Conditions are available for economizer operation but the unit does not use outside air for cooling.
Economizer - econ when not available	Conditions are not available for economizer operation but the unit uses outside air for cooling.
Economizer - excess outside air	Economizer damper is open greater than commanded position.

Fault detection and diagnostics (FDD)

As required per 2016 California Title 24 and ASHRAE 90.1-2019, the fault detection and diagnostics (FDD) system provides the alarms stated in Table 93.

ERW

The ERW can run in both cooling and heating modes. In cooling mode, if the conditions are suitable for economizing, the ERW disables, and the bypass dampers fully open.

The ERW enables when the absolute value of the difference between the return air and the outside air is greater than the energy recovery setpoint. If the OA and RA humidity values are reliable, then enthalpy is compared; other-wise dry bulb temperature is compared.

When the ERW stops for extended periods, it energizes briefly for cleaning and blocking protection.

Single speed ERW

When enabled, the ERW runs at full speed and the ERW bypass dampers fully close. When the exhaust air temperature (EAT) drops below the energy recovery low limit setpoint, the ERW bypass dampers fully open to bypass the ERW.

Variable speed ERW

When enabled, the ERW runs at full speed and the ERW bypass dampers fully close. When the EAT drops below the energy recovery low limit setpoint, the ERW slows to maintain a minimum EAT to prevent freezing. If the minimum speed does not maintain the temperature setpoint, the bypass dampers fully open to bypass the ERW.

Ventilation

Minimum OA damper reset

Fixed minimum

When the control is in the occupied mode and the fan output energizes, the OA damper is positioned to the minimum position setpoint (adjustable) unless another function commands it open or closed. When the control is in the unoccupied mode, the minimum OA position is zero.

Supply fan speed

When the control is in the occupied mode, the fan output energizes, and the output reaches 100%, the OA damper position is the damper_min_pos_high_speed_fan setpoint.

When the VFD output reaches 50% (adjustable) or less, the OA damper position is the damper_min_pos_low_speed_fan setpoint.

When the VFD output is between 50% and 100% (adjustable), the OA damper positions proportionally between damper_min_pos_high_speed_fan setpoint and damper_min_pos_low_speed_fan setpoint.

ⓘ **Note:** Supply fan speed minimum OA damper reset is not available when demand control ventilation (DCV) is enabled.

Outside airflow measurement

This control sequence provides for the dynamic determination of the minimum damper position that meets the ASHRAE building ventilation standards. If the outside airflow input is not reliable, then the damper is positioned based upon the default minimum position.

When the fresh air intake value falls below the fresh air intake setpoint, the OA damper position increases above its minimum position until the fresh air intake value equals the fresh air intake setpoint.

When the fresh air intake value rises above the fresh air intake setpoint, the OA damper position decreases until the fresh air intake value equals the fresh air intake setpoint or it reaches minimum position setpoint.

When DCV is enabled on a unit with an outdoor airflow measurement station installed, the fresh air intake setpoint resets upwards until the CO_2 level falls below the DCV setpoint.

ⓘ **Note:** The low ambient minimum position may force the damper position below the current setpoint and disable the air monitoring station reset.

External control

The OA damper is controlled directly by a third party (communication or voltage signal). External control takes priority over all other damper control except unit protection.

Low MAT limiting

When the control is in the occupied mode, the economizer is not active, the fan output is energized, and MAT is below the MAT cooling limit, the damper begins closing. When the MAT is equal to or above the MAT cooling limit, it exits the low MAT limiting mode.

Demand control ventilation (DCV)

If optional CO_2 sensors are connected to the unit, the controller can reset the minimum OA damper position based on demand. The controller has built-in sequences for indoor CO_2 and comparative CO_2. Demand ventilation remains active as long as the unit is in the occupied mode of operation.

Indoor CO_2

The controller monitors the CO_2 level within the building. If the CO_2 level rises above the CO_2 setpoint, the controller temporarily increases the minimum OA damper position or minimum outside airflow rate to increase ventilation. If the CO_2 level drops below the CO_2 setpoint, the controller decreases the minimum OA damper position or minimum outside airflow rate to decrease ventilation.

Comparative CO_2

If differential DCV is enabled and the outdoor air CO_2 level is greater than or equal to the indoor air CO_2 level by more than the demand ventilation differential setpoint, then the OA damper is overridden closed regardless of economizer suitability. Otherwise, the indoor CO_2 sequence is followed.

Pre-occupancy purge

When the unit is in internal schedule, if pre-occupancy purge is enabled, effective occupancy is unoccupied, the MAT reading is reliable, and the damper is not controlled by higher-priority logic, then pre-occupancy purge mode initiates immediately when the time until occupancy equals the pre-occupancy purge time. Pre-occupancy purge runs until the pre-occupancy purge timer expires or if any of the entry conditions required for pre-occupancy purge mode are violated.

Once in pre-occupancy purge mode, the VAV heat command relay turns on and the system continues to monitor the MAT. If MAT falls below MAT cooling limit, the system limits the outdoor air damper position until the MAT rises above MAT cooling limit. If the outdoor air damper fully opens, the system resumes regular pre-occupancy purge.

If using an indoor CO_2 sensor and low CO_2 control point is satisfied, then the mode terminates.

Continuous ventilation

When enabled, the supply fan runs continuously during occupied mode, even when no heating or cooling is required. When continuous ventilation is disabled, the supply fan cycles OFF when no heating or cooling is required.

Smoke control

If smoke control mode is in one of its active modes, the OA damper sets to 0% or 100%. If smoke control mode is in purge or pressurization mode, the OA damper is 100%. If smoke control mode is in de-pressurization, the OA damper is 0%.

The smoke control sequences are used whenever the safety binary inputs turn ON. It must be noted that the smoke control sequences are not normal operation for rooftop units and are only utilized when smoke or other airborne contaminants must be quickly removed from the conditioned space.

Whenever a smoke control sequence starts, all normal heating and cooling functions stop regardless of the control inputs. This includes all running compressors, condenser fans, heat stages, heating valves, and furnaces.

There are three safety binary inputs available on the customer terminal board. Each input is assigned one of the three sequences. This makes it possible to start and stop any of the three sequences at any time because there are three inputs available for the three sequences. To avoid problems when more than one input is active at the same time, each input has a priority. The sequence assigned to the active input with the highest priority is used over sequences assigned to inputs with a lower priority.

When either one of the safety input mode 1, 2, or 3 is ON, the smoke control mode is assigned one of the following states according to the binary input priorities and the sequence settings:

Note: Johnson Controls rooftop units offer auxiliary smoke control points that are available for smoke purge, pressurization, and de-pressurization sequences. These sequences and functions are not part of a listed UL 864 UUKL Smoke Control System, nor are they intended to be a replacement for a listed UL 864 UUKL Smoke Control System. Johnson Controls does offer options for listed UL 864 Smoke Control Systems for example:

a) Metasys Release 8.1 UL cUL 864 10th Ed. UUKL / ORD-C100-13 UUKLC Smoke Control System; and

b) Simplex 4100ES Fire Control Units (Addressable Fire Detection and Control Basic Panel Modules and Accessories) which are listed to UL 864, Fire Detection and Control (UOJZ), Smoke Control Service (UUKL), and Releasing Device Service (SYZV).

Only listed smoke control equipment complying with UL 864 is typically considered acceptable to comply with the associated Sections of the Building Code, Mechanical Code and ANSI NFPA 92 / 92A / 92B requirements, and are intended to perform specific functions related to Smoke Control Systems. If a listed UL 864 Smoke Control System is required, please contact Johnson Controls for further information.

Depressurize

The depressurize mode evacuates, or negatively pressurizes, the building or space to draw air through the walls from adjacent spaces or outside the building envelope. When this sequence starts, the following occurs:

- Sets the supply fan output to OFF.
- Starts the return fan (exhaust fan output) if not already ON.
- Sets exhaust and return fan VFD to 100%.
- Sets OA damper position to 0%.
- Sets RA damper position to 100%.
- Sets EA damper position to 100%.

Pressurization

The pressurization mode pressurizes the building or space to force the air inside the space through the walls to adjacent spaces or outside the building envelope. When this sequence starts, the following occurs:

- Sets supply fan output to ON.

- If the unit type is MZVAV, the supply fan VFD speed maintains the active duct pressure setpoint and the VAV heat command relay turns ON.
- If the unit type is SZVAV, it sets the supply fan VFD speed to 100%.
- Sets exhaust fan output to OFF.
- Sets return fan output to OFF.
- Sets exhaust and return fan VFD to 0%.
- Sets OA damper position to 100%.
- Sets RA damper position to 0%.
- Sets EA damper position to 0%.

Purge

The purge mode displaces the air inside the space with fresh outside air. When this sequence starts, the following occurs:

- Sets supply fan output to ON.
- If the unit type is MZVAV, the supply fan VFD speed maintains the active duct pressure setpoint and the VAV heat command relay turns ON.
- If unit type is SZVAV, it sets the supply fan VFD speed to 100%.
- Starts the return fan (exhaust fan output) if not already ON.
- Sets exhaust and return fan VFD to 100%.
- Sets OA damper position to 100%.
- Sets RA damper position to 0%.
- Sets EA damper position to 100%.

Humidity control

Humidity sensor

The controller compares the humidity against a setpoint to determine whether to place the unit in dehumidification, humidification, or temperature control only mode. Selectable humidity sensors are zone, return, or external control (BAS communication).

Dehumidification

Modulating hot gas reheat (HGRH)

When dehumidification is enabled, the unit enters dehumidification mode if the humidity sensor rises above the dehumidification setpoint while the unit is in cooling or satisfied mode and the outside air temperature is greater than 55.0°F. The compressors are controlled based on humidity and the reheat device is controlled based on temperature. It also allows for dehumidification when the system is in mechanical cooling or when temperature requirements are satisfied.

When humidity requirements are satisfied, the HGRH status is off-idle and the HGRH valve is closed.

When HGRH status is ON, the staging of compressors is controlled so that the air leaving the evaporator coil is controlled to an evaporator temperature setpoint. This ensures that water condenses and is removed from the air. The HGRH valve is then modulated so that it reheats the discharge air to the HGRH temperature setpoint.

While the system is in mechanical cooling, the HGRH temperature setpoint is equal to the operational cooling DAT setpoint. This is necessary because the compressors can be controlled to an evaporator temperature setpoint lower than the operational cooling DAT setpoint to remove additional moisture content from the air. This strategy allows humidity to be removed from the air while attempting to maintain normal cooling DATs.

While the system is satisfied, the HGRH valve is controlled so that the HGRH temperature setpoint resets between the operational cooling DAT setpoint and the occupied cooling setpoint. This strategy attempts to remove humidity from the air without affecting the temperature of the space.

The controller uses the humidity sensor as the basis for the evaporator temperature setpoint. When the humidity first rises above setpoint, the evaporator temperature setpoint is equal to the low cooling DAT setpoint. As humidity continues to rise, the evaporator temperature setpoint is lowered to remove additional moisture from the air.

If the unit enters heating mode or the OAT falls below 54.0°F, dehumidification mode ends regardless of humidity.

A zone temperature sensor must be installed for dehumidification during unoccupied mode. Unoccupied dehumidification is not allowed when only a return air temperature sensor is installed.

HGRH bleed valve

The HGRH bleed valve is a solenoid valve that connects the HGRH coil to the suction line. The purpose of the bleed valve is to bleed off any remaining or trapped liquid in the HGRH coil when dehumidification mode is not active. When the unit enters HGRH on-normal command, the bleed valve opens. When the HGRH transitions to off-idle mode and after a 5-minute delay, the bleed valve closes.

Humidification output

The unit is placed in humidification mode, if the humidity sensor falls below the humidification setpoint and the unit is not in cooling mode. The humidifier is enabled, and the humidifier valve modulates to maintain the humidity setpoint as sensed by the humidity sensor. A humidity high limit overrides the output if necessary, to prevent the DA humidity from exceeding the DA humidity high limit setpoint. Humidification is not allowed in cooling mode.

Twinning

The application of having two or more separate rooftop units tied into one common main duct trunk line is known as twinning. The application is compatible with BACnet. The twinning process allows for redundancy with the cooling or heating load. All twinned units operate concurrently and are not sequenced based on the cooling or heating load. It allows for the user to design the system in an N+1 arrangement, in which three units are sized to handle the load and the fourth is available as a backup. The user is then able to shut down one of the running units and enable the backup via the BAS.

Each rooftop unit requires an isolation damper with an end switch to ensure the damper is fully open before the supply fan can start. Each rooftop unit in a twinning application also requires its own duct static pressure sensor to allow the rooftop units to run independently if one or more units are shut down for maintenance or BAS communication is lost. Each unit has a manual reset high duct pressure switch installed to prevent over pressurization of the duct.

In a twinning system, several features are synchronized while others can operate independently. The method used to synchronize these features is for each twinned unit in the group to broadcast the key values so it can be used by the other rooftop units in the group. The approach is referred to as independent twinning.

The idea is that if each unit has the exact same values, then each rooftop unit can execute the exact same control algorithm. This has the advantage of being more robust to communication errors or sensor failures and less complex than a manager and subordinate unit arrangement. If one unit fails or is manually shut down, the remaining units continue to run without interruption. In the manager and subordinate unit arrangement, if the manager unit disables, the failover logic to a different manager is more complicated and may result in a disruption in the system. Independent twinning prevents such disruptions or failures in the system.

The synchronized features include supply fan control, economizer suitability, occupancy, demand control ventilation, exhaust fan control, and smoke control.

The supply fan control must be synchronized to maintain DA static pressure between all units serving the same duct. This requires all reliable DA static pressure values from the rooftop units to be averaged before passing to the proportional-integral-derivative (PID). The static pressure setpoint must also be synchronized, and any change to one setpoint must be shared with the other rooftop units. This allows the PID in each rooftop unit to calculate the same output value and run all the supply fans at the same speed.

During start-up of a previously shutdown rooftop unit, the supply fan speed slowly ramps up until it matches the fan speeds of the rooftop units currently in operation. When the additional rooftop unit begins ramping, the static pressure increases, causing the other rooftop unit fans speeds to slow down to reach setpoint. Once the rooftop unit fan speed matches the existing rooftop units, it releases into control.

The economizer suitability must also be synchronized to allow all the units to use the OA damper for free cooling when available. This also avoids a situation where one unit is operating in economizer mode while another unit operates in mechanical cooling mode.

The occupancy for the twinned units must also be synchronized to allow all units to switch between occupied and unoccupied modes simultaneously. This includes the occupancy schedule and warm-up/cool down if enabled.

The DCV is synchronized to allow the OA damper minimum position to be reset equally between the units. This requires the indoor CO_2 values from the units to be shared and the maximum to be passed to the reset logic to ensure sufficient outside air is brought in for proper ventilation. The indoor CO_2 setpoint also synchronizes so the reset calculation is the same across all units.

The exhaust fan system must be synchronized to allow for proper building pressurization. This requires the building static pressure values from each unit to be averaged before passing to the PID. The building static pressure setpoint also synchronizes, and any change to one setpoint must be shared with the other rooftop units.

The smoke control feature is synchronized to ensure all units properly switch between the purge, pressurization, or depressurization modes. This requires the three safety binary inputs and their priorities on each unit to be shared so if any binary input is ON, all twinned units respond the same way.

The temperature control, return fan systems, and safety shutdowns can operate individually on each unit. The temperature setpoints synchronize to allow for similar operation between the units. The temperature loops are not required to be as tightly coupled as the supply and exhaust fan systems.

The twinning feature also has the following additional requirements:

- Customer terminal board required for twinning application
- VAV system types only
- All units must be the same tonnage
- Fan systems operate together based upon average duct static pressure
- External control not supported
- The same static pressure probe can be connected to all discharge pressure transducers
- The same static pressure probe can be connected to all building pressure transducers
- A supervisory device must be installed on the network
- Maximum of 4 twinned units on a single MSTP bus. Units can be twinned as one set of four or two sets of two
- Maximum of forty units are twinned on a BACnet IP trunk
- BACnet MSTP or BACnet IP only
- Twinning is not possible on wireless field bus networks or with RTUs that interface to the BAS using Modbus or N2
- Isolation dampers with end switches must be installed on each twinned unit.

Unit protection

The unit utilizes several system safeties to ensure unit protection, including low voltage, compressor system status monitoring, low and high pressure cutout monitoring, suction temperature monitoring, freeze protection, fan overload, shutdown, furnace limit, and MAT low limit.

Conditions that cause the unit to be locked out require user intervention. Refrigeration fault lockouts cannot be reset from the BAS. All other fault lockouts can be reset from network input or local UI.

Refrigerant detection system (RDS)

The RDS provides R-454B leak protection. The system is connected into the unit controls and automatically starts a sequence to dilute refrigerant gas. In addition, the system triggers an alarm when it senses the presence of refrigerant in the cabinet, indicating a leak equal to 25% of the Lower Flammability Limit. The RDS contains factory or field installed sensors that are located to ensure accurate and timely sensing of a leak.

Unit shutdown

Any time the normally closed contact opens, all relay outputs immediately de-energize and the unit shuts down. An alarm generates and displays on the unit controller.

Unit test

Start-up wizard

Field service technicians use the start-up wizard function to complete the unit configuration and start-up. It is a sequence of automatic self-tests and manually verified tests that are required for start-up documentation. For each test, a PASS/FAIL, date/time stamp, and relevant system data are recorded. After the testing sequence completes, a report with the recorded results is stored and downloadable via USB.

Air balancing wizard

The air balancing wizard is a grouping of configuration screens used for testing and calibrating airflow measuring stations, overriding damper positions, overriding fan speeds, and other functions to assist service technicians and air balancers.

Commissioning mode

Commissioning mode is a function used for manually operating the unit during a service call or commissioning demonstration. It allows for the unit components to be operated individually. The unit must not be operated in this mode for an extended period, therefore commissioning mode terminates automatically after 60 minutes unless extended at the local UI for another 60 minutes.

Cooling and heating is disabled unless the supply fan status is proven. Unit protection and safeties remain active and logged during commissioning mode.

Weights and dimensions

Weights

Figure 12: Standard cabinet unit sections

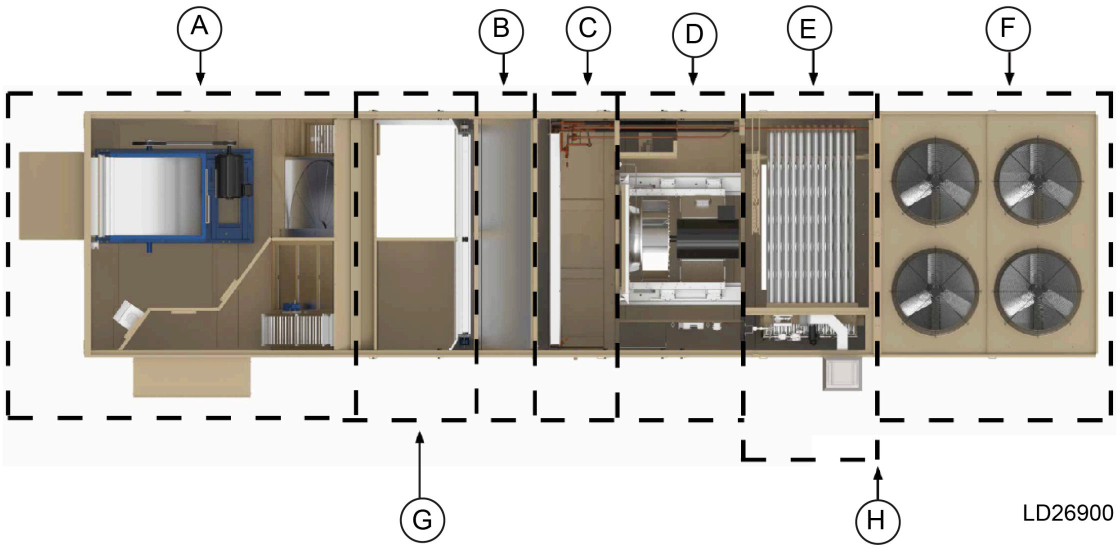


Table 94: Standard cabinet unit sections

Item	Description
A	Economizer cabinet with or without ERW cabinet
B	Draw-through filter cabinet
C	Evaporator cabinet
D	Supply fan cabinet
E	Heat and discharge cabinet[4]
F	Condenser fan cabinet with compressors
G	Pre-evaporator blank cabinet, air blender cabinet: between economizer and draw-through filter
H	Small post-evaporator blank cabinet, large post-evaporator blank cabinet, discharge plenum cabinet: upstream of condenser section

Table 95: Weight data (lb)

Model	60 ton	70 ton	80 ton
Base unit section weights - standard cabinet, see Figure 12			
Economizer cabinet [1]	919	919	919
Energy recovery wheel cabinet	1,869	2,047	2,047
Pre-evaporator blank cabinet [2]	461	461	461
Draw-through filter cabinet - vertical filter, 4 in.	203	203	203
Draw-through filter cabinet - angle, rigid, bag	386	386	386
Evaporator cabinet	487	1,021	1,021
Supply fan cabinet	—	—	—
Supply fan cabinet (55 in. section) [16]	1,368	1,368	1,368
Supply fan cabinet (74 in. section) [16]	1,579	1,579	1,579
Heat section cabinet (requires discharge plenum) [3]	1,113	1,113	1,113
Heat with discharge through heat section [4]	955	955	955
Small post-evaporator blank cabinet [5]	423	423	423
Large post-evaporator blank cabinet [6]	651	651	651
Discharge plenum cabinet [7]	606	686	686

Table 95: Weight data (lb)

Model	60 ton	70 ton	80 ton
Direct drive plenum (DDP) supply fan components [8]			
Supply fan and 5 HP 200-208 V ODP	598	598	598
Supply fan and 7.5 HP 200-208 V ODP	725	725	725
Supply fan and 10 HP 200-208 V ODP	757	757	757
Supply fan and 15 HP 200-208 V ODP	968	968	968
Supply fan and 20 HP 200-208 V ODP	1,076	1,076	1,076
Supply fan and 25 HP 200-208 V ODP	1,294	1,294	1,294
Supply fan and 30 HP 200-208 V ODP	1436	1436	1436
Supply fan and 40 HP 200-208 V ODP	1,463	1,463	1,463
Supply fan and 50 HP 200-208 V ODP	—	—	—
Supply fan and 5 HP 200-208 V TEFC	620	620	620
Supply fan and 7.5 HP 200-208 V TEFC	767	767	767
Supply fan and 10 HP 200-208 V TEFC	831	831	831
Supply fan and 15 HP 200-208 V TEFC	1,044	1,044	1,044
Supply fan and 20 HP 200-208 V TEFC	1,174	1,174	1,174
Supply fan and 25 HP 200-208 V TEFC	1,392	1,392	1,392
Supply fan and 30 HP 200-208 V TEFC	1,566	1,566	1,566
Supply fan and 40 HP 200-208 V TEFC	1,736	1,736	1,736
Supply fan and 50 HP 200-208 V TEFC	—	—	—
Supply fan and 5 HP 230/460 V ODP	598	598	598
Supply fan and 7.5 HP 230/460 V ODP	725	725	725
Supply fan and 10 HP 230/460 V ODP	757	757	757
Supply fan and 15 HP 230/460 V ODP	968	968	968
Supply fan and 20 HP 230/460 V ODP	1,076	1,076	1,076
Supply fan and 25 HP 230/460 V ODP	1,294	1,294	1,294
Supply fan and 30 HP 230/460 V ODP	1,436	1,436	1,436
Supply fan and 40 HP 230/460 V ODP	1,463	1,463	1,463
Supply fan and 50 HP 230/460 V ODP	—	—	—
Supply fan and 5 HP 230/460 V TEFC	620	620	620
Supply fan and 7.5 HP 230/460 V TEFC	767	767	767
Supply fan and 10 HP 230/460 V TEFC	831	831	831
Supply fan and 15 HP 230/460 V TEFC	1,044	1,044	1,044
Supply fan and 20 HP 230/460 V TEFC	1,174	1,174	1,174
Supply fan and 25 HP 230/460 V TEFC	1,392	1,392	1,392
Supply fan and 30 HP 230/460 V TEFC	1,566	1,566	1,566
Supply fan and 40 HP 230/460 V TEFC	1,736	1,736	1,736
Supply fan and 50 HP 230/460 V TEFC	—	—	—
Supply fan and 5 HP 575 V ODP	611	611	611
Supply fan and 7.5 HP 575 V ODP	717	717	717
Supply fan and 10 HP 575 V ODP	743	743	743
Supply fan and 15 HP 575 V ODP	957	957	957
Supply fan and 20 HP 575 V ODP	1,001	1,001	1,001
Supply fan and 25 HP 575 V ODP	1,272	1,272	1,272
Supply fan and 30 HP 575 V ODP	1,360	1,360	1,360
Supply fan and 40 HP 575 V ODP	1,463	1,463	1,463
Supply fan and 50 HP 575 V ODP	—	—	—
Supply fan and 5 HP 575 V TEFC	633	633	633
Supply fan and 7.5 HP 575 V TEFC	759	759	759
Supply fan and 10 HP 575 V TEFC	817	817	817
Supply fan and 15 HP 575 V TEFC	1,033	1,033	1,033
Supply fan and 20 HP 575 V TEFC	1,099	1,099	1,099
Supply fan and 25 HP 575 V TEFC	1,370	1,370	1,370
Supply fan and 30 HP 575 V TEFC	1,490	1,490	1,490

Table 95: Weight data (lb)

Model	60 ton	70 ton	80 ton
Supply fan and 40 HP 575 V TEFC	1,736	1,736	1,736
Supply fan and 50 HP 575 V TEFC	—	—	—
Energy recovery wheel (ERW)			
Low CFM	653	681	681
High CFM	868	1,013	1,013
Exhaust, return, and outside air economizer section [9]			
Economizer - no fan	612	628	628
Economizer with exhaust fan	1,661	1,678	1,678
Economizer with return fan	692	708	708
Motor for exhaust and return fan[10]			
Vertical direct driven return fan and 7.5 HP motor 208-230/460 V ODP	1,382	1,406	1,406
Vertical direct driven return fan and 10 HP motor 208-230/460 V ODP	1,451	1,475	1,475
Vertical direct driven return fan and 15 HP motor 208-230/460 V ODP	1,493	1,517	1,517
Vertical direct driven return fan and 20 HP motor 208-230/460 V ODP	1,551	1,575	1,575
Vertical direct driven return fan and 25 HP motor 208-230/460 V ODP	1,511	1,535	1,535
Vertical direct driven return fan and 30 HP motor 208-230/460 V ODP	1,613	1,637	1,637
Vertical direct driven return fan and 40 HP motor 208-230/460 V ODP	1,792	1,816	1,816
Vertical direct driven return fan and 7.5 HP motor 208-230/460 V TEFC	1,437	1,461	1,461
Vertical direct driven return fan and 10 HP motor 208-230/460 V TEFC	1,462	1,486	1,486
Vertical direct driven return fan and 15 HP motor 208-230/460 V TEFC	1,656	1,680	1,680
Vertical direct driven return fan and 20 HP motor 208-230/460 V TEFC	1,679	1,703	1,703
Vertical direct driven return fan and 25 HP motor 208-230/460 V TEFC	1,669	1,693	1,693
Vertical direct driven return fan and 30 HP motor 208-230/460 V TEFC	1,729	1,753	1,753
Vertical direct driven return fan and 40 HP motor 208-230/460 V TEFC	1,894	1,918	1,918
Vertical direct driven return fan and 7.5 HP motor 575 V ODP	1,379	1,403	1,403
Vertical direct driven return fan and 10 HP motor 575 V ODP	1,448	1,472	1,472
Vertical direct driven return fan and 15 HP motor 575 V ODP	1,498	1,522	1,522
Vertical direct driven return fan and 20 HP motor 575 V ODP	1,523	1,547	1,547
Vertical direct driven return fan and 25 HP motor 575 V ODP	1,511	1,535	1,535
Vertical direct driven return fan and 30 HP motor 575 V ODP	1,588	1,612	1,612
Vertical direct driven return fan and 40 HP motor 575 V ODP	1,767	1,791	1,791
Vertical direct driven return fan and 7.5 HP motor 575 V TEFC	1,444	1,468	1,468
Vertical direct driven return fan and 10 HP motor 575 V TEFC	1,499	1,523	1,523
Vertical direct driven return fan and 15 HP motor 575 V TEFC	1,591	1,615	1,615
Vertical direct driven return fan and 20 HP motor 575 V TEFC	1,681	1,705	1,705
Vertical direct driven return fan and 25 HP motor 575 V TEFC	1,679	1,702	1,702
Vertical direct driven return fan and 30 HP motor 575 V TEFC	1,717	1,741	1,741
Vertical direct driven return fan and 40 HP motor 575 V TEFC	1,909	1,933	1,933
Horizontal direct driven return fan and 7.5 HP motor 208-230/460 V ODP	1,452	1,476	1,476
Horizontal direct driven return fan and 10 HP motor 208-230/460 V ODP	1,521	1,545	1,545
Horizontal direct driven return fan and 15 HP motor 208-230/460 V ODP	1,563	1,587	1,587
Horizontal direct driven return fan and 20 HP motor 208-230/460 V ODP	1,621	1,645	1,645
Horizontal direct driven return fan and 25 HP motor 208-230/460 V ODP	1,581	1,605	1,605
Horizontal direct driven return fan and 30 HP motor 208-230/460 V ODP	1,681	1,707	1,707
Horizontal direct driven return fan and 40 HP motor 208-230/460 V ODP	1,862	1,886	1,886
Horizontal direct driven return fan and 7.5 HP motor 208-230/460 V TEFC	1,507	1,531	1,531
Horizontal direct driven return fan and 10 HP motor 208-230/460 V TEFC	1,532	1,556	1,556
Horizontal direct driven return fan and 15 HP motor 208-230/460 V TEFC	1,726	1,750	1,750
Horizontal direct driven return fan and 20 HP motor 208-230/460 V TEFC	1,749	1,773	1,773
Horizontal direct driven return fan and 25 HP motor 208-230/460 V TEFC	1,739	1,763	1,763
Horizontal direct driven return fan and 30 HP motor 208-230/460 V TEFC	1,799	1,823	1,823
Horizontal direct driven return fan and 40 HP motor 208-230/460 V TEFC	1,964	1,988	1,988
Horizontal direct driven return fan and 7.5 HP motor 575 V ODP	1,449	1,473	1,473

Table 95: Weight data (lb)

Model	60 ton	70 ton	80 ton
Horizontal direct driven return fan and 10 HP motor 575 V ODP	1,514	1,538	1,538
Horizontal direct driven return fan and 15 HP motor 575 V ODP	1,568	1,592	1,592
Horizontal direct driven return fan and 20 HP motor 575 V ODP	1593	1617	1617
Horizontal direct driven return fan and 25 HP motor 575 V ODP	1,581	1,605	1,605
Horizontal direct driven return fan and 30 HP motor 575 V ODP	1,658	1,682	1,682
Horizontal direct driven return fan and 40 HP motor 575 V ODP	1,837	1,861	1,861
Horizontal direct driven return fan and 7.5 HP motor 575 V TEFC	1,514	1,538	1,538
Horizontal direct driven return fan and 10 HP motor 575 V TEFC	1,569	1,593	1,593
Horizontal direct driven return fan and 15 HP motor 575 V TEFC	1,661	1,685	1,685
Horizontal direct driven return fan and 20 HP motor 575 V TEFC	1,751	1,775	1,775
Horizontal direct driven return fan and 25 HP motor 575 V TEFC	1,749	1,773	1,773
Horizontal direct driven return fan and 30 HP motor 575 V TEFC	1,787	1,811	1,811
Horizontal direct driven return fan and 40 HP motor 575 V TEFC	1,979	2,003	2,003
Belt driven exhaust and return fan 5 HP motor 200-208 V ODP	104	104	104
Belt driven exhaust and return fan 7.5 HP motor 200-208 V ODP	142	142	142
Belt driven exhaust and return fan 10 HP motor 200-208 V ODP	155	155	155
Belt driven exhaust and return fan 15 HP motor 200-208 V ODP	281	281	281
Belt driven exhaust and return fan 20 HP motor 200-208 V ODP	314	314	314
Belt driven exhaust and return fan 25 HP motor 200-208 V ODP	414	414	414
Belt driven exhaust and return fan 30 HP motor 200-208 V ODP	—	467	467
Belt driven exhaust and return fan 40 HP motor 200-208 V ODP	—	485	485
Belt driven exhaust and return fan 5 HP motor 200-208 V TEFC	113	113	113
Belt driven exhaust and return fan 7.5 HP motor 200-208 V TEFC	178	178	178
Belt driven exhaust and return fan 10 HP motor 200-208 V TEFC	189	189	189
Belt driven exhaust and return fan 15 HP motor 200-208 V TEFC	301	301	301
Belt driven exhaust and return fan 20 HP motor 200-208 V TEFC	373	373	373
Belt driven exhaust and return fan 25 HP motor 200-208 V TEFC	501	501	501
Belt driven exhaust and return fan 30 HP motor 200-208 V TEFC	—	526	526
Belt driven exhaust and return fan 40 HP motor 200-208 V TEFC	—	685	685
Belt driven exhaust and return fan 5 HP motor 230/460 V ODP	104	104	104
Belt driven exhaust and return fan 7.5 HP motor 230/460 V ODP	142	142	142
Belt driven exhaust and return fan 10 HP motor 230/460 V ODP	155	155	155
Belt driven exhaust and return fan 15 HP motor 230/460 V ODP	281	281	281
Belt driven exhaust and return fan 20 HP motor 230/460 V ODP	314	314	314
Belt driven exhaust and return fan 25 HP motor 230/460 V ODP	414	414	414
Belt driven exhaust and return fan 30 HP motor 230/460 V ODP	—	467	467
Belt driven exhaust and return fan 40 HP motor 230/460 V ODP	—	485	485
Belt driven exhaust and return fan 5 HP motor 230/460 V TEFC	113	113	113
Belt driven exhaust and return fan 7.5 HP motor 230/460 V TEFC	178	178	178
Belt driven exhaust and return fan 10 HP motor 230/460 V TEFC	189	189	189
Belt driven exhaust and return fan 15 HP motor 230/460 V TEFC	301	301	301
Belt driven exhaust and return fan 20 HP motor 230/460 V TEFC	373	373	373
Belt driven exhaust and return fan 25 HP motor 230/460 V TEFC	501	501	501
Belt driven exhaust and return fan 30 HP motor 230/460 V TEFC	—	526	526
Belt driven exhaust and return fan 40 HP motor 230/460 V TEFC	—	685	685
Belt driven exhaust and return fan 5 HP motor 575 V ODP	117	117	117
Belt driven exhaust and return fan 7.5 HP motor 575 V ODP	155	155	155
Belt driven exhaust and return fan 10 HP motor 575 V ODP	157	157	157
Belt driven exhaust and return fan 15 HP motor 575 V ODP	249	249	249
Belt driven exhaust and return fan 20 HP motor 575 V ODP	287	287	287
Belt driven exhaust and return fan 25 HP motor 575 V ODP	394	394	394
Belt driven exhaust and return fan 30 HP motor 575 V ODP	—	442	442
Belt driven exhaust and return fan 40 HP motor 575 V ODP	—	464	464

Table 95: Weight data (lb)

Model	60 ton	70 ton	80 ton
Belt driven exhaust and return fan 5 HP motor 575 V TEFC	114	114	114
Belt driven exhaust and return fan 7.5 HP motor 575 V TEFC	176	176	176
Belt driven exhaust and return fan 10 HP motor 575 V TEFC	190	190	190
Belt driven exhaust and return fan 15 HP motor 575 V TEFC	295	295	295
Belt driven exhaust and return fan 20 HP motor 575 V TEFC	330	330	330
Belt driven exhaust and return fan 25 HP motor 575 V TEFC	499	499	499
Belt driven exhaust and return fan 30 HP motor 575 V TEFC		487	487
Belt driven exhaust and return fan 40 HP motor 575 V TEFC	—	639	639
Dual ERW fan			
Dual ERW fan and dual 5 HP motor 208-230/460 V ODP	1,074	1,399	1,399
Dual ERW fan and dual 7.5 HP motor 208-230/460 V ODP	1,143	1,340	1,340
Dual ERW fan and dual 10 HP motor 208-230/460 V ODP	1,189	1,389	1,389
Dual ERW fan and dual 15 HP motor 208-230/460 V ODP	1,406	1,694	1,694
Dual ERW fan and dual 20 HP motor 208-230/460 V ODP	1470	1758	1758
Dual ERW fan and dual 5 HP motor 208-230/460 V TEFC	1,092	1,269	1,269
Dual ERW fan and dual 7.5 HP motor 208-230/460 V TEFC	1,188	1,385	1,385
Dual ERW fan and dual 10 HP motor 208-230/460 V TEFC	1,230	1,430	1,430
Dual ERW fan and dual 15 HP motor 208-230/460 V TEFC	1,448	1,648	1,648
Dual ERW fan and dual 20 HP motor 208-230/460 V TEFC	1,524	1,724	1,724
Dual ERW fan and dual 5 HP motor 575 V ODP	1,050	1,247	1,247
Dual ERW fan and dual 7.5 HP motor 575 V ODP	1,146	1,343	1,343
Dual ERW fan and dual 10 HP motor 575 V ODP	1,152	1,352	1,352
Dual ERW fan and dual 15 HP motor 575 V ODP	1,330	1,530	1,530
Dual ERW fan and dual 20 HP motor 575 V ODP	1,406	1,606	1,606
Dual ERW fan and dual 5 HP motor 575 V TEFC	1,064	1,261	1,261
Dual ERW fan and dual 7.5 HP motor 575 V TEFC	1,188	1,385	1,385
Dual ERW fan and dual 10 HP motor 575 V TEFC	1,236	1,436	1,436
Dual ERW fan and dual 15 HP motor 575 V TEFC	1,422	1,622	1,622
Dual ERW fan and dual 20 HP motor 575 V TEFC	1,488	1,688	1,688
Motor weights			
5 HP Motor 208-230/460V ODP	104	104	104
7.5 HP Motor 208-230/460V ODP	142	142	142
10 HP Motor 208-230/460V ODP	155	155	155
15 HP Motor 208-230/460V ODP	281	281	281
20 HP Motor 208-230/460V ODP	314	314	314
25 HP Motor 208-230/460V ODP	414	414	414
30 HP Motor 208-230/460V ODP	—	467	467
40 HP Motor 208-230/460V ODP	—	485	485
50 HP Motor 208-230/460V ODP	—	—	—
5 HP Motor 208-230/460V TEFC	113	113	113
7.5 HP Motor 208-230/460V TEFC	178	178	178
10 HP Motor 208-230/460V TEFC	189	189	189
15 HP Motor 208-230/460V TEFC	301	301	301
20 HP Motor 208-230/460V TEFC	373	373	373
25 HP Motor 208-230/460V TEFC	501	501	501
30 HP Motor 208-230/460V TEFC	—	526	526
40 HP Motor 208-230/460V TEFC	—	685	685
50 HP Motor 208-230/460V TEFC	—	—	—
5 HP Motor 575V ODP	116.6	116.6	116.6
7.5 HP Motor 575V ODP	154.6	154.6	154.6
10 HP Motor 575V ODP	156.6	156.6	156.6
15 HP Motor 575V ODP	248.5	248.4	248.4
20 HP Motor 575V ODP	286.5	286.5	286.5

Table 95: Weight data (lb)

Model	60 ton	70 ton	80 ton
25 HP Motor 575V ODP	394	394	394
30 HP Motor 575V ODP	—	442	442
40 HP Motor 575V ODP	—	464	464
50 HP Motor 575V ODP	—	—	—
5 HP Motor 575V TEFC	114	114	114
7.5 HP Motor 575V TEFC	176	176	176
10 HP Motor 575V TEFC	190	190	190
15 HP Motor 575V TEFC	295	295	295
20 HP Motor 575V TEFC	330	330	330
25 HP Motor 575V TEFC	499	499	499
30 HP Motor 575V TEFC	—	487	487
40 HP Motor 575V TEFC	—	639	639
50 HP Motor 575V TEFC	—	—	—
Draw-through filters			
Angled filter rack - 2 in. throwaway filters	142	142	142
Angled filter rack - 2 in. MERV 8 filters	160	160	160
MERV 15 bag filters with 2 in. pre filters	199	199	199
MERV 14 rigid filters with 2 in. pre filters	447	447	447
Vertical filter rack - 4 in. MERV 8 filters	141	141	141
Final filters - post-evaporator coil			
MERV 15 bag filters with 2 in. pre filters	209	209	209
MERV 14 rigid filters with 2 in. pre filters	426	426	426
MERV 17 HEPA filters with 2 in. pre filters	996	996	996
Gas heat, stage or modulating - aluminized or stainless steel			
500 MBH	358	358	358
1000 MBH	610	610	610
1500 MBH	860	860	860
Electric heat[11]			
Low heat	270	270	270
High heat	315	315	315
Hot water heat			
Low heat with valves	192	192	192
High heat with valves	245	245	245
Steam heat			
Low heat with valves	185	185	200
High heat with valves	241	241	241
Refrigeration system			
Standard efficiency cooling system	2,815	4,313	4,427
High capacity cooling system	2,910	4,379	4,508
High efficiency cooling system	2,917	4,405	4,412
Hot gas reheat (HGRH) coil	73	87	87
Louver hail guard	249	286	286
Miscellaneous options			
Humidifier	112	161	161
Air blender	166	185	185
Ultraviolet (UV) lights	115	115	115
Sound attenuator	1,001	1,001	1,001
Internal lights	35	35	35
Smoke detector	7	7	7
Transformer [12]	60	60	60
Split unit	109	47	47
Redundant return fan VFD 7.5-15 HP	21	21	21
Redundant return fan VFD 20-25 HP	54	54	54

Table 95: Weight data (lb)

Model	60 ton	70 ton	80 ton
Redundant return fan VFD 30-40 HP	79	79	79
Belt guard	83	83	83

Notes:

1. Economizer cabinet weight must be included in all 60 ton to 80 ton units.
2. Pre-evaporator blank can be a blank section or include an air blender.
3. Heat section cabinet option for any of the following:
   - Top or right discharge
   - Discharge filters, humidifier or sound attenuator, or blank post-evaporator section
   - Requires discharge plenum. Discharge can be bottom, left, right, or top
4. Heat and discharge through heat option is used unless any of the following are included:
   - Discharge filters
   - Humidifier, sound attenuator, or blank post-evaporator section
   - Top or right discharge
   - Discharge plenum
5. Small post-evaporator blank can contain *blank* or *final filter* or *humidifier* or *sound attenuator*.
6. Large post-evaporator blank can contain *blank* or *sound attenuator and final filter* or sound *attenuator and humidifier* or *humidifier* or *sound attenuator*.
7. Discharge plenum is required with all units that are cooling only. Discharge plenum is required on all units with heat section cabinet.
8. Supply fan weights include the supply fan, motor, isolators, inlet guards, shaft grounding ring, variable frequency drive (VFD), line reactor, disconnect, and VFD bypass. On 60 ton to 80 ton units, supply fan weights are for both fans and motors.
9. Economizer with belt driven exhaust or return fan includes the weight of the fan, isolator, shaft grounding rings, extended lube lines, and inlet guards.
10. For belt driven exhaust or return fan motor includes the weight of the motor, VFD, line reactor, and VFD bypass. For direct driven return fan, weight includes return fan and motor, isolators, inlet guards, shaft grounding ring, VFD, line reactor. Return fan redundant VFD weight included under miscellaneous options.
11. All sizes of electric heat.
12. A transformer is required if convenience outlet, gas heat, lights, or high end display are selected.
13. Roof curb weight is not included with this table.
14. Considering standard cooling unit configuration the Refrigeration system weight includes condenser fan, compressor, piping, condenser coil, refrigerant, wire guards, condenser sheetmetal and evaporator coil.
15. For units that are shipped split please contact your local sales office or use Selection Navigator to get the individual air handler and condenser unit weights.
16. Cooling only units with a no post evaporator blank, a small empty post evaporator blank, a large empty post evaporator blank or a large post evaporator blank equipped with a humidifier or final filter have a 55 in. fan section length, all other units have a 74 in. fan section length.

Dimensions

Figure 13: Rooftop unit component locations

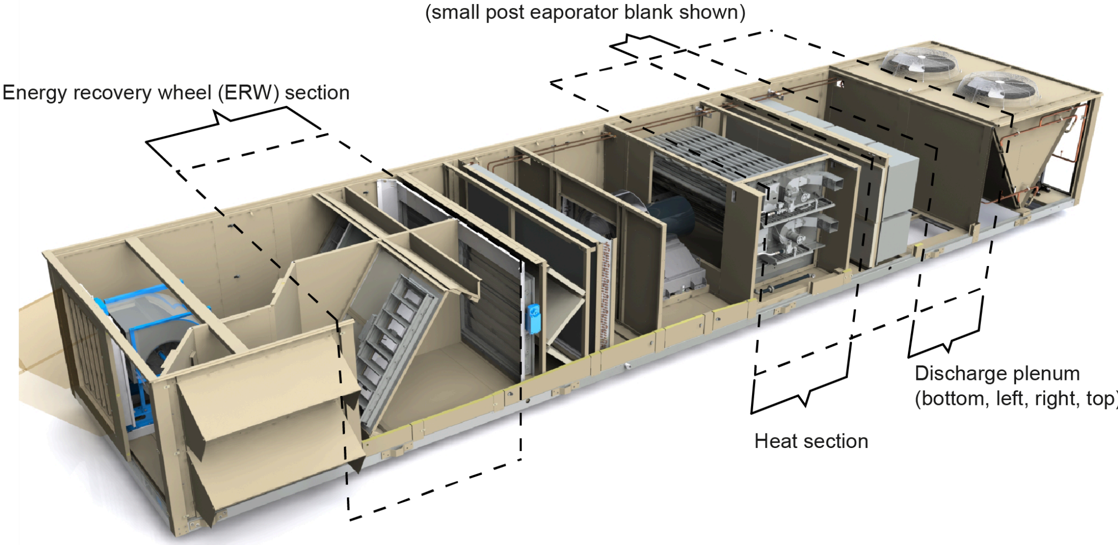


Figure 14: 60 ton dimensional drawing

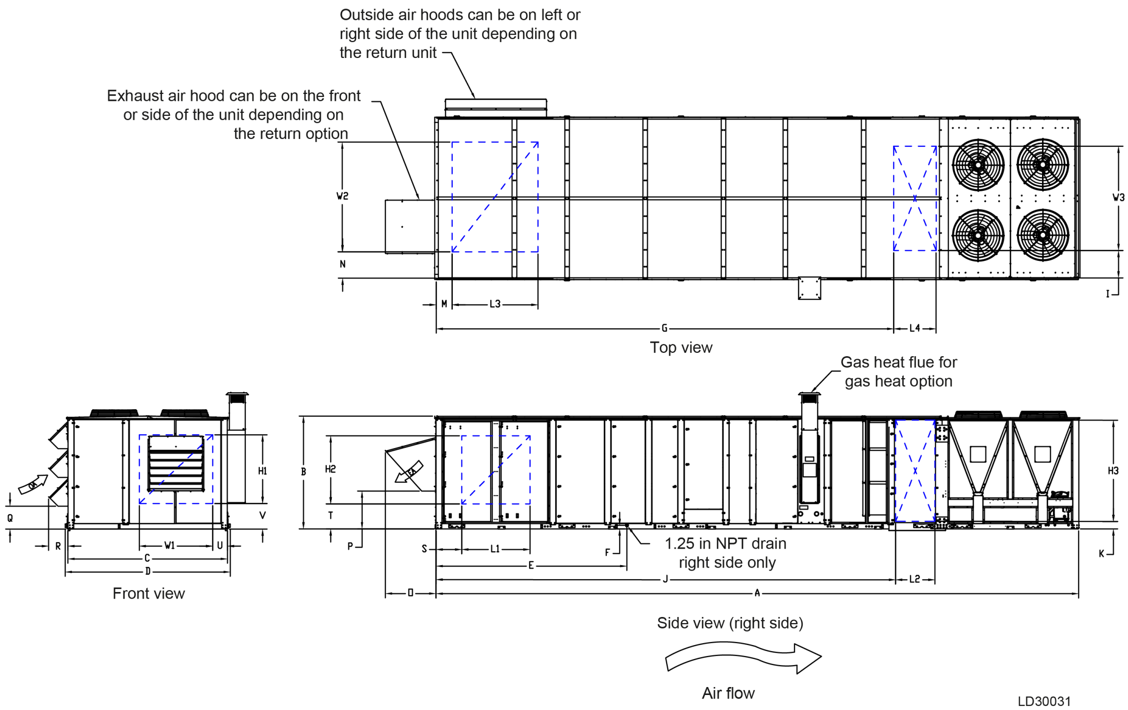


Table 96: 60 ton unit dimensions (in.) for return options with an ERW

Unit configuration	Unit with an exhaust fan with backdraft damper or modulating damper							
Dimension set	**M**	**L3**	**W2**	**N**	**O**	**P**	**Q**	**R**
Bottom return, right OA, side exhaust	111.48	53.92	54.48	54.39	24.29	21.48	16.17	13.93
Bottom return, right OA, front exhaust	111.48	53.92	54.48	54.39	24.29	21.48	16.17	13.93

Table 97: 60 ton unit dimensions (in.) for return options without an ERW

Unit configuration	Unit with a return fan with modulating damper								Unit with an exhaust fan with backdraft damper or modulating damper							
Dimension set	**M**	**L3**	**W2**	**N**	**O**	**P**	**Q**	**R**	**M**	**L3**	**W2**	**N**	**O**	**P**	**Q**	**R**
Bottom return, right OA, side exhaust	13.66	41.63	45.51	54.71	47.06	15.98	16.17	13.93	13.66	56.91	71.31	31.05	35.95	26.92	16.17	13.93
Bottom return, right OA, front exhaust	13.66	41.63	45.51	54.71	47.06	15.98	16.17	13.93	13.66	56.91	77.87	31.05	35.95	26.92	16.17	13.93
Bottom return, left OA, side exhaust	13.66	41.63	45.51	12.83	47.06	15.98	16.17	13.93	13.66	56.91	71.31	10.69	35.95	26.92	16.17	13.93
Bottom return, left OA, front exhaust	13.66	41.63	45.51	12.83	47.06	15.98	16.17	13.93	13.66	56.91	77.87	4.13	35.95	26.92	16.17	13.93
Top return, right OA, side exhaust	—	—	—	—	—	—	—	—	3.23	49.30	82.59	28.69	35.95	26.92	16.17	13.93
Top return, right OA, front exhaust	—	—	—	—	—	—	—	—	3.23	49.30	82.59	28.69	35.95	26.92	16.17	13.93
Top return, left OA, side exhaust	—	—	—	—	—	—	—	—	3.23	49.30	82.59	1.8	35.95	26.92	16.17	13.93
Top return, left OA, front exhaust	—	—	—	—	—	—	—	—	3.23	49.30	82.59	1.8	35.95	26.92	16.17	13.93
Dimension set	**S**	**L1**	**H2**	**T**	**O**	**P**	**Q**	**R**	**S**	**L1**	**H2**	**T**	**O**	**P**	**Q**	**R**
Left return, right OA, front exhaust	18.99	47	47	10.34	47.06	15.98	16.17	13.93	5.09	74.78	66.11	7.64	35.95	26.92	16.17	13.93
Right return, left OA, front exhaust	18.99	47	47	10.34	47.06	15.98	16.17	13.93	5.09	74.78	66.11	7.64	35.95	26.92	16.17	13.93
Dimension set	**S**	**W1**	**H1**	**V**	**O**	**P**	**Q**	**R**	**S**	**W1**	**H1**	**V**	**O**	**P**	**Q**	**R**
Front Return, left OA, right exhaust	12.79	47	47	10.34	47.06	15.98	16.17	13.93	5.09	62.40	66.11	7.64	35.95	26.92	16.17	13.93
Front return, right OA, left exhaust	53.25	47	47	10.34	47.06	15.98	16.17	13.93	45.56	62.40	66.11	7.64	35.95	26.92	16.17	13.93

Table 98: 60 ton unit dimensions (in.) for return options without an ERW

Unit configuration	Unit has no fan and barometric damper							
Dimension set	**M**	**L3**	**W2**	**N**	**O**	**P**	**Q**	**R**
Bottom return, right OA, side exhaust	6.13	54.28	65.00	43.92	33.23	23.08	11.27	13.93
Bottom return, right OA, front exhaust	6.13	54.28	65.00	43.92	33.23	23.08	11.27	13.93
Bottom return, left OA, side exhaust	6.13	54.28	65.00	4.13	33.23	23.08	11.27	13.93
Bottom return, left OA, front exhaust	6.13	54.28	65.00	4.13	33.23	23.08	11.27	13.93
Bottom return, no OA, no EA	6.13	54.28	65.00	43.92	—	—	—	—
Top return, right OA, side exhaust	3.23	49.30	71.29	40.01	33.23	23.08	11.27	17.46
Top return, right OA, front exhaust	3.23	49.30	71.29	40.01	33.23	23.08	11.27	17.46
Top return, left OA, side exhaust	3.23	49.30	71.29	1.76	33.23	23.08	11.27	17.46
Top return, left OA, front exhaust	3.23	49.30	71.29	1.76	33.23	23.08	11.27	17.46
Top return, no OA, no EA	3.23	49.30	71.29	40.01	—	—	—	—
Dimension set	**S**	**L1**	**H2**	**T**	**O**	**P**	**Q**	**R**
Left return, right OA, front exhaust	5.09	50.78	66.05	7.67	33.23	23.08	11.27	13.93
Left return, no OA, no EA	5.09	50.78	66.05	7.67	—	—	—	—
Right return, left OA, front exhaust	5.09	50.78	66.05	7.67	33.23	23.08	11.27	17.46
Right return, no OA, no EA	5.09	50.78	66.05	7.67	—	—	—	—
Dimension set	**U**	**W1**	**H1**	**V**	**O**	**P**	**Q**	**R**
Front return, left OA, right exhaust	5.09	62.40	66.10	7.64	33.23	23.08	11.27	17.46
Front return, right OA, left exhaust	45.56	62.40	66.10	7.64	33.23	23.08	11.27	17.46
Front return, no OA, no EA	45.56	62.40	66.10	7.64	—	—	—	—

Table 99: 60 ton unit dimensions (in.)

		Unit configuration	Unit length (A)	Unit height (B)	Unit width (C)	Unit width lifting lug to lifting lug (D)	Evaporator drain length (E)	Evaporator drain height (F)
Basic economizer - no ERW	No heat - 55 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	298.54	80.00	113.36	117.00	94.50	2.50
		SU with discharge plenum (B,L,R,T)	315.54	80.00	113.36	117.00	111.50	2.50
		SU with small post-evap blank, discharge plenum (B,L,R,T)	360.54	80.00	113.36	117.06	111.50	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T,)	413.54	80.00	113.36	117.06	164.50	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	402.54	80.00	113.36	117.06	111.50	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	455.54	80.00	113.36	117.06	164.50	2.50
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	379.54	80.00	113.36	117.00	111.50	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	432.54	80.00	113.36	117.06	164.50	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	421.54	80.00	113.36	117.06	111.50	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	474.54	80.00	113.36	117.06	164.50	2.50
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	334.54	80.00	113.36	117.00	94.50	2.50
		SU, discharge through heat section (B,L)	351.54	80.00	113.36	117.00	111.50	2.50
		SU with pre-evap blank, discharge through heat section (B,L)	404.54	80.00	113.36	117.06	164.50	2.50
		SU with heat, discharge plenum (B,L,R,T)	387.54	80.00	113.36	117.00	111.50	2.50
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	440.54	80.00	113.36	117.06	164.50	2.50
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	432.54	80.00	113.36	117.00	111.50	2.50
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	485.54	80.00	113.36	117.06	164.50	2.50
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	474.54	80.00	113.36	117.06	111.50	2.50
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	527.54	80.00	113.36	117.06	164.50	2.50

Table 99: 60 ton unit dimensions (in.)

		Unit configuration	Unit length (A)	Unit height (B)	Unit width (C)	Unit width lifting lug to lifting lug (D)	Evaporator drain length (E)	Evaporator drain height (F)
Standard economizer - no ERW	No heat - 55 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	322.54	80.00	113.36	117.00	118.50	2.50
		SU with discharge plenum (B,L,R,T)	339.54	80.00	113.36	117.00	135.50	2.50
		SU with small post-evap blank, discharge plenum (B,L,R,T)	384.54	80.00	113.36	117.06	135.50	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	437.54	80.00	113.36	117.06	188.50	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	426.54	80.00	113.36	117.06	135.50	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	479.54	80.00	113.36	117.06	188.50	2.50
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	403.54	80.00	113.36	117.00	135.50	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	456.54	80.00	113.36	117.06	188.50	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	445.54	80.00	113.36	117.06	135.50	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	498.54	80.00	113.36	117.06	188.50	2.50
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	358.54	80.00	113.36	117.00	118.50	2.50
		SU, discharge through heat section (B,L)	375.54	80.00	113.36	117.00	135.50	2.50
		SU with pre-evap blank, discharge through heat section (B,L)	428.54	80.00	113.36	117.06	188.50	2.50
		SU with heat, discharge plenum (B,L,R,T)	411.54	80.00	113.36	117.00	135.50	2.50
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	464.54	80.00	113.36	117.06	188.50	2.50
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	456.54	80.00	113.36	117.00	135.50	2.50
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	509.54	80.00	113.36	117.06	188.50	2.50
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	498.54	80.00	113.36	117.06	135.50	2.50
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	551.54	80.00	113.36	117.06	188.50	2.50
Standard economizer - ERW	No heat - 55 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	465.54	80.00	113.36	117.06	216.50	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	518.54	80.00	113.36	117.06	269.50	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	507.54	80.00	113.36	117.06	216.50	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	560.54	80.00	113.36	117.06	269.50	2.50
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	484.54	80.00	113.36	117.06	216.50	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	537.54	80.00	113.36	117.06	269.50	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	526.54	80.00	113.36	117.06	216.50	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	579.54	80.00	113.36	117.06	269.50	2.50
	Heat	SU, discharge through heat section (B,L)	456.54	80.00	113.36	117.06	216.50	2.50
		SU with pre-evap blank, discharge through heat section (B,L)	509.54	80.00	113.36	117.06	269.50	2.50
		SU with heat, discharge plenum (B,L,R,T)	492.54	80.00	113.36	117.06	216.50	2.50
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	545.54	80.00	113.36	117.06	269.50	2.50
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	537.54	80.00	113.36	117.06	216.50	2.50
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	590.54	80.00	113.36	117.06	269.50	2.50
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	579.54	80.00	113.36	117.06	216.50	2.50
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	632.54	80.00	113.36	117.06	269.50	2.50

Table 100: 60 ton unit dimensions (in.) with bottom and top discharge

		Unit configuration	Bottom discharge				Top discharge			
			G	L4	I	W3	G	L4	I	W3
Basic economizer - no ERW	No heat - 55 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	167.19	30.00	19.52	74.00	167.19	30.00	19.52	74.00
		SU with discharge plenum (B,L,R,T)	184.19	30.00	19.52	74.00	184.19	30.00	19.52	74.00
		SU with small post-evap blank, discharge plenum (B,L,R,T)	229.19	30.00	19.52	74.00	229.19	30.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	282.19	30.00	19.52	74.00	282.19	30.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	271.19	30.00	19.52	74.00	271.19	30.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	324.19	30.00	19.52	74.00	324.19	30.00	19.52	74.00
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	248.19	30.00	19.52	74.00	248.19	30.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	301.19	30.00	19.52	74.00	301.19	30.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	290.19	30.00	19.52	74.00	290.19	30.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	343.19	30.00	19.52	74.00	343.19	30.00	19.52	74.00
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	190.85	40.00	26.51	60.00	—	—	—	—
		SU, discharge through heat section (B,L)	207.85	40.00	26.51	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	260.85	40.00	26.51	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	256.19	30.00	19.52	74.00	256.19	30.00	19.52	74.00
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	309.19	30.00	19.52	74.00	309.19	30.00	19.52	74.00
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	301.19	30.00	19.52	74.00	301.19	30.00	19.52	74.00
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	354.19	30.00	19.52	74.00	354.19	30.00	19.52	74.00
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	343.19	30.00	19.52	74.00	343.19	30.00	19.52	74.00
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	396.19	30.00	19.52	74.00	396.19	30.00	19.52	74.00
Standard economizer - no ERW	No heat - 55 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	191.19	30.00	19.52	74.00	191.19	30.00	19.52	74.00
		SU with discharge plenum (B,L,R,T)	208.19	30.00	19.52	74.00	208.19	30.00	19.52	74.00
		SU with small post-evap blank, discharge plenum (B,L,R,T)	253.19	30.00	19.52	74.00	253.19	30.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	306.19	30.00	19.52	74.00	306.19	30.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	295.19	30.00	19.52	74.00	295.19	30.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	348.19	30.00	19.52	74.00	348.19	30.00	19.52	74.00
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	272.19	30.00	19.52	74.00	272.19	30.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	325.19	30.00	19.52	74.00	325.19	30.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	314.19	30.00	19.52	74.00	314.19	30.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	367.19	30.00	19.52	74.00	367.19	30.00	19.52	74.00
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	214.85	40.00	26.51	60.00	—	—	—	—
		SU, discharge through heat section (B,L)	231.85	40.00	26.51	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	284.85	40.00	26.51	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	280.19	30.00	19.52	74.00	280.19	30.00	19.52	74.00
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	333.19	30.00	19.52	74.00	333.19	30.00	19.52	74.00
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	325.19	30.00	19.52	74.00	325.19	30.00	19.52	74.00
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	378.19	30.00	19.52	74.00	378.19	30.00	19.52	74.00
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	367.19	30.00	19.52	74.00	367.19	30.00	19.52	74.00
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	420.19	30.00	19.52	74.00	420.19	30.00	19.52	74.00

Table 100: 60 ton unit dimensions (in.) with bottom and top discharge

		Unit configuration	Bottom discharge				Top discharge			
			G	L4	I	W3	G	L4	I	W3
Standard economizer - ERW	No heat - 55 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	334.19	30.00	19.52	74.00	334.19	30.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	387.19	30.00	19.52	74.00	387.19	30.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	376.19	30.00	19.52	74.00	376.19	30.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	429.19	30.00	19.52	74.00	429.19	30.00	19.52	74.00
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	353.19	30.00	19.52	74.00	353.19	30.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	406.19	30.00	19.52	74.00	406.19	30.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	395.19	30.00	19.52	74.00	395.19	30.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	448.19	30.00	19.52	74.00	448.19	30.00	19.52	74.00
	Heat	SU, discharge through heat section (B,L)	312.85	40.00	26.51	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	365.85	40.00	26.51	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	361.19	30.00	19.52	74.00	361.19	30.00	19.52	74.00
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	414.19	30.00	19.52	74.00	414.19	30.00	19.52	74.00
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	406.19	30.00	19.52	74.00	406.19	30.00	19.52	74.00
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	459.19	30.00	19.52	74.00	459.19	30.00	19.52	74.00
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	448.19	30.00	19.52	74.00	448.19	30.00	19.52	74.00
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	501.19	30.00	19.52	74.00	501.19	30.00	19.52	74.00

Table 101: 60 ton unit dimensions (in.) with left and right discharge

		Unit configuration	Left discharge				Right discharge			
			J	L2	K	H3	J	L2	K	H3
Basic economizer - no ERW	No heat - 55 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	168.06	28.21	4.65	72.04	168.06	28.21	4.65	72.04
		SU with discharge plenum (B,L,R,T)	185.06	28.21	4.65	72.04	185.06	28.21	4.65	72.04
		SU with small post-evap blank, discharge plenum (B,L,R,T)	230.06	28.21	4.65	72.04	230.06	28.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	283.06	28.21	4.65	72.04	283.06	28.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	272.06	28.21	4.65	72.04	272.06	28.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	325.06	28.21	4.65	72.04	325.06	28.21	4.65	72.04
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	249.06	28.21	4.65	72.04	249.06	28.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	302.06	28.21	4.65	72.04	302.06	28.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	291.06	28.21	4.65	72.04	291.06	28.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	344.06	28.21	4.65	72.04	344.06	28.21	4.65	72.04
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	190.75	40.00	9.19	60.00	—	—	—	—
		SU, discharge through heat section (B,L)	207.75	40.00	9.19	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	260.75	40.00	9.19	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	257.06	28.21	4.65	72.04	257.06	28.21	4.65	72.04
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	310.06	28.21	4.65	72.04	310.06	28.21	4.65	72.04
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	302.06	28.21	4.65	72.04	302.06	28.21	4.65	72.04
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	355.06	28.21	4.65	72.04	355.06	28.21	4.65	72.04
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	344.06	28.21	4.65	72.04	344.06	28.21	4.65	72.04
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	397.06	28.21	4.65	72.04	397.06	28.21	4.65	72.04

Table 101: 60 ton unit dimensions (in.) with left and right discharge

		Unit configuration	Left discharge				Right discharge			
			J	L2	K	H3	J	L2	K	H3
Standard economizer - no ERW	No heat - 55 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	192.06	28.21	4.65	72.04	192.06	28.21	4.65	72.04
		SU with discharge plenum (B,L,R,T)	209.06	28.21	4.65	72.04	209.06	28.21	4.65	72.04
		SU with small post-evap blank, discharge plenum (B,L,R,T)	254.06	28.21	4.65	72.04	254.06	28.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	307.06	28.21	4.65	72.04	307.06	28.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	296.06	28.21	4.65	72.04	296.06	28.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	349.06	28.21	4.65	72.04	349.06	28.21	4.65	72.04
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	273.06	28.21	4.65	72.04	273.06	28.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	326.06	28.21	4.65	72.04	326.06	28.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	315.06	28.21	4.65	72.04	315.06	28.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	368.06	28.21	4.65	72.04	368.06	28.21	4.65	72.04
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	214.75	40.00	9.19	60.00	—	—	—	—
		SU, discharge through heat section (B,L)	231.75	40.00	9.19	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)					—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	281.06	28.21	4.65	72.04	281.06	28.21	4.65	72.04
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	334.06	28.21	4.65	72.04	334.06	28.21	4.65	72.04
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	326.06	28.21	4.65	72.04	326.06	28.21	4.65	72.04
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	379.06	28.21	4.65	72.04	379.06	28.21	4.65	72.04
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	368.06	28.21	4.65	72.04	368.06	28.21	4.65	72.04
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	421.06	28.21	4.65	72.04	421.06	28.21	4.65	72.04
Standard economizer - ERW	No heat - 55 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	335.06	28.21	4.65	72.04	335.06	28.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	388.06	28.21	4.65	72.04	388.06	28.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	377.06	28.21	4.65	72.04	377.06	28.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	430.06	28.21	4.65	72.04	430.06	28.21	4.65	72.04
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	354.06	28.21	4.65	72.04	354.06	28.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	407.06	28.21	4.65	72.04	407.06	28.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	396.06	28.21	4.65	72.04	396.06	28.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	449.06	28.21	4.65	72.04	449.06	28.21	4.65	72.04
	Heat	SU, discharge through heat section (B,L)	312.85	40.00	9.19	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	365.85	40.00	9.19	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	362.06	28.21	4.65	72.04	362.06	28.21	4.65	72.04
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	415.06	28.21	4.65	72.04	415.06	28.21	4.65	72.04
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	407.06	28.21	4.65	72.04	407.06	28.21	4.65	72.04
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	460.06	28.21	4.65	72.04	460.06	28.21	4.65	72.04
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	449.06	28.21	4.65	72.04	449.06	28.21	4.65	72.04
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	502.06	28.21	4.65	72.04	502.06	28.21	4.65	72.04

Note:

1. A SU includes length for economizer, draw-through filter, cooling coil, supply fan, and condenser. A SU includes a draw-through (pre-evaporator) filter with 2 in. angle filter or 2 in. + 12 in. (or 18 in.) filter. Units with a vertical 4 in. draw-through filter in place of the standard 2 in. angle or 2 in. + 12 in. (or 18 in.) filter are listed as *SU with Vertical Filter 4 in*.
2. Basic economizer unit does not include an exhaust fan or return fan option.
3. The heat section includes either gas, hot water, steam, or electric.
4. SU = Standard Unit, DP = Discharge Plenum, B = Bottom Discharge, L = Left Discharge, R = Right Discharge, T = Top Discharge.
5. Dimensions are ± 1 in.
6. For units that are shipped split, please contact your local sales office or use the Selection Navigator to get the individual air handler and condenser unit lengths.

Figure 15: 70 ton to 80 ton dimensional drawing

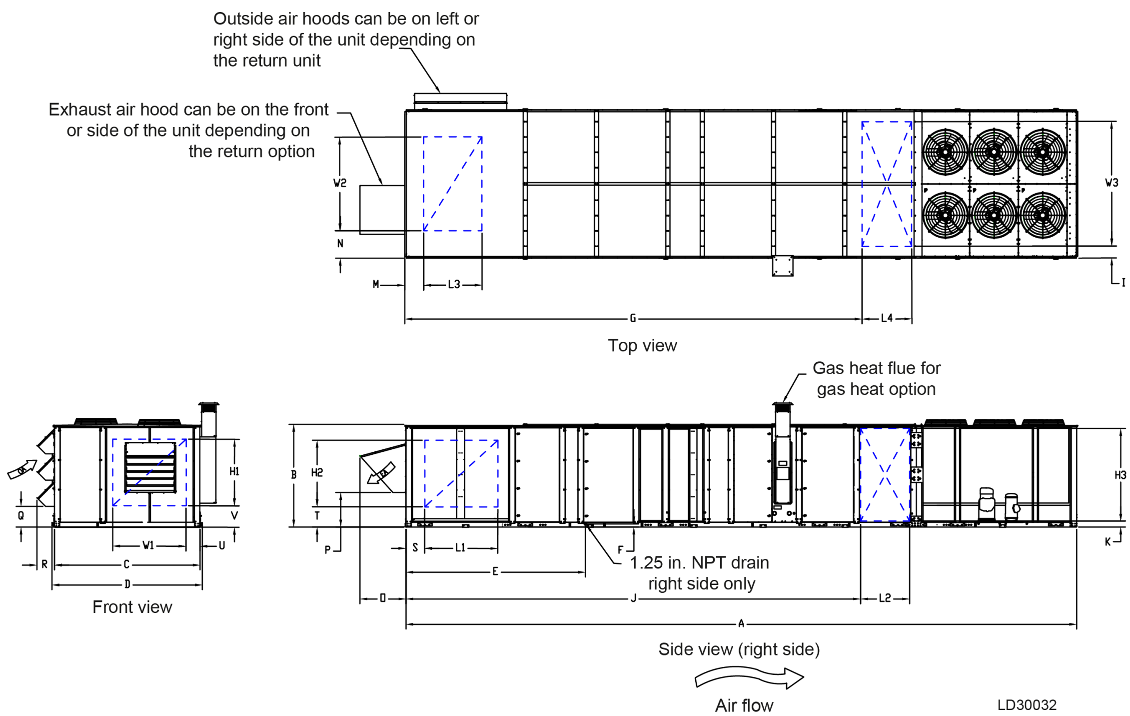


Table 102: 70 ton to 80 ton unit dimensions (in.) for return options with an ERW

Unit configuration	ERW only available with an exhaust fan with backdraft damper or modulating damper							
Dimension set	**M**	**L3**	**W2**	**N**	**O**	**P**	**Q**	**R**
Bottom return, right OA, side exhaust	111.48	72.92	54.50	54.42	26.71	22.48	16.17	13.93
Bottom return, right OA, front exhaust	111.48	72.92	54.50	54.42	26.71	22.48	16.17	13.93

Table 103: 70 ton to 80 ton unit dimensions (in.) for return options without an ERW

Unit configuration	Unit with a return fan with modulating damper								Unit with an exhaust fan with backdraft damper or modulating damper							
Dimension set	**M**	**L3**	**W2**	**N**	**O**	**P**	**Q**	**R**	**M**	**L3**	**W2**	**N**	**O**	**P**	**Q**	**R**
Bottom return, right OA, side exhaust	13.66	41.63	45.51	54.71	52.68	13.07	16.17	13.93	13.66	56.94	71.31	31.05	35.95	26.92	16.17	13.93
Bottom return, right OA, front exhaust	13.66	41.63	45.51	54.71	52.68	13.07	16.17	13.93	13.66	56.91	77.87	31.05	35.95	26.92	16.17	13.93
Bottom return, left OA, side exhaust	13.66	41.63	45.51	12.83	52.68	13.07	16.17	13.93	13.66	56.91	71.31	10.67	35.95	26.92	16.17	13.93
Bottom return, left OA, front exhaust	13.66	41.63	45.51	12.83	52.68	13.07	16.17	13.93	13.66	56.91	77.87	4.13	35.95	26.92	16.17	13.93
Top return, right OA, side exhaust	—	—	—	—	—	—	—	—	3.23	49.30	82.59	28.70	35.95	26.92	16.17	13.93
Top return, right OA, front exhaust	—	—	—	—	—	—	—	—	3.23	49.30	82.59	28.70	35.95	26.92	16.17	13.93
Top return, left OA, side exhaust	—	—	—	—	—	—	—	—	3.23	49.30	82.59	1.76	35.95	26.92	16.17	13.93
Top return, left OA, front exhaust	—	—	—	—	—	—	—	—	3.23	49.30	82.59	1.76	35.95	26.92	16.17	13.93
Dimension set	**S**	**L1**	**H2**	**T**	**O**	**P**	**Q**	**R**	**S**	**L1**	**H2**	**T**	**O**	**P**	**Q**	**R**
Left return, right OA, front exhaust	18.99	47	47	10.34	52.68	13.07	16.17	13.93	5.09	74.78	66.11	7.64	35.95	26.92	16.17	13.93
Right return, left OA, front exhaust	18.99	47	47	10.34	52.68	13.07	16.17	13.93	5.09	74.78	66.11	7.64	35.95	26.92	16.17	13.93
Dimension set	**U**	**W1**	**H1**	**V**	**O**	**P**	**Q**	**R**	**U**	**W1**	**H1**	**V**	**O**	**P**	**Q**	**R**
Front return, left OA, right exhaust	12.80	47	47	10.34	52.68	13.07	16.17	13.93	5.09	62.40	66.10	7.64	35.95	26.92	16.17	13.93
Front return, right OA, left exhaust	53.25	47	47	10.34	52.68	13.07	16.17	13.93	45.56	62.40	66.10	7.64	35.95	26.92	16.17	13.93

Unit configuration	Unit with no fan and barometric damper							
Dimension set	**M**	**L3**	**W2**	**N**	**O**	**P**	**Q**	**R**
Bottom return, right OA, side exhaust	6.13	54.28	65.00	43.92	38.08	20.58	11.27	13.93
Bottom return, right OA, front exhaust	6.13	54.28	65.00	43.92	38.08	20.58	11.27	13.93
Bottom return, left OA, side exhaust	6.13	54.28	65.00	4.13	38.08	20.58	11.27	13.93
Bottom return, left OA, front exhaust	6.13	54.28	65.00	4.13	38.08	20.58	11.27	13.93

Unit configuration	Unit with no fan and barometric damper							
Bottom return, no OA, no EA	6.13	54.28	65.00	43.92	—	—	—	—
Top return, right OA, side exhaust	3.23	49.30	71.29	40.01	38.08	20.58	11.27	17.46
Top return, right OA, front exhaust	3.23	49.30	71.29	40.01	38.08	20.58	11.27	17.46
Top return, left OA, side exhaust	3.23	49.30	71.29	1.76	38.08	20.58	11.27	17.46
Top return, left OA, front exhaust	3.23	49.30	71.29	1.76	38.08	20.58	11.27	17.46
Top return, no OA, no EA	3.23	49.30	71.29	40.01	—	—	—	—
Dimension set	**S**	**L1**	**H2**	**T**	**O**	**P**	**Q**	**R**
Left return, right OA, front exhaust	5.09	50.78	66.05	7.67	38.08	20.58	11.27	13.93
Left return, no OA, no EA	5.09	50.78	66.05	7.67	—	—	—	—
Right return, left OA, front exhaust	5.09	50.78	66.04	7.67	38.08	20.58	11.27	17.46
Right return, no OA, no EA	5.09	50.78	66.04	7.67	—	—	—	—
Dimension set	**U**	**W1**	**H1**	**V**	**O**	**P**	**Q**	**R**
Front return, left OA, right exhaust	5.09	62.40	66.10	7.64	38.08	20.58	11.27	17.46
Front return, right OA, left exhaust	45.56	62.40	66.10	7.64	38.08	20.58	11.27	17.46
Front return, no OA, no EA	45.56	62.40	66.10	7.64	—	—	—	—

Table 104: 70 ton to 80 ton unit dimensions (in.)

		Unit configuration	Unit length (A)	Unit height (B)	Unit width (C)	Unit width lifting lug to lifting lug (D)	Evaporator drain length (E)	Evaporator drain height (F)
Basic economizer - no ERW	No heat - 55 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	365.04	80.00	113.36	117.00	98.80	2.50
		SU with discharge plenum (B,L,R,T)	382.04	80.00	113.36	117.00	115.80	2.50
		SU with small post-evap blank, discharge plenum (B,L,R,T)	427.04	80.00	113.36	117.06	115.80	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	480.04	80.00	113.36	117.06	168.80	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	469.04	80.00	113.36	117.06	115.80	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	522.04	80.00	113.36	117.06	168.80	2.50
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	446.04	80.00	113.36	117.00	115.80	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	499.04	80.00	113.36	117.06	168.80	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	488.04	80.00	113.36	117.06	115.80	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	541.04	80.00	113.36	117.06	168.80	2.50
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	391.04	80.00	113.36	117.00	98.80	2.50
		SU, discharge through heat section (B,L)	408.04	80.00	113.36	117.00	115.80	2.50
		SU with pre-evap blank, discharge through heat section (B,L)	461.04	80.00	113.36	117.06	168.8	2.50
		SU with heat, discharge plenum (B,L,R,T)	454.04	80.00	113.36	117.00	115.80	2.50
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	507.04	80.00	113.36	117.06	168.8	2.50
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	499.04	80.00	113.36	117.00	115.80	2.50
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	552.04	80.00	113.36	117.06	168.8	2.50
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	541.04	80.00	113.36	117.06	115.80	2.50
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	594.04	80.00	113.36	117.06	168.8	2.50

Table 104: 70 ton to 80 ton unit dimensions (in.)

		Unit configuration	Unit length (A)	Unit height (B)	Unit width (C)	Unit width lifting lug to lifting lug (D)	Evaporator or drain length (E)	Evaporator drain height (F)
Standard economizer - no ERW	No heat - 55 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	389.04	80.00	113.36	117.00	122.80	2.50
		SU with discharge plenum (B,L,R,T)	406.04	80.00	113.36	117.00	139.80	2.50
		SU with small post-evap blank, discharge plenum (B,L,R,T)	451.04	80.00	113.36	117.06	139.80	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	504.04	80.00	113.36	117.06	192.80	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	493.04	80.00	113.36	117.06	139.80	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	546.04	80.00	113.36	117.06	192.80	2.50
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	470.04	80.00	113.36	117.00	139.80	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	523.04	80.00	113.36	117.06	192.80	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	512.04	80.00	113.36	117.06	139.80	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	565.04	80.00	113.36	117.06	192.80	2.50
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	415.04	80.00	113.36	117.00	122.80	2.50
		SU, discharge through heat section (B,L)	432.04	80.00	113.36	117.00	139.80	2.50
		SU with pre-evap blank, discharge through heat section (B,L)	485.04	80.00	113.36	117.06	192.80	2.50
		SU with heat, discharge plenum (B,L,R,T)	478.04	80.00	113.36	117.00	139.80	2.50
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	531.04	80.00	113.36	117.06	192.80	2.50
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	523.04	80.00	113.36	117.00	139.80	2.50
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	576.04	80.00	113.36	117.06	192.80	2.50
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	565.04	80.00	113.36	117.06	139.80	2.50
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	618.04	80.00	113.36	117.06	192.80	2.50
Standard economizer - ERW	No heat - 55 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	551.04	80.00	113.36	117.06	239.80	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	604.04	80.00	113.36	117.06	292.80	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	593.04	80.00	113.36	117.06	239.80	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	646.04	80.00	113.36	117.06	292.80	2.50
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	570.04	80.00	113.36	117.06	239.80	2.50
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	623.04	80.00	113.36	117.06	292.80	2.50
		SU with large post-evap blank, discharge plenum (B,L,R,T)	612.04	80.00	113.36	117.06	239.80	2.50
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	665.04	80.00	113.36	117.06	292.80	2.50
	Heat	SU, discharge through heat section (B,L)	532.04	80.00	113.36	117.06	239.80	2.50
		SU with pre-evap blank, discharge through heat section (B,L)	585.04	80.00	113.36	117.06	292.80	2.50
		SU with heat, discharge plenum (B,L,R,T)	578.04	80.00	113.36	117.06	239.80	2.50
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	631.04	80.00	113.36	117.06	292.80	2.50
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	623.04	80.00	113.36	117.06	239.80	2.50
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	676.04	80.00	113.36	117.06	292.80	2.50
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	665.04	80.00	113.36	117.06	239.80	2.50
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	718.04	80.00	113.36	117.06	292.80	2.50

Table 105: 70 ton to 80 ton unit dimensions (in.) with bottom and top discharge

		Unit configuration	Bottom discharge				Top discharge			
			G	L4	I	W3	G	L4	I	W3
Basic economizer - no ERW	No heat - 55 in. supply fanNo heat - 74 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	195.69	40.00	19.52	74.00	195.69	40.00	19.52	74.00
		SU with discharge plenum (B,L,R,T)	212.69	40.00	19.52	74.00	212.69	40.00	19.52	74.00
		SU with small post-evap blank, discharge plenum (B,L,R,T)	257.69	40.00	19.52	74.00	257.69	40.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	310.69	40.00	19.52	74.00	310.69	40.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	299.69	40.00	19.52	74.00	299.69	40.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	352.69	40.00	19.52	74.00	352.69	40.00	19.52	74.00
	No heat - 55 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	276.69	40.00	19.52	74.00	276.69	40.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	329.69	40.00	19.52	74.00	329.69	40.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	318.69	40.00	19.52	74.00	318.69	40.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	371.69	40.00	19.52	74.00	371.69	40.00	19.52	74.00
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	219.35	40.00	26.50	60.00	—	—	—	—
		SU, discharge through heat section (B,L)	236.35	40.00	26.50	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	289.35	40.00	26.50	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	284.69	40.00	19.52	74.00	284.69	40.00	19.52	74.00
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	337.69	40.00	19.52	74.00	337.69	40.00	19.52	74.00
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	329.69	40.00	19.52	74.00	329.69	40.00	19.52	74.00
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	382.69	40.00	19.52	74.00	382.69	40.00	19.52	74.00
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	371.69	40.00	19.52	74.00	371.69	40.00	19.52	74.00
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	424.69	40.00	19.52	74.00	424.69	40.00	19.52	74.00
Standard economizer - no ERW	No heat - 55 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	219.69	40.00	19.52	74.00	219.69	40.00	19.52	74.00
		SU with discharge plenum (B,L,R,T)	236.69	40.00	19.52	74.00	236.69	40.00	19.52	74.00
		SU with small post-evap blank, discharge plenum (B,L,R,T)	281.69	40.00	19.52	74.00	281.69	40.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	334.69	40.00	19.52	74.00	334.69	40.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	323.69	40.00	19.52	74.00	323.69	40.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	376.69	40.00	19.52	74.00	376.69	40.00	19.52	74.00
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	300.69	40.00	19.52	74.00	300.69	40.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	353.69	40.00	19.52	74.00	353.69	40.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	342.69	40.00	19.52	74.00	342.69	40.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	395.69	40.00	19.52	74.00	395.69	40.00	19.52	74.00
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	243.35	40.00	26.50	60.00	—	—	—	—
		SU, discharge through heat section (B,L)	260.35	40.00	26.50	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	313.35	40.00	26.50	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	308.69	40.00	19.52	74.00	308.69	40.00	19.52	74.00
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	361.69	40.00	19.52	74.00	361.69	40.00	19.52	74.00
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	353.69	40.00	19.52	74.00	353.69	40.00	19.52	74.00
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	406.69	40.00	19.52	74.00	406.69	40.00	19.52	74.00
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	395.69	40.00	19.52	74.00	395.69	40.00	19.52	74.00
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	448.69	40.00	19.52	74.00	448.69	40.00	19.52	74.00

Table 105: 70 ton to 80 ton unit dimensions (in.) with bottom and top discharge

		Unit configuration	Bottom discharge				Top discharge			
			G	L4	I	W3	G	L4	I	W3
Standard economizer - ERW	No heat - 55 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	381.69	40.00	19.52	74.00	281.69	40.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	434.69	40.00	19.52	74.00	334.69	40.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	423.69	40.00	19.52	74.00	323.69	40.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	476.69	40.00	19.52	74.00	376.69	40.00	19.52	74.00
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	400.69	40.00	19.52	74.00	400.69	40.00	19.52	74.00
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	453.69	40.00	19.52	74.00	453.69	40.00	19.52	74.00
		SU with large post-evap blank, discharge plenum (B,L,R,T)	442.69	40.00	19.52	74.00	442.69	40.00	19.52	74.00
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	495.69	40.00	19.52	74.00	495.69	40.00	19.52	74.00
	Heat	SU, discharge through heat section (B,L)	360.35	40.00	26.50	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	413.35	40.00	26.50	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	408.35	40.00	26.50	60.00	408.35	40.00	26.50	60.00
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	461.69	40.00	19.52	74.00	461.69	40.00	19.52	74.00
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	453.69	40.00	19.52	74.00	453.69	40.00	19.52	74.00
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	506.69	40.00	19.52	74.00	506.69	40.00	19.52	74.00
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	495.69	40.00	19.52	74.00	495.69	40.00	19.52	74.00
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	548.69	40.00	19.52	74.00	548.69	40.00	19.52	74.00

Table 106: 70 ton to 80 ton unit dimensions (in.) with left and right discharge

		Unit configuration	Left discharge				Right discharge			
			J	L2	K	H3	J	L2	K	H3
Basic economizer - no ERW	No heat - 55 in. supply fanNo heat - 74 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	196.56	38.21	4.65	72.04	196.56	38.21	4.65	72.04
		SU with discharge plenum (B,L,R,T)	213.56	38.21	4.65	72.04	213.56	38.21	4.65	72.04
		SU with small post-evap blank, discharge plenum (B,L,R,T)	258.56	38.21	4.65	72.04	258.56	38.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	311.56	38.21	4.65	72.04	311.56	38.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	300.56	38.21	4.65	72.04	300.56	38.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	353.56	38.21	4.65	72.04	353.56	38.21	4.65	72.04
	No heat - 55 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	277.56	38.21	4.65	72.04	277.56	38.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	330.56	38.21	4.65	72.04	330.56	38.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	319.56	38.21	4.65	72.04	319.56	38.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	372.56	38.21	4.65	72.04	372.56	38.21	4.65	72.04
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	219.25	40.00	9.19	60.00	—	—	—	—
		SU, discharge through heat section (B,L)	236.25	40.00	9.19	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	289.25	40.00	9.19	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	285.56	38.21	4.65	72.04	285.56	38.21	4.65	72.04
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	338.56	38.21	4.65	72.04	338.56	38.21	4.65	72.04
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	330.56	38.21	4.65	72.04	330.56	38.21	4.65	72.04
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	383.56	38.21	4.65	72.04	383.56	38.21	4.65	72.04
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	372.56	38.21	4.65	72.04	372.56	38.21	4.65	72.04
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	425.56	38.21	4.65	72.04	425.56	38.21	4.65	72.04

Table 106: 70 ton to 80 ton unit dimensions (in.) with left and right discharge

		Unit configuration	Left discharge				Right discharge			
			J	L2	K	H3	J	L2	K	H3
Standard economizer - no ERW	No heat - 55 in. supply fan	SU with 4 in. vertical filter, discharge plenum (B,L,R,T)	220.56	38.21	4.65	72.04	220.56	38.21	4.65	72.04
		SU with discharge plenum (B,L,R,T)	237.56	38.21	4.65	72.04	237.56	38.21	4.65	72.04
		SU with small post-evap blank, discharge plenum (B,L,R,T)	282.56	38.21	4.65	72.04	282.56	38.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	335.56	38.21	4.65	72.04	335.56	38.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	324.56	38.21	4.65	72.04	324.56	38.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	377.56	38.21	4.65	72.04	377.56	38.21	4.65	72.04
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	301.56	38.21	4.65	72.04	301.56	38.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	354.56	38.21	4.65	72.04	354.56	38.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	343.56	38.21	4.65	72.04	343.56	38.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	396.56	38.21	4.65	72.04	396.56	38.21	4.65	72.04
	Heat	SU with 4 in. vertical filter, discharge through heat section (B,L)	243.25	40.00	9.19	60.00	—	—	—	—
		SU, discharge through heat section (B,L)	260.25	40.00	9.19	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	313.25	40.00	9.19	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	309.56	38.21	4.65	72.04	309.56	38.21	4.65	72.04
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	362.56	38.21	4.65	72.04	362.56	38.21	4.65	72.04
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	354.56	38.21	4.65	72.04	354.56	38.21	4.65	72.04
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	407.56	38.21	4.65	72.04	407.56	38.21	4.65	72.04
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	396.56	38.21	4.65	72.04	396.56	38.21	4.65	72.04
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	449.56	38.21	4.65	72.04	449.56	38.21	4.65	72.04
Standard economizer - ERW	No heat - 55 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	282.56	38.21	4.65	72.04	282.56	38.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	335.56	38.21	4.65	72.04	335.56	38.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	324.56	38.21	4.65	72.04	324.56	38.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	377.56	38.21	4.65	72.04	377.56	38.21	4.65	72.04
	No heat - 74 in. supply fan	SU with small post-evap blank, discharge plenum (B,L,R,T)	401.56	38.21	4.65	72.04	401.56	38.21	4.65	72.04
		SU with pre-evap blank, small post-evap blank, discharge plenum (B,L,R,T)	454.56	38.21	4.65	72.04	454.56	38.21	4.65	72.04
		SU with large post-evap blank, discharge plenum (B,L,R,T)	443.56	38.21	4.65	72.04	443.56	38.21	4.65	72.04
		SU with pre-evap blank, large post-evap blank, discharge plenum (B,L,R,T)	496.56	38.21	4.65	72.04	496.56	38.21	4.65	72.04
	Heat	SU, discharge through heat section (B,L)	360.25	40.00	9.19	60.00	—	—	—	—
		SU with pre-evap blank, discharge through heat section (B,L)	413.25	40.00	9.19	60.00	—	—	—	—
		SU with heat, discharge plenum (B,L,R,T)	409.56	40.00	9.19	60.00	409.56	40.00	9.19	60.00
		SU with pre-evap blank, heat, discharge plenum (B,L,R,T)	462.56	38.21	4.65	72.04	462.56	38.21	4.65	72.04
		SU with heat, small post-evap blank, discharge plenum (B,L,R,T)	454.56	38.21	4.65	72.04	454.56	38.21	4.65	72.04
		SU with pre-evap blank, heat, small post-evap blank, discharge plenum (B,L,R,T)	407.56	38.21	4.65	72.04	407.56	38.21	4.65	72.04
		SU with heat, large post-evap blank, discharge plenum (B,L,R,T)	496.56	38.21	4.65	72.04	496.56	38.21	4.65	72.04
		SU with pre-evap blank, heat, large post-evap blank, discharge plenum (B,L,R,T)	449.56	38.21	4.65	72.04	449.56	38.21	4.65	72.04

Note:

1. A SU includes length for economizer, draw-through filter, cooling coil, supply fan, and condenser. A SU includes a draw-through (pre-evaporator) filter with 2 in. angle filter or 2 in. + 12 in. (or 18 in.) filter. Units with a vertical 4 in. draw-through filter in place of the standard 2 in. angle or 2 in. + 12 in. (or 18 in.) filter are listed as *SU with Vertical Filter 4 in*.
2. Basic economizer unit does not include an exhaust fan or return fan option.
3. The heat section includes either gas, hot water, steam, or electric.
4. SU = Standard Unit, DP = Discharge Plenum, B = Bottom Discharge, L = Left Discharge, R = Right Discharge, T = Top Discharge.
5. Dimensions are ± 1 in.
6. For units that are shipped split, please contact your local sales office or use the Selection Navigator to get the individual air handler and condenser unit lengths.

Table 107: Blank sections

Post-evaporator blank options		Pre-evaporator blank options
Small blank	Large blank	
Empty	Empty	Empty
Humidifier	Humidifier	Air blender
Sound attenuator	Sound attenuator	—
Final filter	Final filter	—
—	Sound attenuator and final filter	—
—	Sound attenuator and humidifier	—

Example to determine unit length

The following options are critical when determining unit length for any selected unit:

Table 108: Example model number

GV	F	A	C	2	A	1	G	A	1	A	6	0	D	C	2	2	1	A	A	0	A	0	G	2	B	0	D	0	A	A	A	A	0	1	3	0	1	A	0	0	0	W	E	0	0	0	1
1,2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49

Note: The first row displays the example model number. The second row displays the corresponding digit.

Table 109: Model nomenclature description

Digit	Option	Description
3	F	Capacity 60 ton
5	C	Heat source: staged gas stainless steel
11	1	Discharge location: bottom discharge from discharge plenum
12	A	Supply configuration (small or large post evaporator blank sections): none
27	B	Draw-through filter options: angled filter rack, 2 in. throwaway filters
30	0	Energy recovery options: none
41	0	Pre-evap options (pre-evap blank section): none

Using the model number given as an example above, determine the unit length from Table 96 through Table 106.

Unit length = [tonnage of the unit (60 ton) + economizer + draw-through filter (2 in.) + cooling coil + supply fan + condenser]* + no ERW (refer to the no ERW section of the dimensions tables) + no pre-evap blank.

Note:

A standard unit includes length for economizer, draw-through filter, cooling coil, supply fan, and condenser. A standard unit includes a draw-through (pre-evaporator) filter with 2 in. angle filter or 2 in. + 12 in. (or 18 in.) filter. Units with a vertical 4 in. draw-through filter in place of the standard 2 in. angle or 2 in. + 12 in. (or 18 in.) filter are listed as SU with vertical filter 4 in.

Based on the unit options selected, the configuration is a standard unit with heat, discharge plenum (B, L, R, T). For this example, the length of the unit is 387.54 in.

Roof curb dimensions

Figure 16: Roof curbs: left bottom return opening and bottom supply opening with ERW

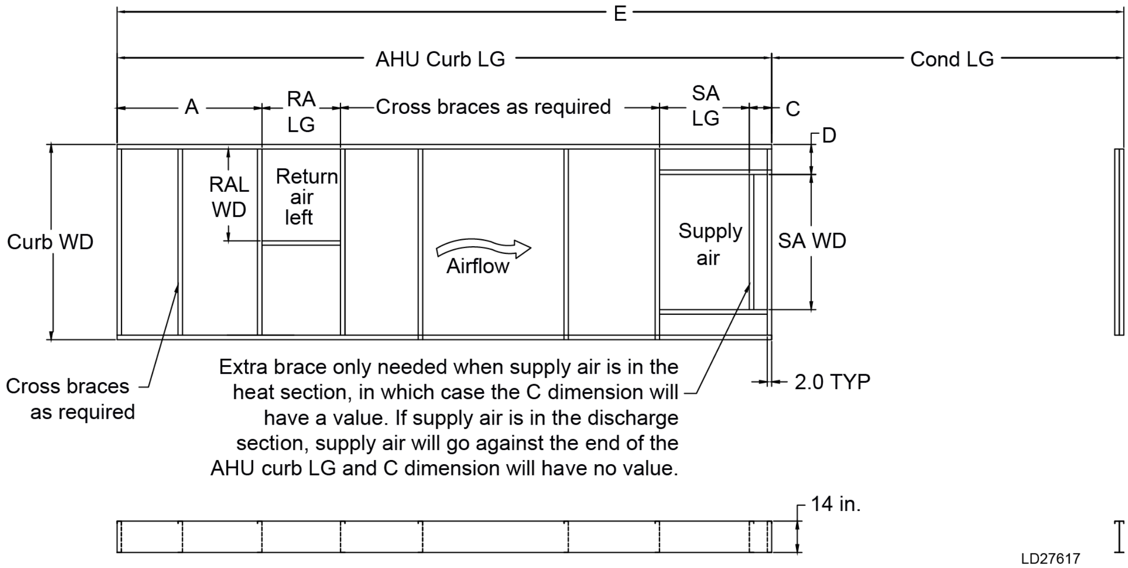


Table 110: 60 ton left bottom return opening with bottom supply opening with ERW (in.)

Unit configuration		AHU LG	Unit WD	AHU curb LG	Curb WD	Cond LG	Total LG (E)	A
No heat + 55 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with ext eco	421.4	113.0	416.8	108.49	97.2	514.0	109.2
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with ext eco	463.4	113.0	458.8	108.49	97.2	556.0	109.2
	SU with small post-evap blank, DP (B, L, R, T) with ext eco	368.4	113.0	363.8	108.49	97.2	461.0	109.2
	SU with large post-evap blank, DP (B, L, R, T) with ext eco	410.4	113.0	405.8	108.49	97.2	503.0	109.2
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with ext eco	440.4	113.0	435.8	108.49	97.2	533.0	109.2
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with ext eco	482.4	113.0	477.8	108.49	97.2	575.0	109.2
	SU with small post-evap blank, DP (B, L, R, T) with ext eco	387.4	113.0	382.8	108.49	97.2	480.0	109.2
	SU with large post-evap blank, DP (B, L, R, T) with ext eco	429.4	113.0	424.8	108.49	97.2	522.0	109.2

Table 110: 60 ton left bottom return opening with bottom supply opening with ERW (in.)

Unit configuration		AHU LG	Unit WD	AHU curb LG	Curb WD	Cond LG	Total LG (E)	A
Heat	SU with discharge through heat section (B, L) with ext eco	359.4	113.0	354.8	108.49	97.2	452.0	109.2
	SU with heat, DP (B, L, R, T) with ext eco	395.4	113.0	390.8	108.49	97.2	488.0	109.2
	SU with pre-evap blank, discharge through heat section (B, L) with ext eco	412.4	113.0	407.8	108.49	97.2	505.0	109.2
	SU with pre-evap blank, heat, DP (B, L, R, T) with ext eco	448.4	113.0	443.8	108.49	97.2	541.0	109.2
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with ext eco	493.4	113.0	488.8	108.49	97.2	586.0	109.2
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with ext eco	535.4	113.0	530.8	108.49	97.2	628.0	109.2
	SU with heat, small post-evap blank, DP (B, L, R, T) with ext eco	440.4	113.0	435.8	108.49	97.2	533.0	109.2
	SU with heat, large post-evap blank, DP (B, L, R, T) with ext eco	482.4	113.0	477.8	108.49	97.2	575.0	109.2

Table 111: 60 ton left bottom return opening with bottom supply opening with ERW (in.)

Unit configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
No heat + 55 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
Heat	SU with discharge through heat section (B, L) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with heat, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with pre-evap blank, discharge through heat section (B, L) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with pre-evap blank, heat, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with heat, small post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0
	SU with heat, large post-evap blank, DP (B, L, R, T) with ext eco	48.6	54.4	53.9	—	17.2	30.0	74.0

Table 112: 70 ton to 80 ton left bottom return opening with bottom supply opening with ERW (in.)

Unit configuration		AHU LG	Unit WD	AHU Curb LG	Curb WD	Cond LG	Total LG (E)	A
No heat + 55 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with ext eco	478.9	113.0	474.3	108.49	125.2	599.5	109.2
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with ext eco	520.9	113.0	516.3	108.49	125.2	641.5	109.2
	SU with small post-evap blank, DP (B, L, R, T) with ext eco	425.9	113.0	421.3	108.49	125.2	546.5	109.2
	SU with large post-evap blank, DP (B, L, R, T) with ext eco	467.9	113.0	463.3	108.49	125.2	588.5	109.2
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with ext eco	497.9	113.0	493.3	108.49	125.2	618.5	109.2
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with ext eco	539.9	113.0	535.3	108.49	125.2	660.5	109.2
	SU with small post-evap blank, DP (B, L, R, T) with ext eco	444.9	113.0	440.3	108.49	125.2	565.5	109.2
	SU with large post-evap blank, DP (B, L, R, T) with ext eco	486.9	113.0	482.3	108.49	125.2	607.5	109.2
Heat	SU with discharge through heat section (B, L) with ext eco	406.9	113.0	402.3	108.49	125.2	527.5	109.2
	SU with Heat, DP (B, L, R, T) with ext eco	452.9	113.0	448.3	108.49	125.2	573.5	109.2
	SU with pre-evap blank, discharge through heat section (B, L) with ext eco	459.9	113.0	455.3	108.49	125.2	580.5	109.2
	SU with pre-evap blank, heat, DP (B, L, R, T) with ext eco	505.9	113.0	501.3	108.49	125.2	626.5	109.2
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with ext eco	550.9	113.0	546.3	108.49	125.2	671.5	109.2
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with ext eco	592.9	113.0	588.3	108.49	125.2	713.5	109.2
	SU with heat, small post-evap blank, DP (B, L, R, T) with ext eco	497.9	113.0	493.3	108.49	125.2	618.5	109.2
	SU with heat, large post-evap blank, DP (B, L, R, T) with ext eco	539.9	113.0	535.3	108.49	125.2	660.5	109.2

Table 113: 70 ton to 80 ton left bottom return opening with bottom supply opening with ERW (in.)

Unit Configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
No heat + 55 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0

Table 113: 70 ton to 80 ton left bottom return opening with bottom supply opening with ERW (in.)

Unit Configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
Heat	SU with discharge through heat section (B, L) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with heat, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with pre-evap blank, discharge through heat section (B, L) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with heat, small post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0
	SU with heat, large post-evap blank, DP (B, L, R, T) with ext eco	62.6	54.4	72.9	—	17.2	40.0	74.0

Note:

1. AHU = Air Handling Unit, LG = Length, WD = Width, Cond = Condenser, RA = Return Air, RAL = RA Left, SA = Supply Air, SU = Standard Unit
2. DP = Discharge Plenum, B = Bottom, L = Left, R = Right, T = Top

Figure 17: 60 ton to 80 ton roof curbs left bottom return opening and bottom supply opening without ERW

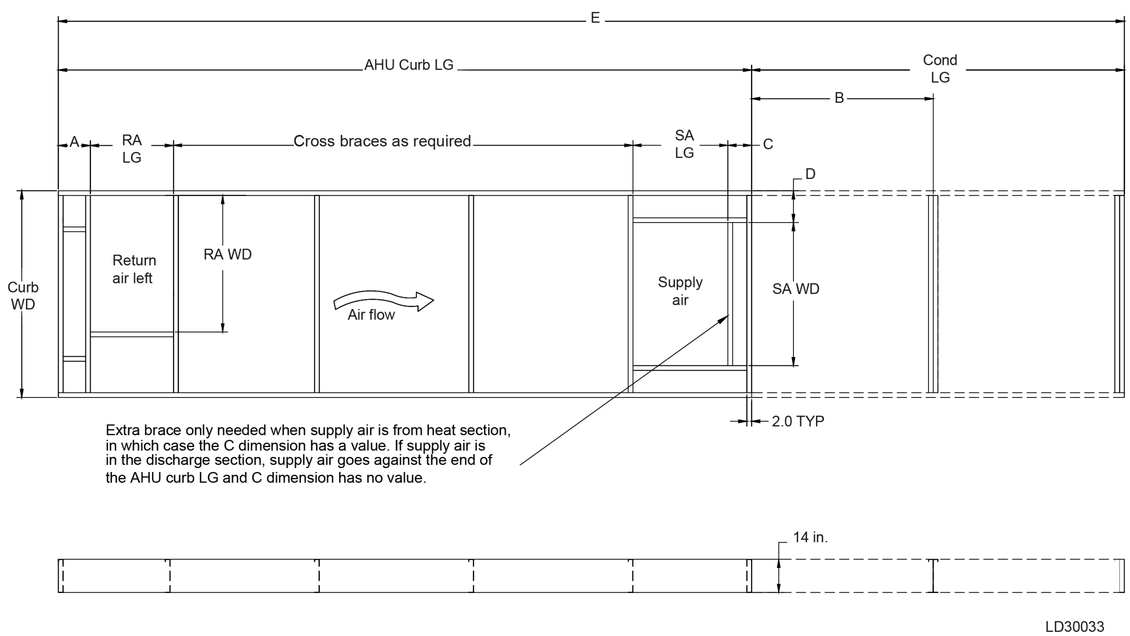


Table 114: 60 ton left bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		AHU LG	Unit WD	AHU Curb LG	Curb WD	Cond LG	Total LG (E)	A
No heat + 55 in. SU	SU with vertical filter 4 in., DP (B, L, R, T) with basic eco	201.4	113.0	196.8	108.49	97.2	294.0	3.8
	SU with vertical filter 4 in., DP (B, L, R, T) with standard eco	225.4	113.0	220.8	108.49	97.2	318.0	11.4
	SU with DP (B, L, R, T) with basic eco	218.4	113.0	213.8	108.49	97.2	311.0	3.8
	SU with DP (B, L, R, T) with standard eco	242.4	113.0	237.8	108.49	97.2	335.0	11.4
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	316.4	113.0	311.8	108.49	97.2	409	3.8
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with std eco	340.4	113.0	335.8	108.49	97.2	433	11.4
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	358.4	113.0	353.8	108.49	97.2	451	3.8
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with std eco	382.4	113.0	377.8	108.49	97.2	475	11.4
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	263.4	113.0	258.8	108.49	97.2	356	3.8
	SU with small post-evap blank, DP (B, L, R, T) with std eco	287.4	113.0	282.8	108.49	97.2	380	11.4
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	305.4	113.0	300.8	108.49	97.2	398	3.8
	SU with large post-evap blank, DP (B, L, R, T) with std eco	329.4	113.0	324.8	108.49	97.2	422	11.4
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	335.4	113.0	330.8	108.49	97.2	428	3.8
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with std eco	359.4	113.0	354.8	108.49	97.2	452	11.4
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	377.4	113.0	372.8	108.49	97.2	470	3.8
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with std eco	401.4	113.0	396.8	108.49	97.2	494	11.4
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	282.4	113.0	277.8	108.49	97.2	375.0	3.8
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	306.4	113.0	301.8	108.49	97.2	399.0	11.4
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	324.4	113.0	319.8	108.49	97.2	417	3.8
	SU with large post-evap blank, DP (B, L, R, T) with std eco	348.4	113.0	343.8	108.49	97.2	441	11.4

Table 114: 60 ton left bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		AHU LG	Unit WD	AHU Curb LG	Curb WD	Cond LG	Total LG (E)	A
Heat	SU with vertical filter 4 in., with discharge through heat section (B, L) with basic eco	237.4	113.0	232.8	108.49	97.2	330.0	3.8
	SU with vertical filter 4 in., with discharge through heat section (B, L) with standard eco	261.4	113.0	256.8	108.49	97.2	354.0	11.4
	SU with discharge through heat section (B, L) with basic eco	254.4	113.0	249.8	108.49	97.2	347.0	3.8
	SU with discharge through heat section (B, L) with standard eco	278.4	113.0	273.8	108.49	97.2	371.0	11.4
	SU with heat, DP (B, L, R, T) with basic eco	290.4	113.0	285.8	108.49	97.2	383.0	3.8
	SU with heat, DP (B, L, R, T) with standard eco	314.4	113.0	309.8	108.49	97.2	407.0	11.4
	SU with pre-evap blank, discharge through heat section (B, L) with basic eco	307.4	113.0	302.8	108.49	97.2	400	3.8
	SU with pre-evap blank, discharge through heat section (B, L) with std eco	331.4	113.0	326.8	108.49	97.2	424	11.4
	SU with pre-evap blank, heat, DP (B, L, R, T) with basic eco	343.4	113.0	338.8	108.49	97.2	436	3.8
	SU with pre-evap blank, heat, DP (B, L, R, T) with std eco	367.4	113.0	362.8	108.49	97.2	460	11.4
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with basic eco	388.4	113.0	383.8	108.49	97.2	481	3.8
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with std eco	412.4	113.0	407.8	108.49	97.2	505	11.4
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with basic eco	430.4	113.0	425.8	108.49	97.2	523	3.8
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with std eco	454.4	113.0	449.8	108.49	97.2	547	11.4
	SU with heat, small post-evap blank, DP (B, L, R, T) with basic eco	335.4	113.0	330.8	108.49	97.2	428.0	3.8
	SU with heat, small post-evap blank, DP (B, L, R, T) with standard eco	359.4	113.0	354.8	108.49	97.2	452.0	11.4
	SU with heat, large post-evap blank, DP (B, L, R, T) with basic eco	377.4	113.0	372.8	108.49	97.2	470	3.8
	SU with heat, large post-evap blank, DP (B, L, R, T) with std eco	401.4	113.0	396.8	108.49	97.2	494	11.4

Table 115: 60 ton left bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
No heat + 55 in. SU	SU with vertical filter 4 in., DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with vertical filter 4 in., DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with std eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with std eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with std eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with std eco	48.6	77.9	56.9	—	17.2	30.0	74.0
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with std eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with std eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with std eco	48.6	77.9	56.9	—	17.2	30.0	74.0

Table 115: 60 ton left bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
Heat	SU with vertical filter 4 in., with discharge through heat section (B, L) with basic eco	48.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with vertical filter 4 in., with discharge through heat section (B, L) with standard eco	48.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with discharge through heat section (B, L) with basic eco	48.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with discharge through heat section (B, L) with standard eco	48.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with heat, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with heat, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with pre-evap blank, discharge through heat section (B, L) with basic eco	48.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with pre-evap blank, discharge through heat section (B, L) with std eco	48.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with pre-evap blank, heat, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with pre-evap blank, heat, DP (B, L, R, T) with std eco	48.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with std eco	48.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with std eco	48.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with heat, small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with heat, small post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with heat, large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with heat, large post-evap blank, DP (B, L, R, T) with std eco	48.6	77.9	56.9	—	17.2	30.0	74.0

Table 116: 70 ton to 80 ton left bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		AHU LG	Unit WD	AHU Curb LG	Curb WD	Cond LG	Total LG (E)	A
No heat + 55 in. SU	SU with vertical filter 4 in., DP (B, L, R, T) with basic eco	239.9	113.0	235.3	108.49	125.2	360.5	3.8
	SU with vertical filter 4 in., DP (B, L, R, T) with standard eco	263.9	113.0	259.3	108.49	125.2	384.5	11.4
	SU with DP (B, L, R, T) with basic eco	256.9	113.0	252.3	108.49	125.2	377.5	3.8
	SU with DP (B, L, R, T) with standard eco	280.9	113.0	276.3	108.49	125.2	401.5	11.4
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	354.9	113.0	350.3	108.49	125.2	475.5	3.8
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with std eco	378.9	113.0	374.3	108.49	125.2	499.5	11.4
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	396.9	113.0	392.3	108.49	125.2	517.5	3.8
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with std eco	420.9	113.0	416.3	108.49	125.2	541.5	11.4
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	301.9	113.0	297.3	108.49	125.2	422.5	3.8
	SU with small post-evap blank, DP (B, L, R, T) with std eco	325.9	113.0	321.3	108.49	125.2	446.5	11.4
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	343.9	113.0	339.3	108.49	125.2	464.5	3.8
	SU with large post-evap blank, DP (B, L, R, T) with std eco	367.9	113.0	363.3	108.49	125.2	488.5	11.4
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	373.9	113.0	369.3	108.49	125.2	494.5	3.8
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with std eco	397.9	113.0	393.3	108.49	125.2	518.5	11.4
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	415.9	113.0	411.3	108.49	125.2	536.5	3.8
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with std eco	439.9	113.0	435.3	108.49	125.2	560.5	11.4
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	320.9	113.0	316.3	108.49	125.2	441.5	3.8
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	344.9	113.0	340.3	108.49	125.2	465.5	11.4
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	362.9	113.0	358.3	108.49	125.2	483.5	3.8
	SU with large post-evap blank, DP (B, L, R, T) with std eco	386.9	113.0	382.3	108.49	125.2	507.5	11.4

Table 116: 70 ton to 80 ton left bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		AHU LG	Unit WD	AHU Curb LG	Curb WD	Cond LG	Total LG (E)	A
Heat	SU with vertical filter 4 in., with discharge through heat section (B, L) with basic eco	265.9	113.0	261.3	108.49	125.2	386.5	3.8
	SU with vertical filter 4 in., with discharge through heat section (B, L) with standard eco	289.9	113.0	285.3	108.49	125.2	410.5	11.4
	SU with discharge through heat section (B, L) with basic eco	282.9	113.0	278.3	108.49	125.2	403.5	3.8
	SU with discharge through heat section (B, L) with standard eco	306.9	113.0	302.3	108.49	125.2	427.5	11.4
	SU with heat, DP (B, L, R, T) with basic eco	328.9	113.0	324.3	108.49	125.2	449.5	3.8
	SU with heat, DP (B, L, R, T) with standard eco	352.9	113.0	348.3	108.49	125.2	473.5	11.4
	SU with pre-evap blank, discharge through heat section (B, L) with basic eco	335.9	113.0	331.3	108.49	125.2	456.5	3.8
	SU with pre-evap blank, discharge through heat section (B, L) with std eco	359.9	113.0	355.3	108.49	125.2	480.5	11.4
	SU with pre-evap blank, heat, DP (B, L, R, T) with basic eco	381.9	113.0	377.3	108.49	125.2	502.5	3.8
	SU with pre-evap blank, heat, DP (B, L, R, T) with std eco	405.9	113.0	401.3	108.49	125.2	526.5	11.4
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with basic eco	426.9	113.0	422.3	108.49	125.2	547.5	3.8
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with std eco	450.9	113.0	446.3	108.49	125.2	571.5	11.4
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with basic eco	468.9	113.0	464.3	108.49	125.2	589.5	3.8
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with std eco	492.9	113.0	488.3	108.49	125.2	613.5	11.4
	SU with heat, small post-evap blank, DP (B, L, R, T) with basic eco	373.9	113.0	369.3	108.49	125.2	494.5	3.8
	SU with heat, small post-evap blank, DP (B, L, R, T) with standard eco	397.9	113.0	393.3	108.49	125.2	518.5	11.4
	SU with heat, large post-evap blank, DP (B, L, R, T) with basic eco	415.9	113.0	411.3	108.49	125.2	536.5	3.8
	SU with heat, large post-evap blank, DP (B, L, R, T) with std eco	439.9	113.0	435.3	108.49	125.2	560.5	11.4

Table 117: 70 ton to 80 ton left bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
No heat + 55 in. SU	SU with vertical filter 4 in., DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with vertical filter 4 in., DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0

Table 117: 70 ton to 80 ton left bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
Heat	SU with vertical filter 4 in., with discharge through heat section (B, L) with basic eco	62.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with vertical filter 4 in., with discharge through heat section (B, L) with standard eco	62.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with discharge through heat section (B, L) with basic eco	62.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with discharge through heat section (B, L) with standard eco	62.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with heat, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with heat, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, discharge through heat section (B, L) with basic eco	62.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with pre-evap blank, discharge through heat section (B, L) with std eco	62.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with pre-evap blank, heat, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with heat, small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with heat, small post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with heat, large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with heat, large post-evap blank, DP (B, L, R, T) with std eco	62.6	77.9	56.9	—	17.2	40.0	74.0

Note:

For the 60 ton to 80 ton units, the basic economizer is available with the barometric damper. The standard economizer is available with the exhaust and return fan.

Note:

1. AHU = Air Handling Unit, LG = Length, WD = Width, Cond = Condenser, RA = Return Air, RAR = RA Right, RAL = RA Left, SA = Supply Air
2. SU = Standard Unit, DP = Discharge Plenum, B = Bottom, L = Left

Figure 18: 60 ton to 80 ton roof curbs right bottom return opening and bottom supply opening without ERW

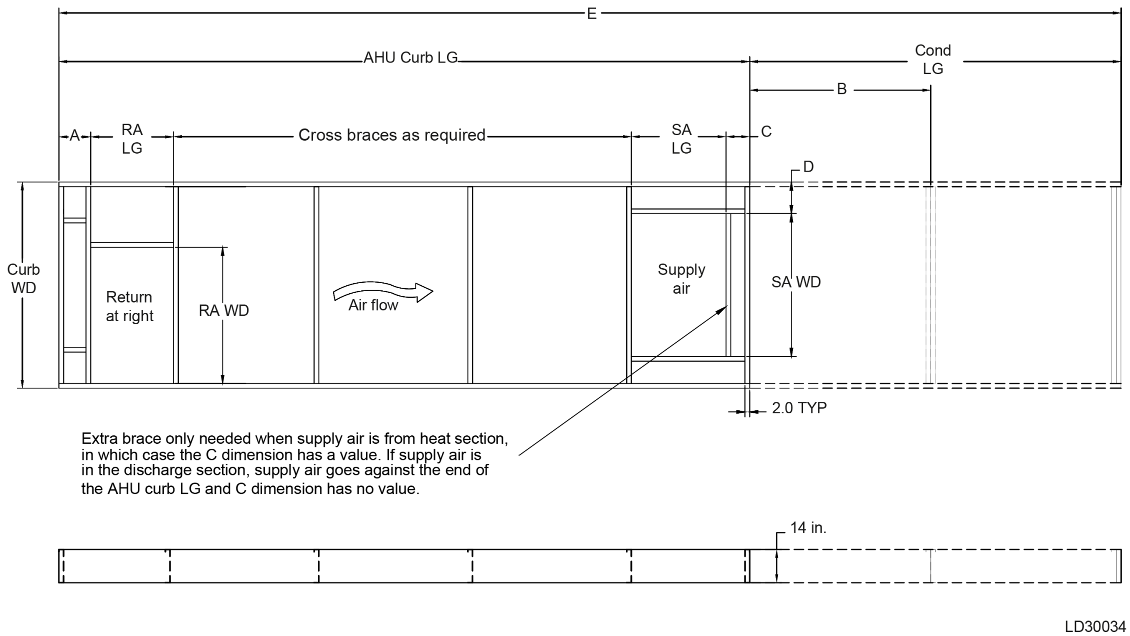


Table 118: 60 ton right bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		AHU LG	Unit WD	AHU Curb LG	Curb WD	Cond LG	Total LG (E)	A
No heat + 55 in. SU	SU with vertical filter 4 in., DP (B, L, R, T) with basic eco	201.4	113.0	196.8	108.49	97.2	294.0	3.8
	SU with vertical filter 4 in., DP (B, L, R, T) with standard eco	225.4	113.0	220.8	108.49	97.2	318.0	11.4
	SU with DP (B, L, R, T) with basic eco	218.4	113.0	213.8	108.49	97.2	311.0	3.8
	SU with DP (B, L, R, T) with standard eco	242.4	113.0	237.8	108.49	97.2	335.0	11.4
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	316.4	113.0	311.8	108.49	97.2	409	3.8
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with standard eco	340.4	113.0	335.8	108.49	97.2	433	11.4
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	358.4	113.0	353.8	108.49	97.2	451	3.8
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with standard eco	382.4	113.0	377.8	108.49	97.2	475	11.4
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	263.4	113.0	258.8	108.49	97.2	356	3.8
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	287.4	113.0	282.8	108.49	97.2	380	11.4
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	305.4	113.0	300.8	108.49	97.2	398	3.8
	SU with large post-evap blank, DP (B, L, R, T) with standard eco	329.4	113.0	324.8	108.49	97.2	422	11.4

Table 118: 60 ton right bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		AHU LG	Unit WD	AHU Curb LG	Curb WD	Cond LG	Total LG (E)	A
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	335.4	113.0	330.8	108.49	97.2	428	3.8
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with standard eco	359.4	113.0	354.8	108.49	97.2	452	11.4
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	377.4	113.0	372.8	108.49	97.2	470	3.8
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with standard eco	401.4	113.0	396.8	108.49	97.2	494	11.4
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	282.4	113.0	277.8	108.49	97.2	375.0	3.8
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	306.4	113.0	301.8	108.49	97.2	399.0	11.4
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	324.4	113.0	319.8	108.49	97.2	417	3.8
	SU with large post-evap blank, DP (B, L, R, T) with standard eco	348.4	113.0	343.8	108.49	97.2	441	11.4

Table 118: 60 ton right bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		AHU LG	Unit WD	AHU Curb LG	Curb WD	Cond LG	Total LG (E)	A
Heat	SU with vertical filter 4 in., with discharge through heat section (B, L) with basic eco	237.4	113.0	232.8	108.49	97.2	330.0	3.8
	SU with vertical filter 4 in., with discharge through heat section (B, L) with standard eco	261.4	113.0	256.8	108.49	97.2	354.0	11.4
	SU with discharge through heat section (B, L) with basic eco	254.4	113.0	249.8	108.49	97.2	347.0	3.8
	SU with discharge through heat section (B, L) with standard eco	278.4	113.0	273.8	108.49	97.2	371.0	11.4
	SU with heat, DP (B, L, R, T) with basic eco	290.4	113.0	285.8	108.49	97.2	383.0	3.8
	SU with heat, DP (B, L, R, T) with standard eco	314.4	113.0	309.8	108.49	97.2	407.0	11.4
	SU with pre-evap blank, discharge through heat section (B, L) with basic eco	307.4	113.0	302.8	108.49	97.2	400	3.8
	SU with pre-evap blank, discharge through heat section (B, L) with standard eco	331.4	113.0	326.8	108.49	97.2	424	11.4
	SU with pre-evap blank, heat, DP (B, L, R, T) with basic eco	343.4	113.0	338.8	108.49	97.2	436	3.8
	SU with pre-evap blank, heat, DP (B, L, R, T) with standard eco	367.4	113.0	362.8	108.49	97.2	460	11.4
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with basic eco	388.4	113.0	383.8	108.49	97.2	481	3.8
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with standard eco	412.4	113.0	407.8	108.49	97.2	505	11.4
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with basic eco	430.4	113.0	425.8	108.49	97.2	523	3.8
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with standard eco	454.4	113.0	449.8	108.49	97.2	547	11.4
	SU with heat, small post-evap blank, DP (B, L, R, T) with basic eco	335.4	113.0	330.8	108.49	97.2	428.0	3.8
	SU with heat, small post-evap blank, DP (B, L, R, T) with standard eco	359.4	113.0	354.8	108.49	97.2	452.0	11.4
	SU with heat, large post-evap blank, DP (B, L, R, T) with basic eco	377.4	113.0	372.8	108.49	97.2	470	3.8
	SU with heat, large post-evap blank, DP (B, L, R, T) with standard eco	401.4	113.0	396.8	108.49	97.2	494	11.4

Table 119: 60 ton right bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
No heat + 55 in. SU	SU with vertical filter 4 in., DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with vertical filter 4 in., DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0

Table 119: 60 ton right bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
Heat	SU with vertical filter 4 in., with discharge through heat section (B, L) with basic eco	48.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with vertical filter 4 in., with discharge through heat section (B, L) with standard eco	48.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with discharge through heat section (B, L) with basic eco	48.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with discharge through heat section (B, L) with standard eco	48.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with heat, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	-	17.2	30.0	74.0
	SU with heat, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with pre-evap blank, discharge through heat section (B, L) with basic eco	48.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with pre-evap blank, discharge through heat section (B, L) with standard eco	48.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with pre-evap blank, heat, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, heat, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with heat, small post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with heat, small post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0
	SU with heat, large post-evap blank, DP (B, L, R, T) with basic eco	48.6	65.0	54.3	—	17.2	30.0	74.0
	SU with heat, large post-evap blank, DP (B, L, R, T) with standard eco	48.6	77.9	56.9	—	17.2	30.0	74.0

Table 120: 70 ton to 80 ton right bottom return opening with bottom supply opening without ERW (in.)

Unit Configuration		AHU LG	Unit WD	AHU Curb LG	Curb WD	Cond LG	Total LG (E)	A
No heat + 55 in. SU	SU with vertical filter 4 in., DP (B, L, R, T) with basic eco	239.9	113.0	235.3	108.49	125.2	360.5	3.8
	SU with vertical filter 4 in., DP (B, L, R, T) with standard eco	263.9	113.0	259.3	108.49	125.2	384.5	11.4
	SU with DP (B, L, R, T) with basic eco	256.9	113.0	252.3	108.49	125.2	377.5	3.8
	SU with DP (B, L, R, T) with standard eco	280.9	113.0	276.3	108.49	125.2	401.5	11.4
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	354.9	113.0	350.3	108.49	125.2	475.5	3.8
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with standard eco	378.9	113.0	374.3	108.49	125.2	499.5	11.4
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	396.9	113.0	392.3	108.49	125.2	517.5	3.8
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with standard eco	420.9	113.0	416.3	108.49	125.2	541.5	11.4
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	301.9	113.0	297.3	108.49	125.2	422.5	3.8
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	325.9	113.0	321.3	108.49	125.2	446.5	11.4
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	343.9	113.0	339.3	108.49	125.2	464.5	3.8
	SU with large post-evap blank, DP (B, L, R, T) with standard eco	367.9	113.0	363.3	108.49	125.2	488.5	11.4
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	373.9	113.0	369.3	108.49	125.2	494.5	3.8
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with standard eco	397.9	113.0	393.3	108.49	125.2	518.5	11.4
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	415.9	113.0	411.3	108.49	125.2	536.5	3.8
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with standard eco	439.9	113.0	435.3	108.49	125.2	560.5	11.4
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	320.9	113.0	316.3	108.49	125.2	441.5	3.8
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	344.9	113.0	340.3	108.49	125.2	465.5	11.4
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	362.9	113.0	358.3	108.49	125.2	483.5	3.8
	SU with large post-evap blank, DP (B, L, R, T) with standard eco	386.9	113.0	382.3	108.49	125.2	507.5	11.4

Table 120: 70 ton to 80 ton right bottom return opening with bottom supply opening without ERW (in.)

Unit Configuration		AHU LG	Unit WD	AHU Curb LG	Curb WD	Cond LG	Total LG (E)	A
Heat	SU with vertical filter 4 in., with discharge through heat section (B, L) with basic eco	265.9	113.0	261.3	108.49	125.2	386.5	3.8
	SU with vertical filter 4 in., with discharge through heat section (B, L) with standard eco	289.9	113.0	285.3	108.49	125.2	410.5	11.4
	SU with discharge through heat section (B, L) with basic eco	282.9	113.0	278.3	108.49	125.2	403.5	3.8
	SU with discharge through heat section (B, L) with standard eco	306.9	113.0	302.3	108.49	125.2	427.5	11.4
	SU with heat, DP (B, L, R, T) with basic eco	328.9	113.0	324.3	108.49	125.2	449.5	3.8
	SU with heat, DP (B, L, R, T) with standard eco	352.9	113.0	348.3	108.49	125.2	473.5	11.4
	SU with pre-evap blank, discharge through heat section (B, L) with basic eco	335.9	113.0	331.3	108.49	125.2	456.5	3.8
	SU with pre-evap blank, discharge through heat section (B, L) with standard eco	359.9	113.0	355.3	108.49	125.2	480.5	11.4
	SU with pre-evap blank, heat, DP (B, L, R, T) with basic eco	381.9	113.0	377.3	108.49	125.2	502.5	3.8
	SU with pre-evap blank, heat, DP (B, L, R, T) with standard eco	405.9	113.0	401.3	108.49	125.2	526.5	11.4
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with basic eco	426.9	113.0	422.3	108.49	125.2	547.5	3.8
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with standard eco	450.9	113.0	446.3	108.49	125.2	571.5	11.4
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with basic eco	468.9	113.0	464.3	108.49	125.2	589.5	3.8
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with standard eco	492.9	113.0	488.3	108.49	125.2	613.5	11.4
	SU with heat, small post-evap blank, DP (B, L, R, T) with basic eco	373.9	113.0	369.3	108.49	125.2	494.5	3.8
	SU with heat, small post-evap blank, DP (B, L, R, T) with standard eco	397.9	113.0	393.3	108.49	125.2	518.5	11.4
	SU with heat, large post-evap blank, DP (B, L, R, T) with basic eco	415.9	113.0	411.3	108.49	125.2	536.5	3.8
	SU with heat, large post-evap blank, DP (B, L, R, T) with standard eco	439.9	113.0	435.3	108.49	125.2	560.5	11.4

Table 121: 70 ton to 80 ton right bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
No heat + 55 in. SU	SU with vertical filter 4 in., DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—		17.2	40.0
	SU with vertical filter 4 in., DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
No heat + 74 in. SU	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, small post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, large post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with small post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with large post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0

Table 121: 70 ton to 80 ton right bottom return opening with bottom supply opening without ERW (in.)

Unit configuration		B	RA WD	RA LG	C	D	SA LG	SA WD
Heat	SU with vertical filter 4 in., with discharge through heat section (B, L) with basic eco	62.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with vertical filter 4 in., with discharge through heat section (B, L) with standard eco	62.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with discharge through heat section (B, L) with basic eco	62.6	65.0	54.3	4.2	24.3	40.0	60.0
	SU with discharge through heat section (B, L) with standard eco	62.6	77.9	56.9	4.2	24.3	40.0	60.0
	SU with heat, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with heat, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, discharge through heat section (B, L) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, discharge through heat section (B, L) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, small post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with pre-evap blank, heat, large post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with heat, small post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with heat, small post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0
	SU with heat, large post-evap blank, DP (B, L, R, T) with basic eco	62.6	65.0	54.3	—	17.2	40.0	74.0
	SU with heat, large post-evap blank, DP (B, L, R, T) with standard eco	62.6	77.9	56.9	—	17.2	40.0	74.0

Note:

For the 60 ton to 80 ton units, the basic economizer is available with the barometric damper. The standard economizer is available with the exhaust and return fan.

Note:

1. AHU = Air Handling Unit, LG = Length, WD = Width, Cond = Condenser, RA = Return Air, RAR = RA Right, RAL = RA Left, SA = Supply Air
2. SU = Standard Unit, DP = Discharge Plenum, B = Bottom, L = Left

Supply and return air duct configurations

Figure 19: Air duct configurations

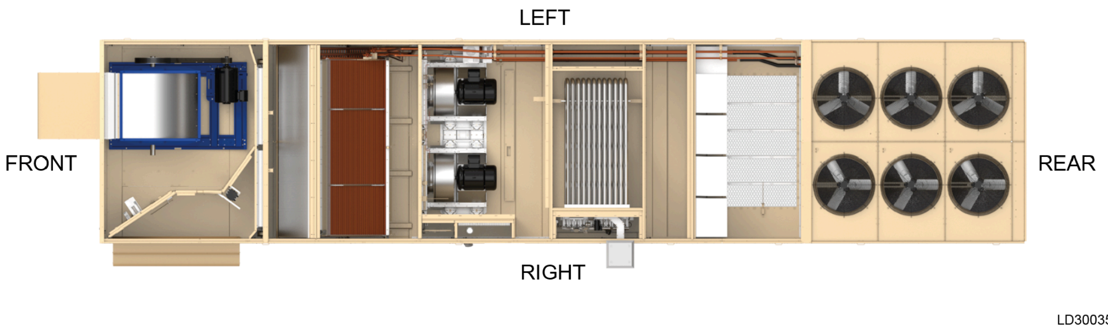


Table 122: Supply and return air duct configurations

Unit configuration	Supply air				Return air					Outside air		Exhaust air	
	Bottom	Left	Right	Top	Bottom	Left	Right	Top	Front	Left	Right	Side*	Front
Cooling only	Yes	Yes	Yes	Yes	—	Yes	—	—	—	—	Yes	—	Yes
	Yes	Yes	Yes	Yes	—	—	Yes	—	—	Yes	—	—	Yes
	Yes	Yes	Yes	Yes	Yes	—	—	Yes	—	Yes	Yes	Yes	Yes
	Yes	Yes	Yes	Yes	—	—	—	—	Yes	Yes	Yes	Yes	—
Cooling with discharge through heat section	Yes	Yes	—	—	—	Yes	—	—	—	—	Yes	—	Yes
	Yes	Yes	—	—	—	—	Yes	—	—	Yes	—	—	Yes
	Yes	Yes	—	—	Yes	—	—	Yes	—	Yes	Yes	Yes	Yes
	Yes	Yes	—	—	—	—	—	—	Yes	Yes	Yes	Yes	—
Cooling with heat and discharge plenum or cooling with downstream section and discharge plenum	Yes	Yes	Yes	Yes	—	Yes	—	—	—	—	Yes	—	Yes
	Yes	Yes	Yes	Yes	—	—	Yes	—	—	Yes	—	—	Yes
	Yes	Yes	Yes	Yes	Yes	—	—	Yes	—	Yes	Yes	Yes	Yes
	Yes	Yes	Yes	Yes	—	—	—	—	Yes	Yes	Yes	Yes	—
Cooling with ERW and discharge through heat section	Yes	Yes	—	—	Yes	—	—	—	—	—	Yes	Yes	Yes
Cooling with heat with ERW with downstream section and discharge plenum	Yes	Yes	Yes	Yes	Yes	—	—	—	—	—	Yes	Yes	Yes

Note:

*Exhaust air side is opposite of outside air side.

Top return with return fan or ERW is not available.

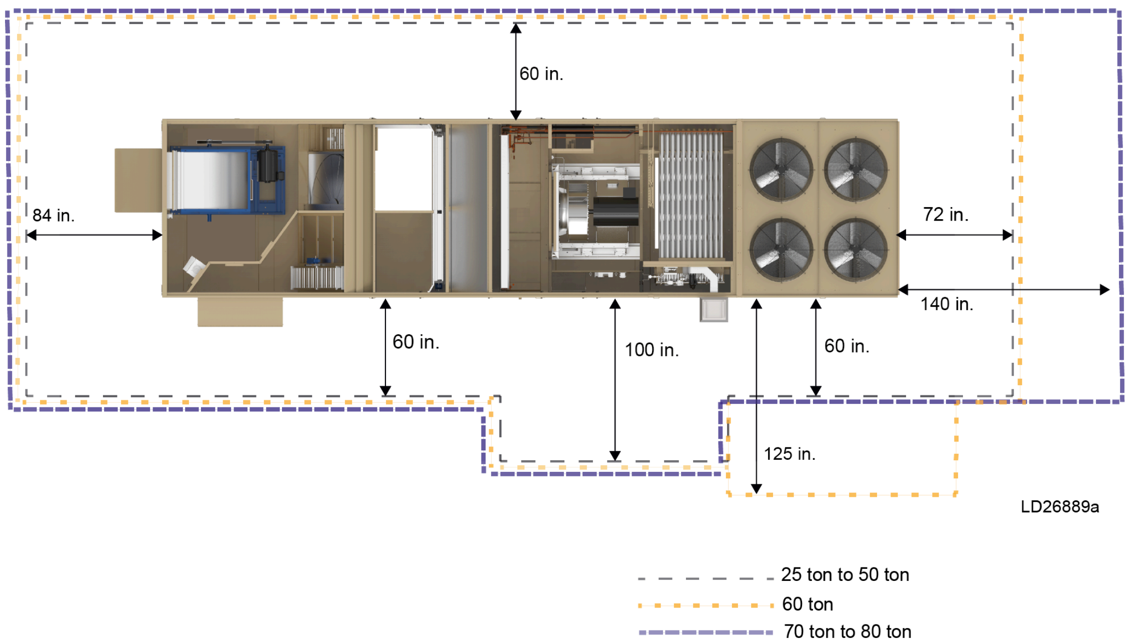
Unit clearances

The Premier units are designed for outdoor installation. When selecting a site for installation, consider the following conditions:

- The unit must be installed on a level surface.
- For units located outside, select a place with minimum sun exposure and an adequate supply of fresh air for the condenser.
- Avoid locations beneath windows or between structures.
- Select optional condenser coil protection for locations in proximity to salt water or other harsh environments.

- Install the unit on a roof that is structurally strong enough to support the weight of the unit with minimal deflection. It is recommended that the unit(s) be installed no more than 15 ft from a main support beam to provide proper structural support and to minimize the transmission of sound and vibration. Ideally, the center of gravity must be located over a structural support or building column.
- Locate unit(s) away from building flue stacks or exhaust ventilators to prevent possible reintroduction of contaminated air through the outside air intakes.
- Ensure the supporting structures do not obstruct the duct, gas, or wiring connections.
- Proper operational and service clearance space around the perimeter of the unit, on one side for coil servicing, and to any adjacent units is required to ensure unit performance, eliminate cross contamination of exhaust and outside air, and for maintenance tasks, such as coil pull and cleaning. There must be no obstructions above the condensing unit section.
- As shown condenser clearance varies based on unit tonnage and coil arrangements.

Figure 20: Unit clearances



Note:

1. 10 ft clearance minimum over the top of the condensing unit.
2. Only one adjacent wall can exceed unit height.
3. 12 ft clearance required to adjacent units.

Location

Of the many factors that can affect the location of equipment, some of the most important to consider are structural, acoustical, and service clearances. Give proper attention during the design stage to ensure proper structural support. In cases where equipment is being replaced, be aware of building design to ensure support is adequate for the application.

Sound from the equipment is an important consideration when applying the unit equipment to a space. Ensure the unit is kept away from sound sensitive areas, such as conference rooms, auditoriums, executive offices, and any other rooms that may potentially occupy tenants. Possible locations to install a single package unit could include above hallways or mechanical or utility rooms. Refer to the *Sound from Rooftop Units Application Guide (Form 5515909-JAD).*

Unit placement

The unit can be placed at ground level or an elevated location.

Elevated locations

Elevated roof curbs or dunnage steel can be used to support the unit in order to raise it to specific heights. The rooftop unit must be supported around the entire perimeter to ensure unit integrity. When this type of placement is required, keep unit access in mind. Cat walks or other forms of unit access may be required to one or both sides of the unit depending on your area of the country and the local codes that are enforced.

Ground level locations

It is important that the units are installed on a substantial base that does not settle or cause strain on the refrigerant lines and sheet metal that could result in possible leaks. A one-piece concrete slab with footers extended below the frost line is highly recommended. Additionally, isolate the slab from the main building foundation to prevent noise and vibration transmission to the building structure.

For ground level installations, take precautions to protect the unit from tampering by or injury to unauthorized persons. Make accommodations that allow for the correct depths of the drain pan trap and humidifier trap. Service clearances must be maintained at all times. For further details, refer to the *Roof Curbs for Premier Rooftop Units Application Guide*.

Guide specifications

General

Available options are shown in brackets.([])

Scope

The requirements of the General Conditions, Supplementary Conditions, Division 1, and drawings apply to all work herein.

Provide specifications for the following features:

- Microprocessor controlled, multiple-scroll compressor.
- Air-cooled, double wall outdoor packaged rooftop air conditioning units.
- Components of the scheduled capacities and performance, as shown and indicated on the drawings, including but not limited to: factory-single packaged air conditioner; charge of refrigerant and oil; roof curb; field duct; power and control connections; and utility connections.

Quality assurance

All units are tested, rated, or certified, as applicable, in accordance with the following standards, guidelines, and codes:

1. All units meet ASHRAE 90.1-2022 minimum energy efficiency ratio/integrated energy efficiency ratio (EER/IEER).
2. All units comply with ASHRAE 189.1-2014 EER/IEER efficiency requirements.
3. All units meet the latest ASHRAE 62 requirements for ventilation and indoor air quality (IAQ).
4. All units are rated in accordance with the AHRI Standard 340/360. Units greater than 63 tons are excluded from the AHRI certification program.
5. All units are certified to UL 60335-2-40.
6. All units meet the 2023 Department of Energy efficiency code.
7. All gas heating units are designed in conformance to ANSI Z83.8-2016/CSA 2.6-2016 standards and carry a CSA listing.
8. All units' outdoor sound data are rated in accordance with ANSI/AHRI 370-2015 *Sound Performance Rating of Large Air-Cooled Outdoor Refrigerating and Air-Conditioning Equipment*.
9. All units' indoor sound data are rated in accordance with the ANSI/AHRI 260-2017 *Sound Rating of Ducted Air Moving and Conditioning Equipment*.

Manufacturers

The design shown on the drawing is based upon products of the scheduled manufacturer. Alternate equipment manufacturers are acceptable if equipment meets the scheduled performance and complies with these specifications. If equipment supplied by manufacturer other than that scheduled is utilized, then the mechanical contractor is responsible for coordinating with the general contractor and all affected subcontractors to ensure proper provisions for installation of the furnished unit. This coordination includes, but is not limited to, the following:

1. Structural supports for units.
2. A prefabricated, heavy gauge galvanized steel mounting curb, designed and supplied by the unit manufacturer, is provided for field assembly on the roof decking prior to unit shipment. The roof curb is a partial [full] perimeter type with complete perimeter support of the air handling section and rail support of the condensing section. Supply and return opening duct frames are provided as part of the curb structure, allowing duct connections to be made directly to the curb prior to unit arrival. The curb is a minimum of 14 in. high and include a nominal 2 in. × 4 in. wood nailing strip. Gasket is provided for field mounting between the unit base and roof curb.
3. Piping size and connection or header locations.
4. Electrical power requirements and wire or conduit and overcurrent protection sizes.
5. All costs incurred to modify the building provisions to accept the furnished units.

Warranty

Unit warranty

The manufacturer warrants all equipment and material of its manufacture against defects in workmanship and material for a period of 12 months from start-up or 18 months from date of shipment ,whichever occurs first. Extended warranty is provided to cover the unit for [24] [60] months from the date of shipment.

- The warranty includes parts [and labor] during this period only.
- The warranty does not include parts associated with routine maintenance, such as belts, air filters, etc.

Compressor warranty

The manufacturer warrants all compressors against defects in workmanship and material for a period of 12 [60] months from date of shipment.

- The warranty includes parts [and labor] during this period only.
- The warranty does not include parts associated with routine maintenance.

Gas heat exchanger warranty [optional]

The manufacturer warrants all stainless steel gas heat exchangers against defects in workmanship and material for a period of 120 months from date of shipment.

- The warranty includes parts during this period only.
- The warranty does not include parts associated with routine maintenance.

Delivery and handling

1. The unit is delivered to the job site fully assembled, wired, and charged with refrigerant and oil by the manufacturer.
2. ***Split ship [optional for units less than 53 feet long]*** Unit is factory tested and delivered to the job site split in two segments, air handling and condensing section, for field assembly and installation. Unit segments ship with valves on each end to recouple the unit and pre-charged with R-454B refrigerant to minimize field setup time. An electrical box is provided in the condensing section to seamlessly integrate power to the unit. The contractor covers the cost of refrigerant, evacuation and charge time for units that are not pre-charged with refrigerant.
3. All handling and storage procedures are per the manufacturer's recommendations.

Submittals

Shop drawings

Shop drawing submittals include, but are not limited to, the following: drawings indicating components, dimensions, weights, required clearances, location, type and size of field connections, and power and control wiring connections.

Product data

Product data includes dimensions, weights, capacities, ratings, fan performance, motor electrical characteristics, and gauges and finishes of materials.

Documentation

1. Fan curves with specified operating point clearly plotted are provided.
2. Product data of filter media, filter performance data, filter assembly, and filter frames is provided.
3. Electrical requirements for power supply wiring, including wiring diagrams for interlock and control wiring are supplied. Factory and field-installed wiring are clearly indicated.

Equipment

Product specification

Summary

Completely factory assembled, unitized construction [delivered to the job site split in two segments], single packaged air conditioning unit including a factory-mounted and wired unit controller and sensors; single point power with terminal block [dual point power with terminal block] [a non-fused electrical disconnect] [a fused electrical disconnect], for main unit power is factory-installed 460 V [208 V to 230 V][575 V] power supply, outside air handling section with return and supply openings, discharge air (DA) plenum, direct-expansion refrigerant condensing section. 5 kA [65 kA] short-circuit current rating (SCCR).

Phase monitor

A phase monitor is provided on unit designed to protect 3 phase equipment from phase loss, reversal, imbalance, and low voltage. The phase monitor fault condition is indicated at the unit control panel, and the unit is placed into an emergency stop condition.

Internal lights [optional]

A 115 V circuit of centrally located marine LED lights is provided with individual wiring run from the single switch located in the low voltage power supply panel common wall. At a minimum lights are provided in the [pre-evaporator blank] [post-evaporator blank] economizer, evaporator coil, and supply fan sections.

Factory test

1. On factory-assembled or split ship units, the refrigerant circuit is pressure-tested, evacuated, and fully charged with refrigerant and oil.
2. On factory-assembled or split ship units, the unit controller is configured and run tested at the factory to minimize field setup time.

3. If the unit is not configured and tested, then the manufacturer provides field start-up and testing to ensure that the controller is functioning correctly.
4. Record test reports [optional] include end of line quality test report to be sent out to the customer.

Unit construction

Base rail - The unit includes an integral design base rail with lifting points clearly marked and visible on the base rail and a 1 1/4 in. female pipe thread (FPT) connection for condensate drainage. The unit base is designed with a recessed curb mounting location. The recessed curb mounting surface provides a continuous surface for field application of curb gasketing to create a weather-tight seal between the curb and unit.

Unit casing - Unit casing has a double wall insulation with injected foam. Insulation application meets NFPA 90 requirements. The insulation system is resistant to mold growth in accordance with UL 181 and ASTM C1338 standardized test methods.

Paint - Exterior champagne color painted surfaces are designed to withstand a minimum of 750 salt spray hours when tested in accordance with ASTM B-117.

Markings and diagrams - All necessary tags and decals to aid in the service or indicating caution areas are provided. Electrical ladder wiring diagrams are attached to the control panel access door.

Documentation - Installation and maintenance manuals are supplied with each unit.

Access doors - Double wall access doors are provided in the fan, [discharge] coil, filter, and inlet sections. Doors are double wall construction with a solid liner and a minimum thickness of 1 in. Doors are attached to the unit with stainless steel hinges. A minimum of three 5/16 in. hex-drive, 90° opening standard finger pull latches are provided. [Viewports are provided in the supply fan, economizer, and cooling coil sections.] [Latches are single handle rotary type with minimum three point contact and an adjustable height handle. Padlockable handle is provided.]

Panels and doors are completely gasketed with a closed-cell, neoprene gasket. Door tiebacks are provided for all doors to secure them while servicing. Doors that provide access to positive pressure segments include a safety latch. The safety latch relieves pressure if inadvertently opened during unit operation. [Doors that provide access to the ultraviolet (UV) light section include an electrical switch that disables UV light operation when the door is opened.]

Burglar bars [optional] - Burglar bars are mounted in the [supply] [return] [supply and return] opening to prevent entry into the building through the ductwork. Burglar bars are provided for accessible sections with a bottom opening.

Economizer section

ⓘ **Note:** Select either no outside air or economizer.

1. **No outside air -** The unit has no provisions for outside ventilation air.
2. **Modulating economizer -** The economizer segment is designed to use outside air for cooling, ventilation, and provides a means of exhausting air from the air handling unit. The segment consists of parallel-acting low leak [ultra low leak] dampers. The outside air (OA) and exhaust air (EA) dampers are sized for 100% of nominal unit airflow. The EA damper assembly has a factory-installed, foldable rain hood permanently attached to the cabinet to prevent windblown precipitation from entering the unit. The rain hood is rotated into the cabinet and secured for shipment so that upon installation, they can be rotated upwards and screwed into place.

OA inlet openings are covered by a factory-installed foldable rain hood permanently attached to the cabinet to prevent windblown precipitation from entering the unit. The rain hoods on the left and right sides of the unit are rotated into the cabinet and secured for shipment so that upon installation they can be rotated upwards and screwed into place. The OA hood contains a removable and cleanable filter with an efficiency rating of 50% based on ASHRAE 52-76. Damper blades are fabricated from a minimum of 16-gauge galvanized steel. Damper shafts are fabricated from solid steel and mounted in the frame with bronze bearings.

Economizer ultra low leakage dampers [optional]Damper assemblies are ultra low leak design. Damper blades are fabricated from a minimum of 16-gauge galvanized steel. Blade edges are covered with vinyl seals. Damper assemblies have a maximum leakage rate of 4 CFM/sq. ft at 1.0 in. of water gauge (iwg) when tested in accordance with AMCA Standard 500, and have a longevity of 60,000 damper opening and closing cycles. The unit provided economizer meets the damper leakage and life cycle requirements for ASHRAE 90.1-2019, 2018 International Energy Conservation Code® (IECC), and 2016 California Title 24 standards.

Outside airflow measurement [optional] - Fresh air is introduced into the unit via an outside airflow measurement station for full airflow measurement from 0–100% outside airflow. The airflow measurement station is designed into the unit and includes all controls, dampers, and components to monitor airflow accurately.

Economizer with fault detection and diagnostics (FDD) - Economizer with FDD is provided when economizer is selected. As required per ASHRAE 90.1-2019 and California Title 24, FDD system provides alarms for:

- Damper not modulating.
- Economizer not available for suitable outside air conditions.
- Economizer available for unfavorable outside air conditions.
- Excess outside air.
- Temperature sensor failure and fault.

ERW - optional

1. ERW performance is AHRI 1060 certified and bears the AHRI certified label. Components that are independently tested or rated in accordance with are not acceptable. Manufacturer membership in AHRI is not an acceptable substitute. Certified components must be listed as active in the AHRI directory, see http://www.ahridirectory.org.
2. The energy recovery cassette is a UL recognized component for fire and electrical safety and bear the UR mark. recognized components are listed in the UL directory, see http://database.ul.com.
3. The energy recovery cassette complies with NFPA 90A standard by virtue of UL 1812 and UL 900 standards for fire tests to determine flammability and smoke density.
4. The energy recovery cassette incorporates a rotary wheel in an insulated cassette frame complete with removable energy transfer media, seals, drive motor, and drive belt. Cassette frame and structural components are constructed of G90 galvanized steel for corrosion resistance.
5. Wheel structure consists of a welded hub, spoke, and continuous rolled rim assembly of stainless steel, and is self-supporting without energy transfer segments present. Wheel structure is connected to the shaft by means of taper lock bushings.
6. Wheel bearings are permanently sealed and selected for a minimum 30 year L10 life of 400,000 hours. Bearings requiring external grease fittings or periodic maintenance are not acceptable.
7. Energy transfer media is constructed of a durable synthetic lightweight polymer. Media is wound continuously with one flat and one structural layer in an ideal parallel plate geometry. Airflow across heat exchanger surface remains laminar. Energy transfer media does not exceed 3 in. in depth. Energy transfer media is suitable for use in corrosive, marine, or coastal environments without the need for additional coatings. Energy transfer media is capable of repeated washings without significant degradation of the desiccant bond as documented by an independent third party.
8. Desiccant is either silica gel or molecular sieve and permanently bonded to the energy transfer media without the use of binders or adhesives, which may degrade desiccant performance. Desiccants not permanently bonded are not acceptable due to potential delamination or erosion of the desiccant from the energy transfer media. Desiccant is non-migrating nor does it dissolve or deliquesce in the presence of water or high humidity.
9. Wheels are provided with removable energy transfer segments. The unit is designed to facilitate side serviceable ERW. Side service allows for all ERW segments to be pulled for removal. Segments are removable without the use of tools to facilitate maintenance and cleaning. Energy recovery segments are cleanable outside of the cabinet with detergent or alkaline coil cleaner and water. Energy transfer segments are capable of submersion in a cleaning solution. Submersion is capable of restoring latent performance to within AHRI certified performance limits.
10. All diameter and perimeter seals are provided as part of the cassette assembly and are factory set. Seals are non-contact nylon pile brush seal oriented in a labyrinth style configuration. Diameter seals are fully adjustable and easily accessible. Perimeter seals are permanently mounted to the wheel rim and not require adjustment. Seals that mount to the frame are not acceptable.
11. The wheel drive motor is a UL recognized component and is mounted in the cassette frame and supplied with a service connector or junction box. Three-phase motors are suitable for use in both standard and inverter rated applications.
12. The unit has 2 in. MERV 8 replaceable filters for the outside air and exhaust air to help keep the wheel clean and reduce maintenance. The ERW is energized briefly for cleaning and blocking protection if it has not been operated for extended periods.

13. The ERW has a bypass damper. The bypass damper has two positions, ON or OFF, according to the heating or cooling demand.
14. The rooftop unit provides frost protection for the ERW. The controller varies the speed of the wheel, by starting and stopping, to prevent frost condition.

Building pressure control

Note: Select one of the following types of building pressure control.

1. **No building exhaust or relief -** The unit has no provisions to exhaust building return air.
2. **Barometric relief damper -** Building air exhaust is accomplished through barometric relief dampers installed in the return air (RA) plenum. The dampers open relative to the building pressure. The opening pressure is adjustable.
3. **Powered exhaust with modulating DA damper -** A double width, double inlet (DWDI) forward-curved centrifugal exhaust fan is provided to exhaust building return air to relieve building static pressure. The fans operates at a constant volume and based on building static pressure. Exhaust airflow is modulated via a parallel-acting control damper. The EA dampers are sized for 100% of the exhaust airflow. An access door is provided on at least one side of the unit for fan and motor access.
4. **Powered exhaust with VFD -** A DWDI forward-curved centrifugal exhaust fan is provided to exhaust building return air to relieve building static pressure. Exhaust airflow is modulated via a factory-installed and commissioned VFD with the same nameplate horsepower as the exhaust fan motor. An access door is provided on at least one side of the unit for fan and motor access.
5. **Dual powered exhaust with VFD (60 ton to 80 ton cabinets with ERW) -** Dual DWDI forward-curved centrifugal exhaust fans provided to exhaust building return air to relieve building static pressure. Exhaust airflow is modulated via a VFD that is provided for each exhaust fan. The VFD driven exhaust fan has a factory-installed and commissioned VFD with the same nameplate horsepower as the exhaust fan motor. An access door is provided on at least one side of the unit for fan or motor access.
6. **Power return fan with VFD and exhaust -** A SWSI plenum fan is provided to draw return air from the building to the rooftop unit. [(60 ton to 80 ton) plenum fan is direct drive.] An access door is provided on at least one side of the unit for fan and motor access. The return fan operates to maintain a constant pressure within the RA plenum. An EA damper is provided to modulate building exhaust. The damper is controlled via building pressure. The RA damper and OA damper independently modulate volumes of return and outside airflows.
7. **Exhaust fan VFD manual bypass [optional with single exhaust fan configurations] -** A two contactor manual bypass is provided to permit fan operation in the event of a fan failure.
8. **Redundant VFD [optional 60 ton to 80 ton] -** Direct drive plenum (DDP) return fans are provided with a 100% redundant VFD. Redundant VFD automatically operates in the event that the primary VFD fails.
9. **Return [exhaust] fan VFD reactor [optional] -** A 3% impedance AC line reactor is provided for return [exhaust] fan VFD.
10. **Airflow measurement station [optional] -** Installed on the return fan. Return fan airflow is viewable on the control panel and able to be communicated across the Building Automation System (BAS).
11. **Belt guards [optional on belt driven fans] -** These are provided to enclose the drive and sheave package on return [exhaust] fan.
12. All VFDs are factory tested and matched with each unit.

For powered exhaust or return fan options above, use the following:

1. **Exhaust or return fan motor -** Fan motors are National Electrical Manufacturers Association (NEMA) design ball-bearing types with electrical characteristics and horsepower as specified. [Shaft grounding rings on motors are provided to prevent electrical bearing fluting damage by safely diverting harmful shaft voltages and bearing currents to ground, increasing the motor longevity.] Motors are nominal 1,800 RPM [or 1,200 RPM], open drip-proof (ODP) type [total enclosed fan-cooled (TEFC) type]. The motor is located within the unit on an adjustable base.
2. **Mountings -** Fan and fan motor are internally mounted and isolated on a full width isolator support channel using 1 in. [2 in.] springs [and seismic restraint]. The fan is connected to the fan cabinet using a flexible connection to insure vibration-free operation.

3. **Bearings and drives [option on belt driven fans] -** Fan bearings are self-aligning, pillow block, or flanged type regreaseable ball bearings and are designed for an average life L50 of at least 200,000 hours. All bearings are factory lubricated and equipped with standard hydraulic grease fittings [and lube lines extended to the motor side of the fan]. Fan drives are selected for a 1.5 service factor and anti-static belts are furnished. [Optional spare belts are shipped loose from the factory.] All drives are fixed pitch. Fan shafts are selected to operate well below the first critical speed and each shaft is factory coated after assembly with an anti-corrosion coating.

Air blender - optional

1. Multiple-blade, air-mixer assembly mix air to prevent stratification.
2. Air blenders have a rotary turbulating design consisting of radially extending blades. Units are completely fixed devices with no moving parts.
3. Static air mixers material are 0.080 in., 5052 H34 alloy aluminum. Static air mixers are welded.
4. Simple mixing devices that do not produce expanding discharge with counter rotational mixing are not acceptable.

Filter section

Draw-through filters

Note: Select a filter rack, filter media, and transducer if desired.

1. **Angled filter rack -** 2 in. filters are provided by others [2 in. throwaway filters are provided in an angled filter rack] [2 in. (MERV 8) pleated filters are provided in an angled filter rack]. Filter efficiencies are rated in accordance with ASHRAE Standard 52.2. All filter holding frames consist of heavy duty galvanized steel designed for industrial applications. All filters are side accessible with access doors provided on both sides of the filter section.
2. **Rigid filter rack -** 2 in. and 12 in. filters are provided by others [12 in. (MERV 14) rigid filters with 2 in. (MERV 8) pleated pre-filters are provided] [18 in. (MERV 15) bag filters with 2 in. (MERV 8) pleated pre-filters are provided]. Filter efficiencies are rated in accordance with ASHRAE Standard 52.2. All filter holding frames consist of heavy duty galvanized steel designed for industrial applications. All filters are side accessible with access doors provided on both sides of the filter section.
3. **Vertical filter rack -** 4 in. filters are provided by others. [4 in. (MERV 8) pleated filters are provided.] Filter efficiencies are rated in accordance with ASHRAE Standard 52.2. All filter holding frames consist of heavy duty galvanized steel designed for industrial applications. All filters are side accessible with access doors provided on both sides of the filter section.
4. **Filter transducer -** A filter transducer is provided and wired to the rooftop's control panel. The unit controller displays the pressure drop across the filter[s]. A single transducer is provided to measure pressure drop across both filter banks. [Transducers are provided to measure pressure drop across each filter bank.]
5. **Differential pressure gauge -** A flush mounted, factory-installed differential pressure gauge is provided to measure pressure drop across both filter banks. [Flush mounted, factory-installed differential pressure gauges are provided to measure pressure drops across each filter bank.] The manufacturer provides fully functional gauges complete with tubing.

Final filters

Note: Select a filter rack, filter media, and transducer if wanted.

1. **Rigid filter rack -** 2 in. and 12 in. filters are provided by others [12 in. (MERV 14) rigid filters with 2 in. (MERV 8) pleated pre-filters] [18 in. (MERV 15) bag filters with 2 in. (MERV 8) pleated pre-filters are provided]. Filter efficiencies are rated in accordance with ASHRAE Standard 52.2. All filter holding frames consist of heavy duty galvanized steel designed for industrial applications. All filters are side accessible with access doors provided on both sides of the filter section.
2. **High efficiency particulate air** (***HEPA) filter rack -*** 2 in. and 12 in. filters are provided by others [12 in. (MERV 17) HEPA filters with 2 in. (MERV 8) pleated pre-filters are provided]. Filter efficiencies are rated in accordance with ASHRAE Standard 52.2. All filter holding frames consist of heavy duty galvanized steel designed for industrial applications. All filters are front loading with access doors provided on both sides of the filter section.
3. **Filter transducer -** A filter transducer is provided and wired to the rooftop's control panel. The unit controller displays the pressure drop across the filter[s]. A single transducer is provided to measure pressure drop across both filter banks. [Transducers are provided to measure pressure drop across each filter bank.]

4. **Differential pressure gauge -** A flush mounted, factory-installed differential pressure gauge is provided to measure pressure drop across both filter banks. [Flush mounted, factory-installed differential pressure gauges are provided to measure pressure drops across each filter bank.] The manufacturer provides fully functional gauges complete with tubing.

Evaporator section

1. **Cooling coil -** Evaporator coils are a direct expansion type with interlaced circuiting to assure complete coil face activity during part-load operation. Coil tubes are copper with internally enhanced tubes. Tubes are enhanced mechanically and expanded to bond with the aluminum [copper fins]. All coils are pressure-tested at a minimum of 450 psig. [Evaporator coils are protected by the E-Coat 10-1 four coat process. Coils are dipped in a phenolic coating, that provides substantial resistance to corrosion of aluminum and copper. E-Coat coils are rated to exceed 6000 hours salt spray in accordance with ASTM B117 / DIN 53167.]
2. **IAQ drain pan -** The main coil drain pan is double-sloped with a condensate connection through the base rail of the unit. Drain pans for cooling coils [and humidifier] meet the requirements of ASHRAE 62.1. Drain pans are constructed of stainless steel. [Provide a condensate overflow switch in the primary drain pan.]
3. **Intermediate drain pan -** Coils with finned height greater than 48 in. have an intermediate drain pan extending the entire finned length of the coil. The intermediate pans have drop tubes to guide condensate to the main drain pan.
4. **UV lights [optional] -** UVC lights provide fly-by virus deactivation and surface decontamination of the coil and drain pan. The UV lights are installed in the downstream side of the evaporator coil or drain pan. UVC lights provide a minimum of 1500 μJ/cm2. UVC lights have a minimum 9,000 hours of service. The UV lights, tubes, are shipped loose in protective packaging for installation by the contractor.

Supply fan section

1. **Fan -** The fan section is equipped with dual SWSI airfoil plenum wheels. Plenum fans are direct drive. An access door is provided on the opposite side of the control panel for fan/motor access. [Dual SWSI supply fans are connected to a single VFD.] An access door is provided on the opposite side of the control panel for fan/motor access.]
2. **Fan motor -** Fan motors are NEMA design ball-bearing types with electrical characteristics and horsepower as specified. Motors are nominal 1,800 RPM, ODP type [TEFC type].
3. **Mountings -** Fan and fan motor are internally mounted and isolated on a full width isolator support channel using 1 in. [2 in.] springs [and seismic restraint].
4. **Variable air volume (VAV) fan control -** VAV supply fan control is accomplished by using [a] VFD[s] and supply fan motor combination matched with supply fan to generate design performance at appropriate fan speed. The VFD[s] includes an integral DC line reactor to reduce harmonic distortion in the incoming and outgoing power feeds. If a DC line reactor is not provided, an AC line reactor[s] is provided.
5. **Single zone VAV (SZVAV) -** In cooling mode, refrigeration capacity or compressor stages are cycled ON or OFF to maintain DA temperature (DAT). In heating mode, additional stages are cycled ON or modulated to maintain DAT setpoint. The supply fan speed is modulated to maintain zone temperature setpoint. The unit uses either a BAS signal, RA temperature (RAT), or a zone temperature sensor to determine zone temperature and deviation from setpoint.
6. **Redundant VFD - optional 60 ton to 80 ton -** Dual DDP supply fans are provided with dual VFDs. [VFDs are supplied to independently operate the two supply fan motors.]
7. All VFDs are factory tested and matched with each unit.
8. **VFD reactor [optional] -** A 3% impedance AC line reactor is provided for the supply fan VFD[s].
9. **Shaft grounding rings [optional] -** on motors are provided to prevent electrical bearing fluting damage by safely diverting harmful shaft voltages and bearing currents to ground, increasing the motor longevity.
10. **Inlet guards [optional] -** are provided on inlet of supply fan.
11. **Supply airflow measurement [optional] -** capability is provided. Supply fan airflow is viewable on the control panel and readable across the BAS.

Discharge air plenum

Note: Select one of the following heat or no heat configurations.

Cooling only

The DAT sensor is located in the DA plenum, such that it accurately measures the DAT.

Staged gas heat

The heating section includes an induced draft furnace in two [four] [six] stages of heating capacity. [Optional high altitude kit is shipped loose from the factory.] [Optional liquid propane (LP) conversion kit is shipped loose from the factory.]

1. **Heat exchanger -** The heat exchanger is constructed of tubular aluminized steel [stainless steel] and flue assembly.
2. **Burner and ignition controls -** The burner includes a direct-driven induced-draft combustion fan with energy efficient direct spark ignition, redundant main gas valves with pressure regulator.
3. **Combustion air fan -** The inducer fan[s] maintains a positive flow of air through each tube to expel the flue gas and to maintain a negative pressure within the heat exchanger relative to the conditioned space.
4. **Safety devices -** A high limit controller with automatic reset to prevent the heat exchanger from operating at an excessive temperature is included. A centrifugal switch on the induced draft fan motor shaft prevents ignition until sufficient airflow is established through the heat exchanger. A rollout switch provides secondary airflow safety protection. The rollout switch discontinues furnace operation if the flue becomes restricted.
5. **Flue -** The furnace flue is shipped loose to protect it from damage during transit. The flue is field-mounted by the installing contractor. The flue outlet is located above the unit to help prevent recycling of combustion gases back through the heat exchanger.
6. **Agency certification -** Gas heating units are designed in conformance with ANSI Z83.8-2016/CSA 2.6-2016 standards and carry a CSA listing. All gas heating units have 81% steady state efficiency rated in accordance to 2023 Department of Energy Efficiency code. All gas heating units meet the requirement of AHRI CFRN Certification program and Code of Federal Regulations, 10 CFR Part 43, which references efficiency testing method in standard ANSI Z21.47.
7. **Valves -** A factory provided gas heater isolation valve is shipped loose for installation in the field by others. Punched holes are provided in the bottom [side] of the unit. The holes are capped for field to make suitable connection. A bottom [side] penetration option is provided with piping to extend out of the unit.

Modulating gas heat

For applications requiring gas heat for morning warm-up, supply air (SA) tempering, or other heating needs, a modulating natural gas furnace is available for finer temperature control. The furnace is located in the DA plenum, downstream of the supply fan. The DAT sensor is located across the face of the supply duct opening in the unit. [Optional high altitude kit is shipped loose from the factory.] [Optional LP conversion kit is shipped loose from the factory.]

1. **Heat exchanger -** The heat exchanger is constructed of tubular stainless steel and flue assembly.
2. **Burner and ignition controls -** The burner includes a direct-driven induced-draft combustion fan with energy efficient direct spark ignition, redundant main gas valves with pressure regulator.
3. **Combustion air fan -** The inducer fan[s] maintains a positive flow of air through each tube to expel the flue gas and to maintain a negative pressure within the heat exchanger relative to the conditioned space.
4. **Safety devices -** A high limit controller with automatic reset to prevent the heat exchanger from operating at an excessive temperature is included. A centrifugal switch on the induced draft fan motor shaft prevents ignition until sufficient airflow is established through the heat exchanger. A rollout switch provides secondary airflow safety protection. The rollout switch discontinues furnace operation if the flue becomes restricted.
5. **Flue -** The furnace flue is shipped loose to protect it from damage during transit. The flue is field-mounted by the installing contractor. The flue outlet is located above the unit to help prevent recycling of combustion gases back through the heat exchanger.
6. **Agency certification -** Gas heating units are designed in conformance with ANSI Z83.8-2016/CSA 2.6-2016 standards and carry a CSA listing. All gas heating units have 81% steady state efficiency rated in accordance to 2023 Department of Energy Efficiency code. All gas heating units meet the requirement of AHRI CFRN Certification Program and Code of Federal Regulations, 10 CFR Part 431, which references efficiency testing method in standard ANSI Z21.47.

7. **Valves -** A factory provided gas heater isolation valve is shipped loose for installation in the field by others. Punched holes are provided in the bottom [side] of the unit. The holes are capped for field to make suitable connection. A bottom [side] penetration option is provided with piping to extend out of the unit.

Electric heat

An electric slip-in heater is installed within the rooftop unit DA plenum to provide the heating requirements per the schedule shown on the plans. The electric heater is wired in such a manner as to provide six [eight] [ten] steps of capacity control. Heating element branch circuits are individually fused to maximum of 48 Amps per National Electrical Code (NEC) requirements. Unit are provided with staged [modulating 0 VDC to 10 VDC silicon controlled rectifier (SCR)] electric heat.

1. **Heat element -** The heating element is an industrial grade design using an open coil made of the highest grade resistance wire containing 80% nickel and 20% chromium. The resistance coils are adequately supported in the air stream using ceramic bushings in the supporting framework. Terminals of the coil are stainless steel with high temperature ceramic bushings.
2. **Safety devices -** The primary high temperature protection is an auto reset type thermal cut out. Secondary protection is provided by a manual reset type thermal cut out.
3. **Agency certification -** The operation of the electric heater is an integral part of the rooftop's control system. Power connection to the heater is through the power panel of the unit. Electric heat is tested and certified in accordance with UL 60335-2-40.

Hot water heat

A hot water coil is installed in the rooftop DA plenum.

1. **Construction -** The hot water coil has one [two] row[s]. Primary surface is 1/2 in. OD copper tube, staggered in direction of airflow. Connections have 1/4 in. FPT drain plug on each connection. A structural galvanized steel casing protects the coil. The coil is circuited to provide free draining and venting through one vent and drain. Freezestat is provided to prevent coil freeze up. Access doors are provided for convenient access to the valve for maintenance and inspection.
2. **Testing -** Completed coil including headers, connections, and return bends are tested with 325 pounds compressed air under water. Coils are designed for operation at 250 psig design working pressure.
3. **Valves -** Factory [field] provides three-way modulating control valve and spring return valve actuator capable of receiving 0–10 VDC signal. The valve actuator is controlled by factory-installed main unit controller. [Field-provided valve matches specifications/ratings as recommended by unit manufacturer.] Unit have hole markings for field to drill holes through the base or side of the unit.

Steam heat

A steam heating coil is installed in the rooftop DA plenum.

1. **Construction -** The steam coil is constructed in the non-freeze style. The steam coil has one row. Tubes are 1 in. OD seamless copper tubing with a minimum wall thickness of 0.035 in. and expanded into the fin collars for maximum fin-tube bond. Inner distributing tubes are 5/8 in. OD seamless copper tubing with a minimum wall thickness of 1/4 in.. All header connections consist of red brass, with male pipe threads and silver braze to headers. Casing is galvanized steel. The core is pitched in the direction of the condensate connection for proper drainage. Freezestat is provided to prevent coil freeze up. Access doors are provided for convenient access to the valve for maintenance and inspection.
2. **Testing -** The completed coil, including headers and connections, are tested underwater with 325 pounds compressed air to ensure a leak-free coil.
3. **Valves -** Factory [field] provides two-way modulating control valve and non-spring return valve actuator capable of receiving 0–10 VDC signal. Steam valve is designed for operation at 2 psig minimum and 15 psig maximum steam pressure. Valve actuator is controlled by factory-installed main unit controller. [Field provided valve must match specifications and ratings as recommended by the unit manufacturer.] The unit has hole markings for field to drill holes through the base or side of the unit.

Humidifier [optional]

Humidifier steam manifold and dispersion panel - Short absorption manifold is designed to distribute pressurized steam from a facility steam boiler, to directly inject the steam into air for humidification.

1. Absorption distance characteristic prevents water accumulation on duct surfaces beyond 12 in. downstream of the steam dispersion panel.
2. Steam dispersion panel consisting of a (one) horizontal 304 stainless steel round header supplying steam to a bank of closely spaced [6 in. for 60 ton, 3 in. for 70 ton to 80 ton] vertical tubes, as necessary to meet absorption distance requirements, and to reduce condensation losses.
3. Steam inlet and condensate return are located on the same side and at the bottom of the header to allow single point entry and floor mounting.
4. Vertical stainless steel distribution tubes provide condensate evacuation. Horizontal distributor tubes are not acceptable.
5. Distribution tubes include threaded standoffs for trouble free attachment to factory-supplied support bracket.
6. All steam tubes are 304 stainless steel construction. 409 stainless steel header and distribution tubes are not acceptable.
7. Stainless steel nozzle inserts ensure condensate free steam is discharged from the center of the distribution tubes. Systems without nozzle inserts, or other than stainless steel, are not acceptable.
8. Stainless steel nozzle inserts have metered orifices, sized to provide even distribution of the discharged steam, spaced for optimum steam absorption.
9. The header functions as an internal steam separator, therefore an external steam separator is not required.
10. Manufacturer provides adjustable 304 stainless steel mounting frame.
11. Tubes and headers accommodate field retrofit of insulation for increased energy efficiency.

Shipped loose items

1. The manufacturer provides steam control valve with electrical actuator normally closed, spring return (24 V), steam control valve with equal percent flow characteristics and positive shut off against steam. The control valve is compliant with EN 13547 standard and ANSI B1.20.1 (NPT) end connection standard. The temperature rating is compliant with ANSI 16.104/FCI 70.2 and for control shut off leakage ANSI/ISA-75.11 flow characteristics standards. The steam control valve has a brass forged body with stainless steel ball and stem assembly.
2. The manufacturer provides cast iron steam traps and iron wye strainer

Sound attenuator [optional]

1. Sound attenuator, or silencer, segments are provided as shown on drawings. Silencers are rectangular, 36 in. long sound attenuators as indicated on drawings and equipment schedule.
2. Outer casings of rectangular silencers are made of 22 gauge type G60 galvanized steel.
3. Interior partitions for rectangular silencers are not less than 26 gauge type G60 galvanized perforated steel.
4. The filler material is inorganic glass fiber of a proper density to obtain the specified acoustic performance and is packed under not less than 5% compression to eliminate voids due to vibration and settling. The material is inert, vermin-proof, and moisture-proof.

Condenser section

1. **Condenser fans -** Condenser fans are matched up with compressors to optimize system control. Condenser fans are propeller type with aluminum blades, directly driven by permanently lubricated three-phase motor.
2. **Condenser coil -** Microchannel condenser coils are constructed of parallel flow aluminum alloy tubes metallurgically brazed to enhanced aluminum alloy fins. Condenser cleaning hatches are provided for access to condenser coil without the removal of condenser fans. [Condenser coils are protected by the E-Coat 10-1 four coat process. Coils are dipped in a phenolic coating, that provides substantial resistance to corrosion of aluminum. E-Coat coils are rated to exceed 6000 hours salt spray in accordance with ASTM B117 / DIN 53167.]
3. **Low ambient [optional] -** Compressors operate down to -10.0°F and resume cooling at -7.0°F by monitoring the refrigeration system liquid line pressure and adjusting condenser airflow to maintain the proper liquid line pressure to protect unit operation. Airflow adjustment occurs through a combination of VFD driven condenser fans and fan cycling. Liquid line pressure transducers are included to provide the pressure information on the unit control display.
4. **Shut-off valves [optional] -** Discharge and suction [and liquid] shut-off valves are included to provide a means of isolating the refrigerant charge in the system so that the refrigeration system is serviced without removing the charge in the unit.

5. **Sight glasses -** are accessible without having to open air handler section access doors or remove panels.
6. **Fixed speed compressors -** Units use hermetic scroll compressors, piped and charged with oil and R-454B refrigerant. Each compressor is protected from over-temperature and over-current conditions. Compressors are vibration-isolated from the unit, and installed in an easily accessible area of the unit. All compressor-to-pipe connections are brazed to minimize potential for leaks. Units are designed to minimize leaving air temperature changes when varying capacity. Steps of capacity control do not exceed 23% of unit capacity when modulating from minimum to maximum. Hot gas bypass operation are not used to meet maximum modulation steps of control. Variable speed drive compressors are used if unit is not capable of meeting the maximum modulation percentage of step capacity.
7. **Variable speed drive compressors [optional] -** Units use variable speed drive scroll compressor, piped and charged with oil and R-454B refrigerant. Units have a minimum capacity of 15% of full load. Each compressor is protected from over-temperature and over-current conditions. Compressors are vibration-isolated from the unit and installed in an easily accessible area of the unit. All compressor-to-pipe connections are brazed to minimize potential for leaks. Variable speed drive scroll compressors are matched and factory tested with VFD. All high efficiency units deliver Tier 2 efficiency levels of the Consortium of Energy Efficiency (CEE) for 2024.
8. **Modulating hot gas reheat (HGRH) [optional] -** Modulating hot gas reheat (HGRH) is provided. The unit includes a three-way stepper motor driven modulating valve and controller, a HGRH coil mounted downstream of evaporator coil, and all associated refrigerant piping. The unit controller modulates the three-way HGRH valve to control the amount of compressor discharge gas to the HGRH coil. [HGRH coils are protected by the E-Coat 10-1 four coat process. Coils are dipped in a phenolic coating that provides substantial resistance to corrosion of aluminum.]
9. **In-line refrigerant filter driers - replaceable core filter driers -** Liquid line filter driers are provided as a standard on the unit. [The replaceable core filter drier on the unit provides a convenient means for maintaining and optimizing the unit's refrigeration system.]
10. **Compressor sound treatment [optional] -** Compressor sound blankets with heat shield are provided to attenuate radiated sound from the compressors.
11. **Condenser enclosure [optional] -** The condenser section is enclosed by a wire grill [louvered] condenser enclosure on the three exposed sides. Paint finish matches the color and salt spray specifications of the unit exterior.
12. **Condenser enclosure [optional] -** The condenser section is enclosed by a louvered hail guard on the end of the unit. Paint finish on the louvered hail guard matches the color and salt spray specifications of the unit exterior. The unit is enclosed by a wire grill on the other two sides.

Controls

1. **Enclosure -** The unit is factory configured, installed, wired, and tested with a rooftop unit controller housed in a rain and dust-tight, powder painted, steel cabinet behind hinged, latched, and gasket-sealed door. VFD control keypads are located in the control cabinet for accessibility and servicing while the unit is operating.
2. **115 V convenience outlet [optional] -** A duplex ground fault circuit interrupt (GFCI) receptacle is factory-mounted in a weatherproof enclosure and wired for a 15 amp load. Electrical power is de-energized to the convenience outlet when the optional unit mounted disconnect has been turned OFF. Unit powered convenience outlet includes transformer to step down to 115 V.
3. **Basic controls -** Control includes automatic start, stop, operating, and protection sequences across the range of scheduled conditions and transients. The rooftop unit controller provides automatic control of compressor start/stop, energy saver delay and anti-recycle timers, condenser fans, and unit alarms. Automatic reset to normal operation after power failure. Software and user programmed setpoints are stored in nonvolatile memory for the 20-year expected life of the microprocessor. The real time clock (RTC) information is retained for a minimum of 72 hours during power loss with a super capacitor-backed circuit.
4. **Smart equipment control -** The unit is supplied with the Smart Equipment control system. All units are factory commissioned, configured, and run tested.

5. **Standard display -** The display is organic light-emitting diode (OLED) with green characters on black background providing high visibility. The display provides a minimum of 175 characters in total with 35 characters × 5 lines. The OLED display is a 256 × 64 dot matrix. The display shows animated text, symbols, and descriptions down to -40.0°F (-40.0°C) without noticeable degradation of contrast or display response time. Numeric data is provided in imperial or metric units. A sealed, membrane style keypad with 20 keys is used to navigate the unit controller and enter data. Specific keys are individually assigned to allow quick selections. The software on the user interface (UI) is upgradable in the field.
6. **Touchscreen display -** The display is a 10 in. WXGA type color wide screen, thin film transistor (TFT) with multi-touch capacitive touchscreen and 1280 × 800 resolution. The screen details all operations and parameters, using an interactive graphical representation of the rooftop unit and its major components. The display does not require a laptop or mobile display to access complete functionality. The display is able to integrate with multiple controllers connected on a common BACnet MS/TP communication bus. The display shows animated text, symbols, and descriptions down to -4.0°F (-20.0°C) without noticeable degradation of contrast or display response time. Numeric data is provided in imperial or metric units. Specific icons are individually assigned to allow quick selections. The software on the UI is upgradable in the field.
7. The following display information is available based on selected unit options:
    a. Unit summary with serial, model number, language, unit status, and enable. Controller hardware, firmware, and network inputs.
    b. Sensor information for all devices connected to the unit and setpoints.
    c. Commissioning menu has options to view key parameters such as occupancy, cooling, heating, economizer, ERW, morning warm-up, staggered start, safety setup, low ambient and demand control ventilation (DCV).
    d. Start-up, air balancing and commissioning wizards are provided for field technicians to complete unit configuration and start-up.
    e. Occupied, unoccupied, coast modes.
    f. Date, time, time zone, schedules.
    g. Cooling enable, status, setpoints, stages, compressor capacity, runtime, anti-short cycle time, rapid start, and morning cool down.
    h. Heating mode, enable, status, stages, capacity, runtime, setpoints, valve commands, and morning warm-up.
    i. Supply fan, return/exhaust fan status, command, air proving switch, VFD speed, runtime, and faults.
    j. Condenser fan staged and modulating command, low ambient.
    k. Economizer mode, damper position, airflow, fresh air, and alarms for economizer FDD.
    l. HGRH status, enable, humidification setpoint, and valve command.
    m. Energy recovery wheel status, command, output, temperature.
    n. Smoke control modes, sensor inputs, smoke detector shutdown.
    o. Unit alarms such as safety switch, phase monitor, compressor safety, filter differential pressure, carbon dioxide (CO_2), low pressure, high pressure, freezestat, UV light, and condensate switch. Alarms mode with up to 5 current alarms and 25 previous alarms with time and date.
8. **Adjustable height display interface -** The display interface has up to three different installation height locations within the panel. The display interface is installed on a hinge with a magnetic catch to allow wiring and electronics behind the display to be easily accessed.
9. **Diagnostics -** Upon start-up of the unit controller, the unit controller runs through a self-diagnostic check to verify proper operation and sequence loading. The rooftop unit controller continually monitors all input and output points to maintain proper operation. The unit continues to operate in a trouble mode or shut down as necessary to prevent an unsafe condition for the building occupants or to prevent damage to the equipment. In the event of a unit shutdown or alarm, the operating conditions, date, and time are stored in the shutdown history to facilitate service and troubleshooting. A minimum of 10 error histories are recorded.
10. **Compressor capacity modulation -** Staging of the multiple fixed [fixed and variable] speed compressors is used to control refrigeration capacity. Upon entering cooling mode from other modes, the unit controller estimates the cooling requirement and match it closely to the capacity in order to reduce the time required to satisfy the cooling requirements. After the initial calculation, the unit controller adds or reduces stage as necessary to establish a balance between the unit capacity and the space cooling load.

11. **Demand control ventilation (DCV) [optional] -** CO_2 sensors are connected to the unit. The unit controller resets the minimum OA damper position based on demand. DCV remains active as long as the unit is in occupied mode. The unit controller has built-in sequences for indoor CO_2 and comparative CO_2.
12. **Smoke detector [optional] -** The supply [return] [supply and return] smoke detector[s] are provided with the unit. A smoke detector ON signal shuts down the unit.
13. **Refrigerant detection system (RDS) -** The RDS has integrated sensors providing R-454B leak detection. The RDS is connected into unit controls and automatically starts a sequence to dilute refrigerant gas as well as raising an alarm when it senses the presence of refrigerant in the cabinet, indicating a leak equal to 25% of the Lower Flammability Limit. The RDS contains factory- or field-installed sensors that are located to ensure accurate and timely sensing of a leak.
14. **Safety inputs -** The smoke control sequences are available through BAS communication [and hard wired connections using the Customer Terminal board]. The smoke control sequences are used whenever the smoke control binary inputs are turned ON to remove, exhaust, or ventilate smoke, fumes, or other airborne contaminates from the occupied space. The unit controller includes purge, pressurization, and evacuation sequences of operation. Smoke control inputs are assigned a higher priority over smoke detector and override a smoke detector shutdown. These sequences and functions are not part of a listed UL 864 UUKL Smoke Control System.
15. **Twinning -** enables control of multiple rooftop units sharing a common supply duct. The control algorithm provides independent logic unit control. The independent logic unit control system does not require switchover of primary logic in case of rooftop unit failure. The unit provides quick response time, shared data for sensors and setpoints, simple configuration, and easy setup. Twinning configuration is available through BACnet. No additional control or communication wiring is required.
16. The control is installed and tested at the factory where the equipment is assembled. The following controls features are available based on selected unit options:
    a. The control uses a wall sensor that has a means of overriding the unoccupied mode for a programmable amount of time.
    b. The unoccupied override time is programmed in minutes up to 4 hours.
    c. The control has a SA sensor as standard.
    d. The control has a RA sensor as standard.
    e. The control has an outside air temperature (OAT) sensor as standard.
    f. The control uses the RA sensor in place of the space sensor if the space sensor fails for any reason.
    g. The control has a 365 day real time clock (RTC).
    h. The control has an occupancy schedule that allows two different occupied schedules per day for each of the seven days of the week individually.
    i. The control has 20 holiday schedules, each capable of 99 days. The control's holiday schedules has a start time associated with each schedule.
    j. The control is capable of operating the economizer using dry bulb, single enthalpy, or dual enthalpy.
    k. The control has the ability to do demand ventilation control using two CO_2 sensors.
    l. The control has a programmable maximum OA damper position for IAQ operation.
    m. The control has the ability to temper the ventilation air during times when heating or cooling is not required.
    n. The control must be able to lock out cooling below a programmable OAT setpoint.
    o. The control is able to lock out heating above a programmable OAT setpoint.
    p. The control is able to do a pre-occupancy purge.
    q. The control has purge, de-pressurization, and pressurization sequence for smoke control.
    r. The control has the ability to read pressure across two filter banks.
    s. The control has the capability of reading a fan proving switch.
    t. The control has an intelligent recovery function that brings the space to the occupied setpoint just before or at the beginning of the first occupied schedule each day. The control learns and applies the minimum runtime required to heat or cool the space to setpoint for the first occupied period of a day. The unit controller is capable of compensating start time based on OAT.
    u. The control satisfies cooling demand using unequal compressor staging combinations.
    v. The control has the ability to enable the ERW.

w. The control monitors the outside airflow.

x. The control provides a low ambient cooling operation.

y. The control enables either dehumidification, humidification, or temperature cool-only mode based on humidity setpoint.

z. The control has compressor safety monitoring.

aa. The control has FDD alarm monitoring for economizer operation.

ab. The control provides UV light replacement warning after certain runtime hours.

ac. The control monitors exhaust air and return airflow.

ad. The control provides rapid start.

ae. The control is capable of initiating warm-up and cool down mode using either occupied command, optimal start or schedule.

af. The control provides staggered start and coast mode.

ag. The control provides condensate overflow switch alarm.

BAS communications

The unit has BACnet MS/TP, Modbus, N2 [BACnet IP] communication protocol integral to the unit controller. The controller is certified for BACnet communication protocol by BACnet Testing Laboratory (BTL) to meet the requirements of the advanced application control profile. A control points list, BACnet interoperability building blocks (BIBBs), and protocol implementation conformance statement (PICS) is provided by the manufacturer to facilitate communications programming with the BAS.

1. **Mobile access portal (MAP) gateway [optional] -** MAP Gateway is a pocket-sized web server that provides a wireless mobile UI to Smart Equipment and manufacturer branded system controllers and thermostats. MAP Gateway provides multi-client connectivity, browser-based interface, WiFi connectivity, browser-based remote building management, permanent audit log, portable size and mobility, and an easy-to-use intuitive UI.
2. **Customer terminal board [optional]** - A customer terminal board is supplied on all units requiring twinning, smoke control sequences, humidifier, or hydronic heat.
3. **Verasys and Metasys Communication interface -** The BAS provides the owner with a simple user experience and configurable controls. The system consists of Smart Equipment technology and is able to interface with Verasys and Metasys systems.

Installation

General

The installing contractor installs the rooftop unit(s), including components and controls required for operation, in accordance with the rooftop unit manufacturer's written instructions and recommendations. Rooftop units are installed as specified.

- All size or shape equipment including electrical components, especially those not built with weatherproof enclosures, VFDs, and end devices are effectively covered for protection against rain, snow, wind, dirt, sun fading, road salt/ chemicals, rust, and corrosion during shipping cycle. Equipment remains clean and dry.

Location

Locate the rooftop unit as indicated on drawings, including cleaning and service maintenance clearance per manufacturer instructions. Adjust and level the rooftop unit on the support structure.

Inspection and start-up supervision - optional

A factory-trained service representative of the manufacturer supervises the unit start-up and application specific calibration of control components.